Portfolio Tracker

User manual

Master G2C · Portfolio tracking and analysis tool

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1. Getting started

This chapter explains what Portfolio Tracker is, how to install it on Windows or Mac, how to start and close it, what works without Internet, how to update or uninstall it, and how to use it online or from its source code. Start here if you are new to the software.

What is Portfolio Tracker and who is it for?

Portfolio Tracker is a tool for tracking and analysing a securities portfolio, built by a student of the Master G2C (Asset Management, Risk Control and Compliance) at IAE Caen (University of Caen Normandy), as part of the programme. It reads a portfolio's transaction history (purchases, sales, dividends), values it at market prices and calculates the indicators used by investment professionals.

What it does

- **Tracking**: cost basis per unit (PRU), unrealised and realised gains, dividends, fees, day-by-day value, foreign-currency holdings converted into euros.
- **Performance and risk**: TWR (time-weighted return), IRR (TRI), comparison with a benchmark index, volatility, drawdown, Sharpe and Sortino ratios, beta, alpha, VaR and CVaR.
- **Advanced analyses**: look-through exposures, Markowitz optimisation, Monte Carlo projection.
- **Wealth advice**: comparative taxation of the CTO (securities account), PEA (French equity savings plan) and assurance-vie (life insurance), stress tests.
- **Asset management**: performance attribution, risk budget, strategy backtesting.
- **Reporting**: interactive dashboard in French or English, and a PDF summary report.

Who it is for

It was designed first for the students and teachers of the Master G2C at IAE Caen: it applies, on a real portfolio, the three strands of the programme: **asset management** (performance measurement and attribution, allocation, Markowitz optimisation, strategy backtesting), **risk control** (volatility, historical, normal and Cornish-Fisher VaR, CVaR, stress tests, risk budget, exposure diagnosis) and **compliance** (checks on the transactions entered, traceability of sources, encrypted data and personal data protection). It can also be used by any individual who wants to understand the performance and risk of their portfolio.

What it does not do

- It does not connect to your bank or your broker: you give it your transactions as a file (CSV, Excel or PDF) or by manual entry.
- It does not place any stock market orders.
- Its results do not constitute investment advice: it is an educational tool, as the footer of every screen reminds you.

What computer do I need to use the software?

The software comes in two installable versions and one online version.

System	Version to use	Conditions
Windows	.exe installer	64-bit Windows (Windows 10 or 11)
Recent Mac	.dmg disk image	Mac with an Apple Silicon chip (M1, M2, M3, M4…) and macOS 12 or later
Intel Mac (before late 2020)	Online version	The Mac application is not designed for these models
Linux or other	Source code	Python installed (see the entry on launching from the source code)

What you do not need to install

The Windows and Mac versions contain their own Python and all the necessary libraries: you have nothing else to install. No administrator password is requested on Windows.

The browser

The dashboard is displayed in a browser window. On Windows, it uses Microsoft Edge (present on Windows 10 and 11) or, failing that, Google Chrome. On Mac, it uses Google Chrome, Microsoft Edge or Brave if installed, otherwise your default browser (usually Safari).

Internet

A connection is only needed to download the software and to obtain the most recent prices. Everything else works offline (see the entry “Using the software without Internet”).

Where do I download the software?

The installers are published on the releases page (“Releases”) of the project’s GitHub repository:

https://github.com/DieuUssop/Python/releases/latest

This address always opens the most recent version. In the “Assets” section of the page, choose the file that matches your computer:

File	For
Installer_Portfolio_Tracker.exe	Windows
Portfolio_Tracker_Mac.dmg	Mac with Apple Silicon

Do not download the “Source code” archives offered by GitHub on the same page: they contain the project’s code, which is only useful for launching it as a developer (see the entry on the source code).

Version number

Each version carries a number made up of its build date (for example “2026.10.05”). The installers are built automatically by GitHub, on Windows and Mac machines, and then published together in the same release.

The software does not update itself

The software does not check by itself whether a new version exists. To get one, come back to this page and install the new version over the old one: your accounts are kept (see the entry “Updating the software”).

Installing the software on Windows

The steps

1. Download Installer_Portfolio_Tracker.exe from the project’s releases page.
2. Double-click the file. If Windows displays “Windows protected your PC”, follow the entry “Windows or Edge blocks the installer”.
3. The installation wizard opens (in French or English). Click Next.
4. If you wish, leave ticked the box that creates a Desktop icon.
5. Click Install, then Finish. The last page offers to launch Portfolio Tracker straight away.

An installation for you alone

The installation is done for the current Windows user, without administrator rights. The software is placed in your personal folder:

C:\Users\<your name>\AppData\Local\Programs\Portfolio Tracker\

If several people use the same computer with different Windows sessions, each must install it in their own session. Conversely, several people can share the same Windows session: each then creates their own encrypted account in the software.

What the installation creates

- a “Portfolio Tracker” shortcut in the Start menu (and on the Desktop if the box was ticked);
- an uninstall shortcut in the same Start menu group;
- the program folder, which contains its own Python, the software’s code and the local securities database (prices usable offline).

Next

Launch the software with the “Portfolio Tracker” icon. A small black window opens, then the dashboard in its own window: see the entry “Launching the software”.

Windows or Edge blocks the installer: what should I do?

The installer is not signed with a publisher certificate (such a certificate costs money every year). Windows and Microsoft Edge therefore display warnings, which are normal for this type of software.

When downloading in Microsoft Edge

Edge may report that the file is not commonly downloaded and block it. Open the downloads list, click the “…” menu next to the file, then choose to keep it (“Keep”, then possibly “Keep anyway”). The exact wording depends on the version of Edge.

On launch: “Windows protected your PC”

This SmartScreen message appears when you double-click the installer.

1. Click the **More info** link.
2. The program name and the line “Publisher: Unknown publisher” appear.
3. Click the **Run anyway** button.

The installation then continues normally.

If your antivirus blocks the file

The cause is the same: the program is not signed. If you downloaded it from the project’s official releases page, you can allow it in your antivirus. If in doubt, ask your teacher or your IT department for advice.

Good habit

Always download the installer from the project’s releases page (https://github.com/DieuUssop/Python/releases/latest) or from the link provided by your teacher, never from a third-party site.

Installing the software on Mac

The steps

1. Download Portfolio_Tracker_Mac.dmg from the project’s releases page.
2. Double-click the .dmg file: a window opens with the “Portfolio Tracker” application, a shortcut to the “Applications” folder and a LISEZ-MOI.txt file (the French “read me” file).
3. Drag “Portfolio Tracker” onto the “Applications” folder.
4. Open the Applications folder and double-click “Portfolio Tracker”.

The very first time, macOS refuses to open the application because it does not come from the App Store: follow the entry “macOS refuses to open the application”. This step only has to be done once.

Conditions

- Mac with an Apple Silicon chip (M1, M2, M3, M4…);
- macOS 12 or later.

Macs with an Intel processor (sold before late 2020) are not supported by this application: use the online version of the dashboard.

What happens at launch

A Terminal window opens (it is the equivalent of the black window on Windows), then the dashboard, in Chrome or Edge in “application” mode if they are installed, otherwise in your default browser. Leave the Terminal window open while you are using the software.

Where your data is

Your encrypted accounts and portfolios are stored in:

~/Library/Application Support/Portfolio Tracker

They are kept when you install a new version.

macOS refuses to open the application on first launch

The application is not distributed through the App Store and is not signed by Apple. At first launch, macOS therefore prevents it from opening. How to allow it depends on your version of macOS.

macOS 15 (Sequoia) and later

1. At the blocking message, click “Done”.
2. Open System Settings, then Privacy & Security.
3. Scroll down to the message about “Portfolio Tracker”.
4. Click “Open Anyway”, then confirm (macOS may ask for your login password).

macOS 12 to 14

1. In the Applications folder, right-click (or Ctrl-click) “Portfolio Tracker”.
2. Choose “Open”.
3. In the window that appears, click “Open” again.

Only once

Once the application has been allowed, it then opens with a simple double-click. You may need to do this again after installing a new version.

The Terminal opens: this is normal

Launching opens a Terminal window that runs the dashboard. Do not close it while you are using the software: closing it stops the software.

Launching the software: the black window and the dashboard window

What happens when you click the icon

1. **A black window opens** (the Terminal on Mac). It shows the software's name, the dashboard address and this message: leave this window open while you are using the software, close it to quit the application. It is the software's "engine".
2. **The dashboard opens in its own window**, as soon as it is ready (allow a few seconds). It is a Microsoft Edge or Google Chrome window in "application" mode: with no address bar or tabs, it looks like a conventional piece of software. If neither of these browsers is found, the dashboard opens in your usual browser.

Why a browser?

The software is a web application that runs on your own computer. Its address begins with http://localhost: followed by a number (usually 8501, or the next free one up to 8510). "localhost" refers to your computer itself: nothing goes through the Internet to display the dashboard, and nobody else on the network (the school Wi-Fi, for example) can connect to it.

What you see first

The dashboard opens on the "Portfolio analysis" space, with the first portfolio in the list. You can immediately choose another one, upload your own file or sign in to your personal space (see the chapter "The screen and navigation").

If you click the icon a second time

If the software is already running, it is not started again: the dashboard window is simply reopened. This is useful if you closed the dashboard window by mistake, as long as the black window is still open.

The dashboard does not open

The black window is open but no window appears

The first start can take a little while. If nothing appears, open your browser yourself and type the address shown in the black window, for example http://localhost:8501. The software waits up to three minutes for the dashboard to be ready before opening the window; after that, it keeps running, but it is up to you to open the address.

Message "No free port between 8501 and 8510"

The software uses the first free number between 8501 and 8510. If all of them are taken by other programs, it cannot start. Close another application (for example another Streamlit application), then relaunch Portfolio Tracker.

Message "The dashboard could not start"

A technical error message is displayed just above. Press Enter to close the window, then relaunch. If the error persists, note down or photograph this message and pass it on to your teacher or to the software's creator.

The window opens but stays blank or shows a connection error

Check that the black window is still open: if it has been closed, the software has stopped. Relaunch it with the icon.

The dashboard shows "Unable to analyse the portfolio"

The software is working, but the chosen portfolio could not be read or valued. Choose another portfolio or see the chapter on importing files.

Closing the software

On Windows

Close the **black window**: it is what runs the software. Closing only the dashboard window does not stop it; you can in fact reopen it by clicking the "Portfolio Tracker" icon again.

On Mac

Close the **Terminal window** opened at launch. macOS asks whether you want to end the running process: click "Close". The dashboard stops with it.

From the source code

If you launched the software with the command `python -m streamlit run app.py`, press Ctrl + C in the terminal.

Is your data saved?

- Portfolios saved in your personal space are written to disk, encrypted, when you save or modify them: nothing is lost on closing.
- By contrast, a file simply uploaded in the sidebar, without being saved to your space, is not kept: you will have to upload it again the next time you open the software.
- Your session (signed-in account, language, screen settings) ends when you close it. The light or dark mode chosen while you were signed in is restored the next time you sign in.

Without closing the software, you are in any case signed out of your account automatically after 30 minutes of inactivity.

Does my data leave my computer?

With the installed version (Windows or Mac), everything stays on your computer.

The dashboard runs locally

The software runs at the address localhost, that is, on your computer itself. It accepts no connection from another device: nobody on the same network can open your dashboard.

Your portfolios are encrypted

Portfolios saved in your personal space are encrypted with a key derived from your password. Other users of the same computer cannot read them. The trade-off: a forgotten password makes the portfolios permanently unreadable (see the chapter on accounts).

What is sent over the Internet

To obtain prices, the software queries Yahoo Finance with the **security codes** (tickers) and, to recognise an unknown security during an import, with its ISIN code or its name. Quantities, amounts and the composition of your portfolio are never sent.

Questions asked to the assistant

The manual's assistant works without Internet. Questions left unanswered are recorded in a file on your computer only; they are passed on to nobody, unless you export that file yourself.

The online version

On the online version, the software runs on a remote server: the files you upload there are processed on that server. For real personal data, prefer the installed version.

Using the software without Internet

The software is designed to work without Internet. It comes with a **local securities database**: the record of several thousand shares, ETFs, indices and exchange rates, with their daily closing prices (since 2015 for securities, since 2007 for indices and exchange rates).

What works offline

- opening the dashboard, signing in to your account, opening your portfolios;
- analysing a portfolio whose securities are in the database or in the cache;
- importing a CSV or Excel file, recognising ISIN codes or names already known to the database or to the securities memory;
- converting currencies, comparing with a benchmark index, and using the other spaces;
- the world map, if its base map has already been saved;
- the assistant and the manual.

How to tell that the prices are not from today

- In the banner at the top of the page, the badge reads “Cached prices (offline)” with an orange dot, instead of “Live prices · Yahoo Finance”.
- At the bottom of the sidebar, the “Prices” line reads “local cache of” followed by the date of the latest known prices, and the note “Yahoo Finance unreachable”.

Offline, prices therefore stop at the latest date known to the database or the cache. The calculations remain correct, but as of that date.

When Internet comes back

Prices are completed automatically at the next analysis. Since results are kept in memory for one hour, use the **Refresh prices** button to force an immediate download.

Limitation

A security that is absent from the local database and has never been encountered before has no price offline: the software flags it.

What requires an Internet connection

Most functions work offline. Internet is still needed in the following cases.

Situation	Why
Downloading the installer or a new version	The files are on the project’s releases page
Getting today’s prices	They come from Yahoo Finance
Analysing a security that is absent from the local database	Its history has to be downloaded the first time (it is then added to the database)
Recognising, at import, an unknown ISIN code or company name	The search is made with Yahoo Finance (the result is then remembered)
Refresh prices button	It downloads the prices again
Displaying the world map if its base map has never been saved	The country outlines are downloaded once, then kept on the computer
Launching the software from the source code for the first time	The Python libraries have to be installed

The database learns as it goes

Each time the software downloads prices or recognises a security, it adds them to its local database or its memory. The next time, this information is available even without Internet.

If Yahoo Finance does not respond

The software does not crash: it uses the latest saved prices and says so in the banner (“Cached prices (offline)”) and at the bottom of the sidebar. If it has no price at all for a security, a clear message flags it.

What never needs Internet

Reading your files, the calculations, your encrypted accounts, the PDF report and the manual's assistant all work entirely on your computer.

Updating the software

The software does not update automatically. To move to a new version, download it from the project's releases page and install it over the old one. **Your accounts and saved portfolios are kept.** There is no need to uninstall first.

On Windows

1. Close the software (close the black window).
2. Download the new Installer_Portfolio_Tracker.exe.
3. Launch it and follow the same steps as the first installation.

The installer completely replaces the embedded Python and the software's code. The accounts folder (data\comptes) is never touched by an update.

On Mac

1. Close the software (close the Terminal window).
2. Open the new Portfolio_Tracker_Mac.dmg.
3. Drag "Portfolio Tracker" onto "Applications" and agree to replace the old application.

At the next launch, the application detects the new version and replaces its code and its securities database, keeping your accounts and the memory of recognised securities, stored in ~/Library/Application Support/Portfolio Tracker. You may need to allow the application again (see the entry "macOS refuses to open the application").

Precaution

Even though updates keep accounts, keep a copy of your important portfolios: the "My account" page lets you download each of them as a CSV file.

Where is my data saved?

On Windows

In the program folder:

C:\Users\<your name>\AppData\Local\Programs\Portfolio Tracker\

- data\comptes\: the encrypted accounts and portfolios of each user of this computer;
- data\base\: the local securities (prices) database and the memory of recognised securities.

The AppData folder is hidden by default in File Explorer: type the path into the address bar to reach it.

On Mac

~/Library/Application Support/Portfolio Tracker

In the Finder, open the Go menu, then “Go to Folder…”, and paste this path.

From the source code

In the project’s data folder (data/comptes and data/base).

The files are encrypted

The portfolios and the list of their names are encrypted, and account folders have random names. The files can only be read through the software, with the right password.

Changing computer or making a backup

The simplest and safest method: sign in, open the “My account” page and download each of your portfolios as CSV. On the new computer, install the software, create an account and add these files. Copying the accounts folder directly from one computer to another is not an operation the software supports.

Uninstalling the software

On Windows

1. Open Windows Settings, then Apps, and choose “Portfolio Tracker”, then Uninstall. You can also use the uninstall shortcut in the Start menu.
2. At the end, if accounts exist, a question is asked: “Also delete the saved accounts and portfolios?”.
   - **No** (the default choice): the accounts and portfolios stay on the computer and will be found again if you reinstall Portfolio Tracker.
   - **Yes**: the software’s folder is erased entirely, accounts included. This deletion is permanent.

On Mac

1. Move “Portfolio Tracker” (in the Applications folder) to the Bin.
2. To also erase the accounts and portfolios, delete the folder ~/Library/Application Support/Portfolio Tracker (Finder, Go menu, then “Go to Folder…”).

If you delete only the application, your data stays on the Mac and will be found again after a reinstallation.

Deleting a single account without uninstalling

Sign in, open the “My account” page and use the “Delete my account” section: the account and all its portfolios are erased permanently (right to erasure), without touching other users.

Before erasing everything

Download a copy of your portfolios from the “My account” page if you think you might need them one day.

The online version and its limits

The dashboard can also be published on Streamlit Community Cloud, a free hosting service. The address of this version is given to you by the software’s creator or by your teacher.

Who it is for

- users of Intel Macs, for whom the Mac application is not designed;
- those who want to try the software without installing anything;
- users of a computer on which they cannot install programs.

Its limits

- **Accounts are temporary.** The online service’s disk is wiped each time the site restarts: saved accounts and portfolios may disappear. A warning reminds you of this in the sign-in form: “Online demo version”. Always keep a copy of your files.
- **Internet is essential**, since the software runs on a remote server.
- **Your files are processed on that server**, not on your computer. For real personal data, prefer the installed application.
- The site may be slower, especially if it was asleep and has to restart.

What is identical

The screens, the calculations, the PDF report and the manual are the same as in the installed version.

Launching the software from the source code

This method is aimed at students and developers who work on the project itself. It requires Python on the computer (the project specifies Python 3.11 or later).

On Windows: the lancer_tableau_de_bord.bat file

Double-click lancer_tableau_de_bord.bat, at the root of the project.

- **The first time**, it installs the necessary libraries (the requirements.txt file), then the character recognition for image PDFs (the requirements-ocr.txt file). Internet is needed at that point. It then creates a small .installe file so as not to repeat this step.
- It then launches the same launcher as the installed version: the dashboard opens in its own window, at the address http://localhost:8501 (or the next free port).
- To quit, close the black window.

In a terminal (Windows, Mac, Linux)

From the project folder:

```
python -m pip install -r requirements.txt       (une seule fois)
python -m pip install -r requirements-ocr.txt   (facultatif : PDF image)
python -m streamlit run app.py                  (tableau de bord)
```

The dashboard opens in the browser, at the address http://localhost:8501. To stop it: Ctrl + C in the terminal.

Other useful commands

- python main.py: command-line analysis and PDF report;
- python construire_base_titres.py --mise-a-jour: adds the latest prices to the local database;
- python -m pytest: runs the automated tests.

In case of the error "No module named streamlit"

The libraries are not installed for the Python being used: run the installation command python -m pip install -r requirements.txt again.

2. The screen and navigation

This chapter describes the software's screen: the sidebar block by block, the choice of language and night mode, the four workspaces, the choice of portfolio and analysis settings (benchmark index, risk-free rate, VaR level), the banner and key figures at the top of the page, the PDF report, refreshing prices, the manual's assistant and chart tips. Read it after the getting-started chapter to find out where each function is.

How is the screen organised?

The screen is split into two parts: the **sidebar**, on the left, which groups all the settings, and the **main area**, on the right, which displays the results.

The sidebar, from top to bottom

1. **The header**: the "PT" monogram, the name "Portfolio Tracker" and the label "Master G2C". Just below, on a single line, two small selectors: language (FR · EN) on the left and display (Light · Dark) on the right; the active option is in bold.
2. **The personal space**: the label "Encrypted personal space" and the **Sign in** link. Once signed in, your initials, your username and the **My account** and **Sign out** links.
3. **Workspace**: the menu of the four workspaces (Portfolio analysis, Wealth advisory, Asset management, Manual and help), each with a grey line of description. The workspace on screen is highlighted in light blue, with a blue bar on the left.
4. **Data**: the list of portfolios, the **Upload a file (CSV, Excel, PDF or image)** area (**Browse** button, or drag and drop the file), then the **Add transactions** and **File template** links.
5. **Settings**: a collapsed panel that contains the benchmark index and the risk-free rate. Its title summarises the current settings. (The VaR confidence level is set in the Risk tab, the only place where it is used.)
6. **Information** about the portfolio being analysed: file, period, number of transactions, source of prices, and the exchange rates folded under "Exchange rates".
7. **The buttons PDF report** and **Refresh prices**.

Blocks 6 and 7 only appear once a portfolio has been analysed: they are absent in the "Manual and help" space and on the "My account" and "Add transactions" pages.

At the bottom right of the screen, on every page, the **Help** button opens a small panel to ask the manual a question (see the article on the help bubble).

The main area

In the analysis spaces, it begins with a blue banner (title, period, benchmark index, source of prices), followed by a row of six key figures, then the tabs of the chosen space. A footer recalls the sources (Yahoo Finance for prices, ECB for the risk-free rate) and that the tool is educational: it does not constitute investment advice.

If the sidebar is hidden

On a narrow screen, the sidebar may be collapsed. The small arrow-shaped button at the top left of the window lets you reopen it.

Switching the software to English (or back to French)

The language selector is at the top of the sidebar, under the software's name, on the left: click **EN** for English, **FR** for French. The change is immediate, without losing the current portfolio or settings. French is the starting language.

What changes

- all the labels, buttons, tabs, help texts and comments of the dashboard;
- data produced by the calculations (regions, sectors, asset classes, scenario names), translated on display;
- the format of numbers and dates: in English, "€1,235", "+12.34%" and "16 Jan 2017"; in French, "1 235 €", "+12,34 %" and "16/01/2017".

What stays in French

- **the PDF report**, always written in French, whatever the screen language;
- the names of securities and of your portfolios, which are proper names;
- the manual, as long as its English version is not installed with your version of the software: the assistant and the contents then use the French manual;
- the command-line analysis and the guides in the docs folder.

An on-screen text that does not yet have a translation is displayed in French, without an error.

How long the choice lasts

The language is kept for the whole session. It is not saved in your account: the next time you open the software, it restarts in French. Clicking a second time on the language already chosen changes nothing.

Turning on night mode (dark background)

The display selector is at the top of the sidebar, under the software's name, on the right (on the same line as the language): click **Dark** for a midnight-blue background, **Light** to return to the white background. Light mode is the starting mode.

What changes

- the background of the page and the sidebar, the figure cards, the boxes, the fields and the menus;
- the charts: texts, grids, hover bubbles and dark curve colours are replaced by shades that are legible on a dark background;
- the data tables, whose colours are inverted so they stay legible.

The choice is remembered in your account

If you are signed in to your personal space, the chosen mode is saved in your preferences, encrypted along with the rest of your account. The next time you sign in, the software restores it automatically. Without an account, the choice only applies to the current session.

For the mode to be saved, choose it **after** signing in.

The PDF report stays light

The PDF report always keeps a white background, whatever mode is chosen on screen: it is designed to be printed.

The four workspaces and what they contain

The **Workspace** menu, in the sidebar, offers four workspaces. Click one to display it; its tabs appear at the top of the main area.

Workspace	Description in the menu	Tabs
Portfolio analysis	Performance, risk, optimisation	Overview, Holdings, Performance, Risk, Exposures, Optimisation, Projection, Transactions
Wealth advisory	Taxation, stress tests	Taxation, Stress tests
Asset management	Attribution, risk budget, backtest	Performance attribution, Risk budget, Strategy backtest
Manual and help	User guide, formulas, questions	“Ask a question” assistant, manual contents

What they have in common

The first three workspaces analyse the same portfolio with the same settings. They all display the blue banner and the six key figures at the top of the page. The software opens on “Portfolio analysis”.

The “Manual and help” workspace opens even if no portfolio could be loaded: you can always look for help there.

Returning to a workspace from “My account” or “Add transactions”

When the **My account** page or the **Add transactions** page is open, no workspace is ticked in the menu. A click on **any workspace**, including the one you were in, closes the page and displays the chosen workspace. You can also:

- click **My account** again to close that page;
- use the **Back to the dashboard** button on the “Add transactions” page.

Choosing the portfolio to analyse

The portfolio to analyse is chosen in the **Data** block of the sidebar, with the drop-down list placed under that title. The software immediately analyses the first portfolio in the list.

What the list contains

1. **Your personal portfolios**, if you are signed in. They appear first, in alphabetical order, preceded by the label “My space ·” (for example “My space · PEA Boursorama”). They are decrypted in memory at the time of analysis.
2. **The sample portfolios** supplied with the software, to discover the functions without your own data: “Diversified portfolio (multi-asset)”, offered first, and “Global equity portfolio”. The small “Sample portfolio” is only offered if it is the only sample file present.

An uploaded file takes priority

If a file is present in the **Upload a file (CSV, Excel, PDF or image)** area, **it** is the one analysed, whatever the choice in the list. To return to a portfolio from the list, remove the uploaded file by clicking the cross next to its name.

My portfolio is not in the list

- Check that you are signed in: without signing in, only the samples are offered.
- A file uploaded without being saved does not appear in the list. To find it next time, save it to your space with the **Save** button that appears under the upload area once the file has been read.

To go further

Uploading a file, the import wizard and adding new transactions are explained in the chapter on importing. Managing your saved portfolios (renaming, downloading, deleting) is done on the **My account** page.

Where do I set the analysis settings?

The analysis settings are in the **Settings** panel of the sidebar, under the Data block. It is collapsed by default: click its title to open it.

The summary in the title

Even when collapsed, the panel’s title recalls the current settings, for example:

Settings · MSCI World · 2.50%

that is, the benchmark index and the annual risk-free rate.

The two settings

Setting	Starting value	Used for
Benchmark index	MSCI World (CW8 ETF, dividends reinvested)	Comparing performance, calculating beta, alpha and tracking error
Risk-free rate (% per year)	2.50%	Sharpe and Sortino ratios, Jensen's alpha

Each has its own detailed entry in this chapter. The VaR confidence level (95% at the start) is not in this panel: it is set at the top of the Risk tab, where the VaR is shown (see the dedicated entry).

Effect of a change

Any change reruns the analysis: the key figures, the tabs of the three analysis workspaces and the PDF report use the new settings. A PDF report already prepared with the old settings is no longer offered for download: it has to be prepared again.

How long the settings last

The settings are kept during the session, even if you change portfolio or workspace. They are not saved in your account: the next time you open the software, the starting values return.

Choosing the benchmark index: the 14 indices offered

The benchmark index serves as a point of comparison: "portfolio versus index" curve in base 100, performance gap, beta, alpha, correlation, tracking error, information ratio. It is chosen in the **Settings** panel, in the **Benchmark index** menu. Each entry in the menu shows its family first, then its name (for example "Equities · CAC 40 (price index, excl. dividends)").

The full list

Family	Index offered	What it represents
Equities	MSCI World (CW8 ETF, dividends reinvested)	Large and mid caps of developed countries. **Default choice**
Equities	MSCI ACWI, world including emerging markets	Developed and emerging countries, dividends reinvested
Equities	S&P 500 (ESE ETF, dividends reinvested)	500 large US companies
Equities	Nasdaq-100	100 large non-financial Nasdaq companies, very technology-heavy, dividends reinvested

Family	Index offered	What it represents
Equities	Stoxx Europe 600	600 European companies, dividends reinvested
Equities	Euro Stoxx 50 (price index, excl. dividends)	50 large eurozone companies
Equities	CAC 40 (price index, excl. dividends)	40 large French companies
Equities	MSCI Emerging Markets	Emerging-country equities, dividends reinvested
Bonds	Euro area government bonds (coupons reinvested)	Debt of eurozone governments
Bonds	Euro corporate bonds (excluding coupons)	Corporate debt issued in euros
Money market	Money market €STR (interest reinvested)	Overnight investment at the ECB's €STR rate
Multi-asset	Conservative mix: 20% world equities / 80% € bonds	Composite index calculated by the software
Multi-asset	Balanced mix: 60% world equities / 40% € bonds	Composite index calculated by the software
Multi-asset	Growth mix: 80% world equities / 20% € bonds	Composite index calculated by the software

How the index is tracked

Most indices are tracked by means of a listed ETF that replicates them. For each one, several codes are provided: the first one whose history is available is used (for example CW8.PA, otherwise IWDA.AS for the MSCI World).

Dividends reinvested or excluding dividends

An accumulating ETF reinvests dividends: its performance is "including dividends". The Euro Stoxx 50 and the CAC 40 are price indices, **excluding dividends**: they put the index at a disadvantage of about 3% a year against a portfolio that does collect its dividends. Prefer a dividends-reinvested index for a fair comparison.

Which index should you choose?

Choose the index that most resembles your portfolio: MSCI World for world equities, CAC 40 or Stoxx Europe 600 for French or European equities, a multi-asset index for a portfolio that mixes equities and bonds.

With a bond or money-market index, the Performance tab displays a warning: beta, alpha and correlation measure sensitivity to an equity market and then make little sense. Mainly compare returns and volatilities.

Limitation

The list is fixed: it is not possible to add another index from the screen.

What are the 20/80, 60/40 and 80/20 multi-asset indices?

The three indices of the "Multi-asset" family do not exist on the stock market: the software **calculates** them from two components.

Index	World equities	€ government bonds	Short label
Conservative mix	20%	80%	Mix 20/80
Balanced mix	60%	40%	Mix 60/40
Growth mix	80%	20%	Mix 80/20

The components

- **World equities**: the MSCI World, tracked by the first available ETF among CW8.PA, IWDA.AS and EUNL.DE;
- **Bonds**: eurozone government bonds, tracked by DBXN.DE (coupons reinvested), otherwise EUNH.DE.

The calculation

The index starts at 100. Each day, its value follows that of the two components, each in proportion to the units held. At the **close of the last day of each month**, the units are recalculated to return exactly to the target weights (for example 60% and 40%): this is the monthly rebalancing.

Example: a 60/40 index is worth 100 at the start of a month. If equities rise by 10% and bonds do not move, it is worth $60 \times 1.10 + 40 = 106$. Equities then weigh $66 / 106 \approx 62.3\%$. At month end, the index returns to 60% equities (63.60) and 40% bonds (42.40).

Why use them

A portfolio that mixes equities and bonds, compared with the MSCI World alone, always looks less profitable in rising markets and less risky in falling ones. A multi-asset index with the same mix gives a fairer comparison. The three mixes correspond to the conservative, balanced and growth profiles.

Limitation

The rebalancing is assumed to be free and perfect, with no fees or tax.

Which risk-free rate should I use?

The risk-free rate is the return on a euro investment regarded as risk-free. It is set in the **Settings** panel, in the **Risk-free rate (% per year)** field.

Default value and source

The starting value is **2.50% per year**: the European Central Bank (ECB) deposit facility rate, in force since 16 Sep 2026, as the field's help text recalls. The €STR, the overnight interbank rate, is very close to it.

What it affects

- the **Sharpe ratio** and the **Sortino ratio**, which measure the return obtained above this rate;
- **Jensen's alpha**;
- the calculations in the Asset management workspace and the PDF report that depend on it.

For daily calculations, the annual rate is converted into a rate per trading day:

daily rate = (1 + annual rate)^(1/252) − 1

With 2.50%: 1.025^(1/252) − 1 ≈ 0.0098% per trading day.

Setting

The field accepts from 0% to 10%, in steps of 0.25 points (the + and − buttons), or any value typed on the keyboard.

A limitation to know about

The same rate is applied to the **whole period** analysed, whereas in reality it has varied. Over a long history, the Sharpe ratio is therefore approximate. For a study of an earlier period, you can enter the average rate for that period.

Setting the VaR confidence level (90, 95 or 99%)

The VaR (Value at Risk) indicates the loss on a bad day. Its confidence level is set **at the top right of the Risk tab** ("Portfolio analysis" workspace), with the **VaR confidence level** selector, which offers three values: **90%**, **95%** (starting value) and **99%**. It is no longer in the sidebar: this is the only screen where the VaR is shown. The chosen level is kept for the whole session, even if you change tab or workspace.

What the level means

With a 1-day VaR at 95%, the loss exceeds this amount only 5% of days, or about one trading day in twenty. At 99%, it is exceeded only one day in a hundred.

For the historical method, the software takes the quantile of daily returns at the threshold 1 − level: the 5th percentile at 95%, the 1st percentile at 99%, the 10th percentile at 90%.

Effect of the setting

The higher the level, the further into the bad days you look: VaR and CVaR **increase**. The setting applies to:

- the VaR card and the CVaR card of the Risk tab;
- the table of historical, normal-distribution and Cornish-Fisher VaR;
- the return distribution chart;
- the PDF report.

Which one to choose?

95% is the most common practice in portfolio management. 99% is that of banking regulation, and is more demanding. Over a short history, the 99% VaR rests on very few days: it is less reliable.

The details of the VaR calculation methods are explained in the chapter on risk.

The banner at the top of the page and the “Live prices” badge

In the analysis spaces, the main area begins with a blue banner.

On the left

- in small type: “Master G2C · Portfolio management”;
- the title “Portfolio tracking”;
- a summary line: the period analysed (“From 15/01/2024 to …”), the number of lines held today and the short name of the chosen benchmark index.

On the right: the source of prices

Badge	Meaning
Green dot, “Live prices · Yahoo Finance”	The latest prices were downloaded from Yahoo Finance during the analysis
Orange dot, “Cached prices (offline)”	Yahoo Finance did not respond: the software is using the latest prices saved on the computer

Under the badge, the “Data as of” line gives the date of the last price in the history: this is the date at which the portfolio is valued.

If the badge is orange

The calculations remain correct, but as of the date of the latest known prices. Check your Internet connection, then click **Refresh prices** at the bottom of the sidebar. The “Prices” line in the sidebar information specifies the date of the cache used.

Good to know

The results of an analysis are kept in memory for one hour. The badge describes the situation at the time of that calculation: a green badge from forty minutes ago does not guarantee up-to-the-minute prices. The **Refresh prices** button forces a new download.

The key figures at the top of the page

Under the banner, six cards summarise the portfolio. They stay displayed in the three analysis workspaces. Hover over the small **?** next to a label to read its definition.

Card	Big figure	Line below
Current value	Quantity × latest price, for each line held, in euros	"Invested at average cost": the capital still invested, at cost basis
Total gain	Unrealised gains + realised gains + dividends	Badge: gain relative to the capital invested
Annualised return	Annualised TWR (time-weighted return)	Badge: total TWR since the start
Volatility	Standard deviation of daily returns × √252	Volatility of the benchmark index
Sharpe	(Return − risk-free rate) / volatility	Sharpe of the benchmark index
Max drawdown	Worst fall from a peak	Date of the lowest point

Reading the colours

The badges are green when the figure is positive, red when it is negative, grey when it is zero.

Example

A portfolio whose capital invested at cost basis is €10,000 and whose total gain is €1,500 shows "+€1,500" in the Total gain card, with the badge 1,500 / 10,000 = +15.00% on the capital invested.

Total gain and performance: why two figures?

The total gain is an amount in euros, which depends on the sum invested and the timing of contributions. The annualised performance (TWR) neutralises contributions and withdrawals: it measures the quality of the investment choices and can be compared with the index. Both are detailed in the chapter on performance.

Comparing with the index

For volatility and Sharpe, the line below gives the value of the benchmark index chosen in the settings: you see at a glance whether your portfolio is more or less risky, and better or worse rewarded for that risk.

Getting the PDF report

The PDF report gathers the complete portfolio analysis in a document ready to print or pass on.

In two steps

1. At the bottom of the sidebar, click **PDF report**. The message “Generating the report…” is displayed during preparation, which may take a few seconds.
2. The button becomes **Download the PDF report**. Click it: the file is saved by your browser, usually in the Downloads folder.

The file is called rapport_portefeuille_ followed by the date of the latest data, for example rapport_portefeuille_20261006.pdf.

The download button has disappeared

Download is only offered if the report matches the current settings exactly. If you change portfolio, benchmark index, risk-free rate or VaR level, the **PDF report** button reappears: click again to prepare an up-to-date report.

Always in French and light

The report is written in French even if the screen is in English, and keeps a white background even in night mode.

Settings used

The report uses the chosen portfolio, benchmark index, risk-free rate and VaR level. By contrast, for optimisation and projection, it uses fixed settings, not those of the on-screen sliders: maximum weight of 30% per security, 10-year horizon, 5,000 scenarios, normal distribution, no monthly contribution.

Where is the button?

It only appears once a portfolio has been analysed, in the Portfolio analysis, Wealth advisory and Asset management workspaces. It is absent from the “Manual and help” workspace and from the “My account” and “Add transactions” pages.

“Install reportlab” message

This message only concerns launching from the source code: the library that produces PDFs is missing. Install it with python -m pip install reportlab. The Windows and Mac versions already contain it.

What does the PDF report contain?

The report is an A4 document, divided into parts that each start on a new page. Each page carries in its footer the line “Rapport de suivi de portefeuille” (Portfolio tracking report), the

date of the day, the reminder “Outil pédagogique, ne constitue pas un conseil en investissement” (Educational tool, not investment advice) and the page number.

The parts

1. **Summary**: banner with the period, the number of lines and the benchmark index; nine key figures (current value, total gain, gain on capital invested, annualised TWR, annual IRR, volatility, Sharpe, max drawdown, 1-day VaR); evolution chart; amount invested, unrealised and realised gains, dividends, fees and source of prices.
2. **Holdings**: table of lines (ticker, security, quantity, cost basis, price, value, gain or loss, weight), then breakdowns by asset class, region and sector, on a look-through basis.
3. **Exposures and diversification**: diagnosis by dimension (geography, sectors, currencies, concentration, interest rates, real diversification), points to monitor or fix, correlations between lines.
4. **Performance**: comparison with the index, return by calendar year, beta, alpha, correlation, tracking error and information ratio.
5. **Risk**: drawdown, volatility, Sharpe, Sortino, the three VaRs, CVaR, return distribution, skewness, kurtosis and Jarque-Bera test.
6. **Markowitz optimisation**: efficient frontier, minimum-variance and maximum-Sharpe portfolios, allocations compared.
7. **Projection**: 10-year Monte Carlo simulation, unfavourable, median and favourable scenarios, probability of loss, distribution of the final value.
8. **Wealth advisory**: taxation of a full sale depending on the wrapper (CTO, PEA, assurance-vie) and stress tests.
9. **Asset management**: performance attribution, risk budget, backtest of rebalancing strategies and comparison between investing €10,000 all at once or gradually.
10. **Methodology**: definition of each indicator and limitations of the analysis.

Parts sometimes missing

If a calculation is impossible (for example optimisation, with too few securities for the 30% maximum weight), the corresponding part is omitted or shortened. The number of pages therefore depends on the portfolio.

Refreshing prices

The **Refresh prices** button is at the very bottom of the sidebar, under the PDF report button.

What it is for

To stay fast, the software keeps the results of each analysis in memory for **one hour**. As long as that time has not elapsed, it does not download the prices again. The button clears all these results from memory and reloads the page: prices are downloaded again from Yahoo Finance and all the calculations are redone.

When to use it

- the banner badge shows “Cached prices (offline)” and the Internet connection has come back;
- you leave the software open for a long time and want the most recent prices;
- a recently added security did not yet have a price.

Internet is needed

Without a connection, refreshing cannot download anything: the software takes the latest saved prices and the badge stays orange.

Good to know

- Refreshing takes a few seconds longer than usual, since everything is recalculated.
- A PDF report that has already been prepared is not changed: prepare it again if you want the latest prices.

The information at the bottom of the sidebar

Once the portfolio has been analysed, a small block of information appears at the bottom of the sidebar, above the report and refresh buttons.

Line	Content
File	The name of the portfolio or file analysed. The label “(updated)” indicates that transactions have been added to it during the session without being saved
Period	First and last date of the valued history
Transactions	The number of transactions read (purchases, sales, dividends)
Prices	The source of the latest prices: “Yahoo Finance (live)”, or “local cache of” followed by the date, with the note “Yahoo Finance unreachable”
Exchange rates (n)	Folded by default: click it to show one line per foreign currency in the portfolio (“€1 in USD”, “€1 in GBP”…), the day’s exchange rate used to convert securities into euros, to four decimal places. The number in brackets is the number of currencies

Reading an exchange rate

“€1 in USD · 1.0850” means that one euro is worth 1.0850 dollars. A share quoted at 200 USD is therefore worth 200 / 1.0850 ≈ €184.33. A portfolio entirely in euros has no rate line, and “Exchange rates” does not appear.

After uploading a file

When an uploaded file has been recognised automatically, an additional note indicates the number of transactions and securities recognised, and flags, for example, that an image PDF was read by character recognition.

Asking the manual’s assistant a question

The assistant is in the **Manual and help** workspace, right at the top, in the **Ask a question** field.

How to do it

1. Choose the “Manual and help” workspace in the sidebar.
2. Type your question in your own words, for example “how do I delete a transaction?”, then press Enter.
3. The manual entry that best answers is displayed in full, with the name of its chapter.

Three clickable examples are offered under the field, such as **How do I delete a transaction?** or **What does the Sharpe ratio mean?**.

What accompanies the answer

- **See also**: up to three other related entries; a click opens the entry.
- **For your portfolio**: some entries display your own figures (for example your Sharpe ratio). These are those of the last portfolio analysed during the session: display an analysis workspace first so that they are available.
- **The “Go to” button**: it opens the relevant workspace. If the entry concerns a specific tab, a message in the sidebar tells you which tab to open, which you still have to click.

It is not artificial intelligence

The assistant writes nothing: it searches the manual for the entry closest to your question (title words, common phrasings, synonyms, text) and displays it as written. It tolerates typing mistakes and missing accents. It works without Internet and sends nothing.

If it does not find anything

It says so and offers the closest entries. Try other words, simpler or more technical. Your question is then recorded on the computer to complete the manual (see the entry on unanswered questions).

The help bubble at the bottom right of the screen

The **Help** button, fixed at the bottom right of the screen, is visible on every page: the three analysis workspaces, “Manual and help”, “My account” and “Add transactions”. It lets you ask the manual a question without leaving the current tab.

How to do it

1. Click **Help**: a small panel opens above the button.
2. Type your question in your own words, then press Enter or click **Send**.
3. The panel shows the title of the article that answers, its beginning (with the formula if there is a short one) and, where the article allows, your own figures.
4. Click outside the panel, or on **Help** again, to close it.

The links under the answer

- **Read the full article**: opens the “Manual and help” space on the whole article.
- **Go to the screen**: opens the workspace concerned; if the article is about a specific tab, a message in the sidebar tells you which tab to open.
- **See also**: up to three related articles; a click shows their beginning in the panel.
- **This is not the answer I was looking for**: records the question on the computer to improve the manual.
- **Your previous questions**: the last five questions of the session; a click shows the answer again. They are forgotten when the software is closed.

Same engine as the manual’s assistant

The bubble uses exactly the same search as the **Ask a question** field in the “Manual and help” space: offline, without artificial intelligence, tolerant of typing errors. A question without a reliable answer is recorded in the same way (see the article on unanswered questions). The button does not appear in the PDF report or when printing.

Browsing the manual’s contents

Under the assistant, the **Manual contents** box gives access to the whole manual. Its subtitle shows the number of chapters and entries.

How to browse it

1. Choose a chapter in the **Chapter** list (chapters are numbered in the recommended reading order).
2. The chapter’s introduction is displayed, followed by its entries, one per question.
3. Click an entry’s title to expand it, and again to collapse it.

An entry opened from “See also” is already expanded in the contents.

The bold elements

In the entries, the names of buttons, tabs or fields are written in bold, exactly as they appear on screen: search for them as they are.

Downloading the full manual

When your version of the software is supplied with the exported manual, “Full manual (Word)” and “Full manual (PDF)” buttons appear under the entries. Otherwise, these buttons are absent and the manual can only be consulted on screen.

Language

The manual is displayed in the screen language if its translation is installed, otherwise in French.

Unanswered questions and exporting them

When the assistant does not find a reliable answer, your question is recorded in a file, **on this computer only**. It is never sent automatically.

Viewing them

At the bottom of the “Manual and help” workspace, a collapsed panel titled “Unanswered questions”, followed by their number in brackets, appears as soon as at least one question has been recorded. Open it to see the 30 most recent, with their date and time.

Exporting them

The **Export the questions (CSV)** button downloads the complete file, questions_sans_reponse.csv, with two columns: date and question. Pass it on to the software’s creator or to your teacher: each question added to the manual improves the assistant.

Details

- Each question is limited to 300 characters.
- The same question typed several times in a row is recorded only once.
- The file is stored in the software’s data folder, under the name questions_sans_reponse.csv. Deleting it erases the history; the software has no button to do this.

Chart tips: zoom, hover, legend

The dashboard’s charts are interactive.

Hover to read values

Move the mouse over a curve, a bar or a country: a bubble shows the exact value. On evolution charts, the bubble gathers all the curves at the same date, which lets you compare the portfolio and the index, or the value and the capital invested, at a glance.

Zoom by click and drag

Click and drag the mouse over the area that interests you: the chart enlarges on that area. You can repeat the action to zoom in further.

Returning to the original view

Double-click the chart.

Choosing a period

The “Portfolio value over time” charts and the comparison with the index have buttons above the curve: **1M** (one month), **6M** (six months), **YTD** (since 1 January), **1Y** (one year) and **All**.

Hiding or isolating a curve

Click a name in the legend, under the chart, to hide the corresponding curve; click again to show it again. A double-click on a name in the legend shows only that curve.

Limitations

- The world map cannot be zoomed: hovering over countries remains possible.
- The charts’ toolbar is hidden; there is therefore no button to save a chart as an image. To keep charts, use the PDF report or a screenshot.
- The charts adapt to the width of the window: enlarge it or collapse the sidebar to see them larger.

3. Your account and the security of your data

This chapter explains how to create your account, sign in, save and manage your portfolios in your encrypted personal space, and use the “My account” page. It then describes precisely how your data is protected, where it is stored, what leaves your computer and what nobody, not even the software’s creator, can see.

What is an account for, and is it mandatory?

An account is **not mandatory** to analyse a portfolio. Without an account, you can choose a sample portfolio or upload your file in the sidebar: all the analyses work.

What the account brings

The account gives you an **encrypted personal space**, where your portfolios are kept from one use to the next:

- your saved portfolios appear at the top of the sidebar’s portfolio list, in the form “My space ·” followed by their name;
- you can add new transactions to them, correct or delete a transaction, then go back;
- the **My account** page lets you rename them, download them or delete them;
- the display mode you chose (light or dark) is restored at each sign-in.

Without an account

A file simply uploaded in the sidebar is **not kept**: when you close the software, you will have to upload it again. Transactions added or corrected without an account only apply to the current session; the software then offers to download the updated file. When a file is uploaded without being signed in, the sidebar in fact reminds you: “To keep this file, sign in”.

Why “encrypted” ?

Everything you save in your space is encrypted with a key derived from your password. Other users of the same computer, and even the administrator, cannot read it (see the entry “How is my data protected?”).

Creating an account: username and password rules

The steps

1. At the top of the sidebar, next to the label **Encrypted personal space**, click **Sign in**. A form opens (the button becomes **Close**).
2. Choose **Create an account**.
3. Fill in **Username**, **Password** and **Confirm password**.
4. Click **Create my account**.

You are signed in straight away, with an empty space.

Username rules

- **3 to 30 characters**;
- only **unaccented lowercase letters** (a to z), **digits** and the three signs **full stop**, **underscore** and **hyphen**;
- no spaces, no accents, no other symbols.

Leading and trailing spaces are removed, and capitals are converted to lowercase: “Kevin.Brule” becomes “kevin.brule”. You can then sign in by typing either one.

Example	Accepted?
etudiant.g2c	Yes
marie_2026	Yes
ab	No: fewer than 3 characters
jean dupont	No: space
kévin	No: accent

Password rule

At least **8 characters**. No maximum, no required type of character: letters, digits, spaces, accents and symbols are accepted. The software does not demand complexity, but your password is the only protection for your data: choose one that is long and impossible to guess (a phrase of several words, for example).

Possible messages

- “Invalid username”: a username rule is not met.
- “The password must be at least 8 characters long.”
- “The two passwords do not match.”
- “This username is already taken.”: an account already has this name on this computer.

Before confirming

The form reminds you: if it is forgotten, the password cannot be recovered, and your portfolios would become permanently unreadable. Write it down somewhere safe.

Signing in, signing out and automatic sign-out

Signing in

1. In the sidebar, click **Sign in**.
2. Leave the choice on **Sign in**, then enter your **Username** and your **Password**.
3. Confirm with the form’s **Sign in** button.

The check takes a short moment (the message “Checking…” is displayed): this delay is deliberate, it makes hacking attempts very slow. Once signed in, the sidebar shows a badge

with your initials, your username, and two links: **My account** and **Sign out**. The light or dark mode you had chosen is restored.

If the username or the password is wrong, the message is always the same: “Incorrect username or password.” The software deliberately does not say which of the two is wrong, so as not to reveal which accounts exist.

Signing out

Click **Sign out**. The encryption key is erased from the session’s memory and your portfolios disappear from the list.

Automatic sign-out

After **30 minutes without any action** in the dashboard, you are signed out. The check takes place at your next action: the sidebar then displays “Automatically signed out after 30 minutes of inactivity.” and you simply need to sign in again. Each click or change of setting resets the counter to zero.

Closing the window or the software

The sign-in is only kept for the open page: it ends if you close the dashboard window, if you reload the page or if you stop the software. Nothing is lost: your saved portfolios are already written to disk, encrypted.

Account locked after several failed attempts

The rule

After **5 wrong passwords** for the same account, that account is **locked for 60 seconds**. During this time, every attempt is refused, even with the right password, with the message “Too many failed attempts: try again in” followed by the number of seconds remaining.

The details

- The failure counter is specific to each account and saved on disk: closing and relaunching the software does not reset it.
- At the moment of lockout, the counter restarts from zero: after the minute’s wait, you have 5 attempts again.
- A successful sign-in also resets the counter to zero.
- A username that does not exist is not locked, but it receives the same message and takes the same calculation time as a real account: it is impossible to guess which accounts exist this way.

Why this lockout?

It prevents someone from very quickly trying a long list of passwords from the sign-in screen. Combined with the calculation time of each attempt, it makes this type of attack very slow.

What should you do?

Wait for the indicated time to end, then check:

- the Caps Lock key (the password is case-sensitive, unlike the username);
- the keyboard layout (AZERTY or QWERTY);
- the exact username of the account.

If the password is truly lost, no unlocking is possible: see the entry “I have forgotten my password”.

Saving a portfolio to my space

You need to be signed in. Two routes are possible.

From the sidebar

1. Upload your file (CSV, Excel or PDF) in the sidebar’s upload area.
2. The software reads it and analyses the portfolio. If the file is not recognised on its own, the import wizard guides you first.
3. Once the analysis is displayed, a **Save to my space** box appears just under the upload area.
4. Check or edit the **Portfolio name** (the file name is suggested), then click **Save**.

The box then confirms that the portfolio is saved in your space, encrypted.

From the “My account” page

1. Click **My account**, then, in the **Add a portfolio** section, choose your file.
2. The software reads it automatically and indicates the number of transactions recognised.
3. Check the name, then click **Save**.

If the file cannot be read automatically, a message invites you to upload it from the sidebar, where the import wizard will help you.

What is saved

The portfolio is saved **after reading and conversion**, in the project’s format (date, type, ticker, name, quantity, price, fees), and not in the form of the original file. It is encrypted before being written to disk. It then appears in the portfolio list, in the form “My space ·” followed by its name.

Beware of identical names

Saving a portfolio under a name **already used** in your space (ignoring capitals) **replaces** the old portfolio of that name; the old version can be recovered with **Undo last change** in “My account”. To keep both, give a different name.

The “My account” page: renaming, downloading, deleting a portfolio

Once signed in, click **My account** in the sidebar. To return to the analyses, click any workspace in the menu (or **My account** again).

The page recalls that your portfolios are encrypted with a key derived from your password. It has four sections: **Add a portfolio**, **My portfolios**, **Change password** and **Delete my account**.

The “My portfolios” section

Each portfolio occupies one row, with its number of transactions and its date of last modification.

Action	How to do it
Rename	Edit the name in the row’s field, then confirm (Enter key). An empty name is ignored.
Download	**Download** button: you get a CSV file bearing the portfolio’s name.
Delete	Tick **Confirm**: the **Delete** button, greyed out until then, becomes active. Click it.
Add transactions	**Add transactions** button (see the next entry).
Go back	**Undo last change** button, present only after an addition or a correction.

Good to know

- **The downloaded file is not encrypted**: it is a CSV readable by Excel, in the project’s format. Store it somewhere safe.
- **Deletion is permanent**: the portfolio and its possible previous version are erased. No undo is possible. Download it first if you have any doubt.
- Renaming a portfolio does not change its content and does not create a previous version.

Adding transactions and undoing the last change

Adding transactions

For a portfolio in your space, two entry points lead to the same page:

- on the **My account** page, the **Add transactions** button on the portfolio’s row;
- in the sidebar, the **Add transactions** link, when that portfolio is selected.

Upload only the **new** movements (PDF contract note, CSV or Excel export) or enter them by hand. The software discards transactions that are already present, flags inconsistencies, then has you check them. Finally click **Save transactions**. The details of this page are explained in the chapter on importing.

What happens when you save

The portfolio is saved again, encrypted, with the new transactions. **The previous version is kept**, also encrypted. The same applies when you correct or delete transactions with **Edit transactions** in the Transactions tab, then **Save changes**.

Undoing the last change

The **Undo last change** button then appears on the portfolio's row, in **My account**, and in the Transactions tab. Its tooltip gives the date of the version you are about to return to. One click puts the portfolio back in exactly that state.

The limits

- **Only one level of undo.** Only the very latest version is kept. After an undo, the button disappears; and each new addition or correction replaces the version previously kept.
- The button does not exist as long as no addition or correction has been saved.
- Saving a file under the same name from the sidebar or the **Add a portfolio** section replaces the portfolio, but keeps the previous version (recoverable with **Undo last change**).

Changing your password

The steps

1. Sign in, then open **My account**.
2. In the **Change password** section, enter the **Current password**, the **New password** (at least 8 characters) and **Confirm password**.
3. Click the **Change password** button.

The message "Password changed." confirms the operation. You stay signed in.

What the software does

Since the encryption key is derived from the password, changing the password also changes the key. The software:

1. checks the old password;
2. decrypts in memory all your portfolios, their previous versions and your preferences;
3. draws two new random "salts" (see the technical entry on encryption);
4. calculates the new fingerprint and the new key;
5. **re-encrypts everything** with the new key.

The old password therefore no longer opens anything. Your portfolios, their names and the ability to undo the last change are kept.

Possible messages

- "Current password is incorrect."

- “The password must be at least 8 characters long.”
- “The two passwords do not match.”

When should you do it?

If you think someone knows your password, or if you chose one that is too short. Changing the password is only possible by knowing the old one: it is not a way of recovering a forgotten password.

Deleting my account and all my data

The steps

1. Sign in and open **My account**.
2. In the **Delete my account** section, enter your **Password**.
3. Click **Permanently delete my account**.

You are signed out and the message “Account and data deleted.” is displayed.

What is erased

- your account’s folder: all your portfolios, their previous versions, the encrypted list of their names and your preferences;
- your row in the accounts register (username, salts, fingerprint).

The computer’s other accounts are not touched. The username becomes free again.

It is permanent

There is no bin and no grace period: a deleted account cannot be restored. Download your portfolios as CSV first if you might need them. Since the files were encrypted, any remnants on the disk would in any case be unreadable without your password.

What is not erased

A few files shared by all users, which contain no portfolio data, stay in place: the memory of recognised securities, the price cache files and the log of questions put to the assistant (see the entry “Several users on the same computer”).

Deleting the whole software

To erase all accounts at once, see the entry on uninstalling in the “Getting started” chapter.

How is my data protected?

Here is the principle, explained simply. The next entry gives the technical detail.

1. Your password is never saved

The software only saves a **fingerprint** (hash) of the password: the result of a one-way calculation. When you sign in, it redoes the calculation with what you type and compares the two results. Reading the fingerprint does not allow the password to be found.

2. Your portfolios are encrypted

When you sign in, the software makes an **encryption key** from your password. Everything you save is encrypted with this key before being written to disk: a file opened without it is only a string of incomprehensible characters.

3. The key is never written

The key exists only in the computer's memory, during your sign-in. It disappears when you sign out, when the page is closed and after 30 minutes of inactivity.

4. Even the names are hidden

The list of your portfolios (their names, their dates) and your display preferences are encrypted too. Account folders have randomly drawn names, unrelated to your username.

5. Modified files are detected

Each encrypted file carries a check code. If someone modifies it, even by a single character, it is rejected instead of being read wrongly.

6. Attempts are slowed down

Each password attempt requires a deliberately long calculation, and the account is locked for a minute after 5 failures.

The consequence

Without your password, **nobody** can read your portfolios: neither the other users of the computer, nor the administrator, nor the software's creator. The trade-off is that a forgotten password makes the data permanently unreadable.

Encryption in detail: PBKDF2, salt and Fernet

The password fingerprint

fingerprint = PBKDF2-HMAC-SHA256(password, fingerprint salt, 600,000 iterations)

- **PBKDF2-HMAC-SHA256**: a standard, one-way derivation function.
- **Salt**: 16 bytes drawn at random by the system's cryptographic generator, different for each account. Two people who choose the same password have different fingerprints, and precomputed fingerprint tables are unusable.

- **600,000 iterations**: the calculation is repeated 600,000 times (the value recommended by OWASP in 2023), which makes it slow: of the order of a few tenths of a second per attempt, depending on the computer.
- The comparison with the saved fingerprint is made in **constant time**, so as to reveal nothing through the duration of the calculation.

The encryption key

key = PBKDF2-HMAC-SHA256(password, key salt, 600,000 iterations)

This is the same calculation, but with a **second, distinct salt**. The fingerprint saved on disk therefore cannot be used as a key. This 32-byte key is **never written**: it exists only in memory, during the sign-in.

File encryption

Data is encrypted with **Fernet** (the Python cryptography library): **128-bit AES** encryption combined with an **HMAC-SHA256** authentication code. The authentication code detects any modification: a file that has been altered, or opened with the wrong key, is rejected.

What is encrypted

File	Content
index.enc	the list of your portfolios (internal identifier, name, dates, number of transactions)
one .enc file per portfolio	the portfolio's transactions
an optional .prec.enc file	the previous version, for the undo
preferences.enc	your display preferences (light or dark mode)

Account folders and portfolio files have random 32-character names. Each write goes through a temporary file that is then renamed: a power cut never leaves a half-written file.

Resistance to an attack

The 60-second lockout after 5 failures protects the sign-in screen. But someone who copied the files could test passwords without going through that screen: only the cost of the calculation slows them down. At half a second per attempt on one processor, a million attempts take about 6 days (1,000,000 × 0.5 s = 500,000 s); an attacker equipped with many processors goes faster. **The real protection is therefore a long, unpredictable password**: a phrase of several words resists far better than a dictionary word followed by a digit.

I have forgotten my password

The frank answer

There is no way to recover a forgotten password, nor the portfolios it protects. There is no "forgotten password" link, no secret question, no backup email address. Neither the computer's administrator nor the software's creator can help you.

Why?

- The password is not saved anywhere: only its fingerprint is, and it does not allow the password to be found.
- The key that decrypts your portfolios is calculated from the password and is never saved.
- Without the password, there is therefore no key, and without a key the files remain unreadable.

This is a deliberate choice: if there were a way to open the data without the password, anyone (another user, a hacker, the administrator) could use it too.

What to do now

1. Try the likely variants (capitals, AZERTY or QWERTY keyboard, old password), keeping in mind the one-minute lockout after 5 failures.
2. If you downloaded your portfolios as CSV, or if you kept your original files (statements, contract notes), create a new account with a different username and save them again.
3. The old account remains unusable. To make it disappear, you have to be able to sign in to it; failing that, only uninstalling with deletion of accounts erases it, but that also erases the accounts of the other users of this computer.

For the future

Write your password down in a password manager or somewhere safe, and regularly download a copy of your portfolios from **My account**.

Where is my account data stored?

Your data stays on the computer where the software is installed. It is not sent to any server.

Location by version

Version	Accounts folder
Windows (installer)	C:\Users\<your name>\AppData\Local\Programs\Portfolio Tracker\data\comptes
Mac (application)	~/Library/Application Support/Portfolio Tracker/app/data/comptes
Source code	data/comptes in the project folder

On Windows, the AppData folder is hidden: type the path into File Explorer's address bar. On Mac, in the Finder, use the Go menu, then "Go to Folder…".

What this folder contains

- comptes.json: the accounts register. For each one: the username, in clear; the two salts; the password fingerprint; the number of iterations; the creation date; the failed-attempt counter and the end of any lockout; the name of the account's folder. **No password and no portfolio data.**

- one folder per account, with a random name, which contains the encrypted files index.enc, preferences.enc and one .enc file per portfolio (plus a .prec.enc for a previous version).

What can be guessed without the password

Someone who opens this folder sees the list of usernames and the accounts' creation date, and can count the encrypted files in a folder. They see neither the name nor the content of the portfolios.

Updates and uninstalling

Accounts are kept when a new version is installed, on Windows as on Mac. When uninstalling on Windows, a question asks whether to delete them too ("No" by default). On Mac, they stay as long as you do not delete the folder ~/Library/Application Support/Portfolio Tracker.

The folder is never published

In the source code, the data/comptes/ folder is excluded from the GitHub repository and from the installers: your accounts are never published, even encrypted.

What the software's creator can and cannot see

With the installed version: nothing

The software runs entirely on your computer, at the address localhost. It has **no server** to send your data to, and the software's creator receives nothing:

- no online account: your account exists only on your computer;
- no usage statistics: the collection of statistics by Streamlit (the library that displays the dashboard) is disabled, both in the project's configuration and at launch;
- the questions put to the assistant are not sent: they stay in a local file;
- no check for updates.

The only outgoing connections are used to obtain market data (see the entry "What leaves my computer").

Even with access to your computer

Your saved portfolios are encrypted with a key derived from your password. The creator, like any administrator, could not read them without this password, and cannot reset it either.

What is readable without a password

A few local files, shared by all the computer's users, are not encrypted: the list of account usernames, the price cache (codes of recently analysed securities and their prices), the memory of recognised securities, the learned trade confirmation models (modeles_pdf.json) and the log of unanswered questions. They contain no quantity or amount, but they reveal which securities have been analysed. The learned models only keep the headings of the values ("Quantité",

“Cours”…) and the document’s vocabulary as fingerprints (hashing): no amount, no name in clear.

The case of a teacher

Your teacher only sees your portfolios if you pass them a file yourself (CSV export, PDF report) or if you show them your screen.

Reporting a misread PDF

To help the creator improve the reading of a type of trade confirmation, you can send them yourself the report produced by **Prepare an anonymised report** (“Complete the transaction” form): names, address lines, e-mail addresses, IBANs, phone numbers and account numbers are masked in it. Nothing is sent without your action.

The online version is different

On the online version, the software runs on a hosted server: the files you upload there are processed on that server, which is administered by the person who published the site. For real data, prefer the installed version.

Several users on the same computer

In the same Windows or Mac session

Each person creates **their own account** in the software. Then:

- each person sees, in the list, only **their** saved portfolios (plus the project’s sample portfolios, shared by everyone);
- other people’s portfolios are encrypted with their own password: they cannot be read;
- deleting an account, or changing its password, requires that account’s password: nobody can do it in your place from the software;
- remember to click **Sign out** before giving up your seat, without waiting for the 30-minute automatic sign-out.

What is shared by everyone

Some files serve everyone and are not encrypted. They contain no quantity or amount:

- the local securities database and the memory of recognised securities (matches between ISIN codes, names and tickers, with their date);
- the price cache files (codes and prices of the latest securities analysed);
- the learned trade confirmation models (modeles_pdf.json): headings of the values and fingerprints of the vocabulary, with no amount or name in clear;
- the assistant’s log of unanswered questions: **all unanswered questions are visible to all users**, in the Manual and help workspace. Do not type any personal information there.

A user who opens the accounts folder also sees the list of existing usernames.

With different Windows sessions

On Windows, installation is done for the current user, in their personal folder. If each person has their own Windows session, each installs the software in their session: accounts, database and caches are then completely separate.

A computer administrator

They can copy or erase the files, like any file on the computer, but they cannot read your encrypted portfolios without your password. Keep a copy of your important portfolios.

What leaves my computer when I use the software

The dashboard runs on your computer (localhost) and accepts no connection from another device. It contacts the Internet only for **market data**, and only when it needs to.

What is sent, and to whom

Recipient	What is sent	When
Yahoo Finance	the **codes** of the securities (tickers), of the benchmark index and of the exchange rates, with the **start date** of the history requested (the date of your first transaction)	at each analysis, when prices are not already in memory, and with **Refresh prices**
Yahoo Finance	a security's code, to find out its trading currency	the first time that security is encountered (the answer is then kept)
Yahoo Finance search engine	an **ISIN code**, a **company name** or a label read from your file, or a ticker without a trading venue	at import, for a security the software does not yet know
GitHub (or, failing that, the jsDelivr service)	a simple request to download the country outlines (Natural Earth)	only once, if the base map is not already on the computer

If the base map could never be saved, it is the charting library, in the dashboard window, that downloads its own base map to draw the world map.

Like any connection to a site, these requests reveal the IP address of your connection to the service contacted.

What is never sent

- your quantities, your purchase and sale prices, your amounts, your fees, your dividends;
- your files, in any form whatsoever;
- the names of your portfolios, your username, your password;

- your PDF reports;
- the questions put to the manual's assistant;
- no usage statistics.

The check of the prices in your file is done on your computer: the software downloads market prices, then compares locally.

What Yahoo Finance can deduce from this

Yahoo sees which security codes are requested, and from what date. It sees neither how many you hold, nor at what price you bought them.

To limit exchanges even further

Offline, the software uses its local database, its securities memory and its cache: it then sends nothing. Securities that are already known are recognised without any online search.

GDPR and data protection

The software was designed taking into account three articles of the General Data Protection Regulation (GDPR). This entry describes the measures taken; it does not constitute legal advice.

Article 25: data protection by design and by default

- Encryption is applied **by default** to everything saved in a personal space: no setting to turn on.
- No password is stored, only a fingerprint.
- No portfolio data is saved in clear in the personal space; even portfolio names are encrypted.
- Minimisation: only security codes are sent to Yahoo Finance, never quantities or amounts.
- The password of a protected PDF is stored nowhere: the decrypted copy of the document stays in memory for the session.
- The trade confirmation models learned on the computer contain no amount or name in clear (headings and fingerprints only).
- To report a misread PDF, an anonymised report masks names, addresses, IBANs, phone numbers and account numbers before you decide to send it.
- Data stays on the user's computer; it is not passed to any third party.

Article 32: security of processing

- authenticated encryption (Fernet: 128-bit AES and HMAC-SHA256);
- salted and deliberately slowed fingerprints (PBKDF2-HMAC-SHA256, 600,000 iterations);
- limiting of attempts: 60-second lockout after 5 failures;
- same error message for an unknown username and a wrong password;
- key never written to disk, automatic sign-out after 30 minutes of inactivity;
- detection of any modification of an encrypted file.

Article 17: right to erasure

The **Delete my account** section of the **My account** page permanently erases the account and all its portfolios, without anyone's intervention. A single portfolio can also be deleted.

Note

With the installed version, you are alone in processing your own data, on your computer. On a shared server (online version, school server), the same protections apply to personal spaces, but the files uploaded are processed on that server.

Accounts on the online version

Temporary accounts

On the online version (Streamlit Community Cloud), the server's disk is wiped each time the site restarts. Saved accounts and portfolios **can therefore disappear at any time**, without prior warning.

The sign-in form reminds you of this with a box that begins "Online demo version". This message only appears on the online version.

What works

Account creation, sign-in, encryption and the **My account** page work exactly as in the installed version, as long as the site has not restarted.

Precautions

- Always download your portfolios (**Download** button in **My account**) before leaving the site.
- Keep your original files.
- For lasting use, or for real personal data, use the application installed on your computer.

The server processes your files

Online, the files you upload are read and analysed on the server. Portfolios saved in a space are encrypted there as elsewhere, but the caches and the assistant's log of unanswered questions are shared there by all the site's visitors.

Backing up my portfolios or changing computer

The recommended method: one CSV per portfolio

1. Sign in and open **My account**.
2. For each portfolio, click **Download**: you get a CSV file in the project's format.
3. Store these files somewhere safe (USB stick, backed-up folder, external drive).

This file contains all the transactions. It is read back directly by the software, on any computer, on Windows as on Mac.

Warning: the CSV is not encrypted

The downloaded file is readable by anyone. If your data is sensitive, store it in a protected location.

On a new computer

1. Install Portfolio Tracker.
2. Create an account (the same username if you wish).
3. Save each of the CSV files in your space, through the **Add a portfolio** section of **My account** or through the sidebar's file upload.

And what about copying the accounts folder?

Copying the data/comptes folder directly from one installation to another is not an operation the software supports. The CSV export is simpler and safer.

What the backup does not contain

- the previous version kept for **Undo last change**;
- your display preferences, to be chosen again;
- your password, which you must remember yourself.

How often?

After each important update of a portfolio, and always before uninstalling the software, reinstalling the system or using the online version.

The log of unanswered questions

What is recorded

When the assistant in the Manual and help workspace does not find a reliable answer to a question typed in **Ask a question**, it says so and records the question in a local file, with the date and time. Questions that found an answer are not recorded. Each question is truncated to 300 characters.

Where is this file?

In the software's data folder, under the name questions_sans_reponse.csv (on Windows, C:\Users\<your name>\AppData\Local\Programs\Portfolio Tracker\data\; on Mac, ~/Library/Application Support/Portfolio Tracker/app/data/). It is in clear, not encrypted, and shared by all the computer's accounts.

It is never sent

The software passes this file to nobody. At the bottom of the Manual and help workspace, the "Unanswered questions" section shows the 30 most recent and offers the **Export the questions (CSV)** button. If you wish, send this file to the software's creator yourself: each question added to the manual improves the assistant.

Visible to all users

This list is not tied to an account: anyone who uses the software on this computer sees it, even without being signed in. On the online version, it is shared by all the site's visitors. **Never type personal information** (amount, password, name) in a question.

Erasing it

The software does not offer a button to empty this log. You can delete the questions_sans_reponse.csv file yourself: it will be recreated at the next unanswered question. It is kept when the software is updated, on Windows as on Mac.

4. Importing and updating your portfolio

This chapter explains how to give your transactions to the software: which files it accepts (CSV, Excel, PDF), the project format and its template, the automatic reading of bank or broker exports and of PDFs (trade confirmations, statements, scanned or password-protected documents, several files at once), the import assistant, the recognition of securities (ticker, ISIN, name, Bloomberg, Google or Reuters codes) and of currencies. It then describes how to update a portfolio: adding new transactions, stock splits, duplicates, checks, correcting or deleting a transaction, and undoing a change. It ends with common errors and a full worked example.

Which files can be imported into the software?

The software reads your transactions from a file that you send to it. It does not connect to your bank or to your broker.

The three accepted formats

Format	Extension	Examples
CSV (text)	.csv	File in the project format, broker export, file saved from Excel
Excel	.xlsx	Personal spreadsheet, bank export
PDF	.pdf	Transaction statement presented as a table, trade confirmation (a confirmation of an executed order), portfolio statement (holdings and cost prices), password-protected PDF

The upload areas accept only these three extensions. They accept several files at once (see the fiche “Uploading several files at once”). The old Excel format .xls is not accepted: open the file in Excel and save it as .xlsx or as CSV. The software recognises the real nature of a file from its content (a PDF starts with %PDF-, an .xlsx is a ZIP archive), then reads it accordingly.

What the file must contain

It must contain a **history of dated transactions**: purchases, sales, and possibly dividends. A simple list of positions (securities and quantities held today, with no purchase dates) does not allow performance to be calculated over the whole history; the one exception is a PDF portfolio statement, which can serve as a starting portfolio, each holding becoming a purchase on the statement date (see the fiche “Importing a PDF portfolio statement”). Three pieces of

information are essential for each row: the **date**, the **security** and the **quantity**, together with the **unit price or the total amount**.

The format does not need to be perfect

The file may have its own column names, in French or English, header lines above the table, ISIN codes instead of tickers, numbers in the French style or amounts in euros for foreign securities: automatic detection takes care of it (see the following fiches). If it has any doubt, the import assistant opens.

What is not possible

- a photo or screenshot in image format (.png, .jpg) is not accepted;
- a scanned PDF (or one whose text is "encoded") is read automatically only if a character recognition (OCR) engine is installed, and only for trade confirmations (see the fiche on image PDFs); otherwise, a form lets you complete the transaction by hand (see the fiche "Completing a transaction when the PDF is not recognised");
- there is no automatic connection to a bank account.

What happens when I upload a file?

Where to upload the file

In the sidebar, under "Data", use the **Upload a file (CSV, Excel, PDF or image)** area: drag the file onto it or click to choose it. You can drop several files at once: their transactions are combined (see the fiche "Uploading several files at once"). The uploaded file takes **priority** over the portfolio selected in the list just above: as long as it is present in the upload area, it is the one being analysed.

What the software does

1. The message "Reading the file and checking prices against market data…" is displayed.
2. The software reads the file, recognises the columns, converts the security codes into Yahoo Finance tickers and checks the prices against market data.
3. **If the result is reliable**, the portfolio is analysed immediately. A box in the sidebar summarises what was understood, for example "File recognised automatically: 12 transaction(s), 5 security(ies).".
4. **If any doubt remains** (unrecognised column, security not found, unreadable row), the **import assistant** is displayed in place of the dashboard, already pre-filled: you just need to check and confirm.

You can restart the reading manually at any time with the **Open the import assistant** button, visible under the upload area as long as a single file is present. A password-protected PDF first shows the **Protected PDF** box (see the dedicated fiche).

The file is not kept automatically

An uploaded file is kept only for the session. To find it again next time:

- if you are signed in, the **Save to my space** block appears under the upload area once the file has been analysed;
- otherwise, the software reminds you: “To keep this file, sign in (”Sign in”, at the top of the sidebar).”.

Going back to another portfolio

Remove the file from the upload area (small cross to the right of its name): the portfolio selected in the “Data” list is analysed again.

The project format: which columns should my file have?

The project format is a CSV file with seven columns, one row per transaction:

```
date,type,ticker,nom,quantite,prix,frais
2024-01-15,ACHAT,CW8.PA,Amundi MSCI World,10,420.00,2.50
2024-02-01,ACHAT,MC.PA,LVMH,3,780.00,2.00
2024-05-22,DIVIDENDE,MC.PA,LVMH,0,39.00,0.00
2024-09-18,VENTE,MC.PA,LVMH,1,700.00,2.00
```

Meaning of each column

Column	Content
date	Transaction date, preferably in YYYY-MM-DD format. A day with no trading is attached to the next trading day.
type	ACHAT (buy), VENTE (sell) or DIVIDENDE (dividend) (see the fiche on transaction types)
ticker	The security’s code on Yahoo Finance: MC.PA (LVMH in Paris), AAPL (Apple), ULVR.L (Unilever in London). An ISIN code is also accepted on upload: it is converted.
nom	Name of the security, free text (used for display)
quantite	Number of securities bought or sold; 0 for a dividend. Fractional units are accepted.
prix	Unit price, **in the security’s trading currency** (dollars for Apple, pence for London); for a dividend, the total amount received
frais	Brokerage fees for the transaction, **always in euros**

The two currency rules to remember

- The **price** is in the currency in which the security is quoted on Yahoo Finance. The software converts it into euros itself, at the exchange rate of the transaction day.
- **Fees** are always in euros, whatever the security.

If your statement gives prices already converted into euros, that is not a problem: on upload, the software compares each price with the actual price of the day and converts back if necessary (see the fiche on currencies).

Useful details

- The order of the rows is free: the software sorts the transactions by date.
- Quantities and prices must be positive: it is the type that indicates the direction. In a broker export, a negative quantity is accepted and made positive.
- Both the dot and the comma are accepted as the decimal separator, and the semicolon as the column separator (see the fiche on broker exports).
- The **File template** button in the sidebar downloads an example ready to be completed.

Buy, sell, dividend: what goes in the price and the fees?

The project format knows only three transaction types. Each uses the columns in a specific way.

Type	quantite	prix	frais
ACHAT	Number of securities bought	Unit purchase price (trading currency)	Brokerage fees in euros
VENTE	Number of securities sold (positive)	Unit selling price (trading currency)	Brokerage fees in euros
DIVIDENDE	0	**Total amount received** for the row (trading currency)	Any fees in euros (often 0)

The case of the dividend

For a dividend, the prix column does not contain the dividend per share but the **total amount** received. Example: 3 LVMH shares with a dividend of €13 per share give the row 2024-05-22,DIVIDENDE,MC.PA,LVMH,0,39.00,0.00 (3 × 13 = 39). The amount is also expressed in the trading currency: in dollars for a US share, in pence for a London share. The software adds this amount, less the frais column, to the security's total dividends. Bond coupons and ETF distributions are entered in the same way.

How fees are counted

- **Purchase**: fees are included in the unit cost basis (PRU, the average purchase cost per unit). New PRU = (quantity held × old PRU + quantity bought × price + fees) / new quantity. Example: 3 LVMH bought at €780 with €2 of fees give a PRU of (3 × 780 + 2) / 3 = €780.67.

- **Sale**: fees reduce the realised gain. Gain = quantity sold × (selling price − PRU) − fees.
- The fees of all transactions are added up in the “Brokerage fees” indicator.

A sale cannot exceed the quantity held

Selling more securities than you own at that date causes an error during analysis (“Sale impossible on …”). Short selling is not supported.

Other transactions

Custody fees, transfers, taxes or corporate actions that are neither purchases, nor sales, nor dividends have no place in the project format. In a broker export, these rows are recognised and ignored (see the fiche on recognised transaction types).

Downloading and filling in the file template

Where to find it

In the sidebar, under “Data”, click **File template**. The file modele_transactions.csv is downloaded. It contains the header of the project format and four example rows:

date	type	ticker	nom	quantite	prix	frais
2024-01-15	ACHAT	CW8.PA	Amundi MSCI World	10	420.00	2.50
2024-02-01	ACHAT	MC.PA	LVMH	3	780.00	2.00
2024-05-22	DIVIDENDE	MC.PA	LVMH	0	39.00	0.00
2024-09-18	VENTE	MC.PA	LVMH	1	700.00	2.00

How to use it

1. Open the file in Excel, LibreOffice or a text editor.
2. Replace the example rows with your own transactions, keeping the first row (the column names).
3. Save, as you prefer, as CSV (comma or semicolon separator) or as an Excel .xlsx workbook.
4. Upload the file with the **Upload a file (CSV, Excel, PDF or image)** area.

Data entry tips

- Find the ticker of each security on Yahoo Finance (for example MC.PA for LVMH in Paris). If you do not know it, you can write the ISIN code in the ticker column: it will be converted on upload.
- Write dates in YYYY-MM-DD or DD/MM/YYYY format; both are recognised.

- Excel often re-saves numbers with a decimal comma and a semicolon between columns: the software accepts both variants.
- If, when opened in Excel, everything appears in the first column, this is not a problem for the software: an entire row placed between quotation marks in column A is also recognised.

No file at all?

You can also start from an example portfolio and enter your orders by hand with **Add transactions**, then download the resulting file (see the fiches on manual entry and on using the software without an account).

Importing my bank or broker export

A broker export does not need to be transformed before uploading. The software reads it as it is, in four stages: reading the table, recognising the columns, converting the securities, checking the currencies.

Reading the table

- **Separator**: for a CSV, the software counts, over the first 40 non-empty lines, the semicolons, commas, tabs and vertical bars |, and keeps the most frequent.
- **Encoding**: the text is read as UTF-8; failing that, as Windows-1252 (the encoding of older versions of Excel), which preserves accents.
- **Excel sheet**: if the workbook has several sheets, the one with the most filled cells is chosen (the transaction table rather than a notes sheet). The assistant lets you choose another one.
- **Column header row**: among the first 30 rows, the software keeps the fullest one that contains at least 60% text. The rows above it (“Statement of transactions”, account number, date of issue) are ignored. If this row contains a date or an ISIN code, there is no header row: the columns are then named “Column 1”, “Column 2”… and recognised by their content.
- **What is discarded**: empty rows, rows with neither a date nor a security (“Total” row, note below the table), and a small table placed beside the main table if it is separated from it by an empty column.

Recognised column names

Names are compared without accents or capital letters. A name is recognised if it is identical to one of the names below, or if it starts with one of them followed by a space (“Quantité exécutée”, “Cours (EUR)”).

Information	Examples of recognised names
Date	date, date opération, date d’exécution, date de valeur, trade date, transaction date, execution date, jour

Information	Examples of recognised names
Type	type, opération, sens, nature, transaction type, side, action, mouvement
Security	ticker, symbole, symbol, code, isin, code isin, bloomberg, bbg, ric, code reuters, valeur, titre, instrument, produit, product, security
Name	nom, name, libellé, désignation, nom du titre, security name, description
Quantity	quantité, qté, qty, quantity, nombre, nombre de titres, nb titres, parts, shares, units
Unit price	prix, prix unitaire, cours, cours d'exécution, price, unit price, execution price
Total amount	montant, montant net, montant brut, montant total, total, amount, net amount, montant en eur
Fees	frais, frais de courtage, courtage, commission, commissions, fees, fee, frais totaux
Currency	devise, currency, monnaie, devise de cotation, ccy, cur
Exchange	place, place de cotation, marché, bourse, exchange, market, mic, venue, lieu d'exécution

Columns named commentaire, remarque, note, observation, memo or info are never interpreted. All other unneeded columns are simply ignored.

What if the names mean nothing?

Columns are also recognised by their **content** and by the **consistency of the figures** (see the next fiche). A file with no header row, or with columns named "A, B, C", can therefore still be understood.

How does the software recognise the columns in my file?

Automatic detection proposes, for each piece of information (date, type, security, name, quantity, unit price, total amount, fees, currency, exchange), the column of the file that corresponds to it. Each column is used only once.

Step 1: names and content

For the date, type, security, name and currency, the software combines the name of the column and what it contains:

Information	Clue drawn from the content
Date	cells written as dates (15/01/2024, 2024-01-15…) and readable
Security	ISIN codes whose check digit is correct, Bloomberg, Google or Reuters codes, short capitalised tickers
Type	transaction words (“Achat”, “Vente”, “Coupon”…), with few distinct values
Currency	codes EUR, USD, GBP, GBX, CHF, JPY, CAD, AUD, HKD, DKK, SEK, NOK, CNY, SGD
Name	fairly long free text that is neither a code, nor a type, nor a currency

A column of numbers (quantity, price, amount, fees) is accepted on the basis of its name only if at least 60% of its cells are numbers.

Step 2: consistency of the figures

Number columns left without a meaningful name are told apart by calculation. The software tries the possible combinations and keeps the one in which, most often:

quantity × price ± fees = amount (within 1%)

It also favours a quantity in whole numbers, and fees that are small relative to the amount (less than 5%).

Example: in a file with no column headers containing 3 ; 740.00 ; 2222.00 ; 2.00, only the reading “3 securities at €740 plus €2 of fees = €2,222” works out: the first column is the quantity, the second the price, the third the amount, the fourth the fees.

When is the result considered reliable?

The analysis starts directly if all these conditions are met:

- the date, security, quantity, and price or amount each have a column;
- the number columns have a recognised name (quantity, and price or amount), **or** the relationship quantity × price ± fees = amount holds on at least 80% of the rows;
- the date column has a recognised name, or at least 90% of its cells are dates;
- all the securities have been identified;
- no row is unreadable (missing date, price, quantity or security).

Otherwise, the import assistant opens; the “Why is the assistant opening?” panel gives the reason, for example “Columns recognised, but not certain about the quantities, prices and amounts.”.

Which date and number formats are recognised?

Dates

Recognised formats: 2024-01-15, 15/01/2024, 15-01-2024, 15.01.2024, as well as dates followed by a time (2024-01-15 00:00:00, or 2024-01-15T10:30). Excel cell dates in date format are read directly.

Day/month or month/day? For a date such as 03/04/2024, the software examines the whole column:

- if even one date has a first number above 12 (for example 13/04/2024), the format is day/month (French);
- if a date has a second number above 12 (04/13/2024), the format is month/day (American);
- if nothing allows a decision, the day/month format is kept, and the summary says so: “Dates read as day/month (DD/MM).”. A detected American format is flagged with “Dates read in US format (MM/DD).”.

Dates written out in words (“15 janv. 2024”) are not recognised: the row is then flagged with the reason “an unreadable date”. Also prefer four-digit years.

Numbers

The software removes spaces (including non-breaking ones), apostrophes and the symbols €, $, £, EUR and USD, then interprets the decimal separator:

Written in the file	Read as
1 234,50 €	1,234.5
1.234,50	1,234.5 (the last separator is the decimal one)
1,234.50	1,234.5
1.234.567 or 1,234,567	1,234,567
(12,5)	−12.5
12,50-	−12.5

Beware of one ambiguous case: a number containing only one separator is always read as a decimal number. 1,234 and 1.234 are therefore both worth 1.234, not one thousand two hundred and thirty-four. If your export writes thousands this way, check the quantities and amounts that were read.

The sign

The sign of a quantity is used to spot sales when the file has no type column; it is then removed. Fees are always made positive.

Which transaction types are recognised in a statement?

The recognised words

Each value in the type column is classified according to the words it contains (ignoring accents and capital letters):

Classified as	Recognised words
Buy	achat, buy, bought, purchase, souscription, acquisition, acheté
Sell	vente, sell, sold, sale, cession, vendu, rachat
Dividend	dividende, dividend, coupon, distribution, détachement, revenu
Ignored	any other value: frais de garde (custody fees), virement (transfer), taxe, impôt (tax)…

So “Achat Comptant” is a purchase, “Coupons/Dividende” a dividend, and “Rachat” a sale (the redemption of fund units).

Ignored rows

Rows of another type are not errors: they do not concern a security and are discarded. The summary says so, for example “2 row(s) ignored (custody fees, transfers…).”. In the assistant, you can change the interpretation of each value (step 3).

Without a type column

If the file has no type column, the software decides according to the quantity:

- positive quantity: purchase;
- negative quantity: sale;
- zero or empty quantity with an amount: dividend.

The assistant reminds you: “Without a”Transaction type” column: a negative quantity is read as a sale, a positive quantity as a purchase.”.

In a PDF trade confirmation

The direction is searched for in the text (achat, vente, souscription, rachat, buy, sell, dividende, coupon…). A reading by headings is accepted only if one of these words is written in the document. For a confirmation read by its content, the direction must also be found (a word, or a quantity preceded by + or −). A confirmation whose direction is only written as a code (“Sens : S”) can be read thanks to a learned model. Otherwise, the **Complete the transaction** form lets you choose the direction.

My statement gives the total amount and not the unit price

The unit price is not mandatory: a total amount column is enough. The software works out the price from it.

The calculation

If the amount is **net** (fees included, that is, the sum actually debited or credited), which is the default setting:

- purchase: price = (amount − fees) / quantity;
- sale: price = (amount + fees) / quantity.

If the amount is **gross** (excluding fees): price = amount / quantity.

Example: purchase of 3 LVMH, net amount debited €2,222, fees €2: price = (2,222 − 2) / 3 = €740. For a sale of one LVMH with €698 credited and €2 of fees: price = (698 + 2) / 1 = €700.

The setting in the assistant

At step 2 of the assistant, as soon as a column is matched to “Total amount”, the **The total amount includes fees (net amount debited or credited)** box appears, ticked by default. Untick it if your amount column excludes fees. Automatic import, for its part, always treats the amount as net.

Price and amount both present

The unit price in the file is then used as it is; the amount serves only to check the consistency of the columns and, for a dividend, to know the sum received.

For a dividend

The quantity is set to 0 and the total amount is placed in the price column, as the project format requires. With no amount column, the software takes price × quantity.

If the price does not come out right

Check in the assistant that the right column is matched to the amount (net or gross, debit or credit) and that the fees box matches your statement. Once the portfolio has been analysed, a price can also be corrected in the Transactions tab.

The import assistant step by step

The assistant is displayed in place of the dashboard when automatic detection has any doubt, or when you click **Open the import assistant**. If it opens automatically, the **Why is the assistant opening?** panel gives the reason. It has four steps, all pre-filled.

1. The file

- **Excel sheet**: only for a workbook with several sheets.

- **Header row**: the number of the row in the file that contains the column names, detected automatically. Set 0 if the file has no header row.
- A preview shows the first eight rows and the total number of rows.

2. Column mapping

For each piece of information, a menu shows the column of your file that contains it: **Transaction date**, **Transaction type**, **Security (ticker, ISIN or name)**, **Security name**, **Quantity**, **Unit price**, **Total amount**, **Fees**, **Currency**, **Exchange**. Fields marked with an asterisk are mandatory; choose "— none —" for a missing piece of information. For the price, a "Unit price" or "Total amount" column is enough. If a mandatory field is missing, the assistant stops and says what remains to be done.

3. Transaction types and securities

- On the left (if a type column is matched), each distinct value in the file, its number of rows and its **Interpretation**: Buy, Sell, Dividend or **Ignore the row**. Change as needed.
- On the right, each security in the file, the proposed **Yahoo Finance ticker** (editable, for example MC.PA for LVMH in Paris), the **Name found** and the **Status**: "as is", "converted (Bloomberg, Google, Reuters)", "found" or "not found". For a security that is not found, type in its ticker by hand, otherwise its rows are likely to be ignored.

4. Result

- **Currency of the prices in the file**: **Automatic detection (recommended)**, **Trading currency of each security** or **Everything is in euros** (see the fiche on currencies).
- Notes flag ignored rows, currency conversions and prices far from the price of the day.
- The table shows the transactions in the project format.

Finally click **Analyse this portfolio**. The **Download the converted file (project format)** button saves the result under the name transactions_converties.csv: uploaded again later, this file is read directly.

Good to know

The converted file is remembered for the session: uploading the same file again does not reopen the assistant. When signed in, save it with **Save to my space** so you do not have to redo these steps.

How does the software find a security's ticker (ISIN, name)?

For each security, the software needs its **Yahoo Finance ticker**, that is, the code that lets it download its prices: MC.PA (LVMH, Paris), AAPL (Apple, New York), SAP.DE (SAP, Xetra), ULVR.L (Unilever, London). The suffix indicates the exchange (.PA Paris, .DE Xetra, .AS Amsterdam, .MI Milan, .L London, .SW Switzerland, no suffix for the United States).

What you can put in the security column

Form	Example	Handling
Yahoo ticker	MC.PA, AAPL	Used as is
ISIN code	FR0000121014	Looked up (see below)
Company name	Air Liquide	Looked up
Bloomberg, Google, Reuters code	MC FP, EPA:MC, AAPL.O	Translated offline (see the dedicated fiche)
Ticker without an exchange	MC, AIR, TSLA	Exchange found thanks to prices (see the dedicated fiche)

A 12-character code starting with two letters and ending with a digit is treated as an ISIN; its check digit (Luhn algorithm) is verified during column recognition and in PDFs.

The search order for an ISIN or a name

1. **Offline first**: the built-in table of common ETF ISINs, then the one for CAC 40 shares and large US stocks, then the memory of securities already recognised, then the local securities database (by ticker, by ISIN, or by name).
2. **Otherwise, the Yahoo Finance search engine** (Internet required), with the ISIN and then, if the file contains one, the name of the security.

Among the answers, only equities, ETFs, funds and indices are kept, and the preferred listing is that of the **stock exchange of the ISIN's country**: FR → Paris, DE → Xetra then Frankfurt, NL → Amsterdam, IT → Milan, ES → Madrid, GB → London, CH → Switzerland, US → New York, etc. For an Irish (IE) or Luxembourg (LU) ISIN, typical of ETFs, a euro listing is preferred (Paris, Xetra, Amsterdam, Milan), then London and New York.

The answer is remembered

A security found online is kept in the local memory: next time, it is recognised immediately, even without Internet.

If the chosen exchange does not suit you

In the assistant (step 3), replace the proposed ticker, for example SAP.DE, with another ticker for the same security. The choice of exchange changes the currency and the prices used, not the quantity held.

Bloomberg, Google Finance, Reuters codes and the "Exchange" column

The security codes used by professional terminals are translated into Yahoo Finance tickers **offline**, thanks to a table of exchange codes built into the software.

Translation examples

Source	Written in the file	Becomes
Bloomberg	MC FP, MC FP Equity	MC.PA
Bloomberg	AAPL US Equity	AAPL
Bloomberg	SAP GY	SAP.DE
Bloomberg	700 HK	0700.HK (padded to 4 digits)
Bloomberg	BRK/B US	BRK-B
Google Finance	EPA:MC, NASDAQ:AAPL	MC.PA, AAPL
Reversed form	MC:EPA, MC:FP	MC.PA
Reuters (RIC)	AAPL.O	AAPL
Reuters (RIC)	NESN.S	NESN.SW

For Bloomberg, the word Equity (the "yellow key") is optional. The recognised Bloomberg exchange codes include in particular FP (Paris), GY and GR (Xetra), NA (Amsterdam), IM (Milan), SM (Madrid), LN (London), SW (Switzerland), US, UN, UW and UQ (United States), JP (Tokyo), CN (Toronto), HK (Hong Kong). On the Reuters side, the suffixes .O, .N, .OQ, .A, .P and .K point to the United States, .S and .VX to Switzerland, .MA to Madrid and .I to Dublin.

Limitation: a Reuters code whose suffix is identical to Yahoo's (for example LVMH.PA) is taken as it is, whereas LVMH's Yahoo ticker is MC.PA. Correct it in the assistant if the security has no prices.

A separate "Exchange" column

If the file gives the ticker without an exchange in one column and the exchange in another (a column named place, marché, bourse, exchange, MIC…), the two are combined: MC + XPAR gives MC.PA. The exchange can be written:

- as a MIC code: XPAR, XETR, XAMS, XMIL, XLON, XSWX, XNAS, XNYS…;
- as a Bloomberg or Google code: FP, GY, EPA, ETR, LON…;
- in full words: Euronext Paris, Paris, Xetra, Frankfurt, Amsterdam, Milan, London, Zurich, New York, Tokyo, Toronto…

The import summary says how many codes were converted: "… ISIN, Bloomberg code(s) or name(s) converted into tickers.".

My file gives a ticker without an exchange (MC, AIR, TSLA)

Many files write MC for LVMH or AIR for Airbus, without indicating the exchange. But MC alone designates, on Yahoo Finance, a US company (Moelis); LVMH is MC.PA. The software resolves the ambiguity **using the prices in your file**.

The method

For each short ticker (1 to 6 characters, with no suffix) that is not already a known ticker:

1. it gathers candidates: the securities in the local database that have this root, the first answers from the Yahoo Finance search engine, and the code followed by the suffixes of the main exchanges (New York, Paris, Xetra, Amsterdam, Milan, Madrid, London, Switzerland, Toronto);
2. it retrieves the prices of these candidates;
3. for each one, it compares the purchase and sale prices in your file with the price of the day of each transaction, reading the price in the trading currency, in the main currency or in euros;
4. it keeps the candidate whose **median gap** is the smallest, provided it is **below 15%**. If the gaps are almost equal (the same security listed in Paris and Frankfurt, for example), the order of preference is kept.

Example: MC bought at €740 in January 2024 matches the LVMH price in Paris, not that of Moelis (around $50): the software keeps MC.PA.

What you need to know

- The method uses only purchases and sales (not dividends).
- It needs the candidates' prices: without Internet, only securities already in the local database can be compared.
- If no candidate fits within 15%, the code goes through the search engine; if it is still not found, the assistant opens so that you can enter the ticker.
- The summary says "… ticker(s) without an exchange identified from market prices.".
- If your file also contains an exchange column (XPAR, Euronext Paris…), it is used first.

Securities recognised without Internet: the ISIN tables and the memory

The ETF ISIN table

Trade confirmations and statements often give only an ISIN code and an abbreviated label ("AM.C.C.40 UC.ETF C"). To recognise the most common ETFs **without Internet**, the software contains a table of 31 ISINs, including:

ISIN	Ticker	ETF
FR0013380607	CACC.PA	Amundi CAC 40 UCITS ETF Acc
LU1681043599	CW8.PA	Amundi MSCI World UCITS ETF Acc
FR0011869353	EWLD.PA	Amundi PEA MSCI World UCITS ETF
IE0002XZSHO1	WPEA.PA	iShares MSCI World

ISIN	Ticker	ETF
		Swap PEA UCITS ETF
FR0011871128	PE500.PA	Amundi PEA S&P 500 UCITS ETF
IE00B4L5Y983	IWDA.AS	iShares Core MSCI World UCITS ETF
IE00BK5BQT80	VWCE.DE	Vanguard FTSE All-World UCITS ETF Acc
IE00B5BMR087	SXR8.DE	iShares Core S&P 500 UCITS ETF
DE000A0S9GB0	4GLD.DE	Xetra-Gold

The table also covers Amundi ETFs (Euro Stoxx 50, Nasdaq-100, emerging markets, Stoxx Europe 600, money market), iShares, Vanguard, SPDR, Invesco and Xtrackers, equity as well as bond funds. It is consulted **first**, before the memory: it thus corrects a wrong match that might have been learned earlier.

The share ISIN table

A second table, consulted just after the ETF one, recognises without Internet the shares of the **CAC 40** (with a few former members) and seven large US stocks: Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), Alphabet (GOOGL), Meta (META), NVIDIA (NVDA) and Tesla (TSLA). Examples: FR0000121014 becomes MC.PA (LVMH), FR0000120578 SAN.PA (Sanofi), US0378331005 AAPL. A trade confirmation for one of these shares is therefore read entirely offline.

The memory of recognised securities

Whenever an ISIN, a name or a code is found through the Yahoo Finance search engine, the match is recorded in a memory file (memoire.csv, in the securities database folder). Next time, it is found instantly, even offline.

- The memory is shared by all users of the software on this computer.
- It contains **no portfolio data**: only matches between codes (for example FR0000121014 → MC.PA), with a name and a date.

The local securities database

After the table and the memory, the software searches the local securities database supplied with the software: by ticker, by ISIN code, then by name (“LVMH” finds “LVMH Moët Hennessy Louis Vuitton”).

Without Internet, in practice

An ETF or share ISIN from the two tables, a security already encountered or a security in the local database are recognised. A completely new security cannot be: the assistant opens and you can enter its ticker by hand.

A security is not found: what should I do?

The message

During an automatic import, an unidentified security prevents direct analysis: the assistant opens and displays “Security(ies) not found: …” followed by the codes concerned. At step 3, these securities have the status “not found”.

The solution

1. Look the security up on the Yahoo Finance website (by name or ISIN) and note its ticker, for example AI.PA for Air Liquide.
2. In the assistant, step 3, type this ticker in the **Yahoo Finance ticker** column of the row concerned.
3. Check the result at step 4, then click **Analyse this portfolio**.

Without a ticker, the rows of an ISIN that is not found are ignored (reason “a missing security”). For a name that is not found, the name itself is kept as the code, which will make it impossible to find prices: correct it.

Frequent causes

- **No connection**: online search is impossible; only securities from the ISIN tables (ETFs, CAC 40, large US stocks), the memory and the local database are recognised.
- **Unlisted fund** (some UCITS funds, euro funds, structured products): with no price on Yahoo Finance, it cannot be tracked by the software.
- **Name too vague or abbreviated**: a label such as “AM.C.C.40 UC.ETF C” yields nothing; it is the ISIN that allows recognition.

If the proposed ticker is wrong

Replace it in the same way at step 3. The price check helps you spot an error: a security whose prices differ by more than 25% from the price of the day is flagged (see the fiche on checking the import).

Foreign securities: in which currency should I put the price?

The project rule

The price of a purchase, sale or dividend is expressed in the security’s **trading currency** on Yahoo Finance: dollars for Apple, Swiss francs for Nestlé, pence for London. **Fees stay in euros**. The software then converts into euros at the exchange rate of the day of each transaction.

But many statements give euros

A French bank statement often shows a price already converted into euros. On upload, the software detects this: for each security, it compares the purchase and sale prices in the file with the **actual closing price of the day**, using three possible readings:

Reading	The price in the file is…
Trading currency	in the trading currency (e.g. $185)
Main currency	in pounds instead of pence (London securities only)
Euros	converted into euros (e.g. €169.72)

It calculates the median gap of each reading against the market and keeps the closest one. If it is the euro reading, the security's prices (dividends included) are converted back into the trading currency: price in currency = price in euros × EUR/currency rate of the day.

Example: a purchase of Apple recorded at €169.72 on a day when €1 is worth $1.09 and the share is quoted at $185. Read in euros, 169.72 × 1.09 = $184.99: the reading matches the market, and the price is converted back to $184.99, or about $185.

The safeguards

- The conversion is made only if it clearly improves the agreement with the market (gap reduced by more than 2 points, measured as a logarithmic gap); otherwise the price is kept as it is. This is useful for the Swiss franc, which is worth almost one euro.
- If even the best reading differs by more than 25% from the price, nothing is converted (unless the file's Currency column imposes the euro reading).
- If the file has a **Currency** column, it restricts the readings: "EUR" on all the rows of a foreign security imposes the euro reading; the security's own currency rules out the euro reading.
- Without a connection, prices are kept as they are and the message "Prices not checked against market data (no connection)." is displayed.

Choosing yourself in the assistant

At step 4, **Currency of the prices in the file** offers **Automatic detection (recommended)**, **Trading currency of each security** (no conversion) or **Everything is in euros** (conversion of all foreign securities).

Limitation

The rate used is the market rate of the day; your bank's rate includes its foreign-exchange margin. Small gaps are therefore normal.

London shares: why pence (GBp)?

The principle

On Yahoo Finance, shares on the London Stock Exchange (tickers ending in .L) are quoted in **pence** (GBp or GBX), not pounds: 1 pound = 100 pence. A share shown at 4,000 GBp is therefore worth £40.00. The software knows this: for these securities, the currency is the pound (GBP) with a factor of 0.01.

In the project format

The price of a London share is written **in pence**, as on Yahoo Finance: 2024-01-15,ACHAT,ULVR.L,Unilever,20,4000,1.00 for 20 shares at £40. A dividend is also written in pence. Fees stay in euros.

If your file gives pounds or euros

On upload, currency detection compares the price with the price in pence using three readings:

- in pence: 4,000 → 4,000 GBp;
- in pounds: 40.00 → 40.00 / 0.01 = 4,000 GBp;
- in euros: €46.50 with €1 = £0.86 → 46.50 × 0.86 / 0.01 = 3,999 GBp.

The reading closest to the price is kept and the price is brought back to pence. The summary shows for example “ULVR.L: prices in GBP converted into the quotation unit”.

In a Currency column

The value GBX is read as GBP. It rules out the euro reading, but leaves the choice between pence and pounds.

Symptom of an error

A London row valued 100 times too high or too low betrays a confusion between pounds and pence. Correct the price in the Transactions tab (**Price (trading currency)** column, in pence), or re-import with automatic detection.

Importing a PDF statement of transactions

How the PDF is read

The software reads the text and tables of the PDF (pdfplumber library), page by page.

- A table is kept if it has at least two rows, three columns, and dates on at least two rows: that is the mark of a genuine transaction table. If the page has no table lines, the software also tries to rebuild the columns from the alignment of the text.
- Tables of the same width are joined end to end (statement over several pages); the header repeated at the top of each page is kept only once.

- The resulting table then follows exactly the same path as an Excel file: header row, column recognition, ISIN, dates, numbers, currencies.

When the PDF is rather a trade confirmation

If the text contains “avis d’opéré”, “avis d’exécution”, “confirmation d’exécution”, “confirmation d’ordre” or “trade confirmation”, the software first reads the document as a trade confirmation (one transaction per page). If there is no usable transaction table, it also tries this reading.

The order of the readings

1. No text, or “encoded” text (it displays correctly but extracts as unintelligible characters): the PDF is treated as a scan (see the fiche on image PDFs).
2. Portfolio statement (holdings and cost prices): each holding becomes a purchase on the statement date (see the fiche “Importing a PDF portfolio statement”).
3. Reading of the trade confirmation by its headings, and reading of the statement’s tables. A transaction read by its headings is accepted only if a direction word (achat, vente, buy, sell, dividende…) is written in the document and a price (or, for a dividend, an amount) was found; otherwise the software moves on to the next reading.
4. Reading **by content**, whatever the broker: ISIN code, execution date and the relation quantity × price = amount (see the fiche “A trade confirmation from another bank or broker”).
5. Reading with a **learned model**: this type of document has already been completed once in the form (see the fiche “The software learns your trade confirmations”).
6. If nothing is certain: “Transaction not recognised automatically in this PDF: complete it in the”Complete the transaction” form (the values found are suggested).”.

A password-protected PDF first asks for that password (see the fiche “My PDF is password-protected”).

If nothing is recognised

Instead of a plain refusal, the **Complete the transaction** form opens with the dates, codes and numbers found in the document (see the fiche “Completing a transaction when the PDF is not recognised”). For a long statement in table form, it is often quicker to:

- download the Excel or CSV export of the transactions from your online banking instead;
- or use the diagnostic tool to see what the software reads (see the fiche on diagnostic_pdf.py).

Tip

On your bank’s website, prefer the document’s download button (“PDF format”, “Download”) to the browser’s “Print” function: a downloaded PDF contains the real text, read instantly and exactly; a page “printed to PDF” is sometimes only an image.

Importing a PDF trade confirmation (Bourse Direct example)

A **trade confirmation** (avis d'opéré) is the confirmation your broker sends after each executed order. The software extracts one transaction per page from it.

The confirmation can come from **any bank or broker**: Bourse Direct is only an example. The software first reads the usual headings (table below); if that fails, it reads the confirmation by its content (see the fiche "A trade confirmation from another bank or broker").

What is read

Information	Where the software looks for it
ISIN code	First 12-character code whose check digit is correct, which starts with an existing country code and contains at least 4 digits
Date	Execution, transaction or trade date, "exécuté le", "trade date"; failing that a date close to the word exécution or followed by "achat"/"vente"; issue, settlement, value or delivery dates are discarded
Direction	Achat, vente, souscription, rachat, buy, sell, dividende, coupon…: one of these words must be written in the document
Quantity	"Quantité", "Qté", "Nombre de titres", "Quantity", "Nominal"
Price and currency	"Cours", "Prix unitaire", "Price", with EUR, USD, GBP, €, $, £…
Fees	Sum of all the "Courtage", "Commission", "Frais", "TTF" (financial transaction tax), "Fees" lines
Amount	"Montant net", "Net à débiter/créditer", otherwise "Montant" or "Brut"
Label	After "Libellé :" or "Valeur :", otherwise the name written just after the ISIN

Headings and values may be presented as text ("Quantité : 15"), arranged in a table, or set out in columns without lines, with the heading above the value: the software then pairs each value with the heading placed above it, from the position of the words on the page ("Quantité" above "12" gives "Quantité : 12").

A transaction is kept by this reading only if the ISIN, quantity and date are found, if a direction word is written in the document, and if the price (or, for a dividend, the amount) is found. Otherwise, the software moves on to the reading by content.

Example: a Bourse Direct confirmation downloaded with "PDF format"

The confirmation contains in particular:

28/09/2026 VENTE COMPTANT FR0013380607 AM.C.C.40 UC.ETF C 2 473,90
QUANTITE : -60
COURS : +41,295 BRUT : +2 477,70
COURTAGE : +3,80 TVA : +0,00

The software reads: date 28/09/2026, direction VENTE (sale), ISIN FR0013380607, label "AM.C.C.40 UC.ETF C", quantity −60 (made positive: 60), price 41.295, fees €3.80. The ISIN is in the ETF table: it becomes CACC.PA without Internet, and the abbreviated label is replaced by the official name "Amundi CAC 40 UCITS ETF Acc". The resulting transaction is: 2026-09-28, VENTE, CACC.PA, 60 securities at €41.295, fees €3.80. Check: 60 × 41.295 = €2,477.70 gross, minus €3.80 of brokerage = €2,473.90 credited.

Several confirmations

A multi-page PDF containing one confirmation per page gives one transaction per page. Several confirmations in separate files can be dropped together, in the sidebar upload area as well as on the **Add transactions** page (see the fiche "Uploading several files at once").

After reading

The sidebar summary says "PDF trade confirmation read." (or, for a reading by content, "PDF trade confirmation read from its content (quantity × price = amount): please check the transaction ("Transactions" tab)."). It also reports a price replaced thanks to the market price (see the fiche "Checking the price read against the market price") and the points to check found after reading (see the fiche "Checks after reading a PDF"). Always check the transaction: an unusual confirmation can be misinterpreted.

A trade confirmation from another bank or broker

Every broker lays out its trade confirmations in its own way ("Quantité :", "Nombre de titres", "Qty", "Buy 15 … at 85.12"…). When the usual headings are not enough, the software reads the confirmation **by its content**, without relying on the layout.

What the software looks for

Every trade confirmation contains the same elements:

- an **ISIN code**, recognised by its check digit, with an existing country code and at least 4 digits;
- an **execution date**, written in digits or with the month in letters, in French or English; issue, settlement or value dates are set aside, and when two words compete for a date, the closest one wins;
- **three linked numbers**: quantity × price = gross amount, to the cent, or failing that quantity × price ± fees = net amount;

- the **fees**: the numbers that follow “courtage”, “commission”, “frais”, “costs”, “TTF”…;
- the **direction**: achat, vente, buy, sell, bought, sold, dividende…;
- the **currency** written next to the price.

Several transactions on the same page are read separately: one per ISIN code.

Example

An English confirmation contains:

Date 07-05-2024 14:32
Buy 15 VANGUARD S&P 500 UCITS ETF IE00B3XXRP09 at 85.12 EUR
Value EUR 1,276.80
Transaction costs EUR 2.00
Total EUR 1,278.80

There is no “Quantity” or “Price” heading. The software tries the numbers in pairs and finds 15 × 85.12 = 1,276.80, which appears in the text. It keeps: date 07/05/2024, purchase (“Buy”), ISIN IE00B3XXRP09, 15 securities at EUR 85.12, fees 2.00 (the number after “costs”), and checks that 1,276.80 + 2.00 = 1,278.80, the total written.

When several readings fit

Sometimes two pairs of numbers give the same amount. The software keeps the most likely reading (whole quantity, numbers placed near their headings), but also remembers the others: with Internet, the day’s closing price then decides between them (see the fiche “Checking the price read against the market price”).

After reading

The summary says “PDF trade confirmation read from its content (quantity × price = amount): please check the transaction (”Transactions” tab).”. Check the date, quantity and price in the **Transactions** tab.

Limits

The transaction is kept automatically only if the ISIN, the date, a consistent trio of numbers and the direction are all found (the direction may also come from a quantity preceded by + or −). No reader can guarantee to understand 100% of existing documents: if one of these elements is missing, the software still tries a learned model (see the fiche “The software learns your trade confirmations”), then the **Complete the transaction** form takes over with the values found (see the next fiche).

A transaction statement with one transaction per line (e.g. Interactive Brokers)

Some brokers produce a statement where **each line holds a complete transaction**: a code (ISIN or a plain ticker symbol such as “ESE”, “PAEJ”, “RMS”), the date, the direction (buy, sell), then

the quantity, price, amount and commission. This is the case of Interactive Brokers' "Trade Confirmation Report".

How the software reads it

- It keeps each line that contains a date, a direction word and a code; "Total" lines are ignored.
- On each line, it looks for the three numbers that satisfy **quantity × price = amount** to the cent. Example: 7 × 33.4540 = 234.18.
- The commission is the small decimal number after the amount (less than 5% of the amount).
- A sale is recognised by the word SELL / vente, or by the "−" sign of the quantity.
- A symbol followed by a lowercase letter added by the broker ("LYSXd") is reduced to the symbol ("LYSX"). The security is then identified by its prices (the exchange whose prices match those of the statement), or by the search engine.
- The statement is used only if **all** its transaction lines are consistent; otherwise the "Complete the transaction" form takes over.

Statement printed in "dark mode"

A page printed from a browser in dark mode (light text on a black background, dark table rows) is turned back into "black on white" area by area, and the table borders are erased before character recognition. Decimal points lost by the reading ("-23418" for -234.18) are recovered thanks to the consistency quantity × price = amount.

Limits

- A commission misread on a line may remain wrong: check the Fees column in the **Transactions** tab.
- The safest option is still to download the statement as CSV or as a "text" PDF from the broker's client area rather than printing it.

Foreign brokers' notes and special cases (German, Spanish, Italian, Dutch…)

Content reading does not depend on the document's language: it always looks for an ISIN, a date and three numbers such that **quantity × price = amount**. It also knows the useful words of several languages.

What is recognised

- **The direction**: achat / vente, buy / sell, Kauf / Verkauf, compra / venta, acquisto / vendita, aankoop / verkoop, as well as savings plans (« investissement programmé », Sparplan, piano d'investimento, plan de inversión, Round up, Saveback) and fund redemptions (« REPRISE SICAV », « REPRISE F.C.P. ») read as sales.
- **The execution date**: « Kauf am », « Handelszeit », « Trade time », « Datum en tijdstip van uitvoering »… The value date (Valuta, Value Date, Valutadatum), the booking date and the document's print date are set aside.

- **The quantity**: a number followed by « Stk. », « titre(s) », « Pz. », « tít. », « shares »… is the quantity; a number followed by a currency (« 55.13 EUR ») is a price or an amount.
- **Fees**: brokerage, commission, Fremdkostenzuschlag, Provision, Makelaarsloon, Beurstaks, entry charges… including when written under their label (« Montant total des frais » with the value on the next line).
- **Fractional shares** (« 0.4534 Stk. at 55.13 = 25.00 »): since the displayed price is rounded, a small gap between quantity × price and the amount is accepted.
- **Several transactions in one document**: a Bourse Direct statement with several securities, a Fortuneo order filled in several parts (« Exécution fractionnée »), a DEGIRO transactions statement (one line per transaction, a sale being recognised by the « − » sign of the quantity). A tax written on a separate line (« TAXE TRANSACT FINANCIERES … TTF 1,45 ») is added to the fees of the transaction on the same security.
- **Crédit Agricole notes** (several transactions per page, letter layout): the date and direction (« 22.03.2024 ACHAT COMPTANT ») are written above the security line (« DE 3 ACTIONS ENGIE (FR0010208488) »); each transaction is attached to its own block. Fees include the VAT on fees (« FRAIS DE NEGOCIATION SOUMIS A TVA 0,85 » + « TVA 20,00 % 0,17 » = 1.02). The page's decorative frames are not taken for a table (they used to cut words into pieces).
- **Prices with 3 decimals or more** (« 3 × 15.392 = 46.18 »): rounding of the amount to the cent is accepted.
- Account numbers (IBAN), postcodes, page numbers and legal mentions (« § 4 Nr. 8 ») are not taken for amounts.

The test bench

The application is checked against **160 real trade notes** from 10 European brokers, published (anonymised) by the free software Portfolio Performance under the EPL licence. Current result: 103 notes read entirely right, 13 more read with the right date, direction and quantity but a price in another currency, only 1 read wrong, the others handing over to the « Complete the transaction » form. All Bourse Direct and BoursoBank notes of the bench are read.

Limits

- A note without an ISIN code (crypto-currencies, options, some derivatives) is not read automatically: the **Complete the transaction** form takes over.
- For a transaction in a foreign currency, the price kept is the one written next to the quantity; check the currency in the **Transactions** tab.
- Corporate actions (merger, redemption) are not read automatically yet. Bonds and funds have their own fiche.

Bond or fund (UCITS) trade confirmations

Content reading always checks an equation between the numbers of the document. For a share it is quantity × price = gross amount. Bonds and funds each have their own variant.

Bond: nominal × price in % = gross amount

A bond's price is expressed **as a percentage of the nominal** (of par). Example: a nominal of €5,000 bought at 101.25% costs 5,000 × 101.25 / 100 = €5,062.50. The software recognises this case when the price is followed by « % », or when the document talks about a bond (obligation, OAT, bond, Anleihe, nominal…).

- **Quantity recorded**: the nominal divided by 100, i.e. 50 « units » of €100 at a price of 101.25. So quantity × price = amount remains true in the portfolio, and the price stays readable as a percentage of par.
- **Accrued interest** (« coupon couru », « intérêts courus », Stückzinsen, accrued interest): the share of the next coupon that belongs to the seller; the buyer advances it and gets it back when the coupon is paid. It is **not a fee**: it is not added to the fees, but it is used to check the net amount (gross + fees + accrued interest = net). The price kept is therefore the clean price (excluding accrued interest).
- Bonds rarely have a price on Yahoo Finance: the security may require you to give a ticker or to be entered by hand, and its valuation will then follow the last known price.

Fund (UCITS, SICAV, FCP): units × net asset value = gross amount

- **Units** are often fractional (« 12.3456 units ») and rounded on the confirmation: a small gap due to this rounding is accepted.
- The **net asset value** (NAV, « valeur liquidative ») is read as the price.
- **Entry charges** (« Droits d'entrée (2,00 %) : 37,62 EUR ») are fees: gross + entry charges = net.
- A subscription is a purchase; a redemption (« REPRISE SICAV », « REPRISE F.C.P. », « rachat ») is a sale.

Checking the price read against the market price

In a trade confirmation, several pairs of numbers can satisfy quantity × price = amount. To choose the right reading, the software compares the price read with the **security's closing price on the day of the transaction**, downloaded from Yahoo Finance.

The rule

For a purchase or sale read in a trade confirmation:

- if the price kept differs by **more than 15%** from the day's closing price,
- and another consistent reading of the same document gives a price **within 5%** of that closing price,

then that other reading replaces the first: the quantity, the price and, if known, the fees of that reading are used. Otherwise, nothing changes.

Example

Two readings fit: 10 × 65.02 = 650.20 and 5 × 130.04 = 650.20. The day's closing price is €130.50. The first reading is 50% away from it (more than 15%), the second 0.4% (less than 5%): the software keeps 5 securities at €130.04.

The sidebar summary reports it, in the form: "[ticker] ([date]): price read [old] replaced by [new], the reading that matches the day's closing price ([price])."

In the "Complete the transaction" form

When the **Complete the transaction** form opens and the security and date are known, a line gives that day's closing price, for example "Closing price of MC.PA on 04/03/2024: … (Yahoo Finance) — the unit price should be close to it.". If no price suggestion was found, the **Unit price** list puts the numbers closest to that closing price first.

Limits

- The day's price is needed, so usually an Internet connection: without it, the reading is neither checked nor changed.
- Only purchases and sales are concerned, not dividends.
- The tie-break only chooses between readings already present in the document: it never invents a price.
- The closing price is not your execution price: a gap of a few per cent is normal.

Completing a transaction when the PDF is not recognised

When a PDF gives no certain transaction, the software does not just show an error message: it opens the **Complete the transaction** form, prefilled with what it found in the document.

When the form appears

- On the **Add transactions** page, **From a file** tab: for any PDF whose automatic import is not certain (no transaction read, security not found…).
- After an upload with **Upload a file (CSV, Excel, PDF or image)** in the sidebar: when the PDF gives no transaction, whether the file is sent alone or with others.

The reason is shown under the form's title, for example "Transaction not recognised automatically in this PDF…".

Filling in the form, step by step

1. ****Type**** (above the form): BUY, SELL or DIVIDEND, with the direction found in the document selected in advance. For a dividend, the quantity and price are replaced by the **Total amount received** field.
2. ****Execution date****: prefilled with the probable execution date; the field's help lists the dates found in the document.
3. ****Security****: the ISIN code found, followed by the name; you can also type a ticker (e.g. MC.PA), an ISIN or a name.

4. ****Quantity****, ****Unit price**** and ****Fees (€)****: each list contains the numbers of the document, each followed by the words around it (for example “Quantity: [60]”). The best suggestion is already selected; you can choose another or type a value.
5. With Internet, a line gives the security’s closing price on that day: the unit price should be close to it (see the fiche “Checking the price read against the market price”).
6. Check the control line, for example “Check: 60 × 41.295 = 2477.7”: it must match the gross amount of your confirmation.
7. Click **Add this transaction**.

The **Text read in the document** box shows the extracted text, so you can check a value. It also contains the **Prepare an anonymised report** button (see the fiche “Sending a misread PDF without personal data”).

The software learns from it

When you confirm a purchase or a sale, you teach the software to read this type of document: it remembers, on the computer, the heading placed before each value you chose. The next confirmation with the same layout will be read automatically (see the fiche “The software learns your trade confirmations”).

Next

- **Add transactions** page: the transaction joins the **Check before saving** table. There, the form adds one transaction per file; for a second transaction from the same PDF, use the **Manual entry** tab.
- Sidebar, file sent alone: the number of transactions ready is displayed; add others if the PDF contains several, then click **Analyse these transactions**.

Nothing is invented

Every suggested value appears in the document. If the PDF contains no readable text (a scan with no character recognition engine), the form says so: “No readable text in this document: enter the transaction by hand.”.

The software learns your trade confirmations: learned models

Some trade confirmations use headings the software does not know (“Nominal exécuté”, “Px moyen”, “Sens : S”…). The first time, the **Complete the transaction** form asks you to complete the transaction. The software takes the opportunity to **learn the layout** of the document.

What is remembered

When you click **Add this transaction** for a purchase or a sale, the software saves a “model” of this type of document:

- for the quantity, the price, the fees and the amount (quantity × price, if it is written): the **heading** written just before the value you chose (at most three words) and its position on the line;
- the heading written just before the date kept;
- if the direction is only written as a code (“Sens : S”, “Op. : A”) and no word such as achat or vente appears in the document: the link between this code and the type chosen (for example “s” for SELL);
- the document’s **vocabulary**, but only as fingerprints (hashing): the words themselves cannot be read in the file.

No amount, no quantity and no name is stored in clear. The model is kept only if the headings of the quantity and the price were found. Dividends are not learned.

The next time

If no other reading succeeds, the software compares the vocabulary of the new document with that of the models. If the similarity reaches **0.6** (60% of words in common, measured over all the words of both documents), it reads the values placed after the learned headings. The transaction is accepted only if an ISIN, a quantity, a price and a date are found; if the model knows the amount’s heading, quantity × price must also match that amount (within 1%, or within a few cents once the fees are added or subtracted). The summary then says: “PDF trade confirmation read with the model learned from a previous entry: please check the transaction (”Transactions” tab).”.

The direction is read in the document (achat, vente…); failing that, in the learned direction code; if the document has neither a direction word nor a code, the type chosen when learning is used. If the document carries a direction code never seen before (for example “Sens : R” when only “A” and “S” have been learned), the form appears again: you choose the type, and the new code is learned in turn.

Where the models are

In the modeles_pdf.json file of the securities database folder (data/base). The last 50 models are kept; a document very similar to an existing model replaces it, keeping the direction codes already learned. The file is shared by all users of the computer. The software offers no screen to erase it: to forget all models, delete this file.

A scanned or “printed” PDF: character recognition

Text PDFs and image PDFs

A PDF may contain **real text** (each character is stored, as in a PDF downloaded from the bank) or only an **image** of the page (scan, photo, or web page printed with “Microsoft Print to PDF”). The software considers a PDF to be an image when it contains fewer than 20 characters of text.

There are also PDFs with **“encoded”** text: the page displays correctly on screen, but the extracted text is only a string of unintelligible signs (for example (cid:12)). The software

recognises them (strange characters, very few letters or no common word) and treats them exactly like a scan.

Character recognition

For an image PDF, the software "looks at" each page and recognises the letters and digits in it, if an engine is installed:

1. **RapidOCR**, a library that works offline, installed with the requirements-ocr.txt file and built into the installers when their build succeeded in installing it;
2. failing that, **Tesseract**, if it is installed on the computer (in French and English if the French language is available).

Three attempts, from the fastest to the most careful

1. A low-resolution **glance** at each page. It serves three purposes: finding the right way up (portrait or landscape), spotting pages of no interest (terms and conditions, letters: running text with no date, amount or ISIN code), which are then not read, and spotting the **useful band** of each page (the lines with dates, amounts, ISINs, direction words or short labels). The glance's figures are never kept: at low resolution a date can be misread (« 2028 » for « 2026 ») without any calculation revealing it.
2. Then a **fine reading** at high resolution, but only of the useful band of the useful pages. Example: a 5-page scan (the note then 4 pages of terms and conditions) is read in 14 seconds instead of 24; a note followed by long text on the same page, in 7 seconds instead of 12.
3. If it gives no certain transaction, a slower **second reading** on a cleaned image at a higher resolution (scale 4 instead of 3): conversion to greyscale, stretched contrast, page straightened (the tilt is searched between −3° and +3°, in half-degree steps), then conversion to pure black and white (Otsu threshold, which automatically separates the ink from the background). A pale, grainy or slightly crooked scan thus becomes readable.

The precautions taken

- **Page with a dark background** (printed in "dark mode", dark table rows): the page is turned back into black on white area by area, at a higher resolution, and table borders are erased before reading.
- **Rotation**: the page is first read upright; the three other orientations (90°, 270°, 180°) are tried only if it gives almost nothing useful (no ISIN code, nor a long text with expected words: quantity, price, buy, sell, quantité, cours…).
- **Several pages**: they are read at the same time, which divides the waiting time. In a PDF where some pages have real text, only the pages without readable text go through recognition.
- **Repair of misread ISINs**: the letter O read in place of the digit 0 (FRO013380607), an I or an l in place of 1, S for 5, B for 8, Z for 2, or a character read twice (FRO0013380607). A correction is kept only if the corrected ISIN has a correct check digit, an existing country

code and at least 4 digits: the software never invents a code. A word like "EURONEXTPARIS" is never mistaken for an ISIN.

- **Repair of numbers**: in a number, a letter read in place of a digit is corrected (O or o to 0, I, l or | to 1, S to 5, B to 8): "65O,2O" becomes "650,20". The correction only happens in a group that already contains at least two real digits and a decimal comma or point, with no other letter: words are never changed.
- **Run-together words**: reading tolerates words stuck together ("VENTECOMPTANT").
- **Date in the future**: a transaction date later than today is necessarily misread; the transaction is not kept automatically and the **Complete the transaction** form takes over.

The recognised text is then read as a "one transaction per line" statement (see the dedicated entry), then as a trade confirmation: by its headings, by its content (ISIN, date, quantity × price = amount), then with a learned model. A scanned portfolio statement is generally not usable. If no transaction is recognised, the **Complete the transaction** form suggests the values read.

Always check

Reading an image takes time: about 10 to 30 seconds per page depending on the computer (more if the second attempt is needed); the waiting message says so, with the number of pages. Meanwhile, the previous dashboard stays greyed out: this is normal. It remains less reliable than a text PDF. The summary says so: "Image PDF read by character recognition: please check the transactions ("Transactions" tab).". Check the date, quantity and price.

A photo or screenshot of a trade confirmation (JPG, PNG)

In addition to CSV, Excel and PDF files, the software accepts a **photo** or a **screenshot** of a trade confirmation: JPG, PNG and WEBP formats, as well as HEIC (iPhone photos) if the pillow-heif library is installed. Upload it like any other file, from **Upload a file (CSV, Excel, PDF or image)** in the sidebar or from the **Add transactions** page.

What the software does

The image is first **prepared**, then read exactly like a scanned PDF (see the fiche « A scanned or printed PDF: character recognition »):

1. **The right way up**: a phone held upright often records the photo lying on its side, with a separate rotation flag (EXIF data); the photo is turned upright.
2. **The sheet flattened**: the 4 corners of the sheet are searched automatically (outline of the sheet on the table, OpenCV library), then the perspective is corrected: a sheet photographed at an angle becomes a rectangle again.
3. **Even lighting**: on a photo, the shadow of a hand or the gradient of a lamp is removed (the background is estimated around each point then erased). A screenshot, evenly lit, is left unchanged.
4. **The right size**: a small screenshot is enlarged (×2 or ×3) so that digits are big enough; a 12-megapixel photo is reduced to the equivalent of an A4 page at 300 dots per inch (bigger does not read better, only more slowly).

The transaction is kept automatically only if it is consistent (quantity × price = amount, plausible date); otherwise the **Complete the transaction** form opens with the values read.

If the photo is misread: indicate the 4 corners

When the sheet is not found (a corner outside the frame, a light table under a white sheet…), the form offers **Photo misread? Indicate the 4 corners of the sheet**:

- click the 4 corners of the sheet, in this order: top left, top right, bottom right, bottom left (if the streamlit-image-coordinates library is installed; otherwise, set each corner with two sliders, horizontal and vertical, following the red outline on the preview);
- click **Straighten and read again**: the photo is flattened using your corners then read again;
- **Start again** clears the corners placed.

Tips for a good photo

- Lay the sheet flat on a surface darker than it, and frame the whole sheet.
- Avoid the flash (glare); indirect daylight is ideal.
- A screenshot of your bank's app reads better than a photo of the screen.
- The safest remains the PDF downloaded from your online account: it contains real text and is read without character recognition.

Why is my scanned PDF rejected?

An image PDF is no longer simply refused: when it is not read automatically, one of the messages below is shown at the top of the **Complete the transaction** form (see the fiche "Completing a transaction when the PDF is not recognised").

"Scanned PDF (image): it cannot be read automatically…"

No character recognition engine is installed: the PDF contains no text to read. The form then says "No readable text in this document: enter the transaction by hand.": you have to type the values yourself.

If you launch the software from the source code, you can install the engine with the command python -m pip install -r requirements-ocr.txt.

"The text of this PDF is"encoded"…"

The PDF contains text, but it extracts as unintelligible signs, and no character recognition engine is installed. Same solution: install the engine, or complete the transaction by hand in the form.

"Image PDF (scan, photo or page printed with"Print to PDF"): the text was read by character recognition, but no transaction was recognised…"

The engine did read the page, twice (the second time on a cleaned and straightened image), but no reading found a certain transaction. Possible causes: blurred or very low-resolution

image, unreadable ISIN, a document that is not a trade confirmation (statement in table form, portfolio summary). The form then suggests the dates, codes and numbers read on the image.

Why the software does not guess

A misread transaction (a quantity of 60 read as 80, a price shifted by one decimal place) would distort all the calculations without you noticing. The software therefore adds automatically only what it is sure of; for the rest, you confirm the values in the form.

The solutions, from the most reliable to the least

1. **Download the real PDF** from your online banking, with the "PDF format" or "Download" button rather than "Print": the text is then read exactly.
2. **Export the transactions as Excel or CSV**, if your bank offers it.
3. **Complete the transaction** in the suggested form, or enter it by hand: **Add transactions** page, **Manual entry** tab.
4. Run the PDF diagnostic to understand what was read, or prepare an anonymised report to send (see the following fiches).

Checks after reading a PDF

After reading a PDF uploaded from the sidebar, the software runs each transaction through four checks. They block nothing: each point found is added to the sidebar summary, so that you can check it in the **Transactions** tab.

The four checks

Check	Message shown in the summary
The amount written in the confirmation does not match quantity × price, with the fees added or subtracted, within 1%	"[ISIN]: the amount written ([amount]) does not match quantity × price ± fees ([calculation]): check the quantity and the price."
Purchase or sale dated on a Saturday or a Sunday	"[ticker]: transaction dated on a Saturday ([date]), a non-trading day: check the execution date."
Fees above 3% of the amount quantity × price	"[ticker]: fees of [fees] for an amount of [amount] (more than 3%): check the fees."
Same transaction twice (same date, type, security, quantity and price)	"[ticker]: the same transaction appears twice ([date]): confirmation sent twice?"

What each message most often points to

- **Inconsistent amount**: a misread quantity or price (a misplaced decimal comma, a digit misrecognised on a scan). This check applies to trade confirmations, not to statements in table form, and ignores dividends.
- **Non-trading day**: the issue or settlement date was taken for the execution date.

- **High fees**: an amount taken for fees, or genuinely high fees on a small order. Example: 2 securities at €50 (€100) with €4.90 of brokerage: 4.9% of the amount, the message appears. For 30 securities at €172.46 (€5,173.80) with €20.69 of fees, i.e. 0.4%, nothing is reported.
- **Duplicate**: the same confirmation present twice in the PDF.

Several files

When several files are uploaded together, the checks of each file are combined, and a transaction present in two files is counted only once; the summary says so: "… transaction(s) found in two files counted only once.".

What to do

Open the **Transactions** tab and compare the transaction with your confirmation. An error is corrected with **Edit transactions**.

These checks are not shown on the **Add transactions** page, which has its own checks before saving (see the fiche "Which checks block saving?").

Importing a PDF portfolio statement (holdings and cost price)

A **portfolio statement** (or "portfolio valuation", "relevé de portefeuille") lists the securities held on a date, with their quantity, price, valuation and often the average cost price (PRU). The software can use it as a **starting portfolio**.

How it is recognised

The PDF must contain a title such as "Relevé de portefeuille", "Relevé de compte-titres", "Portefeuille titres", "État du portefeuille", "Valorisation du portefeuille", "Inventaire du portefeuille", "Estimation du portefeuille", "Positions au", "Portfolio statement" or "Statement of holdings", and at least one ISIN code.

For each security line, the software looks for:

- quantity × price = valuation, to the cent;
- the **cost price (PRU)**: the number preceded by "PRU", "prix de revient", "prix moyen", "PAM" or "cost price"; otherwise, a number on the line between a third and three times the price, confirmed if quantity × (price − PRU) equals the unrealised gain written;
- the **statement date**: a date preceded by "au", "arrêté au", "en date du", "situation", "as of"…, otherwise the first date in the document.

If a single line is not certain, or the date is missing, the document is not read as a portfolio statement: the software moves on to the other readings.

What each holding becomes

Each holding becomes a **purchase** of the quantity held, at the PRU (or, without a PRU, at the statement price), dated on the statement day, with no fees.

Example: the line LVMH FR0000121014 10 650,20 731,00 7 310,00 808,00 of a statement "Positions au 31/12/2023" reads: 10 × 731.00 = 7,310.00 (valuation); PRU 650.20, confirmed because 10 × (731.00 − 650.20) = 808.00, the unrealised gain written. It becomes: 2023-12-31, ACHAT, MC.PA, 10 securities at €650.20.

What you should know

The sidebar summary reminds you: "PDF portfolio statement: each holding is entered as a purchase at its cost price (PRU), on the statement date; performance is therefore measured from that date.".

- The earlier history (real purchase dates, sales, past dividends) is not known.
- As the PRU may be far from that day's price, a "price(s) far from that day's market price" note may appear: that is normal here.
- The reading only works for a text PDF, not for a scanned statement.
- Then add your new transactions with **Add transactions**.

My PDF is password-protected

Some banks send their statements and trade confirmations as password-protected PDFs. The software can open them, provided you give it the password.

What is displayed

When you upload such a PDF (**Upload a file (CSV, Excel, PDF or image)** area of the sidebar, or **Add transactions** page), a **Protected PDF** box appears with the file name and the message:

"Password-protected PDF: enter the password to open it (often given in the bank's e-mail: date of birth, customer ID…). It is not stored."

1. Type the password in the **PDF password** field.
2. Click **Open the PDF**.
3. The document is decrypted, then read normally (automatic reading, otherwise the **Complete the transaction** form).

If the password is wrong, the message "Incorrect password." is shown: try again.

What happens to the password?

- The **password** is kept nowhere: it is used once, to produce an unprotected copy of the document.

- This **decrypted copy** is kept in memory during the session, so you are not asked for the password at every action. It is not written to disk and disappears when the software is closed.
- If you then save the portfolio to your space, it is the transactions read that are saved, encrypted, not the PDF.

In "My account"

The **Add a portfolio** section of **My account** does not ask for a password: upload a protected PDF from the sidebar.

Uploading several files at once

The **Upload a file (CSV, Excel, PDF or image)** area of the sidebar accepts several files at once: for example all the trade confirmations of a year, or a CSV export and a few PDF confirmations. Their transactions are combined into one portfolio.

What happens

A **Files sent** box is displayed in place of the dashboard and processes each file:

- a file read automatically shows "[name]: [n] transaction(s) read.";
- a protected PDF shows the **Protected PDF** box to enter its password (see the dedicated fiche);
- an unrecognised PDF opens the **Complete the transaction** form; once confirmed, its transaction joins the others;
- an unrecognised CSV or Excel file shows its reason and the advice "Send this file on its own to open the import assistant.": the import assistant does not open for a batch.

When all the files have been read, the analysis starts. While some are waiting, the "Waiting: …" list names them, and the "Analyse the [n] transaction(s) already read" button lets you analyse without them.

Duplicates between files

A transaction present in several files (same date, same type, same security, same quantity, same price) is counted only **once**. The sidebar summary says so: "… transaction(s) found in two files counted only once.". It also includes the checks of each file (see the fiche "Checks after reading a PDF").

Keeping the result

As for a single file, when signed in, the **Save to my space** block appears under the upload area; the suggested name is of the form "3 files". Without an account, the result only holds for the session.

To complete an existing portfolio

To add confirmations to a portfolio already saved, use the **Add transactions** page instead: its **Files (CSV, Excel, PDF or image)** area also accepts several files and spots the transactions already in the portfolio.

Diagnosing a misread PDF with diagnostic_pdf.py

The diagnostic_pdf.py script, at the root of the project, shows exactly what the software reads in a PDF. It is mainly aimed at those who have the source code (students, developers) and is used to understand, or to get corrected, a faulty reading.

Running it

In a terminal opened in the project folder (for example the VS Code terminal):

```
python diagnostic_pdf.py "C:\path\to\confirmation.pdf"
python diagnostic_pdf.py "C:\path\to\confirmation.pdf" --anonyme
```

The second form produces a report without personal data (see below).

What it displays

- the file name and size, the number of pages and the number of characters of embedded text;
- **text PDF** (at least 20 characters): the mention “PDF TEXTE (lecture exacte)”, then the text of each page and each detected table, with cells separated by |;
- **image PDF** or “encoded” text: the recognition engine used (RapidOCR, Tesseract or AUCUN, meaning none), then, for the first page, the text read in each of the four orientations with its score, and the text finally kept after correction of the ISINs and numbers;
- the **reading by content**: ISIN codes, dates, suggested date, suggested transaction and numbers found with their context (what the form would suggest);
- finally the **result**: the table of transactions obtained and its nature, or the error message.

Nature	Reading
pdf_tableau	statement of transactions in table form
pdf_avis	trade confirmation read by its headings
pdf_contenu	trade confirmation read by its content
pdf_modele	trade confirmation read with a learned model
pdf_positions	portfolio statement (holdings and PRU)

Nature	Reading
pdf_ocr	reading by character recognition

A password-protected PDF cannot be diagnosed this way: the script stops on the protected-PDF message.

The report

The result is also saved in the diagnostic_pdf.txt file, next to the script. Note: this file contains all the text of your document.

With --anonyme, the report is saved in diagnostic_pdf_anonyme.txt: names, address lines, e-mail addresses, IBANs, phone numbers and account or reference numbers are masked; ISIN codes, dates and amounts are kept. This is the file to send in order to get a faulty reading corrected (see the fiche “Sending a misread PDF without personal data”).

Sending a misread PDF without personal data: the anonymised report

For a misread type of confirmation to be better recognised in a future version, the software’s creator needs its text, not your personal data. The software therefore prepares an **anonymised report**.

From the software

In the **Complete the transaction** form, open the **Text read in the document** box and click **Prepare an anonymised report**. The file rapport_lecture_pdf_anonymise.txt is downloaded. It contains: the file name, whether the text is “encoded”, the ISIN codes and dates found, the suggested date and transaction, then the text read, anonymised.

Nothing is sent automatically: you send this file yourself, if you wish.

From the source code

python diagnostic_pdf.py file.pdf --anonyme produces the same report, followed by the result of the reading, in diagnostic_pdf_anonyme.txt.

What is masked, what is kept

Masked	Kept
lines containing a name or an address (Monsieur, Madame, titulaire, client, address, street number and name, postcode followed by a town)	ISIN codes
e-mail addresses, replaced by “[e-mail]”	dates
IBANs, replaced by “[IBAN]”	amounts, prices, quantities
phone numbers, replaced by “[téléphone]”	headings (“Quantité”, “Cours”, “Courtage”…)
account, reference, order, SIREN… numbers,	

Masked	Kept
and long strings of digits, replaced by “[numéro]”	

Masking is automatic: read the file again before sending it, as data written in an unusual way may slip through.

The test bench of real confirmations

The project checks its PDF reading on 20 fictitious confirmations with very different layouts (various headings, English, columns without lines, pale, grainy and crooked scans, “encoded” text, portfolio statement…), in tests/avis_fictifs.py and tests/test_pdf_universel.py. The tests/donnees/vrais_avis/ folder can also hold **real** anonymised confirmations. Its README.md file explains the procedure:

1. mask the personal data (anonymised report above; for the PDF itself, black it out with PDF software);
2. copy the PDF into this folder;
3. add one line per transaction to attendus.csv, in the format fichier;date;sens;isin;quantite;cours;frais (date DD/MM/YYYY, numbers with a decimal point, fees empty if unknown);
4. run python -m pytest tests/test_pdf_universel.py: each document must be read exactly as stated.

How do I check what the software has read?

After an import, take a minute to check the result. The software helps you in three ways.

1. The sidebar summary

After an automatic import, a box summarises what was understood: number of transactions and securities, PDF read (and how), prices replaced thanks to the market price, points to check after reading a PDF, columns identified without a header row, tickers and ISINs converted, order of the dates, ignored rows, currency conversions. At the bottom of the sidebar, the “Transactions” line reminds you of the number of transactions analysed. Compare it with the number of rows in your statement.

2. The price alerts

Each purchase and sale price is compared with the closing price of the day. If prices differ from it by more than 25%, a note flags it, for example: “AAPL: 2 price(s) far from that day’s market price (median gap +38%) — check the ticker, currency or a stock split.”.

Possible causes: - **wrong ticker** (namesake on another exchange); - **wrong currency** (euros read as dollars, pounds instead of pence); - **stock split**: after a split, the price history may no longer match the price paid at the time (see the fiche “Stock split or reverse split”); - a simple data entry error in the file.

An alert does not stop the analysis: it is up to you to judge.

3. The Transactions tab

In **Transaction history**, all transactions are listed, from most recent to oldest, with the type, ticker, security, quantity, price converted into euros, fees, currency and the **Price in currency** as read. The **Type** and **Securities** filters help you find a row. An error is corrected directly with **Edit transactions** (see the dedicated fiche).

Recommended checkpoints

- the quantities held (Holdings tab) match your portfolio statement;
- the dates are in the right day/month order;
- dividends are total amounts, with a quantity of 0;
- foreign securities have a consistent price in their currency.

Saving an imported portfolio to my space

An uploaded file is kept only for the session. To find it again later, save it to your encrypted personal space. You must be signed in (the **Sign in** button at the top of the sidebar).

From the sidebar

1. Upload the file with **Upload a file (CSV, Excel, PDF or image)** and let the software analyse it (assistant included, if needed).
2. Under the upload area the **Save to my space** block appears, with a **Portfolio name** field pre-filled with the file name.
3. Change the name if you wish, then click **Save**.
4. The message ""…" has been saved to your space (encrypted)." confirms the operation.

What is saved is the portfolio **after reading and conversion**, in the project format (Yahoo tickers, prices in the trading currency): it will be reopened without going through detection again.

The block appears only once the portfolio has been analysed successfully. To then work on the saved version, remove the file from the upload area and choose "My space · portfolio name" in the "Data" list.

From the "My account" page

Click **My account**, then use the **Add a portfolio** section: choose a CSV, Excel or PDF file. If it is recognised automatically, the number of transactions is displayed; enter the name and click **Save**. Otherwise, a message invites you to go through the sidebar, where the import assistant will guide you.

Beware of identical names

Saving under the name of a portfolio already in your space (ignoring capital letters) **replaces** that portfolio; the old version can still be recovered with **Undo last change** in “My account”. Choose a different name if you want to keep both.

Adding new transactions to an existing portfolio

There is no need to resend the whole history with every new order: send only the new movements, and the software merges them with the existing ones.

Where the page is

- in the sidebar, under “Data”, the **Add transactions** link below the upload area: it applies to the portfolio displayed;
- in **My account**, under each portfolio in **My portfolios**, the **Add transactions** button.

The page shows which portfolio it concerns. **Back to the dashboard** closes it without changing anything.

Three ways to add

- ****From a file** tab**: the** Files (CSV, Excel, PDF or image)** area, which accepts several files at once (several trade confirmations, an export of the latest transactions…). Each file goes through the same automatic reading as the main upload; the message gives the number of transactions read, for example “avis.pdf: 1 transaction(s) read.”. A file already read on the page is not read again. A protected PDF first asks for its password (**Protected PDF** box). A split or reverse split notice opens the **Corporate action** box (see the fiche “Stock split or reverse split”).
- ****Manual entry** tab**: an order typed in by hand (see the dedicated fiche).
- The two can be combined: all the transactions accumulate in the same list.

If a file is not understood automatically, the reason is displayed; for a PDF, the **Complete the transaction** form opens with the values found (see the dedicated fiche). This page has no import assistant: for an unusual format, first upload the file from the sidebar, download the converted file, then add it here.

The check before saving

The **Check before saving** box lists the transactions to be added. The cells are editable. The **Add** column lets you untick a row; the **Status** column shows “new” or “already in the portfolio” (duplicates are unticked by default). A summary line gives the number of transactions added and the total after adding. Blocking checks are displayed in red (see the dedicated fiche).

Saving

Click **Save transactions**. The selected transactions are merged with the existing ones and sorted by date. - Portfolio in your space: it is saved again, encrypted, and the previous version is kept

(**Undo last change** in **My account**). - Uploaded file or example portfolio: the addition holds for the session; download the updated file to keep it.

Clear all empties the current list without saving anything.

Stock split or reverse split: adjusting earlier transactions

After a **stock split** or a **reverse split** (share consolidation), Yahoo Finance corrects its prices retroactively. Your old transactions, however, are still expressed in the old units: they must be adjusted. The software does it from the corporate action notice sent by your broker.

How to do it

1. Open the portfolio's **Add transactions** page, **From a file** tab.
2. Drop the split or reverse split notice (text PDF) into the **Files (CSV, Excel, PDF or image)** area.
3. The **Corporate action** box summarises what was read, for example "Split of Michelin (ML.PA) on 16/06/2023: 1 old → 4 new.", then the number of transactions that will be converted.
4. Click **Apply to earlier transactions**.

What the software recognises

- the words "division" (du nominal, d'actions), "split", "fractionnement", "regroupement", "reverse split";
- an ISIN code;
- the ratio: "divisé par 4", "1 action ancienne pour 4 actions nouvelles", "10 actions anciennes pour 1 action nouvelle", "parité : 1 pour 4", "1 old for 4 new";
- the effective date: "date d'effet", "ex-date", "détachement", "à compter du", "effective"..., otherwise the execution date.

The calculation

The factor is the number of new shares for one old share: 4 for a 4-for-1 split, 0.1 for a 10-to-1 reverse split. For each **purchase or sale of this security dated before the effective date**:

new quantity = quantity × factor
new price = price ÷ factor

The amount (quantity × price) does not change, and Yahoo Finance prices, already adjusted, match your prices again. **Dividends** (total amounts) do not change, nor do transactions after the effective date.

Example: 10 Michelin shares bought at €120 on 01/03/2022, 4-for-1 split on 16/06/2023. The purchase becomes 40 shares at €30 (10 × 120 = 40 × 30 = €1,200). A €45 dividend received in 2023 and a purchase on 01/09/2023 stay as they are.

Saving

The adjustment is saved like other changes: the message "… transaction(s) of … adjusted." confirms it.

- Portfolio in your space: saved again, encrypted, the previous version can still be recovered with **Undo last change**.
- Uploaded file or example portfolio: the adjustment holds for the session; download the updated file.

If the notice is not recognised

With no transaction of the security before the effective date, the box says there is nothing to adjust. If the notice is not read (scan, ratio missing), correct it yourself in the Transactions tab with **Edit transactions**: quantity multiplied and price divided by the same factor. Other corporate actions (mergers, spin-offs, free share allotments) are not handled automatically.

Entering an order by hand

Manual entry is found on the **Add transactions** page, **Manual entry** tab.

The fields

Field	Content
Date	Transaction date (today by default), in DD/MM/YYYY format
Type	Buy, Sell or Dividend
Security	Yahoo ticker, ISIN code, company name or Bloomberg code
Quantity	Number of securities (1 by default); 0 for a dividend
Unit price (security currency) or dividend amount	Price of one security in its trading currency (dollars for Apple, pence for London); for a dividend, the total amount received
Fees (€)	Brokerage fees, in euros

Click **Add to the list**: the transaction joins the check table. The form is emptied for the next entry.

Security recognition

- A known ticker or one with a Yahoo exchange suffix (MC.PA, SAP.DE) is taken as it is.
- An ISIN, a name, a Bloomberg code or a short ticker are looked up as during an import: ETF table, memory, local database, then Yahoo Finance.
- In case of failure: "Security not found: …. Enter its Yahoo Finance ticker (e.g. MC.PA).".

When the security is found through a search, its official name is used in the list; otherwise, what you typed serves as the name.

Mind the currency and the price

Unlike an imported file, a price typed by hand **is not checked** against market prices: it must be in the security's trading currency. For 10 Apple shares bought at $185, enter 185, even if your bank debited you in euros.

A price left at 0 for a purchase or a sale blocks saving ("Zero quantity or price for …"). Likewise, a future date is refused.

Example

Purchase of 5 Air Liquide at €172.10 on 12/09/2024, €1.99 of fees: Date 12/09/2024, Type Buy, Security AI.PA (or Air Liquide, or FR0000120073), Quantity 5, Price 172.10, Fees 1.99. Then **Add to the list** and **Save transactions**.

How are duplicates detected?

When transactions are added, each new transaction is compared with those in the portfolio and with the other new transactions in the list.

The exact rule

Two transactions are considered identical if they have:

- the **same date**;
- the **same ticker**;
- the **same type** (buy, sell, dividend);
- the **same quantity** (to within one millionth);
- a **price equal to within 0.5%**: $|\text{price 1} - \text{price 2}| \,/\, \max(\text{price 1}, \text{price 2}) \leq 0.5\%$.

Fees and the name of the security are not compared.

Example: a purchase of 5 Air Liquide on 01/03/2024 at €170.00 is already in the portfolio; a statement gives the same transaction again at €170.40. Gap: 0.40 / 170.40 = 0.23%, below 0.5%: it is a duplicate. At €171.00, the gap would be 0.58% and the transaction would be considered new.

What the software does

A duplicate appears in the check table with the status "already in the portfolio" and its **Add** box unticked. It is therefore not added, unless you tick the box again.

Practical consequence: adding the same file twice changes nothing. You can therefore upload an export that overlaps the previous one without fear of double counting.

Two genuinely identical orders

If you really placed two identical orders on the same day at the same price, the second is taken for a duplicate: tick its **Add** box again before saving.

Outside the add page

This detection applies only to the **Add transactions** page. A complete file uploaded alone from the sidebar is read as it is: if it contains the same row twice, it will be counted twice (for a PDF, the summary reports it: "… the same transaction appears twice …"). Then delete the extra row in the Transactions tab. When several files are uploaded together, an identical transaction (same date, type, security, quantity and price) is counted only once.

Which checks block saving?

Before saving an addition or a modification, the software checks the consistency of the **whole resulting portfolio** (old and new transactions). As long as a blocking check fails, the message is displayed in red and the save button stays inactive.

The three blocking checks

Check	Message
Future date	"Transaction dated in the future: [security], on [date]."
Zero quantity or price (purchase or sale)	"Zero quantity or price for [security], on [date]."
Sale greater than securities held	"Sale of [quantity] [security] on [date], but only [n] held on that date."

A transaction dated today is accepted. A dividend may have a zero quantity (that is the rule of the format).

How the sale is checked

The transactions are replayed in date order; on the same day, a purchase comes before a dividend, which comes before a sale. The software tracks the quantity held of each security; if a sale makes it negative, the check fails. Example: 3 LVMH bought on 16/01/2024; a sale of 10 LVMH on 02/05/2024 gives "Sale of 10 LVMH on 02/05/2024, but only 3 held on that date.".

The other alerts in the Transactions tab

- Deleting a purchase on which a later sale depends triggers the sale check; the software then advises: also delete the sale(s) that depend on it (same security, later date).
- "The portfolio cannot be empty: keep at least one transaction." also blocks.
- If a security is no longer held after the modification, a warning flags it ("After this change, these holdings are no longer held: …"), without blocking.

What to do

Correct the row concerned directly in the table (date, quantity, price) or untick it. If the error comes from an older transaction in the portfolio, correct it in the Transactions tab.

Deleting or correcting a transaction (Transactions tab)

Any transaction in the portfolio can be deleted or corrected, in the “Portfolio analysis” workspace, Transactions tab.

The steps

1. Click **Edit transactions** (to the right of **Transaction history**). The table becomes editable; the button becomes **Done**.
2. Find the row using the **Type** and **Securities** filters; rows hidden by the filters are not modified.
3. To **delete**, tick the **Delete** box on the row.
4. To **correct**, edit the date, the quantity, the **Price (trading currency)** or the fees directly.
5. A **Summary** lists the “Deleted” and “Corrected” transactions, with the old and the new value (for example “quantity: 10 → 1”).
6. The checks are displayed (see the fiche on blocking checks).
7. Tick **I confirm these changes**, then click **Save changes**.

Cancel all abandons the changes in progress and leaves edit mode.

What is modified

It is the portfolio file itself that is modified, in the project format: the price is that of the security’s **trading currency** (dollars, pence…), not the price converted into euros shown when viewing. The “Currency” column reminds you of this.

What cannot be edited here

The type, ticker and name of a transaction cannot be edited, and no row can be added in this table. To change the security or the type of a transaction: delete it here, then add the correct transaction with **Add transactions**.

Where the modification goes

- **Portfolio in your space**: it is saved again, encrypted; the previous version is kept and **Undo last change** lets you go back to it.
- **Uploaded file or example portfolio**: the modification holds for the session; the message invites you to download the updated file.

Undoing the last change

For a portfolio in your space

With each addition of transactions or saved modification, the software keeps the previous version of the portfolio (also encrypted). The **Undo last change** button restores it. It is found:

- in the Transactions tab, below the table, outside edit mode (or in edit mode as long as no change has been entered);
- in **My account**, under the portfolio concerned.

A tooltip reminds you of the date of the version that will be restored ("Go back to the version of …").

A single level of undo

Only the **last** previous version is kept. After an undo, the button disappears: you cannot go back further. Likewise, a new modification replaces the version kept. Saving a file under a name already used with **Save to my space** replaces the portfolio of that name, and the replaced version becomes the recoverable version.

For an uploaded file or an example portfolio

After a modification made in the Transactions tab, the **Undo last change** button also appears in that tab and returns to the previous state for the session. On the other hand, an addition made with **Add transactions** without an account does not offer an undo button: delete the added rows in the Transactions tab. Session changes disappear anyway when the software is closed.

Tip

Before a series of important corrections, download a copy of the portfolio: the **Download** button in **My account**, or **Download the updated file** without an account.

Without an account: retrieving the updated file

Without being signed in, you can do everything: upload a file, add transactions, correct or delete rows. But **nothing is saved on the computer**: changes hold only for the session and disappear when the software is closed.

The updated file

After an addition or a modification, a message reminds you: "Download the updated file to keep it.". In the sidebar, next to the **Add transactions** link, the **Download the updated file** button saves the complete portfolio, in the project format, under the name <original name>_mis_a_jour.csv. The "File" line of the sidebar information then shows "… (updated)".

Next time, simply upload this file: it is read directly.

The other downloads

- **Download the converted file (project format)** in the import assistant: the file as read, before any addition.
- **Download the selection (CSV)** in the Transactions tab: the transactions displayed, with **prices converted into euros** and extra columns (currency, price in currency). This file is for viewing or analysing in a spreadsheet; to resume the portfolio later, prefer the updated file.

One update at a time

Session changes apply to one portfolio at a time: saving additions or corrections on another portfolio replaces the changes in progress. Download the updated file before moving on to another portfolio.

The simplest way

Create an account: your portfolios are saved encrypted, additions and corrections are kept, and undoing is possible.

The example portfolios: what are they for?

What they are

The software comes with example portfolios, offered in the “Data” list, after your personal portfolios. Depending on the files present in the data folder:

Displayed name	File
Diversified portfolio (multi-asset)	transactions_diversifie.csv
Global equity portfolio	transactions_mondial.csv
Example portfolio	transactions.csv (offered only if it is the only example file)

The diversified portfolio is offered first when it exists. The first two are produced by the project’s scripts generer_portefeuille_diversifie.py and generer_portefeuille_mondial.py.

What they are for

- discovering the software without preparing a file;
- practising reading the indicators on a realistic portfolio;
- testing the addition, correction and deletion of transactions without risk;
- seeing what a file in the project format looks like.

Modifying them

You can add transactions to them or correct them: as for a file uploaded without an account, the changes hold only for the session. The original file is never modified. Download the updated

file if you want to keep the result, or save it to your space by uploading it afterwards from the upload area.

Removing them from the list

The list does not offer a way to hide them; they remain available, after your personal portfolios.

Common import errors and solutions

Message or symptom	Probable cause	Solution
“The file is empty.”	File with no content, or wrong Excel sheet	Check the file; in the assistant, choose the right **Excel sheet**
“Unrecognised column(s): …”	Unknown column name, or header row wrongly detected	In the assistant, match the column by hand and check the **Header row**
“Columns recognised, but not certain about the quantities, prices and amounts.”	Number columns without a meaningful name	Check step 2 of the assistant, then confirm
“Security(ies) not found: …”	Unknown ISIN or name, or no Internet	Enter the ticker at step 3 of the assistant
“… row(s) with an unreadable date (rows …): ignored”	Date written in words, empty or damaged cell	Correct the date in the file
“… with a missing price or amount”	Purchase or sale with neither price nor amount	Complete the file, or match the amount column
“… with a zero quantity”	Purchase or sale of 0 securities	Check the quantity column
“… with a missing security”	Empty security cell, or ISIN without a ticker	Complete the file or enter the ticker
“No transaction read.”	All rows ignored or in error	Check the types (step 3) and the mapping (step 2)
“Unable to analyse the portfolio: Sale impossible on …”	Sale of more securities than held (missing purchase in the history)	Add the missing purchase or correct the quantity
“Prices not checked against market data (no connection).”	No Internet during the import	Normal when offline; check the foreign securities
“… price(s) far from that day’s market price …”	Wrong ticker, currency or stock split	See the fiche on checking the import
Messages about scanned PDFs or PDFs with “encoded” text	Image PDF, or text that cannot be extracted	See the fiches on image PDFs and rejected scans

Message or symptom	Probable cause	Solution
“Transaction not recognised automatically in this PDF: complete it…”	Readable PDF, but no certain transaction	Complete the transaction in the **Complete the transaction** form
“Password-protected PDF: enter the password to open it…”	PDF encrypted by the bank	Enter the password in **PDF password**, then **Open the PDF**
“Incorrect password.”	Wrong PDF password	Check it in the bank’s e-mail (date of birth, customer ID…)
“… the amount written (…) does not match quantity × price ± fees …”, “… a non-trading day …”, “… (more than 3%) …”, “… appears twice …”	Points found after reading a PDF	See the fiche “Checks after reading a PDF”
Values 1,000 times too small	Thousands separator read as a decimal (1.234)	Remove the thousands separator in the file
London shares 100 times too expensive	Confusion between pounds and pence	See the fiche on pence

Row numbers

In the messages “row(s) with … (rows 3, 7)”, rows are numbered from the first data row below the column headers, not counting empty rows. Beyond five rows, the list ends with “…”.

When the analysis fails after the import

The message “Unable to analyse the portfolio: …” is followed, for an uploaded file, by the **Open the import assistant** button: it lets you re-read the file differently. For a portfolio in your space, correct the transaction concerned.

Missing libraries (source code)

If you launch the software from the source code without all the libraries, the messages “To read a PDF, install pdfplumber…” or “To read an Excel file (.xlsx), install openpyxl…” give the command to run. The Windows and Mac installers already contain them.

Full example: from the broker export to an up-to-date portfolio

Here is a complete case: a statement exported as CSV, its import, its saving, then the addition of a trade confirmation and a correction.

1. The broker’s file

Relevé des opérations - Compte titres n° 12345678
Édité le 05/10/2026

```
Date opération;Opération;Code ISIN;Libellé;Qté;Cours;Frais;Montant net
15/01/2024;Achat Comptant;FR0000121014;LVMH;3;740,00;2,00;2 222,00
12/03/2024;Achat Comptant;US0378331005;APPLE INC;10;157,50;5,00;1 580,00
22/05/2024;Coupon;FR0000121014;LVMH;0;;;39,00
28/06/2024;Frais de garde;;;;;;-12,00
18/09/2024;Vente Comptant;FR0000121014;LVMH;-1;700,00;2,00;698,00
```

2. Upload and automatic reading

Upload the file with **Upload a file (CSV, Excel, PDF or image)**. The software:

- detects the semicolon as the separator and ignores the first three lines: the header row is the fourth;
- recognises the columns by their name: “Date opération” (date), “Opération” (type), “Code ISIN” (security), “Libellé” (name), “Qté” (quantity), “Cours” (price), “Frais” (fees), “Montant net” (amount);
- reads the dates as day/month (15/01 cannot be a month);
- classifies “Achat Comptant” as a purchase, “Coupon” as a dividend, “Vente Comptant” as a sale, and ignores “Frais de garde”;
- makes the quantity −1 of the sale positive;
- converts the ISINs into tickers (FR0000121014 into MC.PA, US0378331005 into AAPL);
- checks the currencies: LVMH is in euros; for Apple, if €157.50 multiplied by the EUR/USD rate of 12/03/2024 gives a price close to Apple’s price that day, the price is converted back into dollars.

Consistency check: 3 × 740 + 2 = 2,222; 10 × 157.50 + 5 = 1,580; 1 × 700 − 2 = 698. The columns are consistent.

Result, in the project format (before the currency check):

```
date,type,ticker,nom,quantite,prix,frais
2024-01-15,ACHAT,MC.PA,LVMH,3,740,2
2024-03-12,ACHAT,AAPL,APPLE INC,10,157.5,5
2024-05-22,DIVIDENDE,MC.PA,LVMH,0,39,0
2024-09-18,VENTE,MC.PA,LVMH,1,700,2
```

The sidebar summary states in particular the number of transactions and “1 row(s) ignored (custody fees, transfers…).”. If a security had not been found, the assistant would have opened at step 3.

3. The check

Open the Transactions tab: four transactions. In the Holdings tab: 2 LVMH and 10 Apple.

4. Saving

When signed in, enter “Compte titres” in **Portfolio name** under **Save to my space**, then click **Save**. Remove the file from the upload area and choose “My space · Compte titres” in the list.

5. Adding a trade confirmation

In October, you buy 4 LVMH. Click **Add transactions**, **From a file** tab, and drop in the PDF trade confirmation. The **Check before saving** table shows the purchase with the status “new”. You mistakenly drop in a second copy of the same confirmation: its row is marked “already in the portfolio” and unticked. Click **Save transactions**.

6. A correction

You realise that the October purchase was for 3 securities, not 4. Transactions tab, **Edit transactions**, quantity 4 replaced by 3. The summary shows “quantity: 4 → 3”. Tick **I confirm these changes**, then **Save changes**. In case of error, **Undo last change** restores the previous version.

5. Portfolio analysis

This chapter explains, tab by tab, the “Portfolio analysis” workspace: the overview and the world map, holdings, performance, risk, exposures and real diversification, and then the viewing of transactions. Each indicator has its own entry: the exact formula used by the software, the intuition, a worked example, how to read it and its limits. The Optimisation and Projection tabs are covered in a separate chapter.

The eight analysis tabs: what is where?

The “Portfolio analysis” workspace has eight tabs, displayed at the top of the main area below the six key figures.

Tab	What you find there
Overview	Change in value and invested capital over time, allocation by holding, world map, breakdowns by asset class, region and sector, gains, dividends and fees
Holdings	The table of positions held (quantity, cost basis (PRU), price, value, gain, weight) and the chart of unrealised gains by holding
Performance	TWR, IRR (TRI), gap versus the benchmark, base-100 curve, returns by calendar year, drawdown, beta, alpha, correlation, tracking error, information ratio
Risk	Sharpe, Sortino, VaR and CVaR, distribution of daily returns compared with the normal distribution, skewness, kurtosis, Jarque-Bera test
Exposures	Diagnosis across six dimensions (geography, sectors, currencies, concentration, interest rates, real diversification), findings and suggestions, correlations between holdings
Optimisation	Markowitz efficient frontier and compared allocations (chapter devoted to optimisation and projection)
Projection	Monte Carlo simulation of future value (same chapter)
Transactions	The transaction history, filterable and downloadable; editing transactions is described in the chapter on importing

Good to know

- All tabs analyse the same portfolio, with the benchmark index and the risk-free rate chosen in the **Settings** panel of the sidebar; the VaR level is set in the Risk tab.
- All amounts are expressed **in euros**, including for securities quoted in another currency.
- Correlations between holdings are not in the Risk tab: they are in the **Exposures** tab, **Correlations** sub-tab.

What does the Overview tab show?

The **Overview** tab gives a snapshot of the portfolio in four blocks, from top to bottom.

1. Change in value and allocation

- On the left, **Portfolio value over time**: market value and invested capital over time (see the dedicated entry).
- On the right, **Allocation**: one bar per holding, from largest to smallest, with its weight in the total value. Beyond 15 holdings, the smallest are grouped into a grey "Others" bar. Hovering shows the value in euros.

2. Presence around the world

The map colours each country according to its weight in the **equity portion** (dedicated entry). It only appears if the portfolio contains equities whose country is known.

3. Three rings

Ring	Basis of calculation
By asset class	As a % of the **total value** (equities, bonds, gold, etc.)
By region	As a % of the **equity portion** only
By sector	As a % of the **equity portion** only

The three rings are calculated **on a look-through basis**: an ETF is split between the countries and sectors of the index it tracks. They are all the same size; under each one, a legend gives every share with its percentage (one decimal place), from largest to smallest, and the centre of the ring shows the largest share. Hovering also gives the amount in euros.

Shares below 3% are grouped into "Other (n)", n being the number of groups combined; a single small share keeps its own name (there is never "Other (1)"). Each ring shows at most eight named shares. Asset classes always have the same colours throughout the software: equities in blue, bonds in green, gold in golden yellow, money market in light blue, as in the "Allocations compared" chart of the **Optimisation** tab. A ring that would contain only one group (for example a 100% equity portfolio for asset classes) is not displayed.

4. Four cards

Card	Content
Unrealised gains	Gain or loss not yet cashed in on the holdings

Card	Content
	still held (“unrealised”)
Realised gains	Gain or loss cashed in on sales (“realised”)
Dividends and coupons	Total dividends and coupons entered, since inception
Brokerage fees	Total fees entered, since inception

The sum of the first three cards gives the **total gain** shown at the top of the page.

Reading the “Portfolio value over time” chart

This chart, at the top of the **Overview** tab, superimposes two curves, trading day by trading day.

Curve	Calculation
Portfolio value (blue)	Σ quantity held that day × the day’s closing price, in euros
Money invested (net contributions) (orange, stepped)	Cumulative sum of the money that came out of your pocket

How net contributions are calculated

Each transaction creates a cash flow:

- **purchase**: + (quantity × price + fees);
- **sale**: − (quantity × price − fees);
- **dividend**: − (amount − fees).

Net contributions are the sum of these flows since the start. A sale or a dividend therefore makes the orange line **go down**: that money has come back to you. If you have taken out more than you put in, the line can even fall below zero.

Reading the gap between the two curves

The vertical gap, shaded light blue, is the **gain at that date**: value − net contributions. On the last day it matches the total gain in the key figures, give or take a small difference: the card uses the latest downloaded price, the curve uses the last closing price in the history. A blue curve below the orange curve means that at that date the portfolio was worth less than the net money contributed.

Example

You buy €1,000 of securities (fees included), then receive €30 in dividends. Net contributions go to €1,000, then to €970. If the portfolio is worth €1,100, the gain is 1,100 − 970 = €130.

Tips

The 1M, 6M, YTD, 1Y and All buttons choose the period; hovering shows both values at the same date. A transaction entered on a day without trading (a Saturday, for example) is attached to the next trading day.

The “Worldwide presence” map

The map in the **Overview** tab shows **where the companies are** whose shares you hold, directly or through ETFs.

What is shown

- Each country is coloured according to its weight in the **equity portion** (not in the whole portfolio): the darker the blue, the heavier the country. The scale runs from 0 to the weight of the top country.
- Countries with no exposure are light grey.
- On hover: the percentage of the equity portion, the amount in euros, the number of securities involved and up to six of their names.

Why the United States is often dark

An MSCI World ETF is split according to the composition of its index. For €10,000 invested, the software counts €7,294 in the United States, €591 in Japan, €341 in the United Kingdom, €329 in Canada, €224 in France, and so on. A portfolio with no directly held American shares can therefore be heavily exposed to the United States.

What is not on the map

Bonds, gold and securities whose country is unknown (“Unclassified”) are not shown. When they weigh more than 0.5% of the value, a note under the map gives their share, for example “Not on the map: Bonds 25%, Gold 5%”.

Limits

- The map is an approximation (next entry).
- It cannot be zoomed; hovering still works.
- It disappears if no equity has a recognised country.

Why is the world map an approximation?

The note under the map states: “ETFs split according to the composition of their index (approximation as of 30/09/2026)”. This approximation has several causes, all visible in the code.

1. A typical composition per index, not that of your fund

The software does not read the actual holdings of your ETF. It recognises the **index tracked** (by the security's code, otherwise by words in its name such as "World", "S&P 500", "Emerging") and applies that index's composition.

2. Simplified sources

- MSCI World, ACWI, Emerging Markets and Europe: the top five countries and the sectors come from the MSCI factsheets as of 30/09/2026; the following countries are orders of magnitude, scaled to the "other countries" total in the factsheet.
- S&P 500, Nasdaq-100, Euro Stoxx 50, CAC 40, DAX and bond indices: 2025-2026 orders of magnitude.
- The MSCI ACWI is rebuilt by combining 88% MSCI World and 12% Emerging Markets.

3. An assumption about sectors

For an equity ETF, each country receives the **same sector mix** as the whole index. In reality, Japanese companies do not have the same sector breakdown as American ones.

4. Unrecognised ETFs

A fund whose index cannot be identified from either its code or its name keeps the country on its record; if that country is not a real country ("World", for example), it is counted as "Unclassified" and does not appear on the map.

Is this a problem?

The weights of the major indices change little from one year to the next: the gap stays within a few points at most, which is enough to judge an exposure. For an exact figure, consult your ETF's monthly factsheet from its issuer.

What does "look-through" mean?

An ETF is recorded as **a single security**: for example an MSCI World ETF listed in Paris, in euros. But it contains hundreds of shares from many countries. "Look-through" analysis looks **through** the fund to measure the true exposures.

How the software proceeds

For each holding in the portfolio:

1. **Directly held share**: it keeps its country, its sector and its trading currency.
2. **ETF or equity fund**: its value is split between the countries of the tracked index, then, within each country, between the sectors of the index. The currency used is that of each country (dollar for the United States, yen for Japan, etc.), unless the fund's name indicates currency hedging ("Hedged", "couvert"), in which case everything is counted in euros.

3. **Bond fund**: split between the countries of the bond index; euro currency for a euro-zone index, dollar otherwise.
4. **Gold**: counted separately, with no country or sector, with “Gold” as its “currency”.

Example

A €10,000 MSCI World ETF becomes about €7,294 of American shares (in dollars), €591 of Japanese shares (in yen), €341 of British shares (in pounds), €224 of French shares (in euros), and so on.

Where look-through is used

- the world map and the region and sector rings of the overview;
- the whole **Exposures** tab: geography, sectors, currencies, likely duplicates;
- the “Exposures and diversification” part of the PDF report.

By contrast, the table in the **Holdings** tab shows the classification **of the holding itself** (for example region “World” for a world ETF), without breaking it down. This is why its regions differ from those of the rings.

Current value, invested capital and cost basis (PRU)

Current value

Value of a holding = quantity held × latest price (converted into euros)

The current value is the sum of the values of the holdings still held. The latest price is the one downloaded from Yahoo Finance during the analysis, or the last stored price if the Internet is unavailable.

The PRU (cost basis, “prix de revient unitaire”)

The software replays all your transactions in date order, using the weighted average price method used in France.

- **Purchase**: new PRU = (quantity held × old PRU + quantity bought × price + fees) / (quantity held + quantity bought). Purchase fees are **included** in the PRU.
- **Sale**: the PRU **does not change**. If you sell everything, it starts again from zero.
- **Dividend**: no effect on the PRU.

Invested capital at cost (PRU)

Amount invested = Σ quantity held × PRU

This is the cost basis of the securities still in the portfolio. It appears under the Current value card (“Invested at average cost”). It falls after a sale, because the securities sold drop out of the calculation.

Example

Transaction	Calculation	PRU
Buy 10 shares at €100, €5 fees	(10 × 100 + 5) / 10	€100.50
Buy 10 shares at €120, €5 fees	(1,005 + 1,205) / 20	€110.50
Sell 5 shares at €130	PRU unchanged	€110.50

15 shares remain: amount invested 15 × 110.50 = €1,657.50. At a price of €125, the value is 15 × 125 = €1,875.

Limits

- If your broker calculates a PRU excluding fees, theirs will be lower.
- For a security in a foreign currency, the PRU is in euros, each purchase being converted at the exchange rate of its day (see the entry on securities in foreign currencies).

Unrealised gains

An **unrealised** gain is a gain that has not yet been cashed in: it is what you would gain (or lose) by selling today.

Formulas

Unrealised gain = value − amount invested = quantity × (price − PRU)
Unrealised gain % = unrealised gain / amount invested × 100

Example

15 shares at a PRU of €110.50, current price €125: unrealised gain 15 × (125 − 110.50) = €217.50, or 217.50 / 1,657.50 = +13.12%.

Where to see it

- **Unrealised gains** card in the **Overview** tab: the total, all holdings combined.
- "Gain / loss" and "Gain / loss %" columns of the table in the **Holdings** tab.
- **Unrealised gains by holding** chart in the **Holdings** tab: one bar per holding, green for a gain, red for a loss, sorted from the largest loss to the largest gain. Hovering gives the amount and the percentage.

What it includes and does not include

- **Purchase fees** are already in it, since they are included in the PRU.
- **Fees on a future sale** are not deducted.
- **Taxes** are not deducted (the tax on a sale is simulated in the Wealth advisory workspace).
- For a security in a foreign currency, it includes the effect of the exchange rate.
- Dividends already received are not part of the unrealised gain: they are counted separately.

Realised gains (and sold holdings)

A **realised** gain is the gain (or loss) cashed in on a sale.

Formula

Realised gain = quantity sold × (sale price − PRU) − selling fees

The PRU used is the one at the time of the sale; it remains unchanged after a partial sale.

Example

You hold 20 shares at a PRU of €110.50 and sell 5 at €130, with €4 in fees: 5 × (130 − 110.50) − 4 = €93.50. You have 15 shares left, still at a PRU of €110.50.

Where to see it

The **Realised gains** card (“realised”) in the **Overview** tab adds up the gains and losses from all sales, since inception.

A holding sold in full

A holding sold in full disappears from the table in the **Holdings** tab, which only shows securities still held. It is not forgotten, though:

- its realised gain and its dividends remain counted in the total gain;
- its history remains used for the value curve and the performance indicators;
- its transactions remain visible in the **Transactions** tab.

On the other hand, it no longer enters the allocations, exposures or correlations, which cover the holdings held today.

Limit

The calculation follows the weighted average PRU method. It deducts no tax.

Total gain: why is it not the same as unrealised gains alone?

The formula

Total gain = unrealised gains + realised gains + dividends

The total gain measures everything the portfolio has earned since the start. Unrealised gains are only a part of it: they ignore past sales and dividends received. Fees do not appear as a separate term: they are already deducted, in the PRU (purchase fees), in realised gains (selling fees) and in dividends (fees entered on the dividend line).

Full example

Item	Amount
Unrealised gains (15 shares, PRU	+€217.50

Item	Amount
€110.50, price €125)	
Realised gain (sale of 5 shares at €130)	+€93.50
Dividends	+€30.00
Total gain	**+€341.00**

Check using cash flows: purchases 1,005 + 1,205 = €2,210, net sale 650 − 4 = €646, dividend €30. Net contributions: 2,210 − 646 − 30 = €1,534. Value €1,875. Gain: 1,875 − 1,534 = €341. Both methods give the same result.

The "on invested capital" percentage

Gain on invested capital = total gain / amount invested at PRU

In the example: 341 / 1,657.50 = +20.57%. Note: the denominator is the cost of the securities **still held**. After large sales it becomes small and the percentage can look very high. To measure the quality of your investments, prefer the TWR from the **Performance** tab.

My holdings are at a loss, but the total gain is positive

This is possible: profitable past sales or dividends can offset unrealised losses.

Dividends, coupons and fees: how are they counted?

Dividends and coupons

- They are entered as DIVIDENDE (dividend) transactions, with the **total amount received** in the price.
- The **Dividends and coupons** card in the **Overview** tab adds up these amounts, less any fees entered on the same line. Its subtitle "Gross amounts received" means that the software deducts **no tax**: the figure is the one you entered.
- The "Dividends" column of the holdings table gives the total per holding still held.
- Dividends from a holding sold since then remain counted in the total.

They enter the performance in two ways: in the total gain, and in the TWR and IRR, where a dividend is a cash flow coming back to you. A distributing portfolio is therefore not at a disadvantage against an accumulating ETF.

Brokerage fees

The **Brokerage fees** card ("Since inception") adds up the fees on all transactions: purchases, sales and dividends. They are always entered **in euros**, even for a security in dollars.

They are not deducted a second time: they are already taken into account in the PRU (purchases), in the realised gain (sales) and in the net amount of dividends. In daily returns, the

fees on a purchase appear on the day of the transaction, since the money paid out (fees included) exceeds the value of the securities received.

What is not counted

The software only knows what is in your transactions: custody fees, account-keeping fees, taxes and social contributions are not taken into account unless they are entered.

Securities quoted in dollars or another currency: how are they valued?

All calculations are done **in euros**. A security quoted in another currency is converted as follows:

- each **transaction** is converted at the exchange rate of its day (last known rate at that date);
- each **historical price** is converted at the rate of the same day;
- the **current price** is converted at the day's rate; this rate appears in the information at the bottom of the sidebar (for example "€1 in USD").

Price in euros = price in currency / rate (number of currency units for €1)

London shares, quoted in pence, are first divided by 100. Fees remain in euros.

Two effects in performance

The performance of a foreign security, seen in euros, combines the change in its price **and** that of its currency.

Example: you buy a share at USD 100 when €1 is worth USD 1.00, i.e. €100. A year later, it is quoted at USD 110 (+10%) but €1 is worth USD 1.10: it is worth 110 / 1.10 = €100. In euros, the gain is nil.

Where to see it

- "Currency" column of the **Holdings** tab: the trading currency. The PRU, price, value and gains are in euros.
- **Transactions** tab: the "Currency" and "Price in currency" columns show the price entered before conversion.
- **Exposures** tab, **Currencies and rates** sub-tab: the real currency exposure, ETFs included.

Limit

The rate used is a daily market rate, not the one actually applied by your broker, which often adds a margin. The details of the conversion are explained in the chapter on sources and reliability.

Why do my figures differ from my broker's?

A difference from your broker does not necessarily point to an error. Here are the most frequent causes, in the order in which to check them.

1. **A transaction is missing or entered incorrectly.** Compare the **Transactions** tab with your statements: quantity, price, fees, date. This is the most common cause.
2. **The PRU includes purchase fees.** If your broker shows a PRU excluding fees, theirs is lower and their gain higher.
3. **The price is not taken at the same moment.** The software uses the latest price downloaded during the analysis, kept in memory for one hour; your broker may show a more recent price or the previous evening's.
4. **The exchange rate differs.** The software converts at the day's market rate; your broker applied its own rate, with its margin.
5. **Dividends.** The software counts the amounts entered, without tax. A broker may show amounts net of withholding, or may not include them in the gain.
6. **The notion of performance.** The TWR neutralises your contributions; the return shown by a broker or an insurer is often a different calculation (gain relative to payments, return for the year, etc.).
7. **Fees not entered** (custody fees, account-keeping fees) are not known to the software.

What to do

Correct the erroneous transactions in the **Transactions** tab (see the chapter on importing), then click **Refresh prices** to start again from the most recent prices.

The holdings table, column by column

The **Holdings** tab shows one row per security **still held**, from the largest value to the smallest. Click a column header to sort.

Column	Content
Ticker	The security's Yahoo Finance code
Security	The name
Asset class	Equities, Bonds, Gold, etc. (a security unknown to the reference file is classed as an equity)
Region, Sector	The classification of the holding itself, without ETF look-through; "Unclassified" if the security is not known
Currency	The trading currency; the amounts in the other columns are converted into euros
Quantity	The number of securities held

Column	Content
Avg. cost (PRU)	Cost basis per unit, purchase fees included, in euros
Price	The latest price, in euros
Value	Quantity × price
Gain / loss, Gain / loss %	Unrealised gain in euros and as a percentage of the amount invested
Weight	Share of the holding in the total value, with a proportional bar
Dividends	Dividends received on this holding since inception

The weight

Weight = value of the holding / total value × 100

The blue bar is at full length for the largest holding; the others are proportional. Example: a €15,000 holding in a €100,000 portfolio weighs 15.0%.

The chart under the table

Unrealised gains by holding: one horizontal bar per security, green for a gain, red for a loss, in euros, at the last known price.

Region and sector different from the overview?

The table classes a world ETF under the region “World”; the overview rings and the **Exposures** tab split it **on a look-through basis** between the countries of its index. Both are correct: they answer two different questions.

What does the Performance tab contain?

The **Performance** tab answers two questions: how much have you earned, and have you done better than the benchmark index?

From top to bottom

1. **Four cards**: Total TWR (“Since” + start date), Annualised TWR (“365-day basis”), Annual IRR (“Money-weighted return”) and Difference vs the index (an “outperformance” or “underperformance” badge).
2. **Portfolio and index**: the two curves rebased to 100 at the first common date.
3. **Calendar-year returns**: the TWR for each year, portfolio and index side by side.
4. **Drawdown**: the fall from the last peak, with, in the subtitle, the dates of the peak, the trough and the return to the peak.
5. **Five cards** comparing with the index: Beta, Jensen’s alpha, Correlation, Tracking error and Information ratio.

The orange warning

If the index chosen in **Benchmark index** is a bond or money-market index, a box points out that beta, alpha and correlation measure sensitivity to an equity market: against such an index they have little meaning. In that case, mainly compare returns and volatilities.

Suggested reading order

Start with the annualised TWR and the difference vs the index, look at the curve to see **when** the gap opened up, then the drawdown to measure the worst ordeal endured. The following entries detail each indicator.

"Contribution-neutral" daily returns

Almost all indicators (TWR, volatility, Sharpe, VaR, beta, etc.) rest on one series: the return of each trading day, **without the effect of your contributions and withdrawals**.

The problem

If you buy €1,000 of securities today, the portfolio's value rises by €1,000, but you have gained nothing. The effect of that money must be removed.

The software's formula

$r(t) = (value(t) - flow(t)) / value(t-1) - 1$

where flow(t) = money contributed that day (positive for a purchase, negative for a sale or a dividend). Transactions are assumed to be made **at the end of the day**, at the day's price: money contributed today has not yet "worked".

First day (or restart after selling everything): the previous day was worth 0, so the software compares the end-of-day value with the money contributed: $r = value(t) / flow(t) - 1$. This return captures purchase fees and the gap between the price paid and the closing price: it is often slightly negative.

Days when nothing was invested are not counted.

Example

The portfolio was worth €10,000 last night. You buy €1,000 of securities today, and it is worth €11,150 tonight. $r = (11{,}150 - 1{,}000) / 10{,}000 - 1 = +1.50\%$. Without neutralisation, you would have read +11.5%.

The base-100 curve

Chaining these returns gives an index starting at 100: $index(t) = 100 \times (1 + r1) \times (1 + r2) \times \ldots \times (1 + rt)$. It is the "pure performance" of your choices, comparable to a stock market index; the drawdown is calculated on it.

Limit

Transactions are attached to a trading day and assumed to be made at the closing price. A purchase made during the session at a different price creates a small gap on the same day.

The TWR (time-weighted return) and its annualisation

The total TWR

TWR = (1 + r1) × (1 + r2) × … × (1 + rn) − 1

where the r are the contribution-neutral daily returns. The TWR measures the **quality of investment choices**, independently of the timing and amount of your payments. It is the measure used by fund managers (GIPS standard): a manager does not decide when clients deposit money.

Example: +10% then −5% gives 1.10 × 0.95 − 1 = +4.50%, not +5%.

The annualised TWR

Annualised TWR = (1 + total TWR) ^ (365 / number of days) − 1

The number of days is calendar days, between the first and last day of the return series. Examples:

- +21% over 730 days: 1.21 ^ (1/2) − 1 = +10.00% per year (not 10.5%: gains compound);
- +4.50% over 200 days: 1.045 ^ (365/200) − 1 = +8.36% per year.

Why 365 and not 252? Return is annualised in **calendar** days; 252 trading days are used only to annualise volatility.

Where to see it

“Total TWR” and “Annualised TWR” cards in the **Performance** tab; the annualised TWR is also the “Annualised return” card in the key figures.

Limits

- **Short period**: annualisation amplifies everything. +5% over 91 days gives 1.05 ^ (365/91) − 1 ≈ +21.6% per year, which does not mean the year will deliver +21.6%. Below one year, look at the total TWR instead.
- The TWR ignores your actual money: an excellent TWR on a small sum can coexist with a loss in euros if you invested heavily just before a fall. That is the role of the IRR.

The IRR (internal rate of return)

The IRR (TRI in French) is the annual return **on your money**, as you experienced it, taking into account the dates and amounts of your payments.

The formula

The IRR is the annual rate i that brings the present value of all cash flows to zero:

Σ CF_k / (1 + i) ^ (days_k / 365) = 0

From the investor's point of view:

- a purchase is money **leaving** your pocket: negative flow (fees included);
- a sale or a dividend is money **coming in**: positive flow;
- on the last day, the software acts as if you sold everything: + final value.

days_k is the number of days between the first flow and flow k; flows on the same day are added together.

The calculation

There is no direct formula. The software searches for i by **bisection** between −99% and +1,000% per year, halving the interval 200 times. If there is no solution in this interval, the IRR is not calculated. It is the same principle as the XIRR function of a spreadsheet.

Example

You invest €1,000 on 02/01/2023; a year later it is worth €1,200 and you add €10,000. The following year, the portfolio falls by 10% and is worth €10,080 on 01/01/2025. Flows: −€1,000 (day 0), −€10,000 (day 365), +€10,080 (day 730). The IRR is **−7.72% per year**: you lost money, because the largest sum suffered the fall.

Where to see it

"Annual IRR" card in the **Performance** tab.

Limits

The IRR depends on your payment schedule: it cannot be used to judge a manager or to compare yourself with an index. Over a very short period, it can take extreme values.

Why are the TWR and the IRR different?

The two indicators answer two different questions.

	TWR	IRR
Question	Were my investment choices good?	How much has my money earned?
Effect of payments	Neutralised	Taken into account (amounts and dates)
Used for	Comparing with an index or a fund	Measuring your personal result

Example: positive TWR, negative IRR

- Year 1: €1,000 invested, +20%: it is worth €1,200.
- Start of year 2: you add €10,000. The portfolio is worth €11,200.
- Year 2: −10%: it is worth €10,080.

TWR: 1.20 × 0.90 − 1 = +8.00% over two years. Your investments, on average, worked well. **IRR**: −7.72% per year. You contributed €11,000 and €10,080 remains: the bad year hit the large payment, the good year benefited only the small one.

Reading the gap

- **IRR > TWR**: you invested more before the rising periods (good timing, or luck).
- **IRR < TWR**: you invested more before the falls.
- **IRR ≈ TWR**: few payments, or regular payments without a marked timing effect.

Note: the IRR is already annual, whereas the TWR exists in total and annualised versions. Compare the IRR with the **annualised TWR**.

Which one to believe?

Both are correct. To compare yourself with the index, use the TWR, as the “Difference vs” the index card does. To know what your savings have really earned, look at the IRR and the total gain.

Comparing your portfolio with the index (base 100 and gap)

The “Portfolio and index” chart

Two curves start at 100 on the **first common date**: “My portfolio” (blue) and the chosen index (grey). A dotted line marks the level 100. If your curve ends at 130 and the index’s at 125, your portfolio made +30% and the index +25% over the period.

The portfolio curve chains the contribution-neutral daily returns: your payments do not make it jump. The 1M, 6M, YTD, 1Y and All buttons choose the period displayed, but both curves stay rebased to 100 at the start of the history.

The “Difference vs” the index card

Gap = portfolio TWR − index TWR, over the same period

Both TWRs are **total** (not annualised) and calculated only on the days when both returns exist. The very first day of investment, which has no corresponding index return, is excluded: the TWR used may therefore differ slightly from the total TWR displayed next to it. The badge shows “outperformance” if the gap is positive or zero, “underperformance” otherwise.

Example: portfolio +30.0%, index +25.0%: gap **+5.00%**.

A fair comparison

- Choose an index that resembles the portfolio (a blended index for an equity and bond portfolio).
- "Price-only" indices (CAC 40, Euro Stoxx 50) put the index at a disadvantage, since your dividends are counted.
- The index is converted into euros, like the portfolio.

Limit

A gap over a short period is often due to chance. Also look at calendar-year returns and the information ratio.

Calendar-year returns

The **Calendar-year returns** chart in the **Performance** tab shows, for each year, the portfolio's TWR (blue bars, "My portfolio") and the index's (grey bars).

The calculation

Daily returns are grouped by calendar year, then compounded:

Return for the year = Π (1 + r_day) − 1, over the days of the year

Example: a year with a first half at +6% and a second half at −2% gives $1.06 \times 0.98 - 1 = +3.88\%$.

Good to know

- The **first year** only starts at your first investment: it is partial.
- The **current year** runs from 1 January to the last price date: it is a year-to-date return, not a full year.
- The bars are not annualised.
- Hovering gives the return to two decimal places.

How to read it

Steady outperformance, year after year, is more convincing than a single excellent year. A year very different from the index in either direction signals a portfolio far from its reference: the tracking error confirms it.

Max drawdown (worst fall)

The drawdown measures, each day, the fall from the highest level reached so far. The max drawdown is the worst of these falls.

Formulas

drawdown(t) = index(t) / peak of the index up to t − 1 (always ≤ 0)
max drawdown = the smallest drawdown over the period

The calculation is done on the **base-100 curve** (contribution-neutral performance), not on the value in euros: otherwise a withdrawal would look like a loss, and a contribution would mask a genuine fall.

Example

The curve goes through 100, 120, 90, 110, then 130. The peak before the fall is 120; the trough is 90. Max drawdown: 90 / 120 − 1 = −25.00%. The peak is recovered when the curve returns to 120 or more: here, at the 130 point (110 is not enough).

Reading the chart

In the **Performance** tab, the red area shows the drawdown day by day; a dot marks the lowest point, labelled “Max drawdown”. The subtitle gives the date of the peak, that of the trough, then “recovered on” and the date of the return to the peak, or “peak not yet recovered”. The “Max drawdown” card in the key figures shows the date of the trough.

Interpretation

It is the most telling risk indicator: “at the worst moment, I would have lost 25% from the peak”. A 25% fall needs a rise of 1 / 0.75 − 1 = 33.3% to be wiped out.

Limits

It depends on the period observed: a two-year history may not have seen any real crisis. It does not tell you how many times or for how long you were in a decline.

Beta and correlation with the index

Beta

Beta = covariance(portfolio returns, index returns) / variance(index returns)

It is calculated on the daily returns of the days common to the portfolio and the index.

Intuition: when the index makes +1%, the portfolio makes on average beta × 1%.

Beta	Reading
1	The portfolio moves like the index (“1 = same as the index”)
1.2	It amplifies movements: +1.2% when the index makes +1%, and vice versa on the downside
0.8	More defensive than the index
Close to 0	Little linked to the index

Example: over five days, the index makes +1%, −1%, +2%, −2%, 0% and the portfolio +1.2%, −1.1%, +2.5%, −2.4%, +0.1%. Covariance 0.0003025, index variance 0.00025: beta = 0.0003025 / 0.00025 = 1.21.

Correlation

The "Correlation" card ("With" the index) measures, between −1 and +1, how far the two move **in the same direction**, regardless of magnitude. A correlation of 0.95 means the index explains almost all of the portfolio's movements; 0.5, that a large part of its movements is its own.

Beta = correlation × portfolio volatility / index volatility

Example: correlation 0.9, volatility 18% against 15% for the index: beta 0.9 × 18 / 15 = 1.08.

Limits

- Beta only makes sense against an index that resembles the portfolio; with a bond or money-market index, a warning is displayed.
- With a low correlation, beta explains little.
- It is estimated on the past and changes over time.

Jensen's alpha

Alpha measures the performance that is **not explained** by exposure to the market (CAPM model).

The software's formula

daily alpha = mean(r_portfolio − rf) − beta × mean(r_index − rf)
annual alpha = daily alpha × 252

rf is the daily risk-free rate: (1 + annual rate) ^ (1/252) − 1, about 0.0098% per day with 2.50% per year. The means are arithmetic, over the days common to the portfolio and the index.

Intuition

A portfolio with a beta of 1.1 "should" make 1.1 times the index's excess return. Alpha is what it made on top (or fell short).

Example

Portfolio's average annualised excess return: 8%; the index's: 6%; beta: 1.1. Alpha = 8% − 1.1 × 6% = +1.40% per year.

Reading

- **Alpha > 0** (green "annual" badge): your choices created value beyond the market risk taken.
- **Alpha < 0**: for the same market risk, the index did better.

Alpha differs from the gap with the index: a portfolio with a beta of 1.3 that beats the index in a rising market can have a zero alpha, because it only took more risk.

Limits

- The risk-free rate is assumed constant over the whole period.
- Over less than two or three years, alpha is very unstable.
- It only makes sense against an equity index comparable to the portfolio.

Tracking error and information ratio

Tracking error

Tracking error = standard deviation(r_portfolio − r_index) × √252

It is the volatility of the daily return **gap** with the index, scaled to the year ("Tracking error" card, "Annualised").

Tracking error	Reading
Less than 2%	The portfolio sticks to the index (passive management)
More than 5%	Management very different from the index

Information ratio

Information ratio = mean(r_portfolio − r_index) × 252 / tracking error

It answers the question: has the gap with the index been "paid for"? It is the equivalent of the Sharpe ratio, but relative to the index. A ratio above 0.5 is considered good.

Example

Average daily gap of 0.004% and standard deviation of the gap of 0.30%:

- tracking error: 0.30% × √252 = 4.76%;
- information ratio: 0.004% × 252 / 4.76% = 1.008% / 4.76% = 0.21.

The portfolio departs markedly from the index, but that gap has earned little.

Good to know

The mean used by the information ratio is **arithmetic** (daily mean × 252). It therefore does not exactly match the "Difference vs" the index card, which compares compounded, non-annualised TWRs.

Limits

Same reservations as for beta: choose a comparable index, and beware of short periods.

What does the Risk tab contain?

The **Risk** tab measures the risk taken and the shape of bad days.

At the top: four cards

Card	Line below
Sharpe ratio	Sharpe of the benchmark index
Sortino ratio	"Only penalises downside moves"
VaR (chosen level) · 1 day, in euros	The same VaR in %, "historical method"
CVaR · Expected Shortfall, in euros	The CVaR in %, "beyond the VaR"

In the centre: the distribution of daily returns

- On the left, the histogram of daily returns, the normal distribution curve and the VaR lines (dedicated entry). The subtitle gives the number of days used.
- On the right, four cards: Skewness, Excess kurtosis, Days beyond 3 standard deviations and Jarque-Bera test; then the **Loss on a bad day** table, with the historical VaR, the normal VaR, the Cornish-Fisher VaR and the CVaR, in % and in euros.
- "n/a" (not available) means the Cornish-Fisher VaR was not calculated because skewness or kurtosis are too extreme: a note explains this under the table.

At the bottom: the automatic reading

A few sentences interpret the distribution: skewness, fat tails, the result of the Jarque-Bera test and a comparison of the historical and normal VaRs. A final note reminds you that correlations and diversification are in the **Exposures** tab.

The setting that matters

The level of the VaRs and the CVaR is set at the top right of this tab, with the **VaR confidence level** selector (90, 95 or 99%).

Volatility

The formula

Annual volatility = standard deviation of daily returns × √252

The standard deviation is the sample one (division by n − 1), calculated on the contribution-neutral daily returns.

Why √252

There are 252 trading days in a year. If daily returns are independent, their variances add up: annual variance = 252 × daily variance, so annual standard deviation = √252 × daily standard deviation.

Example

A daily standard deviation of 1.00% gives 1% × √252 = 1% × 15.87 = 15.87% per year.

Interpretation

With a volatility of 15%, a year's return "typically" deviates by ± 15% from its mean. The most useful thing is the comparison with the benchmark index, whose volatility is shown under the Volatility card in the key figures.

Limits

- Volatility counts rises as risk, just like falls (the Sortino ratio corrects this flaw).
- The √252 rule assumes returns are independent from one day to the next.
- It describes usual deviations, not crashes: for extreme losses, look at the VaR, the CVaR and the max drawdown.

The Sharpe ratio

The Sharpe ratio answers the question: **was the risk taken well rewarded?**

The software's formula

daily excess = r_day − rf_day, with rf_day = (1 + risk-free rate) ^ (1/252) − 1
Sharpe = mean(excess) × 252 / (standard deviation(excess) × √252)

The return is the **arithmetic** mean of the daily returns, annualised by 252; it is not the annualised TWR. The risk-free rate is the one in the **Settings** panel (2.50% by default).

Example

Average daily return of 0.040%, daily standard deviation of 1.00%, risk-free rate of 2.50% (i.e. 0.0098% per day):

- annualised excess: (0.040% − 0.0098%) × 252 = 7.61%;
- volatility: 1% × √252 = 15.87%;
- Sharpe: 7.61 / 15.87 = 0.48.

Orders of magnitude

Sharpe	Reading
Less than 0	The portfolio did worse than a risk-free investment
Around 0.5	Decent
More than 1	Very good

Above all, compare with the index's Sharpe, shown under the card.

Limits

- It penalises large rises as much as large falls (see the Sortino).

- It assumes a constant risk-free rate over the whole period.
- Over a short period, it varies a lot; a rise in the risk-free rate lowers it.

The Sortino ratio

The Sortino ratio is a Sharpe that **only penalises falls**: an investor does not complain about large rises.

The software's formula

excess = r_day − rf_day
semi-deviation = √(mean(min(excess, 0)²)) × √252
Sortino = mean(excess) × 252 / semi-deviation

Days above the risk-free rate are replaced by 0, then the mean of the squares is taken over **all** days.

Example (risk-free rate at 0% for simplicity)

Six days: +1.2%, −2.0%, +0.6%, −0.8%, +1.5%, +0.3%.

- mean: 0.1333% per day, i.e. 33.60% annualised;
- volatility: 20.91%, so Sharpe = 33.60 / 20.91 = 1.61;
- semi-deviation: √((0.020² + 0.008²) / 6) = 0.879% per day, i.e. 13.96% annualised;
- Sortino: 33.60 / 13.96 = 2.41.

The Sortino is higher than the Sharpe: part of the volatility came from rises.

Reading

- Sortino clearly above the Sharpe: the portfolio's swings are mostly rises.
- Sortino close to the Sharpe: falls and rises are of comparable size.

Limits

With few down days, the semi-deviation rests on few observations and the ratio becomes unstable. It also inherits the limits of the Sharpe (constant risk-free rate, period observed).

Historical VaR (and VaR in euros)

The VaR (Value at Risk) answers: **how much can I lose on a bad day?**

The formula

Historical VaR = − quantile of daily returns at the threshold (1 − level)

At 95%, it is the 5th percentile of past returns, with the sign changed so that it reads as a loss. No assumption is made about the shape of the distribution: the software takes the actual days.

VaR in euros = historical VaR × current portfolio value

Example

Over 1,000 days, the 5th percentile lies between the 50th and 51st worst day. If it is −1.60%, the 95% historical VaR is 1.60%. For a €100,000 portfolio, the card shows **€1,600**.

Reading: “on 95% of days, the loss does not exceed €1,600”; in other words, about one trading day in twenty, roughly once a month, the loss is larger.

Where to see it

- “VaR (level) · 1 day” card in the **Risk** tab: the amount in euros, and the badge in % “historical method”.
- “Historical VaR” row of the **Loss on a bad day** table.
- Dashed orange line on the distribution chart.

What the VaR is not

It is **not** a maximum loss: it says where the zone of bad days begins, not how far it extends. For that, look at the CVaR.

Limits

- It assumes the past repeats itself: a calm history gives a low VaR.
- At 99%, it rests on 1% of days, which is very few observations over a short history.
- It is a **one-day** VaR, calculated on the current value.

The “normal distribution” (parametric) VaR

The formula

Normal VaR = −(mean of daily returns + z × standard deviation)

where z is the quantile of the normal distribution at the threshold 1 − level:

Level	z
90%	−1.2816
95%	−1.6449
99%	−2.3263

Intuition

We assume daily returns follow a normal distribution (a “bell curve”) with the same mean and standard deviation as yours, and read the loss at the desired threshold.

Example

Daily mean 0.04%, standard deviation 1.00%:

- at 95%: −(0.04% − 1.6449 × 1%) = 1.60%;

- at 99%: −(0.04% − 2.3263 × 1%) = 2.29%;
- at 90%: 1.24%.

Reading

It is the simplest method, and it can be calculated even with little data. But real returns have **fat tails**: large falls are more frequent than the normal distribution predicts. At 99%, the normal VaR therefore often underestimates the risk.

The software automatically compares the two methods: if the historical VaR exceeds the normal VaR by more than 5%, the reading under the chart says so ("the normal distribution underestimates the loss on a bad day here").

Limits

- The normality assumption is almost always rejected by the Jarque-Bera test on market data.
- The Cornish-Fisher VaR partly corrects this flaw.

The Cornish-Fisher VaR and its domain of validity

The Cornish-Fisher VaR keeps the normal distribution formula, but **corrects the quantile z** using the skewness S and the excess kurtosis K of the returns.

The formula

$z_cf = z + (z^2 - 1) \times S/6 + (z^3 - 3z) \times K/24 - (2z^3 - 5z) \times S^2/36$

Cornish-Fisher VaR = −(mean + z_cf × standard deviation)

This is the method used for the regulatory risk indicator of investment products (PRIIPs).

Examples (mean 0.04%, standard deviation 1%)

Level	S	K	z_cf	Cornish-Fisher VaR	Normal VaR
95%	−0.5	3	−1.7217	1.68%	1.60%
95%	0	3	−1.5843	1.54%	1.60%
99%	−0.5	3	−3.3013	3.26%	2.29%
99%	0	3	−3.0277	2.99%	2.29%

Reading

- A **negative skewness** always makes the VaR more cautious.
- **Fat tails** (K > 0) increase the 99% VaR, but may lower it slightly at 95%: a fat-tailed distribution also has a more peaked centre, with more calm days.

The domain of validity: why "n/a"

The correction only makes sense if it respects the order of the quantiles: the smaller z is, the smaller z_cf must be. The software checks that the derivative of z_cf stays strictly positive for z from −4 to +4, in steps of 0.1:

derivative = $1 + z \times S/3 + (3z^2 - 3) \times K/24 - (6z^2 - 5) \times S^2/36$

If this is not the case, the VaR is not calculated: the table shows "n/a", the purple line disappears from the chart, and a note states "skewness or kurtosis too high, the correction is no longer reliable". The VaR is also not calculated with fewer than four returns.

Examples of valid pairs (checked on a grid of K in steps of 0.1):

Skewness S	Valid excess kurtosis K
0	from 0 to 7.9
−0.5	from 0.2 to 8.2
−1	from 1.6 to 8.8
−2	from 6.6 to 11.2

A strong skewness with tails that are too **thin** is therefore also outside the domain.

Limit

It is an approximation: with a distribution very far from the normal distribution, prefer the historical VaR and the CVaR.

The CVaR (Expected Shortfall)

The CVaR (Conditional Value at Risk), or Expected Shortfall, answers: **and when things go badly, how badly?**

The formula

CVaR = − mean of daily returns less than or equal to −historical VaR

It is the **average** loss on the days when the historical VaR is reached or exceeded.

CVaR in euros = CVaR × current value

Example (twenty days, for illustration)

Ranked returns: −3.1%, −2.2%, −1.5%, −1.2%, … , +2.0%.

- The 95% historical VaR is interpolated between the two worst days: −(−3.1% + 0.95 × 0.9%) = 2.245%.
- Only one day is below −2.245%: −3.1%. The CVaR is therefore **3.10%**.

On a real history of 1,000 days, the 95% CVaR is the average of about 50 days.

Where to see it

- "CVaR · Expected Shortfall" card in the **Risk** tab, in euros, with the badge in % "beyond the VaR".
- "CVaR (Expected Shortfall)" row of the table and the solid red line on the chart.

Reading

The CVaR is always at least equal to the historical VaR. A large gap between the two signals a fat distribution tail: bad days are rare but violent. It is the measure favoured by banking regulators (Basel III), because it looks beyond the threshold.

Limits

It rests on few days, especially at 99%: over a year of data (about 252 days), the 99% CVaR is the average of two or three days.

Three different VaRs: which one to believe?

The **Loss on a bad day** table in the **Risk** tab shows three VaRs and the CVaR, in % and in euros. They answer the same question under three assumptions.

Method	Assumption	Strength	Weakness
Historical VaR	The past repeats itself	No assumption about shape	Depends on the period observed
Normal VaR	Bell-shaped returns	Simple, stable	Underestimates crashes
Cornish-Fisher VaR	Bell curve corrected for skewness and kurtosis	Takes the real shape into account	Approximation, sometimes "n/a"
CVaR	The past repeats itself	Measures the severity of bad days	Rests on few days

What the automatic reading says

Under the chart, a sentence compares the historical VaR and the normal VaR:

- historical **more than 5% higher**: the normal distribution underestimates the loss on a bad day here;
- historical **more than 5% lower**: extreme losses are concentrated on a few days, which the CVaR measures better;
- otherwise: the two are close.

How to decide

1. Look at the Jarque-Bera test: if normality is rejected, treat the normal VaR with caution.

2. Take the **higher** of the historical and Cornish-Fisher VaRs as a cautious order of magnitude.
3. Complete with the CVaR to see how far bad days go, and with the max drawdown for a fall over several days.

No VaR is a maximum loss. All are one-day estimates, drawn from the portfolio's history.

Reading the return distribution chart

The **Distribution of daily returns** chart in the **Risk** tab compares your actual days with what a normal distribution would predict.

The blue bars: "Observed days"

Each bar counts the number of days whose return falls in a class. All classes have the same width; their number is √n × 1.2, bounded between 20 and 80 (n = number of days). Example: 1,000 days give 37 classes. Hovering shows "Return …" and the number of days.

The dark curve: "Normal distribution (same mean and volatility)"

It is the bell that would be obtained if your returns followed a normal distribution with the **same mean and standard deviation**. It is scaled to the bars:

expected days = normal density × number of days × class width

Hovering gives the number of expected days.

The vertical lines

Each is placed at −VaR (loss side) and appears in the legend above the chart.

Line	Style
Historical VaR (level)	Orange, dashed
Normal VaR (level)	Dark grey, dotted
Cornish-Fisher VaR (level)	Purple, dash-dot
CVaR (Expected Shortfall)	Red, solid line

A missing line corresponds to a value that was not calculated (Cornish-Fisher VaR "n/a"). Clicking a legend name hides or shows the line.

What to look at

- **The centre**: bars that exceed the bell around zero indicate more calm days than expected.
- **The extremes**: isolated bars far to the left, where the bell is almost zero, are the crashes the normal distribution does not predict ("fat tails").
- **The gap between the lines**: an orange line to the left of the grey line means the historical VaR exceeds the normal VaR.

The sentences under the chart summarise these observations.

Skewness

Skewness measures whether the distribution of daily returns leans to one side.

The calculation

The software uses the sample skewness coefficient, bias-corrected (third moment of the deviations from the mean, divided by the cube of the standard deviation). It is **0 for a normal distribution**.

Reading (thresholds of the automatic reading)

Skewness	Sentence displayed
Below −0.3	Large falls are more frequent or more violent than large rises
Between −0.3 and +0.3	The distribution is roughly symmetric
Above +0.3	Large rises outweigh large falls

Example

An equity portfolio often shows a negative skewness, for example −0.5: markets rise in small steps and fall in jolts. At 95%, with an excess kurtosis of 3, the Cornish-Fisher VaR then goes from 1.54% (zero skewness) to 1.68% (skewness −0.5), for a mean of 0.04% and a standard deviation of 1%.

Why it matters

A negative skewness means the risk is concentrated on the bad side: volatility and the normal VaR, which assume a symmetric distribution, then underestimate losses.

Limits

Skewness is very sensitive to a few extreme days: a single crash in the history can tip it. It needs a long history to be reliable.

Excess kurtosis and extreme days

Excess kurtosis

It measures the thickness of the distribution's tails: the frequency of extreme days, in both directions. The software calculates the sample **excess** kurtosis, bias-corrected: it is **0 for a normal distribution**.

Excess kurtosis	Automatic reading sentence
Above 1	"Fat tails", with the number of extreme days observed and expected

Excess kurtosis	Automatic reading sentence
Between 0 and 1	Tails slightly fatter than the normal distribution
0 or less	No more extreme days than the normal distribution predicts

Days beyond 3 standard deviations

Share of extreme days = share of days where $|\text{return} - \text{mean}| > 3 \times \text{standard deviation}$

Under the normal distribution, this share is $2 \times (1 - \Phi(3)) = 0.27\%$, about one day in 370. The "Days beyond 3 standard deviations" card compares the observed share with this theoretical figure.

Example

Over 1,000 days, the normal distribution predicts 2 to 3 extreme days. If you observe 12 (1.20%), there are about 4.4 times more than predicted: the automatic reading says so ("×4.4").

Why it matters

Financial markets almost always have a positive kurtosis. Volatility and the normal VaR describe ordinary days well, but underestimate crashes: at 99%, prefer the historical VaR, the Cornish-Fisher VaR and the CVaR.

Limits

Like skewness, kurtosis is dominated by a few days. A short history without a crisis can show a low kurtosis that guarantees nothing.

The Jarque-Bera test

The Jarque-Bera test checks whether your daily returns can reasonably follow a normal distribution.

The formula

$JB = n / 6 \times (S^2 + K^2 / 4)$
$\text{p-value} = \exp(-JB / 2)$

with n the number of days, S the skewness and K the excess kurtosis. Under the normality hypothesis, JB follows a chi-squared distribution with 2 degrees of freedom, whose exceedance probability is exactly $\exp(-JB/2)$.

Reading

The "Jarque-Bera test" card shows "p =" followed by the p-value ("< 0.001" when it is tiny), with a badge:

- **p < 0.05**: red “Normality rejected” badge. VaRs based on the normal distribution should be treated with caution.
- **p ≥ 0.05**: green “Normality not rejected” badge. This does not prove the distribution is normal: the data simply do not allow it to be ruled out.

Examples

- 1,000 days, S = −0.5, K = 3: JB = 1,000 / 6 × (0.25 + 2.25) = 416.7; p-value almost zero, displayed “< 0.001”: normality rejected.
- 250 days, S = −0.1, K = 0.3: JB = 250 / 6 × (0.01 + 0.0225) = 1.35; p = exp(−0.677) = 0.508: normality not rejected.

Why normality is almost always rejected

With many days, the slightest departure in skewness or kurtosis becomes significant. On several years of market data, rejection is the rule: this is precisely why the software offers the historical VaR and the Cornish-Fisher VaR.

Limit

With fewer than four returns, the test is not calculated (p-value set to 1).

What does the Exposures tab contain?

The **Exposures** tab answers two questions: what is the portfolio **really** exposed to, and is it **truly** diversified? Everything in it is calculated on a look-through basis.

1. The header and the profile

The title “Exposures and diversification” is a reminder that ETFs are split according to their index (approximation as of 30/09/2026). On the right, the **Thresholds for profile** menu chooses the diagnosis thresholds: Cautious, Balanced (default) or Dynamic.

2. The summary: six boxes

Dimension	Figure shown under the badge
Geography	Top country of the equity portion and its weight
Sectors	Top sector of the equity portion and its weight
Currencies	Share outside the euro
Concentration	Weight of the largest holding
Interest rates	Average duration, or “No bonds”
Real diversification	Average correlation between holdings

The badge gives the most serious finding for the dimension: green “Good”, orange “To monitor”, red “To fix”, or grey “Not applicable” (for example, the Interest rates dimension with no bonds).

3. Findings and suggestions

Orange and red points are listed from most to least serious. Each finding gives the observed figure, the associated risk and suggestions. Green points are placed in a “Strengths” panel, open by default if there is no point of attention.

4. Four detail sub-tabs

- **Geography and sectors**: regions and sectors compared with the index, main differences by country, share of value and share of risk by region;
- **Currencies and rates**: real currency exposure, interest-rate sensitivity;
- **Concentration**: number of holdings, weight of the largest ones, 5/10/40 rule;
- **Correlations**: matrix, blocks, pairs, holdings that diversify.

Is the exposures diagnosis advice?

No. As the note in the **Exposures** tab says, it is an “educational analysis based on simple rules and past data: it does not constitute investment advice”.

How the findings are produced

The software applies fixed rules, written in advance, to the portfolio’s figures: weight of a country compared with the world market (MSCI ACWI), weight of a sector, share outside the euro, weight of a share, duration, correlations. Each rule has a threshold; depending on the observed value, it produces a green, orange or red finding, accompanied by a sentence on the risk and general suggestions. The thresholds are detailed in the following entries.

What the diagnosis does not know

- your personal situation: wealth outside this portfolio, income, horizon, objectives, taxation;
- your reasons: an overweight may be a deliberate choice;
- the future: correlations and compositions are drawn from the past and from index factsheets.

How to use it

Read each finding as a question to ask yourself: “is this bias intended?”. The suggestions illustrate common solutions (world ETF, currency hedging, shorter duration, etc.), without taking your constraints into account. Before any decision, especially a sale that would trigger tax, consult a qualified professional.

The PDF report

The “Exposures and diversification” part of the report reuses this diagnosis with the **Balanced** profile thresholds, whatever the profile chosen on screen.

Thresholds by profile: Cautious, Balanced, Dynamic

The **Thresholds for profile** menu, at the top of the **Exposures** tab, sets the severity of the diagnosis. It changes no calculation: only the colour of the findings.

Thresholds that depend on the profile

For each rule, the first value turns it orange (“To monitor”), the second red (“To fix”); you must **exceed** it.

Rule	Cautious	Balanced	Dynamic
Share of portfolio outside the euro	30% / 50%	50% / 70%	70% / 85%
Emerging countries in the equity portion	10% / 20%	20% / 30%	25% / 40%
Largest directly held share	5% / 10%	7% / 10%	10% / 15%
Average duration of bonds	5 / 8 years	7 / 10 years	8 / 12 years

Rules identical for all profiles

Geography (excluding emerging markets), sectors, the 5/10/40 rule, likely duplicates and correlations have the same thresholds whatever the profile (see the corresponding entries).

Example

A €100,000 portfolio: a €60,000 MSCI World ETF and four French shares (LVMH €15,000, TotalEnergies €10,000, Airbus €8,000, Sanofi €7,000). Share outside the euro: 55% (of which 44% in dollars, via the ETF). Largest share: LVMH, 15%.

Profile	Currencies (55%)	LVMH (15%)
Cautious	To fix (> 50%)	To fix (> 10%)
Balanced	To monitor (> 50%)	To fix (> 10%)
Dynamic	Good (≤ 70%)	To monitor (> 10%)

How long the choice lasts

The profile is kept for the session; it is not saved in your account. The PDF report always uses the Balanced profile.

Geography and sectors: the rules and the gaps to the index

The geography rules (equity portion)

They apply if equities weigh more than 5% of the portfolio. The top three countries are compared with their weight in the **MSCI ACWI** (world market):

- **To fix**: a country exceeds its world weight by more than 25 points, or a country other than the United States weighs more than 40%;
- **To monitor**: it exceeds it by more than 15 points;
- **Home bias** (to monitor): France weighs more than 15% **and** more than four times its world weight (about 2%);
- **Emerging countries**: thresholds depend on the profile.

The sector rules (equity portion)

They apply under the same conditions. For the top three sectors: to fix above 40% or 15 points above the world market; to monitor above 30% or 8 points. A “few defensive sectors” finding appears if healthcare, consumer staples and utilities total less than 10%.

Example

MSCI World ETF €60,000, LVMH €15,000, TotalEnergies €10,000, Airbus €8,000, Sanofi €7,000. On a look-through basis, France weighs 41% of the equity portion (against 2% in the world market): red finding, because a country other than the United States exceeds 40%, and an orange home-bias finding. Consumer discretionary weighs 20% against 8% worldwide (+12 points): to monitor.

*The **Geography and sectors** sub-tab*

- **Regions (equity allocation)** and **Sectors (equity allocation)**: two bars per group, “Portfolio” (dark blue) and the index (blue-grey). The index is the one chosen in the settings if it contains equities; otherwise (bond or money-market index), it is the MSCI ACWI.
- **Main differences by country**: the ten largest overweights and underweights, in points.
- **Share of value and share of risk by region**: a region that brings more risk than value is more volatile or more correlated with the rest. The risk share of each holding is explained in the Asset management workspace, Risk budget tab.

Limit

The weights of countries and sectors in ETFs and in the MSCI ACWI are approximations (see the entry on the world map).

Real currency exposure

An MSCI World ETF quoted in euros remains exposed to the **dollar**, the yen, the pound, etc.: it holds shares whose value is formed in those currencies. The **Currencies and rates** sub-tab measures this real exposure.

The calculation

On a look-through basis, each piece of an equity ETF receives the currency of its country; a directly held share keeps its trading currency; a fund whose name indicates hedging ("Hedged", "couvert") is counted in euros; bonds are in euros for a euro-zone index, in dollars otherwise; gold is counted separately, with "Gold" as its "currency". The **Real currency exposure** ring covers the **whole** portfolio.

Share outside the euro = sum of the weights of all currencies except the euro and gold

Example

MSCI World ETF €60,000 and €40,000 of French shares. Dollar: 60% × 72.94% = 43.8%; yen 3.5%; pound 2.0%; Canadian dollar 2.0%; Swiss franc 1.4%; etc. Share outside the euro: **55%**. The finding estimates the effect of a 10% fall in the dollar: 43.8% × 10% ≈ 4% of the portfolio.

The finding

It depends on the profile: above 30% outside the euro (Cautious), 50% (Balanced) or 70% (Dynamic), it turns orange, then red above 50%, 70% or 85%. Otherwise: "Limited currency risk".

Reading

Currency risk is not only negative: the dollar often rises in times of crisis, which cushions falls. Hedging part of the portion is a common compromise; that is the suggestion made.

Limits

A country's currency is an approximation: an American company earns part of its revenue in other currencies. Hedging is only recognised if it appears in the fund's name.

Duration and interest-rate sensitivity

Duration

It is the weighted average life of a bond's cash flows, in years. It measures **sensitivity to rates**: the longer it is, the more the price falls when rates rise (and rises when they fall).

The software reads the duration of each bond fund from its descriptive record, then calculates:

Average duration = Σ (duration × value) / Σ value, over the bond holdings whose duration is known
Loss if rates rise by 1 point ≈ Σ (duration × value) × 1%

Example

€100,000 portfolio; bonds A: €30,000, duration 4 years; bonds B: €10,000, duration 8 years.

- average duration: (4 × 30,000 + 8 × 10,000) / 40,000 = 5.0 years;
- estimated loss: (4 × 30,000 + 8 × 10,000) × 1% = €2,000, i.e. **−2.0%** of the portfolio.

Where to see it

Currencies and rates sub-tab, **Interest-rate sensitivity** block: “Average duration” and “If rates rise by 1 point” cards (in % of the portfolio and in euros). With no bonds, the block says the portfolio contains none, and the Interest rates box in the summary shows “Not applicable”.

The finding

The colour depends on the profile: to monitor above 5 years (Cautious), 7 years (Balanced) or 8 years (Dynamic); to fix above 8, 10 or 12 years. In the example (5.0 years), the finding is green for all profiles.

Limits

- First-order approximation: price change ≈ − duration × change in rates, valid for small changes.
- A bond whose duration is not known is excluded from the average and the estimated loss.
- After a rise in rates, bonds then offer a higher yield, which gradually offsets the loss.

Concentration, effective number of holdings and the 5/10/40 rule

The effective number of holdings

Effective number = 1 / Σ weight² (inverse of the Herfindahl index)

It tells you how many **equal-weight** holdings your portfolio is equivalent to. Example: four holdings of 40%, 30%, 20% and 10%: 1 / (0.16 + 0.09 + 0.04 + 0.01) = 3.33, displayed as “equivalent to 3 equal-weight positions”.

*The **Concentration** sub-tab*

Four cards: “Rows” (with the effective number), “Top 5 positions”, “Top 10 positions” and “Stocks above 5%” (“UCITS limit: 40%”); then the **Largest positions** table (15 at most), with weight and cumulative weight.

The 5/10/40 rule

A UCITS (European mutual fund) rule: no holding exceeds 10%, and holdings above 5% do not total more than 40%. The software applies it only to **directly held shares**: a diversified ETF is not a concentration.

- Shares above 5% total more than 40%: red finding “5/10/40 rule exceeded”.

- The largest share exceeds the profile threshold (5, 7 or 10% for orange; 10, 10 or 15% for red): a finding that quantifies the effect of a 30% fall in the stock.

Example: LVMH 15%, TotalEnergies 10%, Airbus 8%, Sanofi 7% total **exactly 40%**: the rule is not exceeded (it must be more than 40%). However, LVMH at 15% is "To fix" under the Balanced profile; a 30% fall in the stock would cost 15% × 30% ≈ 4% of the portfolio.

Likely duplicates

A directly held share may also be in an ETF that you hold: your exposure to that company is then greater than its line alone. The software flags it (orange finding "held directly AND probably also via your ETF") when:

- the ETF tracks the MSCI World, the ACWI or the EAFE and the share's country is part of the index, unless the security's record indicates it belongs to no major index;
- the ETF tracks the MSCI Europe and the share is European, British or Swiss, with the same reservation;
- the ETF tracks the S&P 500 or the US market and the share is American, unless the security's record lists indices without mentioning the S&P 500;
- or the security's record explicitly mentions the ETF's index.

This is a **probability**: the software does not read the fund's actual holdings.

Reading the correlation matrix

The matrix is in the **Correlations** sub-tab of the **Exposures** tab.

Correlation

It measures, between −1 and +1, how far the daily returns of two securities move together:

Value	Reading	Qualifier displayed
0.8 to 1	Almost always rise and fall together	very strong
0.6 to 0.8	Often move together	strong
0.3 to 0.6	Partial link	moderate
Less than 0.3	Little link	weak
Negative	When one rises, the other tends to fall	(same scale, in absolute value)

It is calculated on the euro prices of the holdings still held, over the days when all have a return. The note under the chart gives this number of days.

Reading the map

- **Colours**: red = move together; white = no link; blue = opposite directions.

- **Half of the table**: only the lower triangle is displayed, without the diagonal (always 1); the other half would repeat the same information.
- **Order**: securities are arranged in blocks that move together (hierarchical clustering, distance = 1 − correlation). Blocks appear as red squares.
- **Values**: written in the cells up to 13 securities; beyond that, hover over a cell to read the two names, the value and its qualifier.

Changing view

The **Show** selector offers **By security**, **By asset class**, **By region** and **By sector** (group views only appear if the portfolio has several groups). A group view shows the correlation between the returns of the sub-portfolios, each holding weighted by its weight; groups follow the classification of the holdings, without ETF look-through. Beyond 20 securities, a group view opens by default.

The tables on the right

- **Most correlated pairs**: the five most closely linked pairs of securities.
- **Best diversifiers**: the five holdings least correlated with the **rest** of the portfolio (the portfolio without them).
- **Blocks of correlated securities**: groups with an average correlation above 0.7 (dedicated entry).

Limit

Correlations are historical: they are not guaranteed in the future and often rise during crises.

Average correlation, in normal times and on sharp-decline days

The weighted average correlation

Average correlation = $\Sigma_{i \neq j} w_i \times w_j \times \rho_{ij} / \Sigma_{i \neq j} w_i \times w_j$

where w are the holdings' weights and ρ their correlations. It is the "typical" correlation between two euros invested in two different holdings: large holdings count for more.

Example: three holdings of 50%, 30% and 20%; correlations 0.8 (A-B), 0.2 (A-C), 0.3 (B-C). $(0.15 \times 0.8 + 0.10 \times 0.2 + 0.06 \times 0.3) / (0.15 + 0.10 + 0.06) = 0.158 / 0.31 = 0.51$.

Findings: above 0.45, "To monitor"; above 0.6, "To fix": the portfolio behaves almost like a single asset.

Sharp-decline days

The software reconstructs the portfolio's daily return **with its current weights**, keeps its worst 10% of days, and recalculates the weighted average correlation on those days alone.

- "On sharp-decline days" card in the **Correlations** sub-tab ("average correlation (worst 10% of days)").

- At least 30 sharp-decline days are needed, i.e. about 300 days of common history: otherwise the card shows a dash.
- If it exceeds the average correlation by more than 0.10, an orange finding says so: “diversification protects less when you need it most”.

Why correlation rises in a crisis

In a crash, investors sell a bit of everything at the same time: assets that are usually independent fall together. Diversification that looks good in normal times can then evaporate. The stress tests in the Wealth advisory workspace let you measure the impact of a crisis.

Limit

Current weights are applied to the whole history: it is the behaviour of **today’s** portfolio in past crises, not that of your real portfolio at the time.

The diversification ratio

The formula

Diversification ratio = $\Sigma\, w_i \times \sigma_i \,/\, \sigma_p$

- $\Sigma\, w_i \times \sigma_i$: the weighted average of the holdings’ volatilities, i.e. the volatility the portfolio would have if all holdings moved exactly together;
- σ_p: the portfolio’s real volatility, which takes correlations into account.

Both are calculated on daily returns in euros, with current weights.

Intuition

The ratio measures **how far risk offsets itself** between holdings. It equals 1 when there is no diversification (perfectly correlated holdings) and increases as holdings offset each other.

Example

Two holdings of 50%, each with volatility 20%:

Correlation	Portfolio volatility	Ratio
1	20.00%	1.00
0.5	17.32%	1.15
0	14.14%	1.41

With a correlation of 0.5: $\sigma_p = \sqrt{(0.25 \times 0.04 + 0.25 \times 0.04 + 2 \times 0.25 \times 0.5 \times 0.04)} = \sqrt{0.03} = 17.32\%$, and $20 / 17.32 = 1.15$.

Where to see it

"Diversification ratio" card in the **Correlations** sub-tab ("1 = no diversification"); it is also cited in the green finding "Satisfactory real diversification".

Limits

- The ratio has no alert threshold in the diagnosis: it is read by comparison, from one version of the portfolio to another.
- It depends on past correlations.
- Adding a very low-risk holding (money market) changes it little: it measures the offsetting between holdings, not the level of risk.

Blocks of correlated securities (independent blocks)

Ten holdings that rise and fall together are worth only **one bet**. Blocks measure this real diversification.

How blocks are formed

1. Distance between two securities = 1 − correlation.
2. Hierarchical clustering by average linkage: the closest securities are grouped, then the groups with each other.
3. Grouping stops at a distance of 0.3: a block gathers securities whose average correlation exceeds about **0.7**.

A security that resembles no other forms a block on its own.

Where to see it

- "Independent blocks" card ("for" n holdings) in the **Correlations** sub-tab: the total number of blocks, isolated securities included.
- **Blocks of correlated securities** block: the five largest blocks of at least two securities, with their weight, their names and their average correlation.
- Blocks also appear as red squares in the matrix.

Example

Suppose an 8-holding portfolio that shows "5 independent blocks". Three semiconductor shares, correlated at 0.78 on average and weighing 28%, form a block: the diagnosis flags "3 holdings highly correlated with each other (0.78 on average) weigh 28% … 3 holdings, but in practice a single bet".

The findings

The three largest blocks are examined:

- weight below 15%: no finding;
- from 15 to 25%: "To monitor";

- 25% or more: “To fix”.

The suggestions made: replace part of the block with weakly correlated assets, or group these holdings into a single ETF on the same theme.

Limits

The 0.7 threshold is a convention. Blocks rest on past correlations and may change from one period to another.

Viewing the transaction history (Transactions tab)

The **Transactions** tab shows all the portfolio’s transactions, from most recent to oldest.

Filtering

- **Type**: remove or add ACHAT (purchase), VENTE (sale) and DIVIDENDE (dividend) (all are selected at the start).
- **Securities**: choose one or more securities; left empty, it shows “All securities”.

A line gives the number of transactions displayed out of the total.

The columns

Column	Content
Date	Transaction date
Type	ACHAT, VENTE or DIVIDENDE
Ticker, Security	Code and name of the security
Quantity	Number of securities (0 for a dividend)
Price / amount (€)	Unit price, or total amount for a dividend, **converted into euros**
Fees	Fees in euros
Currency, Price in currency	Present if the portfolio contains securities in foreign currencies: the trading currency and the price entered before conversion

Exporting

The **Download the selection (CSV)** button saves the transactions displayed in the file transactions_selection.csv. The types remain written ACHAT, VENTE, DIVIDENDE, but the price column is the one **converted into euros**; the currency and price-in-currency columns are added where relevant. To re-import a portfolio, use the original file or the updated file instead.

Editing a transaction

The **Edit transactions** button lets you delete or correct any transaction. This function, its checks and undoing are described in the chapter on importing and updating the portfolio.

6. Optimisation and projection

This chapter explains the two forward-looking tabs of the "Portfolio analysis" area. The **Optimisation** tab applies Markowitz theory to your holdings: efficient frontier, minimum-variance and maximum-Sharpe portfolios, and a "from → to" table of the changes to make. The **Projection** tab simulates thousands of possible paths for your portfolio using the Monte Carlo method. Each fiche gives the exact formula used by the software, a worked example and the limitations to keep in mind.

What does the Optimisation tab do? The Markowitz principle

The **Optimisation** tab looks for other allocations, **among the securities you already hold**, that would offer a better return / risk trade-off. It adds no security: it only changes the weights.

Markowitz's idea (1952)

An investor looks not only at return but also at risk. Yet the risk of a portfolio is not the average of the risks of its securities: when two securities do not rise and fall exactly together (correlation below 1), their movements partly offset each other. Risk can therefore be reduced without reducing return.

The two formulas at the heart of the calculation

Expected portfolio return = $\Sigma\, w_i \times \mu_i$ ($w^T \mu$)
Portfolio variance = $\Sigma\Sigma\, w_i \times w_j \times cov_{ij}$ ($w^T \Sigma\, w$)
Volatility = $\sqrt{}$variance

where w_i is the weight of security i, μ_i its expected annual return and cov_{ij} the annual covariance between securities i and j.

Worked example

Two securities: A (return 8%, volatility 20%) and B (return 4%, volatility 10%), correlation 0.2.

- 50 / 50 portfolio: return = 0.5 × 8% + 0.5 × 4% = **6%**. Variance = 0.25 × 0.04 + 0.25 × 0.01 + 2 × 0.25 × 0.2 × 0.20 × 0.10 = 0.0145, i.e. a volatility of **12.04%**, clearly less than the average of the volatilities (15%).
- Minimum-variance portfolio: 14.3% of A and 85.7% of B, volatility **9.56%**, lower than that of B alone (10%) even though it contains a security twice as risky.

That is the whole point of diversification.

What the software calculates

1. the expected return and the covariance matrix of your securities;
2. the **minimum-variance** portfolio (the least risky possible);
3. the **maximum-Sharpe** portfolio (the best return per unit of risk);

4. the **efficient frontier** (40 points);
5. a cloud of 4,000 randomly drawn portfolios, to visualise the full set of possibilities.

Three cards, below the maximum-weight slider, give for the current portfolio, the minimum variance and the maximum Sharpe, the Sharpe ratio, the return and the volatility.

The constraints applied

- no short selling: each weight is positive or zero;
- all the capital is invested: the weights add up to 100%;
- a maximum weight per security, adjustable with the **Maximum weight per security** slider.

The numerical optimisation uses the SLSQP method from the scipy library (at most 500 iterations), starting from equal weights. The whole exercise remains academic: the software says so under the results, and it is not investment advice.

How are each security's expected return and risk estimated?

The optimisation needs two ingredients, estimated from the **history** of your securities.

The exact formulas

μ (expected annual return) = average daily return × 252
Σ (annual covariance) = covariance of daily returns × 252
Daily return = today's price / previous day's price − 1

252 is the conventional number of trading days in a year. The diagonal of Σ holds the variance of each security; elsewhere it shows how two securities move together. The volatility of a single security is √(annual variance).

Example

A security gains 0.04% a day on average, with a daily standard deviation of 1.2%:

- μ = 0.0004 × 252 = **10.08%** a year;
- volatility = 1.2% × √252 = **19.05%** a year.

Over what period?

- The prices used are those of the portfolio analysis period, which begins at your first transaction.
- Only the days on which **all** your current securities have a return are kept. A recently listed security therefore shortens the period for all the others.
- Prices are converted into euros and are not adjusted for dividends (see the chapter on sources): for a security that pays out a lot, μ is slightly underestimated.

Interpretation

μ is not a forecast: it is the average observed in the past, extrapolated to one year. Over a short, favourable period, a security can show a μ of 40% or more, which says nothing about the future. This is the first source of fragility in the optimisation (see the fiche on limitations).

Reading the efficient frontier chart

The chart in the "Efficient frontier" section places each portfolio according to its **annual volatility** (horizontal axis) and its **expected annual return** (vertical axis). The best place is at the top left: plenty of return for little risk.

The elements of the chart

Element	What it represents
Blue dots ("Random portfolios")	4,000 randomly drawn allocations; the darker the blue, the higher the Sharpe (scale on the right)
Thick black curve ("Efficient frontier")	for each level of return, the least risky portfolio
Grey dots with a code	each security on its own (100% of the portfolio in that security); codes are only shown up to 20 securities
Orange circle ("My portfolio")	your current allocation
Diamond ("Minimum variance")	the least risky portfolio
Green star ("Maximum Sharpe")	the best return per unit of risk
Dashed line ("Capital market line")	combinations of the risk-free investment and the maximum-Sharpe portfolio

Each notable portfolio shows its Sharpe in the legend. Hover over a point to read its volatility and return.

How the frontier is built

The software sets 40 target returns, evenly spaced between the return of the minimum-variance portfolio and the maximum achievable return given the per-security cap (securities are filled from the most profitable to the least profitable, each up to the cap). For each target, it looks for the weights that minimise the variance $w^{\intercal} \Sigma w$. An impossible target is simply ignored.

Below the minimum variance, portfolios are "inefficient": more return can be obtained for the same risk.

The random portfolios

The weights are drawn from a Dirichlet distribution (all positive, summing to 100%), with a fixed random seed: the cloud is identical from one display to the next. These portfolios do **not**

respect the per-security cap. As the frontier is calculated with that cap, a few points may slightly exceed a very constrained frontier; without a cap, none exceeds it, which visually checks the optimisation.

The capital market line

Return = risk-free rate + slope × volatility
slope = (maximum-Sharpe return − risk-free rate) / maximum-Sharpe volatility

It starts from the risk-free rate (set in **Risk-free rate (% per year)**) and touches the frontier at the maximum-Sharpe portfolio, hence its other name, the "tangency portfolio".

The "Maximum weight per security" setting

The **Maximum weight per security** slider, at the top of the Optimisation tab, sets the maximum share that a security can take in the optimised portfolios (minimum variance, maximum Sharpe and frontier).

Why a cap?

Without a limit, the optimiser often concentrates everything in 2 or 3 securities: those that have done best in the past. The cap imposes a minimum level of diversification. The default value is **30%**. For reference, fund regulations (UCITS) limit each holding to 10%, with a tolerance of up to 40% for the total of all holdings above 5%.

The values offered

10%, 15%, 20%, 25%, 30%, 40%, 50% and "no limit". Only the values that allow 100% of the capital to be invested appear:

condition: number of securities × maximum weight ≥ 100%

Examples:

- with 3 securities, 30% × 3 = 90%: impossible. The slider only offers 40%, 50% and no limit;
- with 5 securities, all values from 20% upwards are offered;
- with 10 or more securities, all values are offered.

If 30% is not possible (fewer than 4 securities), the slider starts on "no limit".

Effect on the results

The lower the cap, the more the optimised portfolios resemble a balanced portfolio, and the more their Sharpe falls slightly. The cap appears in the compared allocations chart as a dashed vertical line, labelled "Cap 30% per holding" (or the value chosen).

In the two-security example A (8%, 20%) and B (4%, 10%), correlation 0.2, risk-free rate 2.5%: with no limit, the maximum Sharpe puts 56.3% in A (Sharpe 0.292); with a 50% cap, it is held at 50 / 50 (Sharpe 0.291).

Maximum Sharpe or minimum variance: which one should I look at?

The software calculates two notable portfolios, with the same constraints (positive weights, sum 100%, per-security cap).

Minimum variance

minimise $w^T \Sigma w$

This is the **least risky portfolio possible** with your securities. It **does not use expected returns**, only volatilities and correlations. That is an advantage: covariances can be estimated much better than returns. It is often more stable over time.

Maximum Sharpe

Sharpe = (expected return − risk-free rate) / volatility
maximise Sharpe (in practice: minimise −Sharpe)

This is the portfolio that offers the **best return per unit of risk**. Combined with the risk-free investment, it gives the best possible line: it is the tangency portfolio. It depends directly on the expected returns, and therefore on the history.

Example

Securities A (8%, 20%) and B (4%, 10%), correlation 0.2, risk-free rate 2.5%, no cap:

Portfolio	Weights A / B	Return	Volatility	Sharpe
Minimum variance	14.3% / 85.7%	4.57%	9.56%	0.22
Maximum Sharpe	56.3% / 43.7%	6.25%	12.87%	0.29

The Sharpe is calculated here on the expected annual return ($w^T \mu$): it may differ from the Sharpe ratio in the Performance tab, which is calculated on the portfolio's actual returns.

Which one should I look at?

- **Minimum variance**: for a cautious investor, or if you doubt past returns.
- **Maximum Sharpe**: for the best theoretical efficiency, accepting that it exploits the securities that have done best.

The **Compare my portfolio with** selector in the "Allocations compared" section lets you display the changes to make towards one or the other. By default, it is the maximum Sharpe.

Reading the "Allocations compared": what would need to change?

The "Allocations compared" section turns the optimisation result into concrete changes, **with the total value unchanged and excluding fees**. First choose the target with **Compare my portfolio with**: **Maximum Sharpe** or **Minimum variance**.

1. The summary sentence

It cites the three largest increases and the three largest reductions, the number of securities that are sold out, the share of the portfolio that changes place, and the asset class that moves the most (if its weight changes by at least 5 points).

2. The bars by asset class

Two stacked bars at 100%: “Current” and the chosen portfolio, split into Equities, Bonds, Gold, Money market. They only appear if the portfolio contains something other than equities. A security missing from the reference database is counted as an equity.

3. The “from → to” chart

For each security: a white circle at the current weight and an arrow to the suggested weight, with a label such as “40% → 34.7%”.

Colour	Meaning	Rule in the code
Green, arrow pointing right	To increase	positive difference
Red, arrow pointing left	To reduce	negative difference
Red	To sell entirely	suggested weight below 0.25%
Not shown	Unchanged	difference below 0.25 points in absolute value

The biggest change is at the top. At most 20 rows are drawn; a note under the chart gives the number of unchanged securities not shown and the number of small adjustments left to the detailed table.

The formulas

difference = suggested weight − current weight
amount = difference × total portfolio value
turnover = Σ |difference| / 2

Turnover is the share of the portfolio to be moved: every euro sold is reinvested elsewhere, hence the division by 2.

Verified example

Portfolio of €50,000, maximum Sharpe target:

Security	Class	Current	Suggested	Amount	Action
Stock X	Equities	30%	0.1%	−€14,950	Sell entirely
Bond fund	Bonds	10%	30%	+€10,000	Increase
Gold	Gold	5%	20%	+€7,500	Increase

Security	Class	Current	Suggested	Amount	Action
World ETF	Equities	40%	34.7%	−€2,650	Reduce
Stock Y	Equities	15%	15.2%	+€100	Unchanged

Turnover = (29.9 + 20 + 15 + 5.3 + 0.2) / 2 = **35.2%**. The summary says: increase Bond fund and Gold; reduce Stock X and World ETF; 1 security is sold out entirely; 35% of the portfolio changes place; the equity share goes from 85% to 50%.

Note that “Sell entirely” is displayed as soon as the suggested weight falls below 0.25%, even if the calculated amount is not quite equal to the value of the holding.

The “Adjustment details” table: how much should I buy or sell?

Under the “from → to” chart, the collapsed panel **Adjustment details (amounts to buy / sell)** lists **all** the securities, including the unchanged ones and those that do not fit in the chart.

The columns

Column	Content
Security	name of the security
Class	asset class (Equities, Bonds, Gold, etc.)
Current	current weight, in %
Suggested	weight in the chosen target portfolio, in %
Buy / sell	(suggested weight − current weight) × total value, in euros, signed
Action	Increase, Reduce, Sell entirely or Unchanged

The rows are sorted from the biggest change to the smallest. The sum of the amounts is zero: sales fund purchases.

Example

Portfolio of €50,000, one holding weighs 10% and the target portfolio gives it 30%: amount = (0.30 − 0.10) × 50,000 = **+€10,000** to buy.

What the amounts do not include

- brokerage **fees**;
- **taxes** on capital gains realised by selling (see the Wealth advisory area, Taxation tab);
- rounding to a whole number of units.

There is no export of stock market orders: the software is not connected to any broker. You have to place the orders yourself with your institution.

The limitations of Markowitz optimisation

Markowitz optimisation is a powerful teaching tool, but its results must be read with caution. The software itself displays this under the results.

1. Estimation error

The expected returns μ are the average of past returns. But an average estimated over a few years is very imprecise. Order of magnitude: for a security with a volatility of 20% observed over 3 years, the standard error on the average annual return is 20% / √3 ≈ **11.5 points**. A μ estimated at 10% is therefore compatible with a true return of between roughly −13% and +33%.

The optimiser takes these figures at face value: it **over-exploits the securities that have done best** and neglects those that have fallen. It is sometimes nicknamed an “error maximiser”.

2. Instability

A small change in the data (one more month, a different cap) can strongly change the maximum-Sharpe weights. The minimum-variance portfolio, which does not use μ, is more stable.

3. The model’s assumptions

- risk boils down to volatility (crashes, which are more frequent than the normal distribution predicts, are not better accounted for);
- correlations are assumed to be stable, whereas they often rise in times of crisis;
- a single horizon, with no transaction costs or taxes.

4. Limitations specific to the software

- only your current securities are considered;
- the estimation period begins at your first transaction and is limited to the days on which all securities have a price;
- prices exclude dividends.

How to use it intelligently

- above all, compare your portfolio with the **minimum variance**;
- lower the **Maximum weight per security** to obtain more reasonable allocations;
- see the differences as food for thought, not as orders;
- complement it with the “Risk budget” tab in the Asset management area, whose risk parity does not use expected returns either.

Why does the Optimisation tab require at least 2 securities?

Optimising means choosing an **allocation between several securities**. With a single security, there is only one possible allocation: 100% in that security. The tab then displays the message:

“The optimisation compares several allocations between securities: at least 2 securities are needed in the portfolio.”

The number of securities taken into account is that of the portfolio’s **current positions**.

If you hold a single ETF

A world ETF contains hundreds of stocks, but the software treats it as **a single security** in the optimisation: it cannot change the fund’s internal composition. To use the tab, you need at least two lines (for example an equity ETF and a bond ETF).

The “Optimisation failed” message

This message, followed by an explanation, appears when the calculation fails. Possible causes:

- the numerical optimiser did not converge (message “The optimisation did not converge”);
- a security does not have enough common history with the others to estimate the covariances.

Try another **Maximum weight per security** or **Refresh prices**. If only the frontier fails while the two notable portfolios are calculated, the software draws an approximate frontier (see the corresponding fiche).

With 2 or 3 securities, the maximum-weight slider only offers caps compatible with 100% investment and starts on “no limit”.

Why is the efficient frontier “approximate”?

When the chart legend says “Efficient frontier (approximate)”, it means the optimiser could not calculate at least two points of the exact frontier. The software then uses a **fallback solution** drawn from the cloud of 4,000 random portfolios.

How the approximate frontier is built

1. the random portfolios are sorted by increasing volatility;
2. only those whose return reaches the best return already seen at a lower volatility (the “running maximum”) are kept.

keep a portfolio if: return ≥ best return of the less risky portfolios

This gives the **upper envelope** of the cloud: for each level of risk, the best return already reached by a randomly drawn portfolio.

Example

Portfolios sorted by volatility: (8%; 4%), (9%; 5%), (10%; 4.5%), (11%; 6%). The third is discarded (4.5% < 5%); the envelope passes through the other three.

Frequent causes

- a security with too short a history;
- very tight constraints (low cap with few securities);
- a numerical failure of the optimiser on certain target returns.

What this changes

The approximate frontier is **slightly below** the true frontier (a random draw never reaches the optimum exactly) and may look like a staircase. It remains useful for placing your portfolio. It ignores the per-security cap, since the random portfolios do not respect it. The "Minimum variance" and "Maximum Sharpe" cards remain calculated exactly.

What is a Monte Carlo projection?

We cannot predict **the** future, but we can simulate **thousands of possible futures**, all consistent with the portfolio's return and risk, and then study the distribution of the results. This is the Monte Carlo method, used in the **Projection** tab.

What the software does

1. It starts from the **current value** of your portfolio.
2. It simulates **5,000 scenarios**, **month by month**, up to the chosen horizon.
3. Each month, the value is multiplied by a randomly drawn growth, then the monthly contribution is added:

Value(month + 1) = Value(month) × exp(log-return of the month) + monthly contribution

4. At each date, it calculates the 5th, 25th, 50th, 75th and 95th percentiles of the 5,000 values.

The time step is the **month** (not the day): this is sufficient for a projection over several years and about 20 times faster.

What is displayed

- four cards: **Adverse scenario** (5th percentile), **Median scenario**, **Favourable scenario** (95th percentile) and **Probability of loss**, plus **Target reached** if a target is entered;
- the fan chart or scatter plot of the paths;
- the distribution of the final value and its reading sentences.

Reproducible results

The randomness is fixed by a "seed": with the same assumptions, you get exactly the same figures each time it is displayed. Changing a setting launches a new simulation.

Example

€100,000, return 7%, volatility 15%, 10 years, no contributions, normal distribution: median **€178,727**, adverse scenario **€81,876**, favourable scenario **€389,613**, probability of loss **10.4%**. Theory gives a median of €179,948: the difference comes from the finite number of scenarios.

The adjustable assumptions of the projection

The “Simulation assumptions” box, at the top of the Projection tab, gives in its subtitle the portfolio’s **historical return and volatility**, then offers six settings.

Setting	Range	Starting value
Horizon (years)	1 to 30 years	10 years
Monthly contribution (€)	€0 to €1,000,000, steps of €100	€0
Target (€, optional)	€0 to €100,000,000, steps of €5,000	0 (no target)
Method	**Normal distribution** or **Historical (bootstrap)**	Normal distribution
Assumed annual return (%)	−5% to 20%, steps of 0.5	historical return
Annual volatility (%)	1% to 50%, steps of 0.5	historical volatility

Where the historical values come from

They are calculated from the portfolio’s daily returns (those of the TWR performance):

historical return = average daily return × 252
historical volatility = standard deviation of daily returns × √252

The sliders start on these values, **rounded to the nearest 0.5 point** and **brought back within the range** of the slider. Example: a historical return of 12.3% gives a slider at 12.5%; a historical return of 35% is brought back to 20%.

Tips

- A historical return obtained over a short, favourable period is too optimistic: **reduce it** for a cautious projection.
- The **monthly contribution** is added at the end of each month. The total amount invested is current value + contribution × number of months.
- The **target** adds a probability card and a dotted horizontal line on the fan chart.

Why the volatility is greyed out

With the historical method, the volatility slider is disabled: the dispersion comes from the portfolio’s actual trading days, drawn at random. The return slider remains active (see the fiche on the two methods).

The amounts are **nominal**: neither inflation, nor taxes, nor fees are deducted.

Normal method or historical method (bootstrap)?

The **Method** button offers two ways of randomly drawing each month’s return.

1. Normal distribution (geometric Brownian motion)

This is the Black-Scholes model. Over one month (dt = 1/12 year), the log-return follows:

monthly log-return ~ Normal(mean = $(\mu - \sigma^2/2) \times 1/12$, standard deviation = $\sigma \times \sqrt{(1/12)}$)

With μ = 7% and σ = 15%: monthly mean = (0.07 − 0.01125) / 12 = **0.490%**, monthly standard deviation = 0.15 × 0.2887 = **4.33%**.

Why "$-\sigma^2/2$"? The return μ is an arithmetic average, but compound growth is lower: +50% then −50% gives an average of 0%, but you end up at 0.75, i.e. −25%. The $-\sigma^2/2$ term corrects this effect (Itô's lemma).

2. Historical (bootstrap)

Each month is built by randomly drawing, **with replacement**, **21 actual daily returns** of the portfolio, and adding their logarithms. This preserves the real shape of the returns: **fat tails** (crashes more frequent than in the normal distribution) and skewness.

The return is then **re-centred** on the **Assumed annual return (%)** slider:

σ = standard deviation of daily returns × √252
target = $(\mu - \sigma^2/2) / 252$ (targeted average daily log-return)
re-centred log-returns = log-returns − their mean + target

Example: μ = 7%, σ = 15%: target = 0.05875 / 252 = 0.0233% per day, i.e. 0.490% over 21 days, as with the normal distribution. Only the shape of the distribution changes.

This method requires at least 20 daily returns in the history.

Which one should I choose?

- **Normal distribution**: simple, standard, adjustable (you choose the volatility). It underestimates crashes.
- **Historical**: more realistic on extremes, but limited to the events of your analysis period. If your history contains no crash, it will not invent one.

Compare the two: an adverse scenario clearly lower with the historical method signals fat tails in your portfolio (see also the Risk tab).

Reading the fan chart and the scatter plot of the projection

The **Display** selector offers three views: **Fan chart** (default), **Scatter plot** or **Both**.

The fan chart

Element	Meaning
Light blue area	90% of scenarios (between the 5th and 95th percentiles)
Darker blue area	50% of scenarios (between the 25th and 75th

Element	Meaning
	percentiles)
Blue curve	median scenario (50th percentile)
Orange dashes	money invested (starting value + cumulative contributions)
Dotted line “Target”	your target, if entered

Hover over the curve to read, at each date, the median and the 90% range.

Warning: the percentiles are calculated **date by date**. The median curve is not a real scenario: no scenario stays in the middle all the time.

The fan **widens with the horizon**: this is uncertainty accumulating. With the normal distribution, the standard deviation of cumulative log-returns grows as σ × √years.

The scatter plot

Each point is **one scenario at a given date**. The software displays the **first 400 scenarios**, every 6 months (every year if the horizon exceeds 15 years), plus the final date. Points on the same date are slightly offset horizontally to stay legible.

Each point is coloured according to its band **at that date**:

Colour	Band
Dark red	Worst 5% (below the 5th percentile)
Light red	Adverse (5 to 25%)
Grey	Central (25 to 75%)
Light blue	Favourable (75 to 95%)
Dark blue	Best 5% (above the 95th percentile)

Reference marks complete the cloud: median, 5% and 95% thresholds (dotted) and money invested. Hover over a point to see its scenario number and its date: the same scenario can change band over time.

The scatter plot shows the **real dispersion** of the results, which the fan chart summarises.

The final-value distribution: why is it not symmetrical?

The “Distribution of the final value” chart is a histogram: the portfolio value at the horizon, for each of the 5,000 scenarios. It carries three vertical markers: **Amount invested** (orange dashes), **Median** (solid line) and **Mean** (dotted), as well as a bluish zone “90% of scenarios (P5 to P95)”.

Why the distribution is log-normal

Returns **compound**: the final value is a **product** of monthly growths. It is therefore its **logarithm** (a sum) that follows a roughly normal distribution. The value itself follows a **log-normal**

distribution: bounded on the left (you cannot lose more than 100%), with a long right tail (compounded gains have no ceiling).

With no contributions and the normal distribution:

median = initial value × exp((μ − σ²/2) × T)
mean = initial value × exp(μ × T)
mean / median = exp(σ² × T / 2)

Example

€100,000, μ = 7%, σ = 15%, T = 10 years:

- theoretical median = 100,000 × exp(0.5875) ≈ **€179,948** (simulated: €178,727);
- theoretical mean = 100,000 × exp(0.70) ≈ **€201,375** (simulated: €199,757);
- ratio = exp(0.1125) ≈ 1.12: the mean exceeds the median by about 12%.

Mean or median?

A few very favourable scenarios pull the **mean** upwards. The **median** (a one-in-two chance of doing better) is the most representative marker. The software says so in the reading sentences as soon as the mean exceeds the median by more than 2%.

P5 and P95

- **P5**: 5% of scenarios end below it ("1 chance in 20 of doing worse"). In the example: €81,876.
- **P95**: 5% of scenarios end above it. In the example: €389,613.

The first reading sentence summarises: "In 90% of scenarios, the value in 10 years lies between … and …".

The logarithmic scale

The **Log scale** switch plots the histogram of the logarithm of the value. The bell shape becomes roughly **symmetrical** again, centred on the median. The tick labels remain in euros (for example €50k, €100k, €200k, €500k).

What is not displayed

For legibility, the axis stops at the 0.5th and 99.5th percentiles (widened if needed to show the amount invested): the most extreme 1% of scenarios are not drawn, but they count in all the figures. With monthly contributions, the distribution is no longer exactly log-normal, but the shape remains the same.

Probability of loss, of doubling or of reaching a target

The cards and sentences of the Projection tab simply count the **share of the 5,000 scenarios** that meet a condition, at the chosen horizon.

The formulas

total invested = current value + monthly contribution × number of months
probability of loss = share of scenarios whose final value < total invested
probability of doubling = share of scenarios whose final value ≥ 2 × total invested
probability of target = share of scenarios whose final value ≥ target
adverse scenario = 5th percentile of the final values
median scenario = 50th percentile
favourable scenario = 95th percentile

Verified example

€100,000, contribution of €200 a month, 10 years, μ = 7%, σ = 15%, normal distribution, target €250,000:

- total invested = 100,000 + 200 × 120 = **€124,000**;
- probability of loss = **10.6%**;
- probability of doubling (finishing above €248,000) = **35.9%**;
- probability of reaching €250,000 = **35.2%**;
- adverse scenario €104,075, median €211,406, favourable €439,552.

Interpretation

- "Adverse scenario" does not mean "worst case": 1 scenario in 20 does even worse.
- The probability of loss compares the final value with the **sum paid in**, without remunerating it: finishing at €124,500 for €124,000 paid in is not a loss in the software's sense, but it is a poor investment.
- It generally decreases with the horizon, because the average rise accumulates faster than the dispersion.

Limitations

All the amounts are **nominal**: neither inflation, nor taxes, nor fees are deducted. A loss of purchasing power is therefore not counted as a loss. These probabilities depend entirely on the return and volatility chosen.

A projection is not a forecast

A Monte Carlo projection answers the question: **"if the assumptions hold, what is the range of possible outcomes?"**. It does not say what will happen. The software reminds you of this in the "How to read this projection" box: the median is not a forecast, it is the middle of the possibilities.

What the projection assumes

- a **constant** return and volatility over the whole horizon;
- months that are **independent** of one another (no cycles, no trend);
- for the normal distribution, returns without fat tails;
- for the historical method, a future that resembles the days of your analysis period;

- a portfolio composition that does not change.

What it ignores

- inflation, taxes and fees: the amounts are nominal and gross;
- changes of strategy, withdrawals or switches;
- events absent from the history or from the model.

Why it can be too optimistic

The default return is the portfolio's **historical** return. Over a short, favourable period, it can be very high. Example: over 10 years with €100,000, going from 7% to 4% assumed return (volatility 15%) brings the theoretical median down from €179,948 to 100,000 × exp((0.04 − 0.01125) × 10) ≈ **€133,309**.

How to use it

- test several returns, including a cautious one;
- compare the normal distribution and the historical method;
- look above all at the P5 – P95 **range** rather than a single figure;
- re-run the analysis regularly, as the history lengthens.

7. Wealth advisory

This chapter describes the "Wealth advisory" workspace and its two tabs. The **Taxation** tab compares what would be left of your gain after tax in a securities account (CTO), a PEA (French equity savings plan) or a life insurance contract (assurance-vie), at 2026 rates, today or at a future horizon, and measures the share of your portfolio that is eligible for a PEA. The **Stress tests** tab replays five past crises and four to five hypothetical shocks on your current portfolio. The tax rules are deliberately simplified: each fiche states what is modelled and what is not.

What does the Wealth advisory workspace contain?

The **Wealth advisory** workspace is selected from the "Workspace" menu in the sidebar. It analyses the same portfolio, with the same settings, as the "Portfolio analysis" workspace. It has two tabs.

Tab	Question it answers
Taxation	How much is left after tax, depending on the wrapper (securities account, PEA, life insurance) and the exit date? What share of the portfolio can go into a PEA?
Stress tests	What would the current portfolio lose if a past crisis happened again, or if a hypothetical shock occurred?

What the workspace does not do

- It contains no client questionnaire, no regulatory risk profile and no suitability test: these functions are not part of the software. The only profile choice (Cautious, Balanced, Dynamic) is found in the Exposures tab of the "Portfolio analysis" workspace.
- It does not replace personalised tax advice: the progressive income tax scale, the contribution ceilings and contract fees are not modelled (see the fiche on simplifications).

The main results of these two tabs also appear in the PDF report; the taxation there is calculated for a single person, the software's default setting.

As the footer reminds you, the tool is educational and does not constitute investment advice.

Which tax rates does the software use (2026)?

The software applies the rules for an **individual who is tax resident in France, in 2026**. The source cited in the code is Law No. 2025-1403 of 30 December 2025 on the financing of social security for 2026, which raised the CSG (a social contribution) on investment income.

Rate	Value	Used for
Income tax (PFU)	12.8%	CTO; PEA before 5 years; life insurance before 8 years
Social contributions, investment income	**18.6%**	CTO and PEA
Social contributions, life insurance	**17.2%**	life insurance (not affected by the increase)
Total flat tax (12.8 + 18.6)	**31.4%**	CTO; PEA before 5 years
Income tax, life insurance after 8 years	7.5%	above the allowance
Annual allowance, life insurance after 8 years	€4,600 (single), €9,200 (couple)	on the part of the gain subject to income tax

The latest change

Before 2026, the flat tax was 30% (12.8% + 17.2%). The CSG increase takes social contributions to 18.6% and the flat tax to **31.4%**. Life insurance stays at 17.2% social contributions: before 8 years it is therefore taxed at 30% in total, slightly less than the securities account.

The subtitle of the “Assumptions” box on the tab shows “2026 rates”, and a note at the bottom of the “PEA eligibility” box summarises these rates.

What is not modelled

The option for the progressive income tax scale, the 12.8% rate on life insurance above €150,000 of premiums, and the partial deductibility of the CSG are not taken into account.

How is tax calculated on a securities account (CTO)?

On an ordinary securities account (CTO), the software applies the **31.4%** flat tax to the taxable base, with no advantage linked to the holding period.

The exact formula

capital gains = total gain − dividends
taxable base = max(0 ; capital gains) + max(0 ; dividends)
income tax = 12.8% × base
social contributions = 18.6% × base

The **total gain** is the one shown in the Overview tab: unrealised gains + realised gains + net dividends.

The idea: a loss does not wipe out dividends

Capital losses are offset against capital gains, **not against dividends**. A portfolio that is down overall may therefore still owe tax on its dividends.

Examples

Gain of €20,000, including €2,000 of dividends: base = 18,000 + 2,000 = €20,000; tax = €2,560 + €3,720 = **€6,280**; net gain = **€13,720** (effective rate 31.4%).

Loss of €5,000, but €2,000 of dividends: capital gains = −€7,000; base = 0 + 2,000 = €2,000; tax = €256 + €372 = **€628**, even though the total gain is negative.

Interpretation

On the "CTO exit" card, the "No holding-period advantage" label is a reminder that the rate never falls over time. This is the benchmark against which to compare the PEA and life insurance.

Limitations

The calculation assumes the whole gain is taxed at once, on exit. In reality, on a CTO, dividends and realised gains are taxed in the year they are received. The option for the progressive scale, sometimes more favourable for low incomes, is not modelled.

How is tax calculated on a PEA?

The **PEA (plan d'épargne en actions, a French equity savings plan)** offers an exemption from income tax after 5 years. The software calculates the tax as if your portfolio had been held in a PEA since your first transaction.

The exact formula

base = max(0 ; total gain)
before 5 years: income tax = 12.8% × base (a withdrawal closes the plan)
after 5 years: income tax = 0
social contributions = 18.6% × base (in all cases)

Age is measured from the date of your **first transaction** to the last date in the history, divided by 365.25 days.

Example

Gain of €20,000:

- age of 3 years: €2,560 + €3,720 = €6,280 of tax, net gain **€13,720** (31.4%, as for the CTO). The card shows "Tax advantage in 2.0 year(s)";
- age of 9 years: €0 + €3,720 = €3,720, net gain **€16,280** (18.6%). The card shows "Tax advantage reached".

Difference after 5 years: **€2,560** more in your pocket, which is the 12.8% income tax.

Interpretation

Unlike the CTO, a loss never creates tax: the base is the total gain, floored at zero if negative, without distinguishing dividends from capital gains.

Limitations

- The €150,000 contribution ceiling is not applied in the calculation; a note flags it if your invested amount exceeds it.
- Only certain securities are eligible: see the fiche "What share of my portfolio can go into a PEA?".
- The calculation applies the PEA to the **whole** portfolio, even to ineligible holdings: it is a theoretical comparison.

How is tax calculated on life insurance (assurance-vie)?

The software treats the portfolio as if it were held in **unit-linked funds** (unités de compte) of a life insurance contract (assurance-vie) opened on the date of your first transaction.

The exact formula

base = max(0 ; total gain)
social contributions = 17.2% × base
before 8 years: income tax = 12.8% × base
after 8 years: income tax = 7.5% × max(0 ; base − allowance)
allowance = €4,600 (single) or €9,200 (couple)

Marital status is chosen with the **Marital status (life insurance allowance)** control at the top of the **Taxation** tab. The default setting is "Single".

Examples

Gain of €20,000:

Situation	Income tax	Social contributions	Net gain	Effective rate
3 years old	€2,560	€3,440	€14,000	30.0%
9 years, single	7.5% × 15,400 = €1,155	€3,440	€15,405	23.0%
9 years, couple	7.5% × 10,800 = €810	€3,440	€15,750	21.3%

Interpretation

- Before 8 years, life insurance (30%) is slightly more favourable than the CTO and the PEA under 5 years (31.4%), because its social contributions remain at 17.2%.
- After 8 years, the allowance makes life insurance very attractive for modest gains: a gain below €4,600 (€9,200 for a couple) bears only social contributions.

- Compared with a PEA over 5 years old (18.6%), life insurance over 8 years old (17.2% + 7.5% above the allowance) is taxed less only for small gains: up to about €5,656 for a single person (0.172 G + 0.075 (G − 4,600) = 0.186 G) and €11,311 for a couple. Beyond that, the PEA wins.

Limitations

- The allowance is annual in reality; here, the full exit is made in one go, so only one allowance applies.
- Premiums are assumed to be below €150,000 (above that, the rate is 12.8% rather than 7.5%); a note flags it if your invested amount exceeds that threshold.
- The contract's management fees are not deducted.
- In reality, the tax on a partial withdrawal applies only to the share of gains contained in the withdrawal; the software simulates a full exit.

Reading "If everything were sold today": cards and wrapper table

In the **Taxation** tab, the "If everything were sold today" section assumes you sell the **whole portfolio** today, and compares the result in the three wrappers. Its subtitle shows the **gross gain** (the total gain from the Overview tab).

The three cards

One card per wrapper: "CTO exit", "PEA exit", "Life insurance exit". Each shows:

- the **net gain** after tax, in euros;
- a "tax x%" badge: the **effective rate**;
- the state of the tax advantage: "No holding-period advantage" (CTO), "Tax advantage in n year(s)" or "Tax advantage reached".

The formulas

net gain = gross gain − income tax − social contributions
effective rate = total tax / gross gain (0 if the gain is negative)
gross return = gross gain / amount invested
net return = net gain / amount invested
years before advantage = max(0 ; wrapper term − age) (5 years PEA, 8 years life insurance)

Example

Amount invested €100,000, gain €20,000 including €2,000 of dividends, age 3 years, single:

Wrapper	Income tax	Social contributions	Net gain	Net return
CTO	€2,560	€3,720	€13,720	13.7%
PEA	€2,560	€3,720	€13,720	13.7%
Life insurance	€2,560	€3,440	€14,000	14.0%

Gross return: 20% for all three. The PEA shows “Tax advantage in 2.0 year(s)”, life insurance “in 5.0 year(s)”.

The table

Below the cards, a table gives for each wrapper: Gross gain, Income tax, Social contributions, Net gain, Gross return and Net return.

Interpretation

This is not your actual tax: it is what **you would have paid** if this same portfolio had been held in each of the wrappers since your first transaction. The tool is for comparing wrappers, not for preparing a tax return.

The “Net gain by exit year” chart

On the left, below the cards in the **Taxation** tab, the “Net gain by exit year” chart shows, for each wrapper, the net gain after tax if you sold everything in 0 to 20 years.

The setting

The **Assumed annual return for future exits (%)** slider (0 to 10%, in steps of 0.5, default 5%) sets the assumed growth of the portfolio.

The formula

gain in n years = current gain + current value × ((1 + return)^n − 1)
age in n years = current age + n
net gain = gain − wrapper tax (calculated with this age)

Dividends stay fixed at their current amount (which matters only for the CTO).

The jumps

The PEA and life insurance curves are **step-shaped**: they jump upwards when the age passes **5 years** (PEA) or **8 years** (life insurance). The CTO curve, with no holding-period advantage, is continuous.

Example

Current value €120,000, current gain €20,000 (including €2,000 of dividends), age 3 years, assumed return 5%, single:

Exit in	Gross gain	CTO	PEA	Life insurance
0 years	€20,000	€13,720	€13,720	€14,000
1 year	€26,000	€17,836	€17,836	€18,200
2 years (PEA at 5 years)	€32,300	€22,158	**€26,292**	€22,610
5 years (life	€53,154	€36,464	€43,267	**€40,370**

Exit in	Gross gain	CTO	PEA	Life insurance
insurance at 8 years)				

Interpretation

The chart shows why an adviser recommends opening these wrappers **early**, to "start the clock": it is the opening date that starts the waiting periods, not the date of the payments. Hover over a curve to read the net gain of each wrapper for a given exit year.

Limitations

Steady, certain growth, with no contract fees; same simplifications as for the immediate exit.

What share of my portfolio can go into a PEA?

The "PEA eligibility" box, to the right of the chart in the **Taxation** tab, shows the "PEA-eligible share" card and the number of ineligible holdings.

The software's rule

A holding is considered eligible if:

- it is a **share** (asset class "Equities") of a company whose country is in the **European Union or the EEA** (Norway, Iceland and Liechtenstein included); or
- it is one of the **index funds recognised as eligible**: CW8.PA (Amundi MSCI World) and ESE.PA (BNP Paribas Easy S&P 500), eligible thanks to their synthetic replication.

eligible share = value of eligible holdings / total portfolio value

What is not eligible

- US, British (since Brexit), Swiss or Japanese shares;
- bond funds and gold (the PEA is reserved for equities);
- ETFs not listed above, even if they are in fact eligible;
- securities whose country is not recorded in the reference data.

Example

Portfolio of €100,000: €40,000 of French and German shares, €30,000 of CW8.PA, €20,000 of US shares, €10,000 of bond ETF. Eligible share = (40,000 + 30,000) / 100,000 = **70%**; 2 ineligible holdings.

The notes displayed

- if the share is below 100%: only the eligible share can go into a PEA; the rest would go into a securities account or into life insurance unit-linked funds;
- if the amount invested exceeds €150,000: a reminder of the PEA contribution ceiling and the life insurance threshold;
- in all cases, a reminder of the 2026 rates.

Limitations

The list of eligible funds is short: check the actual eligibility of your ETFs in their documentation. The wrapper comparison fiche applies the PEA to the whole portfolio, even to ineligible holdings.

The simplifications in the tax calculation

The **Taxation** tab is an **educational comparison tool**. It is not designed to calculate your actual tax or to fill in a tax return.

What is simplified

Simplification	Consequence
Option for the progressive scale ignored	for a lightly taxed household, the actual tax may be lower
Life insurance premiums assumed below €150,000	above that, the actual rate after 8 years is 12.8% rather than 7.5% (a note flags it)
PEA contribution ceiling (€150,000) not applied	a note flags it if the amount invested exceeds it
Contract management fees ignored	the net gain on life insurance is overstated
Dividends assumed to stay in the wrapper	the total gain (unrealised + realised + dividends) is taxed in one go on exit
Full exit in one go	a single life insurance allowance, no staggered partial withdrawals
Same portfolio in all three wrappers	the PEA is applied even to ineligible holdings
Age = since the first transaction	a wrapper opened earlier would already have started its clock

What is not covered at all

Inheritance, gifts, the property wealth tax (IFI), retirement savings plans (PER) and non-resident situations are not covered.

Why keep the model simple?

A simple model makes the essentials visible: the difference in rates between wrappers and the effect of the 5- and 8-year waiting periods. For a real decision, have the calculation validated by a professional.

Historical stress tests: what would my portfolio lose in a crisis?

The **Stress tests** tab applies to **each holding in your current portfolio** the change it actually suffered during a past crisis, from the market peak to the market trough.

The five crises replayed

Crisis	From (peak)	To (trough)
Financial crisis (2008-2009)	01/09/2008	09/03/2009
European debt crisis (2011)	01/07/2011	22/09/2011
Covid crash (2020)	19/02/2020	23/03/2020
Inflation and rate hikes (2022)	03/01/2022	12/10/2022
August 2024 mini-crash	16/07/2024	05/08/2024

Dates are in DD/MM/YYYY format.

The formula

change of a holding = price at end date / price at start date − 1
portfolio change = Σ current weight of the holding × change of the holding
loss in euros = portfolio change × current value

The price used at each date is the last known price at that date.

Example

50% in a holding that lost 45%, 30% in a holding that lost 40% and 20% in a bond fund that gained 5%: change = −22.5% − 12% + 1% = **−33.5%**, i.e. −€33,500 on €100,000.

What is displayed

- three cards: “Worst historical scenario”, “Corresponding loss” (in euros, on the current value) and “Portfolio beta”;
- a bar chart “Past crises replayed on the current portfolio” (green for a gain, red for a loss);
- the table “Historical scenarios in detail”: Scenario, Period, Change, Gain / loss, Share estimated with an index;
- the **Show the detail by holding** list, which shows, for the selected crisis, the change of each security and its source (the security itself or an index).

First display

Prices since 2008 have to be downloaded: the message “Replaying past crises (history since 2008)…” is shown while the calculation runs. The prices are then kept in a separate cache file. Without Internet on the first display, the message “Stress tests unavailable” may appear.

Interpretation

"At the worst point of 2008, this portfolio would have lost 33.5%, or €33,500" speaks to a client more than a volatility figure does. The weights are those of **today**: the test applies to the current composition, not to the portfolio you held at the time.

A security not listed at the time: how does the stress test estimate it?

Many securities did not exist during the older crises. The software then replaces the security's change with that of a **representative index or fund** (a "proxy").

When is a security replaced?

When it has no price history, or when its first known price is more than 7 days after the start of the crisis.

Which substitute?

For a bond or gold (an asset class other than "Equities"), a fund of the same category, listed since 2007 at the latest:

Category	Fund used
Government bonds	IEF
Inflation-linked bonds	TIP
Corporate bonds	LQD
High-yield bonds	HYG
Emerging-market bonds	EMB
Gold	GLD

An equity index would be a poor approximation: in 2008, government bonds rose while equities fell.

For an equity (or if the category is not recognised), the index of its region:

Region	Index
United States, World (ETF)	^GSPC (S&P 500)
Europe	^STOXX50E (Euro Stoxx 50)
United Kingdom	^FTSE
Switzerland	^SSMI
Japan	^N225
Asia-Pacific	^AXJO
Canada	^GSPTSE
Emerging markets	EEM

If the region is unknown, or if the substitute has no price history either, the S&P 500 is used. If no price is available at all, the change of the holding is counted as 0.

The "Share estimated with an index" column

estimated share = Σ current weights of the holdings replaced by an index or a fund

Example: a 30% holding replaced by ^GSPC and a 10% holding replaced by IEF give **40%**. The higher this figure, the more the result is an approximation. The detail by holding shows the source ("security" or "index ^GSPC").

Limitations

A "world" ETF approximated by the S&P 500 ignores the rest of the world; a small-cap approximated by a large index often underestimates its fall.

Hypothetical shocks: a fall in equities and in the dollar

To the right of the past crises, the "Hypothetical shocks" chart applies simple scenarios. Its subtitle summarises the rules: "Equities: beta × shock · Dollar: share invested in dollars · Rates: duration".

Equities falling by 10%, 20% and 35%

portfolio change ≈ beta × shock

The **beta** is the one from the Risk tab: the portfolio's sensitivity to the index chosen with the **Benchmark index** menu in the Settings. A beta of 0.85 means the portfolio moves on average by 0.85% when the index moves by 1%.

Example: beta 0.85, shock −20%: change = 0.85 × −20% = **−17%**, i.e. −€17,000 on €100,000.

The dollar falling by 10%

change = −10% × share of the portfolio quoted in dollars

Example: 35% of the portfolio in USD-quoted securities: change = −10% × 35% = **−3.5%**.

Note: the software looks at the **trading currency**. A euro-quoted ETF that invests in US equities is not counted, even though it does in fact suffer the currency effect. For a full view of currency exposure, see the Exposures tab.

Interpretation

Beta summarises market risk in a single figure. It is convenient but linear: in a crisis, correlations rise and the actual loss may exceed beta × shock. This is why the historical scenarios, which use real changes, complement these shocks.

Limitations

- beta is estimated over your analysis period;
- the dollar shock ignores the effect on the companies themselves (exporters, etc.);
- a shock to an index says nothing about a shock specific to one sector.

The interest-rate rise shock and duration

The “Rates up 1 point” scenario estimates the loss on the **bond funds** in your portfolio if interest rates rose by 1 percentage point (for example from 3% to 4%).

Duration

A bond’s **duration** (in years) measures its sensitivity to rates: a duration of 7 years means that a 1-point rise in rates lowers its price by about 7%. When rates rise, older bonds, which pay less than new ones, lose value.

The formula

portfolio change = −1% × Σ (weight of the holding × duration of the holding)

Holdings with no duration (equities, gold) count as 0. The duration of each fund comes from the software’s securities reference data.

Example

15% of the portfolio in a fund with a duration of 7 years and 5% in a fund with a duration of 3 years:

- change = −1% × (0.15 × 7 + 0.05 × 3) = −1% × 1.2 = **−1.2%**, i.e. −€1,200 on €100,000;
- average bond duration = 1.2 / 0.20 = 6 years.

When does this scenario appear?

Only if at least one holding has a known duration. A 100% equity portfolio, or bond funds absent from the reference data, will not show it.

Limitations

- the relationship is linear: for a large shock, convexity makes the actual loss somewhat smaller;
- the effect of a rate rise on **equities** is not taken into account in this scenario (see instead the “Inflation and rate hikes (2022)” crisis among the historical scenarios);
- the durations in the reference data are fixed values, which in reality change over time.

The limitations of the stress tests

The stress tests give orders of magnitude, not exact figures.

The simplifications

- **Local currency**: in the historical scenarios, changes are calculated in each security’s trading currency. The currency effect for a euro-based investor is ignored: a rise or fall of the dollar during the crisis would have cushioned or worsened the actual loss in euros.
- **Excluding dividends**: prices are not adjusted for dividends. Over a crisis lasting a few months, the difference is small.

- **Current weights**: the test applies to today's composition, not the one you had at the time. It is therefore not what you actually lost.
- **From market peak to market trough**: the dates are those of the market as a whole; a security may have hit its low on a different day.
- **Proxies**: a recent security is replaced by an index or a fund (see the "Share estimated with an index" column).
- **Linear shocks**: beta × shock and duration × rate rise are first-order approximations.

The worst can be worse

The five crises do not cover every risk: a future crisis may be different (lasting inflation, geopolitical shock, sector crash). The worst scenario displayed is not a limit.

If the message "Stress tests unavailable" appears

The most frequent cause is the lack of an Internet connection on the first display: the history since 2008 has not yet been downloaded. Reconnect and reopen the tab. After that, the cache takes over.

8. Asset management

This chapter presents the “Asset management” space, which brings together the tools of a portfolio manager. The **Performance attribution** tab explains the gap with a world index using the Brinson-Fachler model. The **Risk budget** tab measures where the risk comes from and compares your allocation with risk parity. The **Strategy backtest** tab tests rebalancing and gradual investing on real historical data. Each entry gives the exact formula, a worked example and the limitations.

What does the Asset management space contain?

The **Asset management** space is chosen in the “Workspace” menu of the sidebar. It analyses the same portfolio, with the same settings, as the “Portfolio analysis” space. It has three tabs.

Tab	Question it answers
Performance attribution	Why did the portfolio do better (or worse) than a world index? Choice of regions or choice of securities?
Risk budget	Which holdings bring the most risk? What would an allocation in which every holding brings the same amount of risk look like?
Strategy backtest	Would it have been better to rebalance regularly? To invest all at once or gradually?

Good to know

- The attribution downloads regional indices: Internet is needed the first time it is displayed, after which a cache takes over.
- The risk budget and the backtest use the prices already loaded for the analysis.
- The diversification diagnosis (regions, sectors, currencies, correlations) is in the Exposures tab of the “Portfolio analysis” space: the Risk budget tab recalls this with a note.
- The main results also appear in the PDF report (backtest with 0.1% fees and comparison on €10,000 over 12 months).

In case of an error, each tab displays a “… unavailable” message followed by the cause, without blocking the others.

Brinson-Fachler performance attribution

The **Performance attribution** tab breaks down the return gap between your portfolio and a benchmark index, **region by region**, into three effects.

The exact formulas (for one month)

Allocation effect = (wp − wb) × (rb − Rb)
Selection effect = wb × (rp − rb)
Interaction effect = (wp − wb) × (rp − rb)

- wp, wb: weight of the region in the portfolio and in the index;
- rp, rb: return of the region in the portfolio and in the index;
- Rb: total return of the index = Σ wb × rb.

The sum of the three effects across all regions equals the gap Rp − Rb.

The intuition

- **Allocation**: did you overweight the regions that did better than the overall index?
- **Selection**: within each region, did your securities do better than that region's index?
- **Interaction**: cross effect, positive if you overweighted a region where you also chose well.

Verified example

Region	wp	rp	wb	rb
United States	50%	+10%	70%	+8%
Europe	50%	+2%	30%	+4%

Rb = 0.7 × 8% + 0.3 × 4% = **6.8%**; Rp = 0.5 × 10% + 0.5 × 2% = **6.0%**; gap **−0.8 points**.

Region	Allocation	Selection	Interaction
United States	(−0.2) × (8 − 6.8) = −0.24	0.7 × 2 = +1.40	(−0.2) × 2 = −0.40
Europe	0.2 × (4 − 6.8) = −0.56	0.3 × (−2) = −0.60	0.2 × (−2) = −0.40
Total	**−0.80**	**+0.80**	**−0.80**

Sum: −0.80 + 0.80 − 0.80 = **−0.80 points**, exactly the gap. Reading: the selection of US securities was good, but the overweight in Europe, which did worse than the index, cost performance.

What is displayed

Six cards (Portfolio, Benchmark index, Difference, Allocation effect, Selection effect, Interaction effect), the "Effects by region" chart, the "Cumulative effects over time" chart and the "Detail by region" table. In this table, the portfolio weights are **monthly averages** and the regions' returns are **compounded** over the period.

Special rules

- a region absent from the index takes rb = Rb: its allocation and selection effects are zero, everything goes into **interaction**;
- a region absent from the portfolio takes rp = rb: only the allocation effect applies.

Why a “Cariño smoothing”? The month-by-month calculation

The attribution is calculated **month by month**, with the weights at the start of each month, then the months are linked.

The problem

Returns compound: +10% then +10% gives +21%, not +20%. Adding up the monthly gaps therefore does not give the total gap back.

Cariño’s method (1999)

Each monthly effect is multiplied by a factor k_t / K:

$k_t = [\ln(1 + Rp_t) - \ln(1 + Rb_t)] / (Rp_t - Rb_t)$ (returns of month t)
$K = [\ln(1 + Rp) - \ln(1 + Rb)] / (Rp - Rb)$ (compounded returns of the whole period)
if the two returns are equal: $k = 1 / (1 + R)$
linked effect of month t = effect of month t × k_t / K

In this way, the sum of the linked effects is **exactly** equal to the compounded gap Rp − Rb. An automatic test in the software checks it.

Verified example

Two months: portfolio +10% then +10% (Rp = 21%), index +5% then +5% (Rb = 10.25%).

- Actual gap: 21% − 10.25% = **10.75 points**; sum of the monthly gaps: 5 + 5 = 10 points.
- k_t = (ln 1.10 − ln 1.05) / 0.05 = 0.9304; K = (ln 1.21 − ln 1.1025) / 0.1075 = 0.8655.
- Factor = 0.9304 / 0.8655 = **1.075**; each month is worth 5 × 1.075 = 5.375 points; total **10.75 points**. The sum is right.

Splitting into months

The software takes the last trading day of each month of the history (the latest available date counts as the end of the current month) and calculates each month from one month end to the next. At least two month ends are needed; otherwise the message “History too short: at least two month ends are needed” is displayed.

The “Cumulative effects over time” chart

It plots the cumulative sum of the linked allocation, selection and interaction effects, month after month. The three curves end on the values of the cards. A curve that rises steadily signals a persistent effect; a jump, an exceptional month.

Limitation

Weights are frozen at the start of each month: a purchase or sale during the month is only taken into account the following month. The “Portfolio” return is therefore that of the holdings, and may differ slightly from the TWR in the Performance tab.

Which benchmark index for the attribution? Regions and weights

The attribution compares your portfolio with a **reconstituted world equity index**, and **not** with the index chosen with the **Benchmark index** menu in Settings. The "Benchmark index" card says so: "MSCI ACWI (regional weights)".

The index weights

These are the regional weights of the **MSCI ACWI IMI at 30 Jun 2026**, renormalised so that they add up to 100% (the raw sum is 99.6%):

Region	Raw weight	Weight used	Representative index
United States	62.7%	62.95%	S&P 500 (^GSPC)
Emerging markets	12.3%	12.35%	EEM
Europe (excluding the United Kingdom and Switzerland)	8.7%	8.73%	Euro Stoxx 50 (^STOXX50E)
Japan	5.6%	5.62%	Nikkei 225 (^N225)
United Kingdom	3.1%	3.11%	FTSE 100 (^FTSE)
Canada	3.0%	3.01%	^GSPTSE
Asia-Pacific	2.3%	2.31%	^AXJO
Switzerland	1.9%	1.91%	^SSMI

weight used = raw weight / 99.6

The calculation

Each month, the index return is Rb = Σ region weight × return of its index. Index prices are **converted into euros** with exchange rates, to be comparable with your portfolio.

How your securities are classified

Each security is placed in a region according to the software's securities reference file. A security absent from the reference file is classified as "Unclassified".

Limitations

- **Indices excluding dividends**: regional indices ignore dividends. Your prices do not take them into account either, which keeps the comparison consistent, but both returns are underestimated, all the more so as the regions pay out a lot.
- **One index per region**, often large caps (Euro Stoxx 50 for Europe).
- **Attribution by region only**, not by sector.
- The index weights are fixed over the whole period.

Attribution of a diversified portfolio or a world ETF

Only the equity sleeve is analysed

The benchmark index is an **equity** index. For a diversified portfolio (equities, bonds, gold), the software therefore compares only the **equities** with this index, as a manager does for each "sleeve". Bonds and gold are excluded. A security absent from the reference file is considered to be an equity.

If equities represent less than 99.5% of the portfolio, a note says so with their current weight: "the attribution covers the equity sleeve (x% of the portfolio today)". If there are no equities at all, the tab displays "Attribution unavailable" with the note "no equities in the portfolio".

The case of world ETFs

A world ETF is placed in the "World (ETF)" region, which does not exist in the index. The software applies to it the rule for regions absent from the index ($wb = 0$, $rb = Rb$):

allocation = $wp \times (Rb - Rb) = 0$
selection = $0 \times (rp - Rb) = 0$
interaction = $wp \times (rp - Rb)$

The whole gap on this line goes into the **interaction effect**. Example: a world ETF that weighs 50% and returns 9% when the index returns 6.8% gives an interaction of $0.5 \times 2.2 =$ **+1.1 points**. The index regions absent from the portfolio then produce an allocation effect (for example, a Europe at 30% in the index and 0% in your portfolio, returning 4% against 6.8%: $(0 - 0.3) \times (4 - 6.8) =$ **+0.84 points**).

The warning displayed

If more than 20% of the portfolio is not classified in an index region (world ETF, securities absent from the reference file), a note warns that the attribution is mainly relevant for a **portfolio of direct holdings**, such as the "global equity" sample portfolio. For a portfolio made up of one or two ETFs, the Performance tab (comparison with the index, alpha, tracking error) is more telling.

The risk budget: which holdings bring the most risk?

A holding's weight does not tell you what share of the **risk** it brings: a very volatile share that is highly correlated with the rest of the portfolio weighs more in the risk than in the value. The **Risk budget** tab does this decomposition.

The exact formulas

portfolio volatility $\sigma p = \sqrt{(w^T \Sigma w)}$
marginal contribution $MC_i = (\Sigma w)_i / \sigma p$
risk contribution $RC_i = w_i \times MC_i$
share of risk $= RC_i / \sigma p$ (the sum is 100%)

Σ is the annual covariance matrix, estimated as in the Optimisation tab (daily returns × 252, over the common period of your current securities).

Euler's property: the sum of the contributions is exactly equal to the portfolio's volatility ($\Sigma RC_i = \sigma p$). An automatic test checks it.

Verified example

Two securities at 50 / 50: A (volatility 20%), B (volatility 10%), correlation 0.2 (covariance 0.004).

- $\sigma p = \sqrt{(0.25 \times 0.04 + 0.25 \times 0.01 + 2 \times 0.25 \times 0.004)} =$ **12.04%**;
- MC_A = (0.04 × 0.5 + 0.004 × 0.5) / 0.1204 = 0.1827; RC_A = 0.5 × 0.1827 = 0.0914;
- MC_B = (0.004 × 0.5 + 0.01 × 0.5) / 0.1204 = 0.0581; RC_B = 0.0291;
- shares of risk: A **75.9%**, B **24.1%**, for weights of 50 / 50.

What is displayed

- three cards: "Current volatility", "Largest contributor" (its share of risk, its name and its share of value) and "Risk-parity volatility";
- the "Share of value and share of risk" chart: for the 20 largest contributors, a light grey-blue bar (share of value) and a blue bar (share of risk).

Interpretation

A holding whose **share of risk exceeds its share of value** is more volatile, or more correlated with the rest, than average. A negative share of risk is possible for a holding that falls when the others rise (hedge).

Limitations

Risk is measured by volatility alone, over the observed period. Correlations can change, especially in a crisis.

Risk parity

Risk parity is an allocation in which **each holding contributes equally** to the portfolio's risk. It was popularised by Bridgewater's "All Weather" fund.

Why it is interesting

It does **not use expected returns**, which are very poorly estimated, but only volatilities and correlations. It is a direct answer to the main limitation of Markowitz (see the chapter on optimisation).

The calculation (Spinu's method, 2013)

The software minimises the function:

$$\tfrac{1}{2}\, w^T \Sigma\, w - (1/n) \times \Sigma \ln(w_i)$$

starting from weights inversely proportional to the volatilities (scipy's L-BFGS-B method, strictly positive weights). The weights obtained are then renormalised so that they add up to 100%. At the optimum, all the risk contributions are equal.

Verified example

Securities A (volatility 20%) and B (volatility 10%), correlation 0.2, expected returns 8% and 4%, risk-free rate 2.5%:

Allocation	Weights A / B	Volatility	Shares of risk	Sharpe
50 / 50 portfolio	50% / 50%	12.04%	75.9% / 24.1%	0.29
Risk parity	33.3% / 66.7%	10.33%	50% / 50%	0.27

The more volatile security receives less weight. With two securities, risk parity amounts to weighting by the inverse of the volatilities (1/0.20 and 1/0.10, that is 1/3 and 2/3).

Interpretation

Risk parity generally reduces volatility and increases risk diversification. It does not maximise return: its Sharpe can be lower, as in the example. With equities and bonds, it gives a lot of weight to bonds, which are less volatile.

Limitations

- risk-parity funds often use leverage to raise returns: that is not the case here;
- the weights depend on past volatilities and correlations;
- no cap per security is applied.

Comparing four allocations: diversification ratio and effective number of bets

The "Four ways to allocate the same securities" table compares, using **your current securities**:

Allocation	Principle
Current portfolio	your weights today
Equal weight	the same weight for each holding (1/n)
Risk parity	each holding brings the same share of the risk
Minimum variance	the lowest possible volatility, **with no cap per security**

Note: this minimum variance applies no cap, unlike the one in the Optimisation tab (adjustable cap). It can therefore be more concentrated.

The columns and their formulas

Expected return $= \Sigma\, w_i \times \mu_i$ (μ = daily average × 252)
Volatility $= \sqrt{(w^T \Sigma\, w)}$
Sharpe = (expected return − risk-free rate) / volatility
Diversification ratio $= \Sigma\, w_i \times \sigma_i / \sigma p$
Effective number of bets $= 1 / \Sigma\, (\text{share of risk}_i)^2$
Largest contribution = largest share of risk of a single holding

Verified example

Two securities A (20%) and B (10%), correlation 0.2, at 50 / 50:

- diversification ratio = (0.5 × 0.20 + 0.5 × 0.10) / 0.1204 = **1.25**;
- effective number of bets = $1 / (0.759^2 + 0.241^2)$ = **1.58**;
- largest contribution = 75.9%.

In risk parity: ratio 1.29, effective number of bets **2.0**, largest contribution 50%.

Interpretation

- **Diversification ratio**: 1 means no diversification (all securities perfectly correlated); the higher it is, the more the securities offset one another.
- **Effective number of bets**: how many "independent" holdings the portfolio really represents from a risk point of view. It equals n if all holdings contribute equally, 1 if a single holding carries all the risk. A portfolio of 69 holdings may be worth only about thirty bets.
- **Largest contribution**: dependence on a single holding.

The **Show the weights of each allocation** panel displays the weights (in %) of each security in the four allocations.

Limitations

The expected returns are historical (the same fragility as in the Optimisation tab); the other columns depend only on past volatilities and correlations.

The backtest: should you rebalance your portfolio?

The **Strategy backtest** tab replays the real history with **the same securities and the same starting weights** as your current portfolio, according to four strategies.

Strategy	Rule
Buy and hold	you buy once and touch nothing more: weights drift with prices
Monthly rebalancing	back to the starting weights on the first trading day of each month
Quarterly rebalancing	on the first trading day of each quarter

Strategy	Rule
Annual rebalancing	on the first trading day of each year

The first day of the period is not a rebalancing day. All strategies pay the fees on the initial purchase.

This is not your real history

The backtest assumes you had bought the current weights **right from the start**. It ignores your real transactions. The period starts on the first day on which **all** your current securities have a price, within the portfolio's analysis period.

How rebalancing works

amount traded = Σ |target value of the holding − current value of the holding|
fees = fee rate × amount traded
new value = value − fees, split according to the target weights

What is displayed

- the "Rebalance or not?" chart, rebased to 100;
- the "Results" table: Annualised return, Volatility, Max drawdown, Sharpe, Turnover / year.

Annualised return = (final value / initial value)^(365 / number of days) − 1
Volatility = standard deviation of daily returns × √252
Max drawdown = largest fall from a peak
Sharpe = (annualised return − risk-free rate) / volatility

Sharpe example: annualised return 8%, volatility 14%, risk-free rate 2.5%: (8 − 2.5) / 14 = **0.39**.

Interpretation

Rebalancing means **selling what has risen to buy what has fallen**. It maintains the intended risk, but costs fees and holds back performance when a trend persists (winners are trimmed too early). In a trendless market, with ups and downs, rebalancing can on the contrary pay off. Buy and hold lets winners grow: its risk drifts over time.

Backtest fees and turnover

The setting

The **Transaction costs (%)** slider goes from 0 to 0.5%, in steps of 0.05, with **0.10%** to start. Fees apply to the amounts bought or sold, including the initial purchase. The chart's subtitle recalls the chosen rate.

Turnover

turnover of one rebalancing = amount traded / portfolio value
Turnover / year = sum of the turnovers / number of years in the period (days / 365)

The amount traded counts sales **and** purchases: a turnover of 10% corresponds to 5% of the portfolio sold and 5% bought back.

Verified example

Two securities at 50 / 50 on €100. After one month, A has gained 10% and B has lost 10%: the holdings are worth €55 and €45. The rebalancing brings each holding back to €50:

- amount traded = |50 − 55| + |50 − 45| = **€10**;
- fees at 0.1% = **€0.01**;
- turnover = 10 / 100 = **10%**.

With the fees on the initial purchase (0.1%), the portfolio is worth €99.90 before this rebalancing and €99.89 after.

Interpretation

The higher the frequency, the higher the turnover and fees. The “Turnover / year” column lets you estimate the annual cost: turnover × fee rate. For example, 40% turnover per year at 0.1% costs 0.04% per year. The performance gaps between strategies often come more from the trend effect than from fees.

Limitations

Fees are proportional: no fixed fee per order, no bid-ask spread, no tax on gains realised on sales (which weighs on a securities account).

Investing all at once or gradually (DCA)?

The second part of the **Strategy backtest** tab compares two ways of investing the same capital in the same basket of securities (current weights, no rebalancing afterwards).

The settings

- **Capital for the DCA comparison (€)**: from €1,000 to €10,000,000, in steps of 1,000, **€10,000** to start;
- **Length of the gradual investment (months)**: from 3 to 24 months, **12** to start;
- the fees from the **Transaction costs (%)** slider apply to each purchase.

The two strategies

- **All at once**: all the capital is invested on the first day of the period.
- **Gradual**: the capital is divided into equal parts, invested on the first trading day of each of the first months. Money waiting is remunerated at the **risk-free rate** from Settings:

monthly instalment = capital / number of months
daily rate on cash = (1 + risk-free rate)^(1/252) − 1
value = basket units held × basket value + cash

Example

€10,000 over 12 months: €833.33 per month. With 0.1% fees, each instalment buys 833.33 × 0.999 = **€832.50** of securities. With a risk-free rate of 2.5%, cash earns (1.025)^(1/252) − 1 ≈ **0.0098%** per trading day.

What is displayed

The chart of the two value curves, in euros, and the “Comparison” table: Final value, Gain (final value − capital) and Lowest value reached during the period.

Interpretation

In a rising market, investing **all at once** generally earns more: the money works sooner. **Gradual** investing reduces the risk of investing just before a fall: its lowest value is often higher, especially since part of it stays in cash at the start. It is a trade-off between expected return and regret.

Limitations

- a single history: the result depends entirely on the period observed;
- if the period has fewer months than the chosen length, the capital is spread over the months available;
- prices excluding dividends.

The limits of the backtest

A backtest tells you what **would have** worked over a past period. It guarantees nothing for the future.

The software’s own limitations

- **A single period**: that of your portfolio, starting from the day on which all your current securities have a price. If one of your securities is recent, the period is short, and conclusions are fragile.
- **Prices excluding dividends**: returns are slightly underestimated, in the same way for all strategies.
- **Securities chosen today**: the backtest tests securities that you hold **now**, often because they have done well. This is a selection bias.
- **Simplified fees**: proportional, with no tax on sales.
- **Not your real transactions**: the starting weights are your current weights.

Good practice

Compare strategies with one another rather than retaining an absolute figure, and look at return, volatility, max drawdown and turnover together.

In case of the “Backtest unavailable” message

The message is followed by the cause. It appears, for example, if the common history of your securities is empty or too short. Check your prices with **Refresh prices**.

9. All the formulas

This chapter gives, indicator by indicator, the exact formula used by the software, as written in its code, with a small worked example, its interpretation and its limitations. Each fiche can be read on its own. The common conventions (252 trading days, 365-day basis, amounts in euros) are recalled in the first fiche.

The calculation conventions common to all indicators

All indicators rely on the same conventions, written into the code.

Convention	Value used
Trading days per year (volatility, Sharpe, alpha, tracking error)	252
Days per year to annualise a return (TWR, IRR)	365 calendar days
Timing of transactions	at the end of the day, at that day's price
Price used	closing price, not adjusted for dividends
Currency	everything is converted into euros
Risk-free rate	constant over the whole period (2.50% by default)

Two different annualisations

- An **average** of daily returns is multiplied by 252.
- A daily **standard deviation** is multiplied by √252 ≈ 15.87: if the days are independent, variances add up, so the standard deviation grows as the square root of time.
- A **total return** is compounded over 365 days: (1 + R) ^ (365 / number of days) − 1.

Days without a quote

A transaction entered on a day when the market is closed (a Saturday) is attached to the next trading day. On a holiday that affects only one exchange, the security's last known price is carried forward.

How is each day's return calculated, without the effect of contributions?

A purchase of €1,000 raises the portfolio value by €1,000, with no gain at all. The software therefore removes the effect of flows.

The formula

r(t) = (value(t) − flow(t)) / value(t−1) − 1

where, for each day, flow = money paid in (+) or taken out (−):

- BUY: + (quantity × price + fees);
- SELL: − (quantity × price − fees);
- DIVIDEND: − (amount − fees).

On the first day (or after everything has been sold), the previous day's value is 0: r = value(t) / flow(t) − 1. This return captures the purchase fees and the gap between the price paid and the closing price. Days when nothing is invested are excluded.

Example

- Previous day: €10,000. Purchase that day: €1,000. Value in the evening: €11,200. r = (11,200 − 1,000) / 10,000 − 1 = +2.00%.
- First day: 10 securities bought at €500 with €2.50 of fees (flow €5,002.50), close at €505 (value €5,050): r = 5,050 / 5,002.50 − 1 = +0.95%.

Limitation

The transaction is assumed to be made at the closing price: the return on the purchase day includes the gap between your price and that close.

These returns are used for the TWR, volatility, Sharpe, VaR and the projection.

The TWR: time-weighted return

The TWR (*time-weighted return*) measures the quality of investment choices, independently of the timing and size of contributions. It is the measure used by fund managers (GIPS standard): a manager does not choose when clients pay in money.

The formula

TWR = (1 + r1) × (1 + r2) × … × (1 + rn) − 1

where the r are the daily returns neutralised for contributions (see the previous fiche). The "base 100" curve in the Performance tab is the same product, accumulated day after day: index(t) = 100 × Π (1 + r).

Example

+10% one day, then −5% the next: 1.10 × 0.95 − 1 = +4.5%, not +5%.

Interpretation

Two investors who hold the same securities in the same proportions have the same TWR, even if one invested €1,000 and the other €100,000, or on different dates. The TWR can therefore be compared directly with an index.

Limitation

It ignores the effect of the timing of your contributions: to measure what your money has actually earned, see the IRR. The “by calendar year” TWR applies the same formula to the days of each year.

The annualised TWR

This is the “Annualised return” figure in the key figures and the “Annualised TWR” card in the Performance tab.

The formula

Annualised TWR = (1 + total TWR) ^ (365 / number of days) − 1

The number of days is the gap, in calendar days, between the first and the last daily return in the history.

Example

+21% over 730 days: 1.21 ^ (365 / 730) − 1 = +10% a year, not 10.5%: gains compound from one year to the next.

Interpretation

It is the constant return that, each year, would have led to the same result. It allows periods of different lengths to be compared.

Limitations

- Over a short period, it **extrapolates**: +5% over 91 days gives about +21.6% a year, which says nothing about the coming year. Be wary of the annualised TWR over less than a year.
- It is meaningless if the first and last days are the same (the software then displays “n/a”).

The IRR: return on the money invested

The IRR is the annual return as seen by the investor: it takes into account the amount and date of each contribution.

The formula

It is the annual rate i that makes the discounted value of all flows equal to zero:

Σ CF_k / (1 + i) ^ (days_k / 365) = 0

- purchase: money leaving your pocket, CF < 0;
- sale, dividend: money coming in, CF > 0;
- on the last day, everything is treated as sold: + final value;
- days_k: number of days since the first flow.

There is no direct formula: the software searches for i by bisection (200 iterations) between −99% and +1,000% a year. If there is no solution in this interval, it displays “n/a”.

Example: TWR and IRR point in opposite directions

€10,000 invested, +20% in the first year (€12,000). You then add €50,000, and the portfolio then loses 10% (€55,800).

- TWR: $1.20 \times 0.90 - 1 = +8\%$ over two years, i.e. +3.9% a year.
- IRR: −6.05% a year, because the fall hit a sum five times larger than the rise.

Interpretation

An IRR above the TWR means your contributions were well timed; below it, that they arrived before a fall.

Annualised volatility

The formula

Annual volatility = standard deviation of daily returns × √252

The standard deviation is the sample standard deviation (division by n − 1), calculated on daily returns neutralised for contributions.

Why √252?

If daily returns are independent, their variances add up: annual variance = 252 × daily variance, so annual standard deviation = √252 × daily standard deviation.

Example

A daily standard deviation of 1% gives $1\% \times \sqrt{252} = 15.87\%$ a year.

Interpretation

With a volatility of 15%, a year’s return “typically” departs by 15 points from its average. The card also shows the volatility of the benchmark index over the same period.

Limitations

- Volatility counts rises as well as falls: the Sortino ratio corrects this.
- The √252 rule assumes independent days; it underestimates risk when falls come in a row.
- It does not describe extreme losses: see the VaR, the CVaR and the max drawdown.

The max drawdown (worst fall)

The formula

drawdown(t) = index(t) / highest index up to t − 1 (always ≤ 0)
Max drawdown = the lowest of these values

The calculation is based on the **base-100 index** (the cumulative TWR), not on the portfolio value: otherwise a simple withdrawal of money would look like a loss, and a contribution would mask a real fall.

The software also gives the date of the peak, the date of the trough and the date on which the peak was regained ("peak not yet recovered" if that has not happened yet).

Example

The index goes from 100 to 120, falls to 90, then rises to 110. Max drawdown: 90 / 120 − 1 = −25%. The peak of 120 has not been regained.

Interpretation

It is the worst loss that an investor who got in at the worst moment and out at the worst moment would have suffered. After a 25% fall, a 33% rise is needed to get back to the peak.

Limitations

It depends on the period observed: a recent portfolio may not yet have been through a crisis. Stress tests complement this measure.

The Sharpe ratio

The Sharpe ratio answers the question: was the risk taken well paid?

The formula

daily risk-free rate rf_d = (1 + annual rate) ^ (1/252) − 1
excess(t) = r(t) − rf_d
Sharpe = mean(excess) × 252 / (standard deviation(excess) × √252)

The annual rate is the one set in **Settings** (2.50% by default).

Example

Average daily return 0.05%, daily standard deviation 1%, risk-free rate 2.50% (i.e. 0.0098% a day): (0.05% − 0.0098%) × 252 / (1% × √252) ≈ 10.13% / 15.87% ≈ 0.64.

Interpretation

Benchmarks given by the code: negative, poor (worse than a risk-free investment); 0.5, acceptable; above 1, very good. The card also shows the index's Sharpe over the same period.

Limitations

- Volatility also penalises rises (see the Sortino).
- The risk-free rate is constant over the whole period, whereas it has varied: over several years, the Sharpe is approximate.
- Over a short period, it is very unstable.

The Sortino ratio

A criticism of the Sharpe: volatility counts big rises as risk. The Sortino only penalises the days below the risk-free rate.

The formula

excess(t) = r(t) − rf_d
falls(t) = min(excess(t), 0)
semi-deviation = √(mean of falls²) × √252
Sortino = mean(excess) × 252 / semi-deviation

The mean of squares is taken over **all** days: rising days count as 0. The rate rf_d is the same as for the Sharpe.

Example

Average annual excess return of 8%, volatility of 16% and semi-deviation of 10%: Sharpe 8 / 16 = 0.50, Sortino 8 / 10 = 0.80.

Interpretation

A Sortino clearly above the Sharpe indicates that the volatility comes mainly from rises. If they are close, rises and falls are of comparable size.

Limitation

Like the Sharpe, it depends on the risk-free rate used and on the period.

Beta

The formula

Beta = covariance(r_portfolio, r_index) / variance(r_index)

- Daily returns of the portfolio and of the benchmark index, on common days only.
- The index is aligned with the portfolio's calendar (last known price carried forward), then r = price(t) / price(t−1) − 1.
- Sample covariance and variance (division by n − 1).

Example

Daily covariance of 0.00012 and daily index variance of 0.00010: beta = 1.2. When the index gains 1%, the portfolio gains 1.2% on average.

Interpretation

- 1: the portfolio moves like the index;
- above 1: more sensitive, riskier than the market;
- below 1: more defensive.

Beta is also used in the hypothetical stress tests (equities fall by X% ≈ beta × X).

Limitations

Beta measures sensitivity to an equity market: against a bond or money-market index, it has little meaning, and the Performance tab flags this. It depends on the index chosen in **Benchmark index**.

Jensen's alpha

The formula

daily alpha = mean(r_p − rf_d) − beta × mean(r_i − rf_d)
Annual alpha = daily alpha × 252

r_p and r_i are the daily returns of the portfolio and of the index on common days, rf_d the daily risk-free rate (1 + rate) ^ (1/252) − 1.

Example

Daily averages: portfolio 0.06%, index 0.04%, beta 1.2, risk-free rate 2.50%: [(0.06% − 0.0098%) − 1.2 × (0.04% − 0.0098%)] × 252 ≈ +3.5% a year.

Interpretation

Alpha is the performance that is **not** explained by exposure to the market (market model, CAPM). Positive: security selection, or a bias absent from the index, created value. Negative: the market risk taken was not rewarded.

Limitations

- It depends on the index chosen: a poorly suited index (for example the CAC 40 excluding dividends) distorts alpha.
- It depends on the risk-free rate, assumed constant.
- Over a short period, it is mostly statistical noise.

Tracking error

The formula

gap(t) = r_portfolio(t) − r_index(t)
Tracking error = standard deviation(gap) × √252

Calculated on the days common to the portfolio and the index.

Example

A daily gap with a standard deviation of 0.3% gives 0.3% × √252 ≈ 4.76% a year.

Interpretation

It is the volatility of the gap with the index. Benchmarks from the code:

- less than 2%: the portfolio "sticks" to the index (passive management);

- more than 5%: management very different from the index.

A high tracking error is neither good nor bad in itself: it measures how bold the management is. The information ratio says whether that boldness paid off.

Limitations

It depends on the index chosen. A portfolio of European equities compared with the MSCI World will necessarily have a high tracking error.

The information ratio

The formula

Information ratio = mean(r_portfolio − r_index) × 252 / tracking error

It is the average annualised return gap, divided by the tracking error (see the previous fiche).

Example

Average daily gap of 0.02% (i.e. 5.04% a year) and tracking error of 4.76%: 5.04 / 4.76 ≈ 1.06.

Interpretation

It is the equivalent of the Sharpe, but relative to the index: has the gap with the index been "paid for"? The code uses as a benchmark that a ratio above 0.5 is considered good. Negative: the portfolio did worse than the index.

Do not confuse

The "Difference vs" the index card, at the top of the Performance tab, is the gap between the **compounded TWRs** over the same period (portfolio TWR − index TWR), whereas the information ratio uses the **arithmetic mean** of the daily gaps. The two may differ slightly.

Limitation

Like the tracking error, it only makes sense with an index comparable to the portfolio.

Historical VaR

The formula

Historical VaR = − quantile(1 − level) of daily returns
VaR in euros = historical VaR × current portfolio value

The level is set at the top of the Risk tab, with **VaR confidence level**: 90, 95 (default) or 99%. The quantile is calculated by the pandas library, with linear interpolation between two observations. The result is a positive number: a loss.

Example

Over 20 days, the five worst returns are −3.1%, −2.4%, −1.8%, −1.5% and −1.2%. The 5% quantile falls between the worst and the second-worst day: −3.1% + 0.95 × 0.7% ≈ −2.44%. 95% VaR = 2.44%. On €100,000, about €2,440.

Interpretation

“On 95% of days, the loss does not exceed this amount.” The 95% VaR is therefore exceeded about one trading day in twenty. This is the “VaR · 1 day” card in the Risk tab.

Limitations

- No distribution assumption, but the past is assumed to repeat itself: a crisis absent from the history is invisible.
- It says nothing about the size of the loss beyond the threshold: that is the role of the CVaR.
- Over a short history, it relies on very few days.

VaR under the normal distribution

The formula

Normal VaR = − (mean + z × standard deviation)

- mean and standard deviation: those of the daily returns;
- z: quantile of the standard normal distribution at the threshold 1 − level: −1.282 at 90%, −1.645 at 95%, −2.326 at 99%.

Example

Daily mean 0.05%, standard deviation 1.2%, level 95%: −(0.05% − 1.645 × 1.2%) ≈ 1.92%, i.e. about €1,924 on €100,000.

Interpretation

It is the loss on a bad day if returns followed a normal distribution with the same mean and the same volatility. The table in the Risk tab displays it next to the historical VaR, the Cornish-Fisher VaR and the CVaR, as a percentage and in euros.

Limitations

Real returns have “fat tails”: large falls are more frequent than the normal distribution predicts. It therefore often **underestimates** crashes, especially at 99%. The Jarque-Bera test indicates whether the normal distribution is rejected for your portfolio; in that case, the automatic reading advises taking this VaR with caution.

Cornish-Fisher VaR and its domain of validity

The normal-distribution formula is kept, but the quantile z is corrected using the skewness S and the excess kurtosis K (Cornish-Fisher expansion).

The formula

$z_{cf} = z + (z^2 - 1)\cdot S/6 + (z^3 - 3z)\cdot K/24 - (2z^3 - 5z)\cdot S^2/36$
Cornish-Fisher VaR = − (mean + z_{cf} × standard deviation)

Example

Mean 0.05%, standard deviation 1.2%, $S = -0.5$, $K = 3$:

- at 95%: $z_{cf} \approx -1.722$ instead of −1.645, VaR 2.02% instead of 1.92%;
- at 99%: $z_{cf} \approx -3.301$ instead of −2.326, VaR 3.91% instead of 2.74%.

Interpretation

Negative skewness always makes the VaR more cautious. The effect of fat tails depends on the level: at 99%, the VaR increases; at 95%, it can on the contrary fall slightly, because a fat-tailed distribution also has a more “peaked” centre. This is the method used for the regulatory PRIIPs risk indicator.

Domain of validity

The correction only makes sense if it preserves the order of the quantiles. The software checks that the derivative $1 + z\cdot S/3 + (3z^2 - 3)\cdot K/24 - (6z^2 - 5)\cdot S^2/36$ stays positive for z from −4 to +4 (in steps of 0.1). Otherwise, or with fewer than 4 days, the VaR is displayed as “n/a”, with the note “Cornish-Fisher n/a: skewness or kurtosis too large, the adjustment is no longer reliable.” Example: with no skewness, K must stay between about −0.5 and 8.

The CVaR (Expected Shortfall)

The formula

CVaR = − mean of the daily returns r such that r ≤ − historical VaR
CVaR in euros = CVaR × current value

Example

With the 20 days from the “Historical VaR” fiche (95% VaR of 2.44%), only one day is worse than −2.44%: −3.1%. CVaR = 3.1%. Over 1,000 days, it would be the average of roughly the 50 worst days.

Interpretation

The VaR answers “how far does a bad day go?”, the CVaR answers “and when things go badly, how badly do they go?”. It is always at least equal to the historical VaR. It is the measure preferred by banking regulators (Basel III). “CVaR · Expected Shortfall” card in the Risk tab.

Limitations

It relies on few days (5% of the history at 95%, 1% at 99%): over a short history, it is unstable. Like the historical VaR, it ignores crises absent from the period observed.

Skewness

The formula

Skewness is calculated by the pandas library: bias-corrected sample coefficient (adjusted Fisher-Pearson coefficient).

$d = r - \text{mean}(r)$; $m2 = \text{mean}(d^2)$; $m3 = \text{mean}(d^3)$
$S = [m3 / m2^{(3/2)}] \times \sqrt{(n(n-1))} / (n-2)$

Interpretation

- 0 for a normal (symmetrical) distribution;
- negative: big falls are more frequent or more violent than big rises (the usual case for equities);
- positive: big rises dominate.

The automatic reading in the Risk tab considers the distribution “roughly symmetrical” when $|S|$ is below 0.3.

Example

A series of small regular gains punctuated by a few sharp drops has negative skewness, even if its mean is positive.

Limitations

A single extreme day can change the skewness considerably: it is unstable over a short history. At least 4 days are needed; otherwise it is 0 by convention.

Skewness is used for the Jarque-Bera test and for the Cornish-Fisher VaR.

Excess kurtosis (fat tails)

The formula

Calculated by pandas, directly as an **excess** (the normal distribution is 0), corrected for sample bias:

$g2 = \text{mean}(d^4) / \text{mean}(d^2)^2 - 3$
$K = [(n+1) \times g2 + 6] \times (n-1) / [(n-2)(n-3)]$

with $d = r - \text{mean}(r)$.

Interpretation

- 0: as many extreme days as the normal distribution;

- positive: “fat tails”, extreme days (in both directions) are more frequent than expected. The automatic reading speaks of clearly fat tails above 1.

Extreme days

The Risk tab also counts the share of days more than 3 standard deviations from the mean, against 0.27% under the normal distribution ($2 \times (1 - \Phi(3))$). Example: 1.2% of extreme days observed is about 4.4 times more than the normal distribution.

Limitations

Like skewness, it is very sensitive to a few days and unstable over a short history (minimum 4 days, otherwise 0). It is used for the Jarque-Bera test and for the Cornish-Fisher VaR.

The Jarque-Bera test

The formula

$JB = n / 6 \times (S^2 + K^2 / 4)$
$p\text{-value} = \exp(-JB / 2)$

n is the number of days, S the skewness, K the excess kurtosis. Under the hypothesis of normality, JB follows a chi-squared distribution with 2 degrees of freedom, whose exceedance probability is exactly $\exp(-JB/2)$.

Example

1,000 days, $S = -0.4$, $K = 2$: $JB = 1{,}000 / 6 \times (0.16 + 1) \approx 193.3$, p-value $\approx 10^{-42}$, displayed as “p = < 0.001”.

Interpretation

- p-value below 5%: “Normality rejected” badge. VaRs calculated with the normal distribution should then be taken with caution.
- Otherwise: “Normality not rejected”. This is not proof of normality, only the absence of proof to the contrary.

Over several years of daily equity returns, normality is almost always rejected.

Limitations

The test is asymptotic (reliable for a large number of days). Over a very long history, the slightest deviation is enough to reject normality, even if it has no practical consequence.

Correlation between securities

The formula

Pearson coefficient of the daily returns $r = \text{price}(t) / \text{price}(t-1) - 1$ of two securities, on their common days:

$\rho(A, B) = \text{covariance}(r_A, r_B) / (\text{standard deviation}(r_A) \times \text{standard deviation}(r_B))$

The derived measures (Exposures tab)

- **Weighted average correlation**: $\Sigma\, w_i w_j \rho_{ij} / \Sigma\, w_i w_j$, over the pairs $i \neq j$, with w the weight of each holding.
- **Independent blocks**: hierarchical clustering (distance $1 - \rho$, average linkage); securities correlated above 0.7 form a single block, and therefore a single "bet".
- **Days of sharp falls**: weighted average correlation calculated over the portfolio's worst 10% of days (at least 30 days).
- **Qualifiers**: very strong from 0.8, strong from 0.6, moderate from 0.3, weak below.

Example

Annual covariance 0.018, volatilities 20% and 30%: $\rho = 0.018 / (0.20 \times 0.30) = 0.30$, moderate correlation.

Limitations

Correlation only measures a linear link and varies over time: it often rises during crises, when diversification would be most useful. Past correlations are not guaranteed in the future.

The diversification ratio

The formula

Diversification ratio = $\Sigma\, w_i \times \sigma_i / \sigma_p$
$\sigma_p = \sqrt{(w^T \Sigma\, w)}$

w_i is the weight of each holding, σ_i its volatility, Σ the covariance matrix of daily returns, σ_p the portfolio volatility.

Example

Two holdings at 50%, with volatilities of 20% and 30%, correlated at 0.3: $\sigma_p \approx 20.37\%$, hence a ratio of $(0.5 \times 20\% + 0.5 \times 30\%) / 20.37\% \approx 1.23$.

Interpretation

- 1: no diversification, the holdings move like a single asset;
- the higher it is, the more the holdings offset each other.

The ratio is the same whether daily or annualised covariances are used.

The effective number of bets (Risk budget)

Effective number of bets = $1 / \Sigma\, (\text{risk share of each holding})^2$

In the example, the risk shares are 34.9% and 65.1%: $1 / (0.349^2 + 0.651^2) \approx 1.83$ bets, for 2 holdings.

Limitations

It relies on past volatilities and correlations, which change, especially in a crisis.

Holding weight and effective number of holdings

The formulas

value of a holding = quantity × last price (in euros)
weight = value of the holding / total value of the holdings
effective number of holdings = $1 / \Sigma \text{ weight}^2$

The effective number is the inverse of the Herfindahl index.

Example

Four holdings weighing 40%, 30%, 20% and 10%: $1 / (0.16 + 0.09 + 0.04 + 0.01) \approx 3.3$. The portfolio is equivalent to about 3 holdings of equal weight.

Other concentration measures (Exposures tab)

- weight of the top 5 and top 10 holdings;
- the **5/10/40 rule** for UCITS funds: a holding at most 10%, and holdings above 5% at most 40% in total. It applies to directly held stocks: a diversified ETF is not a concentration.

Interpretation

The alert thresholds on the largest stock depend on the profile chosen in **Thresholds for profile**: 5 and 10% for Cautious, 7 and 10% for Balanced, 10 and 15% for Dynamic (first threshold: caution, second: alert).

Limitation

Weight says nothing about risk: a very volatile 10% holding can contribute 30% of the risk (see risk contributions).

The PRU (unit cost basis)

The software replays all the transactions in date order, using the PRU method (weighted average cost basis) used in France.

The rules

BUY: new PRU = (quantity held × PRU + q × price + fees) / (quantity held + q)
SELL: the PRU does not change
Everything sold: the PRU restarts from 0
Amount invested = quantity held × PRU

Purchase fees are included in the PRU: they increase the cost basis. Prices are converted into euros at the rate on the day of the transaction.

Example

1. Purchase of 10 securities at €100 + €5 of fees: PRU (1,000 + 5) / 10 = €100.50.
2. Purchase of 10 securities at €120 + €5: PRU (10 × 100.50 + 1,205) / 20 = €110.50.
3. Sale of 5 securities: the PRU stays at €110.50; 15 securities remain, i.e. €1,657.50 invested.

Why a difference with your bank?

- some banks exclude fees from the PRU;
- a foreign security is converted at the Yahoo Finance exchange rate, not at your bank's;
- a missing transaction (an old purchase, a stock split) changes the PRU.

Unrealised gains, realised gains and total gain

The formulas

Unrealised gain = quantity × (price − PRU) = value − amount invested
Unrealised gain in % = unrealised gain / amount invested
Realised gain (on each sale) = q × (selling price − PRU) − selling fees
Dividends = Σ (amount received − fees)
Total gain = unrealised gains + realised gains + dividends

Fees are already deducted: in the PRU for purchases, in the realised gain for sales. "Unrealised" means not cashed in: what a full sale would earn today, before selling fees and taxes.

Example (continuing from the PRU fiche)

- sale of 5 securities at €130 with €4 of fees: 5 × (130 − 110.50) − 4 = €93.50 realised;
- 15 securities remaining, price €125: 15 × (125 − 110.50) = €217.50 unrealised;
- a dividend of €30 net of fees;
- total gain: 217.50 + 93.50 + 30 = €341.

The percentage "on invested capital"

The "Total gain" card displays total gain / amount invested, where the amount invested is the cost of **only the holdings still held** (here 341 / 1,657.50 ≈ 20.6%). It is neither a TWR nor an IRR.

Cross-check

The day-by-day history also calculates gain = value − net contributions; on the last day, it must land exactly on the total gain, which the project's tests verify.

The Markowitz efficient frontier

The parameters

μ = mean of daily returns × 252 (expected return of each security)
Σ = covariance of daily returns × 252 (covariance matrix)

calculated on the days when **all** securities have a price.

A portfolio with weights w

return = $w^T\mu$ volatility = $\sqrt{(w^T\Sigma w)}$ Sharpe = (return − risk-free rate) / volatility

The optimisations

Constraints: each weight between 0 and the **Maximum weight per security** (30% by default), sum of weights = 100%. SLSQP method (SciPy library), starting from equal weights.

- **Minimum variance**: minimises $w^T\Sigma w$.
- **Maximum Sharpe**: maximises the Sharpe (the "tangency" portfolio).
- **Frontier**: 40 target returns, from the return of the minimum variance to the maximum achievable return; for each, the minimum variance. If fewer than two points succeed, the frontier is approximated by the envelope of 4,000 randomly drawn portfolios (Dirichlet distribution).

Example

Security A (μ 6%, σ 20%), security B (μ 9%, σ 30%), correlation 0.3, no weight limit: minimum variance at 76.6% of A, volatility 18.67%; maximum Sharpe (rate 2.50%) at about 50/50, return 7.50%, volatility 20.36%, Sharpe 0.25.

Allocations compared

amount to buy (+) or sell (−) = (target weight − current weight) × total value. A gap below 0.25 point is considered unchanged.

Limitation

Expected returns are past averages: the optimiser over-exploits the securities that have done best.

The Monte Carlo projection

The parameters

By default, those of the portfolio: μ = daily mean × 252, σ = daily standard deviation × √252. They can be changed with the sliders.

"Normal distribution" method (geometric Brownian motion)

The time step is the month (dt = 1/12):

monthly log-return ~ Normal($(\mu - \sigma^2/2) \times dt$, $\sigma \times \sqrt{dt}$)
value(m + 1) = value(m) × exp(log-return) + monthly contribution

The $-\sigma^2/2$ term corrects the gap between arithmetic mean and compound growth: +50% then −50% gives a zero average, but a loss of 25%.

"Historical (bootstrap)" method

Each month adds up 21 daily log-returns drawn at random, with replacement, from the portfolio's actual days, re-centred so that their mean matches the chosen return. The real fat tails are preserved.

The results

5,000 scenarios (fixed seed, reproducible results); 5th, 25th, 50th, 75th and 95th percentiles; probability of loss = share of scenarios that end below starting value + contributions.

Example

€100,000, μ 7%, σ 15%, 10 years, normal distribution: median ≈ €178,700 (theory: $100{,}000 \times \exp((0.07 - 0.01125) \times 10) \approx €179{,}900$), adverse scenario ≈ €81,900, favourable ≈ €389,600, mean ≈ €199,800, probability of loss ≈ 10.4%.

Interpretation

The final value follows a log-normal distribution: the mean exceeds the median, which is the most representative marker. A projection is not a forecast.

Risk contributions (Euler)

A holding's weight does not say how much of the **risk** it contributes.

The formulas

$\sigma_p = \sqrt{(w^T \Sigma w)}$
marginal contribution $MC_i = (\Sigma w)_i / \sigma_p$
risk contribution $RC_i = w_i \times MC_i$
risk share RC_i / σ_p (the total is 100%)

Euler property: $\Sigma\, RC_i = \sigma_p$, verified by an automatic test. Σ is the annualised covariance matrix (daily returns × 252), estimated on the history.

Example

Two holdings at 50% (volatilities 20% and 30%, correlation 0.3): $\sigma_p \approx 20.37\%$. Contributions: 7.12 and 13.25 points, i.e. **34.9%** and **65.1%** of the risk for 50% of the value each. The risk / weight ratio of the second holding is 1.30.

Interpretation

A holding whose risk share exceeds its value share is more volatile or more correlated with the rest. The **Risk budget** tab shows the 20 largest contributors; the Exposures tab breaks these shares down by region.

Risk parity

The “Risk parity” allocation looks for the weights for which each holding contributes the same share of the risk: it minimises $\frac{1}{2} w^T \Sigma w - (1/n) \times \Sigma \ln(w_i)$, then scales the sum of weights to 100% (Spinu method). In the example, these are 60% and 40%: each holding then contributes 50% of the risk. It does not use expected returns.

Brinson-Fachler performance attribution

The model explains, region by region, the gap between the equity portion and a global equity index.

The formulas (over one period)

Allocation effect = (wp − wb) × (rb − Rb)
Selection effect = wb × (rp − rb)
Interaction effect = (wp − wb) × (rp − rb)
Rb = Σ wb × rb Rp = Σ wp × rp

wp, wb: weight of the region in the portfolio and in the index; rp, rb: returns of the region. The sum of the three effects, all regions included, equals Rp − Rb.

Example

United States: wp 70%, wb 60%, rp 12%, rb 10%. Europe: wp 30%, wb 40%, rp 3%, rb 5%. Rb = 8%, Rp = 9.3%. Allocation: +0.2 + 0.3 = +0.5 point; selection: +1.2 − 0.8 = +0.4; interaction: +0.2 + 0.2 = +0.4. Total: +1.3 points = 9.3% − 8%.

The implementation

- Calculated month by month, with the weights at the end of the previous month.
- Months linked by Cariño smoothing: each month is weighted by k_t / K, with $k = [\ln(1 + Rp) - \ln(1 + Rb)] / (Rp - Rb)$, so that the sum lands exactly on the compounded gap.
- Index: regional weights of the MSCI ACWI IMI as at 30/06/2026 (United States 62.7%, emerging markets 12.3%, Europe 8.7%..., scaled back to 100%), and one equity index per region, converted into euros, excluding dividends.
- Bonds and gold excluded; a region absent from the index takes rb = Rb.

Converting currencies into euros

The formula

price in euros = price in currency × factor / EURcurrency rate

- EURUSD=X (Yahoo Finance) is the number of dollars for 1 euro;
- factor = 0.01 for quotes in pence (GBp, London stocks), 1 otherwise.

Which rate?

Item	Rate used
purchase, sale, dividend	rate on the day of the transaction

Item	Rate used
day-by-day value	rate of each day
current value	last known rate
fees	none: always in euros

A day without a rate takes the last known rate.

Examples

- Apple at 200 USD, €1 = 1.10 USD: 200 / 1.10 = €181.82.
- London stock at 1,250 pence, €1 = 0.85 GBP: 1,250 × 0.01 / 0.85 = €14.71.

What this implies

The performance of a foreign security combines its price and its currency: a US stock that gains 10% in dollars earns almost nothing in euros if the dollar loses about 10%. The rate is Yahoo Finance's, not your bank's. The day's rate appears at the bottom of the sidebar ("€1 in USD", for example).

Duration and interest-rate shock

The formulas

average duration = Σ (duration × value) / Σ value (bond holdings whose duration is known)
loss if rates rise by 1 point ≈ Σ (duration × value) × 1%
as a % of the portfolio = loss / total value

First-order approximation: price change ≈ − duration × change in rates.

Example

€20,000 of a bond fund with a duration of 7 years in a portfolio of €100,000: rates rise by 1 point, loss ≈ 7 × 1% × 20,000 = €1,400, i.e. −1.4% of the portfolio.

Where does the duration come from?

From the project's reference database (data/referentiel.csv), only for the bond funds that appear in it. Without a known duration, a holding is not included in the calculation.

Interpretation

The longer the duration, the more the price falls when rates rise, and rises when they fall. The diagnostic thresholds depend on the profile: 5 and 8 years (Cautious), 7 and 10 years (Balanced), 8 and 12 years (Dynamic).

Limitations

Linear approximation, valid for small shocks (convexity is ignored). After the fall, bonds offer a higher yield again.

Stress tests: scenario formulas

Historical scenarios

change in a holding = price at the low date / price at the high date − 1
change in the portfolio = Σ current weight × change in the holding

Five crises: 2008-2009, European debt (2011), Covid (2020), inflation and rate rises (2022), mini-crash of August 2024. If a security was not listed (first price more than 7 days after the start of the crisis), the index of its region, or a bond or gold fund of the same category, is used as an approximation. Changes in local currency, excluding dividends.

Hypothetical scenarios

Equities fall by X% : change ≈ beta × (−X%) (X = 10, 20, 35)
Dollar falls by 10% : change = −10% × share of holdings quoted in dollars
Rates rise by 1 point : change = −1% × Σ weight × duration

Example

Beta of 0.90 and equities falling by 20%: 0.90 × (−20%) = −18%, i.e. −€18,000 on €100,000.

Limitations

The current portfolio is assumed unchanged throughout the crisis; the currency effect is ignored in the historical scenarios; the “dollar” shock only looks at the quote currency, not the content of ETFs.

Taxation on exit: CTO, PEA, life insurance

The software calculates the tax due if the whole portfolio were sold, under the 2026 rules (individual resident in France). Terms: CTO = compte-titres ordinaire (ordinary securities account); PEA = plan d’épargne en actions (French equity savings plan); assurance-vie = life insurance wrapper; PFU = prélèvement forfaitaire unique (flat tax).

The formulas

Wrapper	Base	Income tax	Social charges
CTO	max(capital gains, 0) + max(dividends, 0)	12.8%	18.6%
PEA, under 5 years	max(gain, 0)	12.8%	18.6%
PEA, 5 years and over	max(gain, 0)	0	18.6%
Life insurance, under 8 years	max(gain, 0)	12.8%	17.2%
Life insurance, 8 years and over	max(gain, 0)	7.5% × max(gain − €4,600, 0) (€9,200 for a couple)	17.2%

In a CTO, a loss is set against capital gains, not against dividends. Age is counted from the first transaction: days / 365.25.

Example, gain of €10,000

- CTO: 31.4% × 10,000 = €3,140;
- 6-year-old PEA: 18.6% × 10,000 = €1,860;
- 9-year-old life insurance, single person: 7.5% × 5,400 + 17.2% × 10,000 = 405 + 1,720 = €2,125.

Simplifications

Option for the progressive income tax scale ignored, life insurance premiums assumed below €150,000, contract management fees ignored, dividends assumed to be kept within the wrapper. This is not an official tax calculation.

10. Where the data comes from and how far to trust it

This chapter states where each piece of data used by the software comes from: share prices, the local securities database, currencies, recognition of ISIN codes, ETF composition, security classification, benchmark indices, the risk-free rate and the world map background. It then sets out, honestly, what the software checks, what it does not check, and what to do if a figure looks wrong.

Where does the software's data come from?

The software has no subscription to a professional data provider (Bloomberg, Refinitiv…) and is not linked to any bank. Its data comes from the following sources.

Data	Source	Updated
Daily prices of securities, indices and exchange rates	Yahoo Finance, through the Python library yfinance	at every analysis with Internet; otherwise local database and cache
Quotation currency of each security	Yahoo Finance, otherwise the local database, otherwise a rule based on the security's code	once per security, then kept
List of securities in the local database	Wikipedia pages of 32 stock indices, a list of 66 ETFs and the project's reference file	when the database is built
Matching an ISIN, name or Bloomberg code to a ticker	built-in table of 31 ETFs, local memory, local database, then the Yahoo Finance search engine	at every import of an unknown security
Country, region, sector, asset class, duration	the project's data/referentiel.csv file, supplemented by the local database	fixed, shipped with the software
ETF composition by country and sector	MSCI factsheets as of 30/09/2026 and 2025-2026 orders of magnitude	fixed, to be reviewed once a year
Regional index weights for performance attribution	MSCI ACWI IMI as of 30/06/2026	fixed
Risk-free rate	fixed value of 2.50% (the ECB deposit facility rate), editable	setting
Country outlines (world map)	Natural Earth, public domain	downloaded once

What comes from you

Transactions (dates, quantities, prices, fees, dividends) come only from your files or your entries. The software checks them, but never corrects them without your knowledge.

Where to see the source of the prices

- In the banner at the top of the page: “Live prices · Yahoo Finance” (green dot) or “Cached prices (offline)” (orange dot), and the “Data as of” note followed by the date of the last price in the history.
- At the bottom of the sidebar, the “Prices” line: “Yahoo Finance (en direct)”, or “cache local du” followed by a date, with the note “Yahoo Finance injoignable”. These two strings come from the program as it stands and are shown in French.

Which prices exactly does the software use?

The closing price, not adjusted for dividends

The software downloads prices with the `auto_adjust=False` setting of the `yfinance` library and keeps the **Close** column (closing price). It **never** uses the `Adj Close` column (price adjusted for dividends). This choice is the same everywhere: analysis history, latest price, building the local database.

Why? To value your portfolio at the **real price shown on the exchange**, the one you see at your broker, and to count dividends only once: through the “DIVIDENDE” lines of your file.

Two uses

Use	What is requested from Yahoo Finance
History (day-by-day value, performance, risk)	the daily closing prices since the date of your first transaction
Current value	the prices of the last 5 days; the software keeps the last known value of each security
Check at import	the closing price on the day of each transaction: currency check, tie-break between readings of a PDF trade confirmation, price reminded in the “Complete the transaction” form

The latest price is therefore the most recent one Yahoo Finance provides: the closing price of the last session, or a value from the current session if the exchange is open. The 5 days are there so that a price can always be found at weekends and on public holidays.

Days without trading

When a market is closed (local public holiday), the value of the security that day is its last known price. A transaction entered on a day without trading (a Saturday, for example) is attached to the next trading day.

Why a difference from your broker?

- different moment (closing price versus price at the moment);
- different trading venue (the same security in Paris and in Amsterdam, or an ETF listed in two currencies);
- conversion into euros at the Yahoo Finance exchange rate, not your bank's;
- a possible error in the source (see the entry "A price looks wrong: what should I do?").

Dividends and stock splits: what an unadjusted price implies

Dividends

With an unadjusted price, a share's price falls by roughly the amount of the dividend on its ex-dividend date. The software compensates for this fall **only if the dividend is in your transactions** (a "DIVIDENDE" line, with the total amount received). It does not download dividends on your behalf.

- **Recorded dividends**: they are added to the total gain and to the performance.
- **Forgotten dividends**: the performance is underestimated by the same amount.
- **Accumulating ETFs**: dividends are reinvested in the fund, so they are already in its price; there is nothing to enter.
- **Distributing ETFs**: they pay out dividends, to be recorded as for a share.

The same principle applies to the benchmark index: an accumulating ETF gives a "dividends reinvested" performance, a price index such as the CAC 40 (^FCHI) gives an "excluding dividends" performance. The name of each index says which.

Stock splits

Yahoo Finance retroactively corrects its closing prices after a stock split. The software, for its part, only knows the transaction types ACHAT (buy), VENTE (sell) and DIVIDENDE (dividend): it **does not detect** stock splits or reverse splits by itself, but it can adjust your transactions from your broker's notice.

Example: you bought 10 shares at €800. The company then splits each share into 10. Yahoo now shows a history of around €80 for the date of your purchase. Your file still says 10 shares: the software would value 10 × €80 = €800 instead of €8,000.

At import, the price check flags it: the price in the file is far from the market price of the day (here, a gap of +900%), with the note "check the ticker, currency or a stock split".

The fix: express the transaction in securities as they stand after the split, without changing the amount. Here, 100 shares at €80 (100 × 80 = €8,000). The simplest way is to drop the split or reverse split notice sent by your broker on the **Add transactions** page: the **Corporate action** box offers **Apply to earlier transactions**, which multiplies the quantity and divides the price of each earlier purchase and sale by the same factor (see the chapter on import, fiche "Stock split or

reverse split"). You can also do it by hand in the Transactions tab with **Edit transactions**. The total cost, fees and cash flows stay the same.

Other corporate actions

Mergers, spin-offs, ticker changes or free share allotments are not handled automatically. You have to translate them yourself into buys and sells.

The local securities database: what does it contain?

The local database is the software's "memory": it lets it work without Internet and faster online. It is stored in the data/base folder.

Its contents

Item	Contents
titres.csv	the record of each security: Yahoo ticker, name, ISIN (when known), country, region, sector, asset class, currency, indices it belongs to
cours/paquet_00.npz to paquet_31.npz	the daily closing prices, split into 32 compressed files
memoire.csv	the matches learned during imports (ISIN, name or code to ticker)
pays.geojson	the country outlines, for the world map

The universe covered

- **Equities**: the composition, read from Wikipedia, of 32 indices: S&P 500, Nasdaq-100, Dow Jones, Russell 1000, CAC 40, SBF 120, CAC Mid 60, DAX, MDAX, SDAX, TecDAX, FTSE 100, FTSE 250, Euro Stoxx 50, AEX, AMX, BEL 20, IBEX 35, FTSE MIB, SMI, SMIM, OMX Stockholm 30, OMX Copenhagen 25, OMX Helsinki 25, OBX, ATX, PSI, ISEQ 20, Nikkei 225, S&P/TSX 60, S&P/ASX 200 and Hang Seng. The project estimates this at about 3,500 shares.
- **ETFs**: a list of 66 common ETFs (equities, bonds, gold, money market), European and American.
- **Project reference file**: the securities in data/referentiel.csv.
- **Market**: 16 stock indices and 12 euro exchange rates (US dollar, pound, Swiss franc, yen, Canadian, Australian, Hong Kong and Singapore dollars, Danish, Swedish and Norwegian krone, zloty).

Depth of history

- from **1 January 2015** for shares and ETFs;
- from **1 January 2007** for indices and exchange rates, which makes it possible to replay the 2008 crisis in the stress tests.

Precision and limits

- Prices are stored with about 7 significant digits, which is more than enough for share prices.
- The list reflects the composition of the indices **at the time of building**: a company that left an index beforehand is not in it. If you hold it, its prices are downloaded at the first online analysis, then added to the database.
- The ISIN is filled in only if the index's Wikipedia page gives it.
- The exact number of securities depends on the version shipped: some Wikipedia pages may be unreadable on the day of building, and codes unknown to Yahoo Finance are discarded.

How the securities database is built and updated

The full build

It is done by the script construire_base_titres.py (on Windows, from the source code, by double-clicking construire_base.bat), with Internet:

1. reading the composition of the 32 indices on Wikipedia;
2. adding the 66 ETFs and the securities of the reference file;
3. downloading the map background;
4. downloading prices: first the indices and exchange rates from 2007, then the securities from 2015, **in batches of 100 securities**, with a 2-second pause between two batches and up to 3 attempts per batch if Yahoo Finance refuses.

The project estimates this at about an hour. If it is interrupted, running the command again resumes where it stopped. Codes that Yahoo Finance does not recognise are listed in data/base/echecs.csv.

Installed versions are shipped with the database that was in the project when they were built.

The update

python construire_base_titres.py --mise-a-jour

The script takes all the securities that have prices in the database, and downloads their prices from the last known date minus 7 days. It takes only a few minutes. The --liste option rebuilds only the list of securities, without the prices.

A cumulative database

The database also fills itself in **on its own**:

- each history downloaded for an online analysis is added to it;
- the new values replace the old ones for the same dates;
- only modified files are rewritten, through a temporary file that is then renamed: an interruption never leaves a damaged file.

So a security analysed once with Internet remains available offline.

With a new version

Installing a new version replaces the database with the one shipped with that version. On Mac, the memory of recognised securities is kept. The prices that your own use had added will be downloaded again at the next online analysis.

The price cache and the “Refresh prices” button

The software keeps prices in two places to avoid downloading them again and again.

1. In memory, for one hour

The results of an analysis are kept in memory for **one hour** as long as the portfolio and the settings do not change: the page stays fast. During this time, prices are not requested again from Yahoo Finance. The result of the automatic reading of an imported file is kept in the same way.

The **Refresh prices** button, at the bottom of the sidebar, clears all this memory and reloads the page: prices are downloaded again (Internet required).

2. In files, for offline use

File (data folder)	Contents
cache_prix.csv	the latest prices from the last analysis, with the date and time of the download
cache_historique.csv	the history from the last analysis
cache_devises.csv	the quotation currency of each security already encountered
cache_import.csv, cache_candidats.csv	the prices used to check imported files
cache_stress.csv, cache_attribution.csv	the long histories for the stress tests and performance attribution

These files are rewritten at every successful download. If Yahoo Finance does not respond, the software reads the cache again, completes it with the local securities database, and says so: “Cached prices (offline)” in the banner, “cache local du” followed by a date in the sidebar (this sidebar text is shown in French). That date is the oldest of the sources used.

A particularity: the currency

The currency of each security, once known, is kept **with no time limit** in cache_devises.csv. If it was deduced offline from the security’s code and is wrong, it stays wrong. Deleting this file forces the software to ask Yahoo Finance for it again.

Clearing the caches

Refresh prices clears the one-hour memory. The cache_*.csv files can be deleted without risk: they are recreated at the next online analysis. Offline, however, they will no longer be available.

The memory of recognised securities

The principle

When an import asks the Yahoo Finance search engine to recognise an ISIN code, a company name or a ticker, the answer is **kept** in the file data/base/memoire.csv. At the next import, the same identifier is recognised instantly, even without Internet.

Each line contains: the identifier (in capitals), the Yahoo Finance ticker found, the name of the security, the origin of the match and the date. No quantity, no amount, no portfolio data.

The order of lookup

For an identifier to be recognised, the software looks in this order:

1. the built-in table of ETF ISINs, then the one of share ISINs (CAC 40 and large US stocks): they take priority over the memory, to repair a wrong match learned earlier;
2. the memory;
3. the local securities database;
4. the Yahoo Finance search engine, as a last resort.

Only answers obtained through the online search are added to the memory.

A wrong match

If an identifier has been matched to the wrong security, the memory will propose it again at every import. Two remedies:

- for one file, in the import assistant (**Open the import assistant**), correct the **Yahoo Finance ticker** column of the securities step; this correction applies to the file but does not change the memory;
- to correct it permanently, delete the relevant line from data/base/memoire.csv (or the whole file). The software does not provide a screen for this.

Shared between users

The memory is common to all users of the software on the computer: a security recognised by one is recognised for everyone.

The built-in tables of ETF and share ISINs

Trade confirmations and statements often give only an ISIN code and an abbreviated name (for example “AM.C.C.40 UC.ETF C”). To recognise the most common securities **without Internet**,

the software contains two tables: **31 ETF ISINs**, and the ISINs of the **CAC 40** shares and of large US stocks, each matched to a Yahoo Finance ticker.

The ETFs in the table

Family	ETFs (ticker)
France and eurozone equities	Amundi CAC 40 Acc (CACC.PA) and Dist (CAC.PA), Amundi Euro Stoxx 50 (MSE.PA), iShares Core DAX (EXS1.DE)
World equities	Amundi MSCI World (CW8.PA), Amundi PEA MSCI World (EWLD.PA), iShares MSCI World Swap PEA (WPEA.PA), iShares Core MSCI World (IWDA.AS), Vanguard FTSE All-World Acc (VWCE.DE) and Dist (VWRL.AS), iShares MSCI ACWI (IUSQ.DE), SPDR MSCI ACWI IMI (SPYI.DE)
US equities	BNP Paribas Easy S&P 500 (ESE.PA), Amundi PEA S&P 500 (PE500.PA), Amundi PEA Nasdaq-100 (PUST.PA), Vanguard S&P 500 (VUSA.AS), iShares Core S&P 500 (SXR8.DE), Invesco Nasdaq-100 (EQQQ.DE)
European and emerging equities	Amundi Stoxx Europe 600 (MEUD.PA), iShares Core MSCI Europe (EUNK.DE), Amundi PEA MSCI Emerging Markets (PAEEM.PA), Amundi MSCI Emerging Markets (AEEM.PA), iShares Core MSCI EM IMI (IS3N.DE)
Bonds	iShares Core Euro Govt Bond (EUNH.DE), iShares Euro Inflation Linked Govt Bond (IBCI.DE), iShares Core Euro Corporate Bond (EUN5.DE), iShares Euro High Yield Corp Bond (EUNW.DE), Xtrackers II Eurozone Government Bond 1C (DBXN.DE)
Money market and gold	Amundi Smart Overnight Return (CSH2.PA), Xtrackers II EUR Overnight Rate Swap (XEON.DE), Xetra-Gold (4GLD.DE)

(PEA = Plan d'Épargne en Actions, the French equity savings plan.)

The shares in the table

Family	Shares (ticker)
CAC 40, with a few former members	Air Liquide (AI.PA), Airbus (AIR.PA), AXA (CS.PA), BNP Paribas (BNP.PA), Danone (BN.PA), Hermès (RMS.PA), L'Oréal (OR.PA),

Family	Shares (ticker)
	LVMH (MC.PA), Michelin (ML.PA), Orange (ORA.PA), Safran (SAF.PA), Sanofi (SAN.PA), Schneider Electric (SU.PA), TotalEnergies (TTE.PA), Vinci (DG.PA)…
Large US stocks	Apple (AAPL), Microsoft (MSFT), Amazon (AMZN), Alphabet (GOOGL), Meta (META), NVIDIA (NVDA), Tesla (TSLA)

Priority

The ETF table is consulted **first**, then the share table, before the securities memory and before any online search. They thus repair any wrong match learned earlier.

A security missing from the tables

It is looked up in the memory, the local database, then, with Internet, through the Yahoo Finance search engine (by its ISIN, then by its name). The answer is then memorised. Offline and unknown to the memory, it cannot be recognised: the import reports it as not found.

The trading venue chosen

The same ETF is often listed on several venues and in several currencies. The tables fix one specific listing for each ISIN: if you bought on another venue, the price used may differ slightly from yours.

How an ISIN, a name or a Bloomberg code becomes a ticker

The software works with **Yahoo Finance tickers** (MC.PA for LVMH in Paris, AAPL for Apple). At import, each identifier is first classified, then translated.

1. Already a Yahoo ticker

A code with an exchange suffix known to Yahoo (.PA, .DE, .L, .AS…), or a security from the reference file, is kept as it is.

2. Bloomberg, Google Finance and Reuters codes

They are converted **by rules, without Internet**:

Written in the file	Becomes
MC FP or MC FP Equity (Bloomberg)	MC.PA
AAPL US Equity	AAPL
EPA:MC (Google Finance)	MC.PA
NESN.S (Reuters)	NESN.SW
700 HK	0700.HK

MIC venue codes (XPAR, XETR…) and usual names ("Euronext Paris") are also recognised in a venue column.

3. ISIN code or company name

The software searches the tables of ETF and share ISINs, the memory, then the local database (by ticker, by ISIN, then by name). Failing that, it queries the Yahoo Finance search engine with the ISIN, then with the label from the file. Among the answers (shares, ETFs, funds, indices), it prefers the venue of the ISIN's country (Paris for an FR ISIN, Frankfurt for DE…), then a listing in euros (Paris, Frankfurt, Amsterdam, Milan), then the United States.

4. Ticker without a venue (MC, AIR, TSLA)

The code may designate several securities. The software tries the possible listings (local database, search results and 9 venues: United States, Paris, Frankfurt, Amsterdam, Milan, Madrid, London, Zurich, Toronto). It keeps the one whose price, on the date of each buy or sell, best matches your prices, provided the median gap stays below 15%. Example taken from the code: "MC" bought at €740 in January 2024 corresponds to MC.PA (LVMH), not to MC (Moelis, around fifty dollars).

If nothing matches

The security is declared not found and the import assistant opens: enter its ticker yourself in the **Yahoo Finance ticker** column.

Converting currencies into euros

Everything is expressed in euros. A security quoted in another currency is converted using the Yahoo Finance exchange rates.

The currency of each security

The software asks Yahoo Finance for it. Offline, it takes the one in the local database, then, failing that, deduces it from the code: no suffix = US dollar, .L = pence, .SW = Swiss franc, .T = yen, and so on. The answer is kept in cache_devises.csv.

The formula

price in euros = price in currency × factor / EURcurrency rate

- The EURUSD=X rate is the number of dollars for 1 euro.
- The factor is 1, except for London shares, quoted in **pence**: factor 0.01.

Examples:

- Apple at 200 USD, with €1 = 1.10 USD: 200 / 1.10 = **€181.82**.
- A London share at 1,250 pence, with €1 = 0.85 GBP: 1,250 × 0.01 / 0.85 = **€14.71**.

Which rate, which day?

- a **buy, sell or dividend** is converted at the rate **of the day of the transaction**;

- the **day-by-day value** uses each day's rate;
- the **current value** uses the last known rate.

A day with no rate takes the last known rate. **Fees** are always considered to be in euros and are not converted.

What this implies

The performance of a foreign security combines the change in its price and that of the currency: this is currency risk. A US share that gains 10% in dollars earns nothing in euros if the dollar loses about 10% against the euro over the same period.

The limits

- The rate is the Yahoo Finance one, not the one applied by your bank.
- Offline, only currencies whose rate is in the local database (12 currencies) or in the cache can be converted.
- If a rate is missing, the analysis stops with the message "Taux de change introuvables" (exchange rates not found).
- Offline, for an uncommon venue, the currency deduced from the code may be wrong (the euro is used by default).

ETF composition: where does the look-through analysis come from?

The Exposures tab "breaks down" each ETF according to the composition of the index it tracks. These compositions **are not downloaded**: they are written into the software.

The sources and their dates

- **MSCI World, MSCI ACWI, MSCI Emerging Markets, MSCI Europe**: the top 5 countries and the sectors come from the MSCI factsheets as of **30/09/2026**. The following countries are usual orders of magnitude, scaled to the factsheet's "other countries" total.
- **S&P 500, Nasdaq-100, Euro Stoxx 50, CAC 40, DAX, Russell 2000**, government and corporate bonds: 2025-2026 orders of magnitude, based on ETF issuers' factsheets.

The MSCI World thus has 72.94% in the United States: a €10,000 MSCI World ETF is counted as about €7,300 of US equities. The screen reminds you: "approximation as of 30/09/2026".

How an ETF is linked to an index

1. through a table of 58 ETF tickers (CW8.PA and IWDA.AS for the MSCI World, ESE.PA for the S&P 500…);
2. otherwise, through words in the name: "All-World" or "ACWI", "Emerging", "World", "S&P 500", "Stoxx Europe 600"…;
3. a name containing "Hedged" or "couvert" is considered hedged against currency risk.

A fund that matches nothing stays "Unclassified".

Approximations to be aware of

- Actual weights change every day; the project estimates the gap at a few points at most, as the weights of the major indices move little from one year to the next.
- The same sector mix is applied to every country of an index: this is a calculation assumption.
- An ETF is assumed to track its index exactly.
- Dated data must be reviewed once a year in the project's code.

Other fixed reference data

Performance attribution uses the regional weights of the MSCI ACWI IMI as of 30/06/2026, also written into the software.

How each security is classified (country, sector, asset class)

Two sources, in this order

1. **The project's reference file**, data/referentiel.csv: 86 securities described by hand, with the columns ticker, name, country, region, sector, class (Equities, Bonds, Gold, Money market) and duration for bond funds. It **always prevails**.
2. **The local securities database** (data/base/titres.csv), for all other securities.

How the local database classifies a security

- **Country and region**: according to the **trading venue** of the ticker (.PA = France, .DE = Germany, no suffix = United States…), not according to the company's head office. A foreign company listed in Paris is therefore classed as "France".
- **Sector**: the sector given by the index's Wikipedia page (GICS sectors, translated), when there is one.
- **Asset class**: "Equities" for shares; for the ETFs on the list, "Bonds" or "Gold" depending on their name.

Default values

- A security with no known country, region or sector is classed as **"Unclassified"**.
- A security with no known asset class is considered to be an **equity**: this is the cautious assumption for measuring risk.
- **Duration** is known only for the bond funds in the reference file; elsewhere, it is absent.

Correcting a classification

The software does not provide a screen for changing the classification of a security. From the source code, you can add or correct a line of data/referentiel.csv, respecting its columns: it will take priority over the local database. In the installed version, this file is in the program's data folder and would be replaced by a new installation.

Benchmark indices and composite indices

The benchmark index is chosen in **Settings**, at the bottom of the sidebar, under **Benchmark index**. By default: the MSCI World, represented by the CW8 ETF.

The four families

Family	Indices	Represented by
Equities	MSCI World, MSCI ACWI, S&P 500, Nasdaq-100, Stoxx Europe 600, MSCI Emerging Markets	accumulating ETFs: dividends reinvested
Equities	Euro Stoxx 50, CAC 40	the price index: **excluding dividends**
Bonds	Eurozone government bonds; euro corporate bonds	an accumulating ETF (coupons reinvested); an "excluding coupons" ETF
Money market	€STR	an accumulating ETF (interest reinvested)
Blended	conservative 20/80, balanced 60/40, dynamic 80/20	calculated by the software

The label of each index says whether it includes dividends. Comparing with an index that excludes dividends flatters the portfolio: the project estimates this at about 3% a year for the CAC 40.

Why an ETF?

An accumulating ETF includes dividends in its price, just as your performance includes your dividends: the comparison is fair. Each index has several "candidate" ETFs; the first one whose history is available is used (for example CW8.PA, otherwise IWDA.AS).

The blended indices

Composite = X% equity sleeve + Y% bond sleeve, rebalanced at every month end

- Equity sleeve: MSCI World (CW8.PA, otherwise IWDA.AS, otherwise EUNL.DE).
- Bond sleeve: eurozone government bonds (DBXN.DE, otherwise EUNH.DE).
- Prices converted into euros, series in base 100.
- On the last trading day of each month, the weights are reset to their target.

Example for the 60/40: over a month in which equities gain 10% and bonds lose 2%, the index gains 0.60 × 10% + 0.40 × (−2%) = **5.2%**. The weights then start again from 60/40.

The limits

An ETF tracks its index with a small gap (fees, replication). The composite includes neither fees nor rebalancing costs.

The risk-free rate

The default value

2.50% per year, which is the deposit facility rate of the European Central Bank. The project's code gives it as in force since 16/09/2026 (source cited: ECB, "Key ECB interest rates"). The €STR, the overnight interbank rate, is very close to it.

This rate is used for the **Sharpe ratio**, the **Sortino ratio** and **alpha**.

It is not downloaded

The software does not fetch this rate from the Internet: it is a **fixed value**, written into the code. When the ECB changes its rates, it is updated only with a new version of the software, or by yourself.

Changing it

In the sidebar, open **Settings** and change **Risk-free rate (% per year)**: from 0 to 10%, in steps of 0.25 points. The tooltip recalls the ECB value. All the indicators concerned are recalculated.

The simplification to be aware of

The rate is **constant over the whole period analysed**, whereas it has varied in the past (the code cites 4% in early 2024 and 2% in mid-2025). Over a long period, the Sharpe ratio and alpha are therefore approximate. The project notes, as a possible improvement, using the historical €STR series.

For comparison

Choosing the benchmark index "Money market €STR (ETF, interest reinvested)" shows what a money-market investment would have earned over the same period, based on the price of a money-market ETF.

The world map background

The world map in the Exposures tab colours each country according to its weight in the equity sleeve.

The source

The country outlines come from **Natural Earth**, a public-domain mapping database, at a scale of 1:110,000,000. The file is downloaded from GitHub, or failing that from the jsDelivr service, then simplified:

- coordinates rounded to about 1 km, for a file of about 400 KB;
- Antarctica removed (no stock exchange);
- for France, only the mainland is kept: otherwise, French Guiana, drawn together with France, would be coloured by a French share.

It is saved in data/base/pays.geojson.

When it is downloaded

Only once: when the database is built, when the installer is made, or, failing that, by the dashboard the first time it needs it (a single attempt each time the software is launched). After that, the map is displayed without Internet.

If the file is missing

The map is then drawn with the map background that the charting library downloads itself: without Internet, it stays empty. The rest of the tab is not affected.

What the map does not show

A note under the map gives the share of the portfolio that is “not on the map”, by asset class: bonds, gold, money market, or equities with no known country. ETFs are allocated there according to the approximate composition of their index.

What the software checks

Here, one by one, are the checks that are actually in place.

When reading the file

- Only the types ACHAT, VENTE and DIVIDENDE (buy, sell, dividend) are accepted; negative prices and quantities are refused.
- **Sale not possible**: selling more securities than are held on that date stops the analysis, with a message giving the date and the quantities.
- The columns of an unknown file are accepted automatically only if their names are meaningful, or if the numbers are consistent (quantity × price, plus or minus fees, equals the amount to within 1% on at least 80% of the rows). Otherwise, the import assistant opens.
- An **ISIN code** is recognised only if its check digit is correct; in a PDF, it must in addition start with an existing country code and contain at least 4 digits.
- Ambiguous dates (day/month or month/day) are detected and the choice made is reported.

When reading a PDF

- A transaction read by its headings is accepted only if a direction word (achat, vente, buy, sell…) is written in the document and a price (or a dividend amount) was found.

- When several readings of a trade confirmation are consistent (quantity × price = amount), the day's Yahoo Finance closing price decides: a reading within 5% of that price replaces a reading more than 15% away from it.
- The summary reports an amount written that does not match quantity × price ± fees (within 1%), a date falling on a Saturday or Sunday, fees above 3% of the amount and the same transaction present twice.

Prices compared with the market

At import, with Internet, each buy or sell price is compared with the security's closing price that day:

- the software recognises a price entered in euros, in currency or in pence, and brings it back to the quotation unit;
- a price more than 25% above (or 20% below) the day's price is flagged: "price(s) far from that day's market price", with the median gap, "check the ticker, currency or a stock split".

Without Internet, the summary says "Prices not checked against market data (no connection)."

When adding transactions

- **Duplicates**: a transaction with the same date, same security, same type, same quantity and the same price to within 0.5% is unticked automatically.
- Transaction dated in the future, zero quantity or price, sale greater than the quantity held: blocking alerts.

During the analysis

- A missing price, a missing history or an exchange rate that cannot be found stops the analysis with a message naming the securities concerned, rather than giving a wrong result.
- The source of the prices (live or cache) and the date of the data are always displayed.

In the project's tests

The automatic tests check, among other things, that the gain calculated day by day falls back exactly on the total gain. These tests are run by the developer, not while you are using the software.

What the software does not check

To be honest about reliability, here is what is **not** checked.

Prices

- **A single source**: all prices come from Yahoo Finance. The software does not compare them with any other source (Euronext, Bloomberg, your broker).

- **No detection of outlier prices** in the history: an abnormal one-day jump, a missing value or a frozen price are not spotted. A day with no price simply takes the last known price.
- **No freshness check per security**: a suspended or delisted security keeps its last known price, with no alert.
- **No detection of corporate actions**: a split or reverse split is adjusted only if you drop the corresponding notice on the **Add transactions** page; spin-offs, mergers and ticker changes are not handled.
- **No download of dividends**: only those in your file count.
- The 25% price check takes place only **at import**, with Internet, and only on buys and sells.

Other data

- The quotation currency, once known, is not checked again.
- Recognition of an ISIN by the search engine may choose a different trading venue from yours.
- ETF composition, security classification and the risk-free rate are fixed, dated values, not continuously updated.
- The country of a share in the local database is that of its trading venue.

Your own data

The software cannot know whether you have forgotten a transaction, a dividend or fees. It calculates correctly from what you give it.

As a result

The results are reliable for analysis and learning, but do not replace the official statements of your financial institution, in particular for a tax return. They do not constitute investment advice.

A price looks wrong: what should I do?

The software does not let you enter a price by hand: you have to find the cause. Proceed in this order.

1. Check the source and the date

Does the banner say “Cached prices (offline)”? If so, the prices date from the day shown in the sidebar. Reconnect to the Internet and click **Refresh prices**.

2. Check the ticker and the venue

In the Holdings or Transactions tab, look at the ticker used. MC (Moelis, in New York) is not MC.PA (LVMH, in Paris); an ETF listed in London in dollars does not have the same price as its equivalent listed in Paris in euros. If the ticker is wrong, correct your file, or import it again with **Open the import assistant** and correct the **Yahoo Finance ticker** column.

3. Check the currency

London shares are quoted in pence (1,250 pence = 12.50 pounds). The price in your file must be in the security's quotation currency, and the fees in euros. The import summary shows the conversions made.

4. Think of a stock split

A loss of about 50%, 67% or 90% that appeared suddenly suggests a stock split. Correct the transaction as explained in the entry "Dividends and stock splits: what an unadjusted price implies".

5. Compare with another source

Compare the price with your broker's site, with that of the exchange concerned, or with the security's page on Yahoo Finance. If Yahoo Finance itself shows a wrong value, the software picks it up: wait for it to be corrected, then click **Refresh prices**. The new values then replace the old ones in the local database.

6. Check your transactions

A wrongly entered quantity or price distorts the value as much as a wrong price. Use **Edit transactions** in the Transactions tab to correct it.

How do you make sure the prices are right?

The honest answer

The software **does not guarantee** the accuracy of the prices: it takes them from Yahoo Finance, a free and widely used source, but with no quality commitment, and **without comparing them with a second source**. Nor does it look for outliers in the history.

What it does to limit errors

1. **A raw, traceable price**: the closing price not adjusted for dividends, which anyone can compare with the one shown by the exchange or by a broker.
2. **A source always displayed**: live or cache, with the date of the data.
3. **A check of your prices at import**: each buy and sell price is compared with the Yahoo Finance closing price of the same day. A gap of more than 25% is flagged. This cross-check reveals an error in your file just as well as a wrong security or an abnormal price.
4. **Recognition of securities through prices**: a ticker without a venue is accepted only if its price matches your prices (median gap below 15%).
5. **Stops rather than approximations**: a missing price, history or exchange rate stops the analysis with a clear message.
6. **A database that corrects itself**: each download replaces the old values with those from Yahoo Finance.
7. **Tested calculations**: the formulas are verified by automatic tests, on examples whose result is known.

What you can say

For academic work, a fair wording is: “Prices are the daily closing prices from Yahoo Finance, not adjusted for dividends; transaction prices were checked against these prices, with an alert threshold of 25%; the prices were not cross-checked against a second source.”

To go further

For an important figure, check it yourself against a second source: the exchange’s site, your broker or your institution’s statement.

Offline: up to what date do the prices go?

There is no fixed date

Offline, prices stop at the **last date known** on your computer. It depends on:

- the database shipped with your version, updated by the creator at the time of building it;
- your own online analyses: each history downloaded is added to the database;
- the cache of the last analysis.

Where to read this date

- **“Data as of”**, in the banner at the top of the page: the date of the last price in the history used;
- the **“Prices”** line at the bottom of the sidebar: “cache local du” followed by a date, the oldest of the sources used for current prices (this text is shown in French);
- the **“Cached prices (offline)”** badge, with an orange dot.

The calculations remain correct, but are stopped at this date: the current value is the value at the date of the last known price.

Getting more recent prices

- Reconnect to the Internet, then click **Refresh prices**: the missing prices are downloaded and added to the database.
- From the source code, `python construire_base_titres.py --mise-a-jour` updates the whole database in a few minutes.
- Installing the latest version of the software brings a more recent database.

A security with no prices at all

A security that is absent from the database, the cache and the memory has no prices offline: the analysis stops with a message that names it. It will be available from the first analysis with Internet.

Not all securities stop on the same day

The prices of all the securities in a portfolio do not necessarily stop on the same day. For each security, the software takes its last known price.

11. Error messages and problems

This chapter lists the error and warning messages the software may display, with their exact text in quotation marks, their cause and what to do. It then covers the problems that occur without a message: software that does not open, blank page, busy port, installation warnings, prices that do not move, empty chart, security not found or slowness. An ellipsis replaces the variable part of a message (a security name, a date, a number). Where a message exists in the English interface, its English text is quoted; a few messages are not yet translated in the program and are still displayed in French: for those, the French text is given first and an English rendering follows.

“No transactions file: upload a CSV file from the sidebar.”

Cause

No portfolio is available: no file uploaded, no portfolio saved in your space, and no sample portfolio found in the software’s data folder (transactions*.csv files).

Solution

1. In the sidebar, under “Data”, use **Upload a file (CSV, Excel, PDF or image)** to load your transactions.
2. Or sign in with **Sign in**: your saved portfolios then appear in the list.
3. If you do not have a file yet, download the **File template**, fill it in and upload it.

The software is shipped with sample portfolios. If the list is empty, the transactions*.csv files in the data folder may have been moved or deleted: reinstalling the software puts them back (your accounts are kept).

The “Manual and help” area remains accessible even without a portfolio.

“Unable to analyse the portfolio: …”

This general message always comes before a more precise explanation: the portfolio could not be read, valued or calculated. The software prefers to stop rather than display a wrong result. The analysis tabs are then not displayed.

Read the rest of the message

Rest of the message	Entry to consult
“Vente impossible le …” (sale not possible on …)	Sale not possible
“Cours manquant pour : …”, “Historique de cours manquant pour : …” (missing price / price history for: …)	Missing price or history
“Impossible de récupérer les cours : pas de	No connection and no cache

Rest of the message	Entry to consult
connexion…" (unable to retrieve prices: no connection…)	
"Taux de change introuvables : …" (exchange rates not found: …)	Exchange rates not found
"Impossible de récupérer l'historique de l'indice …" (unable to retrieve the history of the index …)	Benchmark index unavailable
"Type(s) de transaction inconnu(s) : …" (unknown transaction type(s): …)	Unknown transaction type

What to do right away

- For an uploaded file, the **Open the import assistant** button appears under the message: it lets you read the file again in another way (columns, tickers, currencies).
- To add a missing transaction, the **Add transactions** button remains available in the sidebar.
- To correct an existing transaction, since the Transactions tab is not displayed, correct your file, or download the portfolio from **My account**, correct it in a spreadsheet and add it again.
- For a price problem, connect to the Internet and click **Refresh prices**.

"Vente impossible le … : on vend … mais on n'en détient que …" (sale not possible)

Two wordings exist:

- during the analysis: "Vente impossible le … : on vend … mais on n'en détient que …" (in English: "Sale not possible on …: selling … but only … held");
- when adding or editing transactions: "Sale of … on …, but only … held on that date."

Cause

The software replays the transactions in date order. On the date shown, the sale covers more securities than you hold. Frequent causes:

- an older purchase is missing, because the statement does not cover the whole history;
- a quantity is wrongly entered or wrongly read;
- the purchase is dated after the sale (day and month swapped);
- a stock split has not been translated into a quantity (see the chapter on sources);
- the same security appears under two different tickers.

Solution

1. Add the missing purchase with **Add transactions** (sidebar button, available even when the analysis fails).

2. To correct an existing quantity or date, correct the original file (or the portfolio downloaded from **My account**), then import it again.
3. If you have just deleted a purchase in the Transactions tab, also delete the sales that depend on it: the software reminds you of this under the message.
4. The software does not accept short selling: there is no such thing as a negative position.

"Cours manquant pour : …" or "Historique de cours manquant pour : …" (missing price or price history)

The messages

- "Cours manquant pour : …" (missing price for: …): no latest price for a security that is still held.
- "Historique de cours manquant pour : …" (missing price history for: …): no history for a security held at some point, even if sold since.
- "Aucun cours disponible après la première transaction." (no price available after the first transaction): there is no day of prices after your first transaction.

Causes

- ticker misspelt or unknown to Yahoo Finance (MC instead of MC.PA, typing mistake);
- security removed from the market, or unlisted fund;
- no Internet, and the security is absent from the local database and the cache;
- for the last message: transactions dated in the future, or offline prices that stop before your first purchase.

Solution

1. Check the ticker on the Yahoo Finance website, then correct it: import the file again with **Open the import assistant** and the **Yahoo Finance ticker** column, or correct your file.
2. Connect to the Internet and click **Refresh prices**: the security will then be kept in the local database.
3. A security with no prices at all on Yahoo Finance cannot be tracked by the software.

"Impossible de récupérer les cours : pas de connexion à Yahoo Finance et aucun cache disponible." (unable to retrieve prices)

In English: "Unable to retrieve prices: no connection to Yahoo Finance and no cache available."

Variant for the history: "Impossible de récupérer l'historique : pas de connexion à Yahoo Finance et titres absents du cache et de la base locale." (in English: "Unable to retrieve the history: no connection to Yahoo Finance and securities absent from the cache and the local database.") The message ends with "Détail :" (Detail:) followed by the technical error.

Cause

Yahoo Finance did not respond (no Internet, filtered network, service outage), and the software found these securities neither in its cache nor in its local database. With the local database shipped, this case mostly concerns securities that it does not contain.

Solution

1. Check your connection. On a school or company network, a firewall may block Yahoo Finance: try another network (phone tethering, for example).
2. Click **Refresh prices**.
3. If Yahoo Finance is down, try again later.

Once analysed with Internet, securities remain available offline.

Not to be confused with

When Yahoo Finance does not respond but the cache is sufficient, there is **no** error: the banner simply displays “Cached prices (offline)”.

“Taux de change introuvables : …” (exchange rates not found)

In English: “Exchange rates not found: …”. Variants: “Taux de change actuels introuvables.” (current exchange rates not found) and “Aucun taux de change disponible pour …” (no exchange rate available for …).

Cause

The portfolio contains a security quoted in a currency other than the euro, and the corresponding exchange rate (for example EURUSD=X) could be obtained neither from Yahoo Finance nor from the local database. The local database contains 12 currencies; a rarer currency is available only with Internet.

Solution

1. Connect to the Internet, then click **Refresh prices**.
2. Check the security’s currency: offline, it is read from the local database or, failing that, deduced from the code (no suffix = US dollar). A misspelt ticker can lead to an unexpected currency.
3. If a currency was recorded wrongly, deleting the file data/cache_devises.csv forces the software to ask Yahoo Finance for it again (Internet required).

“Impossible de récupérer l’historique de l’indice …” (unable to retrieve the index history)

In English: “Unable to retrieve the history of the index …”. Variants: “historique introuvable pour la poche … de …” (history not found for the sleeve … of …) and “pas de cours communs pour l’indice composite” (no common prices for the composite index).

Cause

None of the ETFs that represent the chosen index has a history available (no Internet and absent from the database), or, for a blended index, one of its sleeves (equities or bonds) cannot be found.

Solution

1. Open **Settings**, then choose another **Benchmark index**, for example the MSCI World offered by default.
2. Connect to the Internet and click **Refresh prices**.

The index is used mainly for comparisons (beta, alpha, tracking error, base 100 chart): changing it unblocks the analysis without changing anything in your positions.

"Type(s) de transaction inconnu(s) : …" (unknown transaction type)

In English: "Unknown transaction type(s): …". A related message: "Les prix et les quantités doivent être positifs." (prices and quantities must be positive).

Cause

These messages are rare, because the import converts most labels and turns quantities positive. A portfolio can contain only three types: ACHAT (buy), VENTE (sell) and DIVIDENDE (dividend). Any other value ("Frais", "Virement"… that is, fees, transfer), or a negative price or quantity, stops the analysis.

Solution

1. Correct the type column of the file, or delete the lines that are not transactions in securities.
2. Write quantities and prices as positive numbers: it is the type that indicates the direction.
3. For a bank export, upload the file as it is instead: the automatic import recognises many labels (achat, buy, souscription, vente, sell, coupon…) and ignores custody fees or transfers. If in doubt, **Open the import assistant** and check step 3.

"Colonne(s) manquante(s) ou non reconnue(s) : …" (columns missing or not recognised)

In English: "Missing or unrecognised column(s): …". Several wordings, depending on where it appears:

- "Colonne(s) manquante(s) ou non reconnue(s) : …", followed by the columns read and a reminder of the expected format;
- "Colonne(s) non reconnue(s) : …" or "Colonnes reconnues, mais sans certitude sur les quantités, prix et montants." (columns recognised, but no certainty about quantities, prices and amounts), under **Why is the assistant opening?**;

- “Still needed: …”, in the import assistant (in the French interface: “À indiquer : …”), as long as a required field is not matched. The full English text is: “Still needed: {fields}. For the price, either a”Unit price” or a “Total amount” column is enough.”

Cause

The required fields are: the date, the security (ticker, ISIN or name), the quantity, and a unit price **or** a total amount. An unusual column name, a header row that was wrongly detected, or columns of numbers without a meaningful name prevent automatic reading.

Solution

1. In the assistant, check the **Header row** (and the **Excel sheet** for a workbook).
2. In step 2, match each field with the right column. Fields marked with an asterisk are required.
3. Check the result in step 4, then click **Analyse this portfolio**.

“… ligne(s) avec date illisible (lignes …) : ignorée(s)” and “Aucune transaction lue.” (rows ignored)

In English: “… row(s) with an unreadable date (rows …): ignored”. The sentence is built from the cause (“an unreadable date”, and so on) and the row numbers.

The four reasons

Reason	Cause	Solution
unreadable date	empty date, written out in words or damaged	correct the date (15/01/2024 or 2024-01-15)
missing price or amount	buy or sell with no price or amount	complete it, or match the amount column
zero quantity	buy or sell of 0 securities	check the quantity column
missing security	empty cell, or ISIN without ticker	complete it, or enter the ticker in step 3

Rows are numbered from the first data row; beyond five, the list ends with “…”.

Related messages

- “Certaines lignes n’ont pas pu être lues : …” (some rows could not be read: …): for a file in the project format, any row in error prevents direct reading; the assistant opens.
- “Aucune transaction lue.” (no transaction read): all rows were ignored. Check the transaction types (step 3) and the column matching (step 2).
- “… row(s) ignored: other kinds of operations (custody fees, transfers…)” (French: “… ligne(s) ignorée(s) : opérations d’un autre type (frais de garde, virements…)”): this is not an error, these rows are not transactions in securities.

“Unreadable file: …” or “Le fichier est vide.” (file is empty)

Causes

- empty file, or the chosen Excel sheet contains nothing;
- damaged file, or saved in an unaccepted format: the upload accepts only .csv, .xlsx and .pdf (not the old .xls, nor .ods or .numbers).

Solution

1. Open the file in your spreadsheet to check that it does contain the transactions.
2. Save it again in .xlsx or CSV format.
3. For a workbook with several sheets, choose the right **Excel sheet** in the assistant (the software proposes the fullest sheet by default).

“Titre(s) introuvable(s) : …” (securities not found)

Wordings:

- at import: “Titre(s) introuvable(s) : …” (securities not found: …), followed by the codes concerned; the assistant opens with the status “not found” in step 3;
- in manual entry: “Security not found: …. Enter its Yahoo Finance ticker (e.g. MC.PA).”;
- empty field in manual entry: “Enter the security.”.

Solution

1. Look the security up on the Yahoo Finance website and note its ticker (AI.PA for Air Liquide).
2. Enter it in the **Yahoo Finance ticker** column of the assistant, or directly in the “Security” field of **Manual entry**.
3. Without a ticker, the rows concerned are ignored.

The possible causes are detailed in the entry “A security is not found: the possible causes”.

PDF messages: scan, encoded text, password, transaction not recognised

The reading messages

Message	Cause
“Transaction not recognised automatically in this PDF: complete it in the”Complete the transaction” form (the values found are suggested).”	Readable text PDF, but neither a transaction table nor a trade confirmation read with certainty (by its headings or by its content)
“The text of this PDF is”encoded” (it displays correctly but cannot be extracted) and no character recognition engine is installed. Complete the transaction in the form, or install character recognition (requirements-	Text that extracts as unintelligible signs, with no character recognition engine

Message	Cause
ocr.txt)."	
"Scanned PDF (image): it cannot be read automatically…"	PDF with no text and no character recognition engine installed
"Image PDF (scan, photo or page printed with"Print to PDF"): the text was read by character recognition, but no transaction was recognised…"	Image (or encoded text) read, but no certain transaction found

A PDF is treated as an image when it contains fewer than 20 characters of text; "encoded" text is treated the same way.

The password messages

Message	Cause	Solution
"Password-protected PDF: enter the password to open it (often given in the bank's e-mail: date of birth, customer ID…). It is not stored."	The bank has encrypted the PDF: it does not open without a password	Type the password in **PDF password**, then click **Open the PDF** (**Protected PDF** box)
"Incorrect password."	The password entered does not open the document	Check it in your bank's e-mail or terms (often a date of birth or a customer ID) and try again

The password is kept nowhere; the decrypted copy of the PDF stays in memory for the session. In "My account", "Add a portfolio" section, no password is requested: upload the PDF from the sidebar (see the chapter on import, fiche "My PDF is password-protected").

The points to check in the summary

After reading a PDF, the sidebar summary may contain messages that block nothing but call for a check: "…: the amount written (…) does not match quantity × price ± fees (…): check the quantity and the price.", "…: transaction dated on a Saturday (…), a non-trading day: check the execution date.", "…: fees of … for an amount of … (more than 3%): check the fees.", "…: the same transaction appears twice (…): confirmation sent twice?". It may also say that a price read was replaced by the reading that matches the day's closing price. See the chapter on import, fiches "Checks after reading a PDF" and "Checking the price read against the market price".

What is displayed

These messages no longer block the import of a PDF: they appear at the top of the **Complete the transaction** form, which suggests the date, the ISIN code and the numbers found in the document. Check them, correct them if needed, then click **Add this transaction** (see the chapter on import, fiche "Completing a transaction when the PDF is not recognised").

Solution, from the most reliable to the least reliable

1. Download the real PDF from your online banking (the “PDF format” or “Download” button, not “Print”).
2. Export the transactions to Excel or CSV.
3. Complete the transaction in the suggested form, or enter it by hand: **Add transactions**, **Manual entry** tab.
4. For a scan or encoded text, install character recognition from the source code: `python -m pip install -r requirements-ocr.txt`.
5. From the source code, `python diagnostic_pdf.py` shows what the software has read; with `--anonyme`, the report contains no name, address or account number. In the form, **Prepare an anonymised report** produces the same kind of report (see the chapter on import).

“To read a PDF, install pdfplumber…” and other missing libraries

The messages

- “To read a PDF, install pdfplumber: python -m pip install pdfplumber”
- “Pour lire un fichier Excel (.xlsx), installer openpyxl : python -m pip install openpyxl. Ou l’enregistrer au format CSV.” (to read an Excel file (.xlsx), install openpyxl: python -m pip install openpyxl; or save it as CSV)
- “Install reportlab: python -m pip install reportlab” (when clicking **PDF report**)

Cause

These messages concern only launching **from the source code**: a Python library is missing. The Windows and Mac installers already contain all the libraries.

Solution

From the project folder, install everything at once:

```
python -m pip install -r requirements.txt
```

Then launch the dashboard again. To read image PDFs, add `python -m pip install -r requirements-ocr.txt` (optional).

“… price(s) far from that day’s market price …” and “Prices not checked…”

The messages

- “…: … price(s) far from that day’s market price (median gap …) — check the ticker, currency or a stock split.”
- “Prices not checked against market data (no connection).”

They are displayed in the import summary (sidebar) or in step 4 of the assistant. They are warnings: the import is not blocked.

Cause of the first

With Internet, each buy and sell price is compared with the closing price of the same day. A price more than 25% above (or 20% below) is flagged. Causes: wrong ticker (another company, another venue), currency or pence wrongly interpreted, stock split, typing mistake.

Solution

1. Check the ticker used; correct it in **Yahoo Finance ticker**.
2. In step 4, try another choice of **Currency of the prices in the file**.
3. For a stock split, express the transaction in securities as they stand after the split.

The second message only means that the check could not take place: check foreign securities yourself.

"Transaction dated in the future" and other blocking checks

These messages appear on the **Add transactions** page or in the Transactions tab, during an edit.

Message	Cause	Solution
"Transaction dated in the future: …, on …"	date later than today	correct the date (often day and month swapped)
"Zero quantity or price for …, on …"	buy or sell of 0 securities or at €0	correct the quantity or the price
"Sale of … on …, but only … held on that date."	sale greater than the quantity held	see the entry "Sale not possible"
"The portfolio cannot be empty: keep at least one transaction."	all transactions are ticked "Delete"	untick at least one row

As long as one of these messages is displayed, the **Save transactions** button (add page) or **Save changes** (Transactions tab) remains greyed out; in the Transactions tab, the **I confirm these changes** box is also inactive.

A simple warning

"After this change, these holdings are no longer held: …" does not prevent saving: it tells you that a line is disappearing from the portfolio. Check that this is intended.

A file refused when adding transactions or in "My account"

The messages

- On the **Add transactions** page: the file name followed by the reason (for example for a CSV file whose columns are not recognised). A PDF is not refused this way: the **Complete the transaction** form opens instead, with the reason at the top (for example "Transaction not recognised automatically in this PDF…").

- In **My account**, under “Add a portfolio”: “This file could not be read automatically. Upload it from the sidebar…”.

Cause

These two pages use only **automatic** reading. They do not open the import assistant: a file in an unusual format, with ambiguous columns or securities that cannot be found, is refused there.

Solution

1. Upload the file from the sidebar, with **Upload a file (CSV, Excel, PDF or image)**: the import assistant will guide you.
2. Once the file has been read, save it with the “Save to my space” button that appears under the upload.
3. For just a few transactions, use **Manual entry**.

“Incorrect username or password.” and “Too many failed attempts…”

“Incorrect username or password.”

The same message is displayed for an unknown username and for a wrong password: the software does not reveal which accounts exist. Check:

- the username (capitals are converted to lowercase, spaces around it are removed);
- the password, which is case-sensitive (Caps Lock key);
- that you are on the right computer and the right session: accounts are specific to each installation.

“Too many failed attempts: try again in … seconds.”

After 5 failed attempts, the account is locked for one minute. Wait for the delay shown.

Forgotten password

There is no way to recover it: portfolios are encrypted with it. Create a new account and import your files again (see the chapter on the account).

Messages when creating the account or changing the password

Message	Cause	Solution
“Invalid username: 3 to 30 characters among lowercase letters, digits,”.”, “_” and “-”.”	space, accent or special character, or length outside the limits	choose, for example, marie.dupont
“The password must be at least 8 characters long.”	password too short	make the password longer
“The two passwords do not match.”	typing mistake in the confirmation	enter both fields again

Message	Cause	Solution
“This username is already taken.”	account already exists on this computer	choose another username, or sign in
“Current password is incorrect.”	error in the old password, in **My account**	enter the old password again
“Incorrect password.”	error when deleting the account	enter the password again

Other account messages

- “Automatically signed out after 30 minutes of inactivity.”: sign in again; your saved portfolios are not lost.
- “Portfolio not found.” or “Nothing to undo.”: the portfolio has been deleted, or there is no previous version. Reload the page.

“Optimisation compares allocations across holdings…” and “Optimisation failed: …”

“Optimisation compares allocations across holdings: the portfolio needs at least 2 holdings.”

With a single line, there is nothing to allocate. Add securities to use the tab.

“Optimisation failed: …”

The message is followed by the cause, most often “L’optimisation n’a pas convergé : …” (the optimisation did not converge: …): the optimiser did not find a solution that respects the constraints. Frequent causes: a security with a very short history (the calculations use only the days on which all securities have a price), almost identical securities, or very tight constraints.

Solution

1. Change the **Maximum weight per security** (only the values that allow 100% to be invested are offered).
2. Wait until recent securities have more history, or analyse the portfolio without them.

“Approximate” frontier

If the optimiser finds no exact point on the frontier, the software replaces it with the envelope of 4,000 randomly drawn portfolios: this is not an error.

“Stress tests unavailable: …”, “Attribution unavailable: …”

These yellow warnings concern the “Wealth advisory” and “Asset management” areas. The rest of the software works normally.

Message	Frequent causes	Solution
“Stress tests unavailable: …”	history since 2008 cannot be	connect, then **Refresh prices**

Message	Frequent causes	Solution
	obtained (no Internet, no cache)	
“Attribution unavailable: …”	“Historique trop court : il faut au moins deux fins de mois.” (history too short: at least two month ends are needed); “aucune action dans le portefeuille…” (no equities in the portfolio…); regional indices not downloaded	wait for a full month of history; attribution covers only the equity sleeve
“Risk budget unavailable: …”	calculation impossible; the rest of the message gives the reason	check the history of recent securities
“Backtest unavailable: …”	not enough days on which all securities have a price	same

The PDF report calculates these analyses separately: if one fails, the others still appear in it.

“n.d.” in place of a figure

“n.d.” (French for “non disponible”) means “not available”: the calculation makes no sense or did not succeed. It is not a crash. In the English interface, the same condition is written “n/a”.

Common cases

- **Cornish-Fisher VaR**: the note “Cornish-Fisher n/a: skewness or kurtosis too large, the adjustment is no longer reliable.” appears under the table. The formula no longer preserves the ordering of losses: use the historical VaR and the CVaR.
- **IRR (TRI)**: no rate between −99% and +1,000% per year cancels out the cash flows (extreme cases, or a very short history).
- **Annualised TWR**: first and last day identical.
- **Shape indicators** (skewness, kurtosis): fewer than 4 days of history; they are then 0 by convention.

Most often, these indicators become available with a longer history (see the chapter “All the formulas”).

The assistant answers “I could not find a reliable answer in the manual.”

Cause

The assistant does not write anything itself: it looks for the manual entry closest to your question. If none matches well enough, it says so rather than answering off the point, and offers the “Closest entries”.

Solution

1. Rephrase with other words: simpler (“remove a purchase”) or more technical (“delete a transaction”).
2. Use the exact name of an indicator or a button (“VaR”, “PRU”, “Refresh prices”).
3. Browse the **Manual contents**, chapter by chapter.
4. Your question is recorded on this computer; you can send it to the creator with **Export the questions (CSV)** to complete the manual.

“The manual cannot be found (docs/manuel folder).”

The manual files are missing from the software’s folder: reinstall it.

The software does not open

Nothing happens

1. Wait a few seconds: the first start is longer (see the entry on slowness).
2. Look at the taskbar (Windows) or the Dock (Mac): the black window or the Terminal may be hidden behind another window.
3. On Mac, at first launch, macOS blocks the application: see the entry “Windows or macOS warnings at installation”.

The black window shows “Le tableau de bord n’a pas pu démarrer (voir le message ci-dessus).”

(In English: “The dashboard could not start (see the message above).”) The server has stopped. The technical message above gives the reason. Press Enter to close, launch again, and if the error persists, take a photo of the message and pass it on to your teacher or to the software’s creator. You can also reinstall the latest version: your accounts are kept.

The black window is open, but not the dashboard

The launcher waits up to three minutes for the dashboard to be ready. After that, open your browser yourself at the address shown in the black window, for example http://localhost:8501.

The software was already open

Clicking the icon again does not launch it a second time: it reopens the dashboard window (“Portfolio Tracker est déjà ouvert : ouverture de la fenêtre…”, that is, “Portfolio Tracker is already open: opening the window…”).

The page stays blank or shows a connection error

Causes and solutions

1. **The black window (or the Terminal) was closed**: the software is stopped, the page can no longer load. Launch it again with the “Portfolio Tracker” icon.
2. **The dashboard is still starting**: be patient, then reload the page (F5 or Ctrl + R; Cmd + R on Mac).

3. **Wrong port number**: the address must be the one displayed in the black window (8501, or a following number up to 8510).
4. **A long calculation is under way**: a waiting message is then displayed at the top of the page ("Fetching prices and computing indicators…", for example). Let it finish.

If nothing works

Close the black window, then launch the software again, after restarting the computer if necessary. The dashboard needs no Internet connection to be displayed: localhost means your own computer.

"Aucun port libre entre 8501 et 8510": the port is busy

How the software chooses its port

At launch, it goes through ports 8501 to 8510:

- if the dashboard is already running on one of them, it simply reopens its window;
- otherwise, it starts on the first free port.

The message

"Aucun port libre entre 8501 et 8510 : fermez une autre application puis réessayez." (in English: "No free port between 8501 and 8510: close another application then try again."), followed by "Appuyez sur Entrée pour fermer." ("Press Enter to close."). All ten ports are occupied by other programs, often other Streamlit applications, or old windows of the software left open.

Solution

1. Close the other black windows or Terminals (old launches, other Streamlit projects).
2. Launch Portfolio Tracker again.
3. If the problem persists, restart the computer.

Windows or macOS warnings at installation

The Windows installer and the Mac application are not signed with a paid publisher certificate nor distributed through the App Store. The warnings are therefore normal.

Windows

- **Edge** may block the download: in the downloads list, "…" menu, choose to keep the file.
- **"Windows protected your PC"** (SmartScreen): click "More info", then "Run anyway".
- **Antivirus**: if the file comes from the official releases page, you can allow it.

Mac

- **macOS 15 and later**: at the blocking message, click "Done", open System Settings, Privacy & Security, then "Open Anyway".
- **macOS 12 to 14**: right-click the application, "Open", then "Open".

- The application requires an Apple Silicon Mac (M1 and later): on an Intel Mac, use the online version.

Download only from https://github.com/DieuUssop/Python/releases/latest or the link from your teacher. The details are in the chapter “Getting started”.

The prices are not updating

Check the source first

- Banner: “Live prices · Yahoo Finance” (green dot) or “Cached prices (offline)” (orange dot).
- “Data as of”: the date of the last price in the history.
- Sidebar, “Prices” line: “Yahoo Finance (en direct)” (live), or “cache local du” (local cache of) followed by a date. These two sidebar texts are displayed in French.

The causes

1. **The one-hour memory**: as long as the portfolio and the settings do not change, results are kept for an hour and prices are not requested again. Click **Refresh prices**.
2. **No Internet, or Yahoo Finance unreachable**: the software uses its cache and its local database. Reconnect, then refresh.
3. **Market closed**: at weekends or on a public holiday, the latest price is that of the last session. This is normal.
4. **Suspended or delisted security**: it keeps its last known price, with no alert.

What the software uses

The closing price (or a value from the current session if the exchange is open), never a real-time feed.

A chart is empty or missing

The cases the software provides for

What you see	Cause
Empty world map	map background never saved and no Internet
World map missing	no share with a known country (100% bond, gold or money-market portfolio)
Donut by region or by sector missing	only one group: the donut is drawn only from two groups
“At least two positions are needed.”	correlations with a single line
“No equity allocation to analyse.”	regions and sectors with no equities
“The portfolio holds no bonds.”	rate sensitivity without any bonds
Optimisation tab without a chart	fewer than 2 securities, or optimisation impossible

Other causes

- **A click in the legend** hides a curve: click its name again to show it again.
- **A zoom** that is too tight: double-click in the chart to return to the overall view.
- **Very short history**: with only a few days, the curves and histograms are almost empty.

A security is not found: the possible causes

The software knows only the securities that have a **Yahoo Finance ticker** with daily prices.

The causes, in order of frequency

1. **No Internet**: only the 31 ETFs of the built-in table, the memory of recognised securities and the local database are consulted.
2. **Abbreviated name**: "AM.C.C.40 UC.ETF C" gives nothing; it is the ISIN that allows recognition.
3. **Ticker without a venue** (MC, AIR): the software looks for the listing whose price matches your prices (median gap below 15%); failing that, it queries the search engine, which may find nothing.
4. **Unlisted product**: euro funds of life insurance (assurance-vie), some UCITS funds (OPCVM), structured products, bonds with no listing on Yahoo Finance. They cannot be tracked.
5. **Delisted or merged security**: Yahoo Finance sometimes no longer has its history.

Finding the right ticker

On the Yahoo Finance website, search for the name or the ISIN, then note the full code with its venue: .PA (Paris), .DE (Frankfurt), .AS (Amsterdam), .L (London), nothing for the United States. Enter it in **Yahoo Finance ticker** (import assistant) or in **Manual entry**.

The software is slow, especially at first launch

At first launch

- **On Mac**, at the first launch of a new version, the application copies its code and its securities database into ~/Library/Application Support/Portfolio Tracker before starting.
- **Everywhere**, the dashboard server has to start: allow a few seconds.

At the first display of a portfolio

Prices are downloaded from Yahoo Finance ("Fetching prices and computing indicators…"). After that, the results are kept for an hour: navigation becomes fast.

The longest calculations

- **stress tests**: download of the history since 2008, then kept in cache;
- **performance attribution**: download of the regional indices;
- **reading an uploaded file**: checking prices, searching for unknown securities;
- **image PDF**: character recognition, several seconds;

- **PDF report**: all the analyses recalculated.

Tips

- Avoid **Refresh prices** without a reason: it clears all the memory.
- The online version may be slower, especially if the site was asleep.

12. Glossary

This glossary defines, in one to three sentences, the finance, statistics and computing terms used by the software and by this manual. The definitions are those adopted by Portfolio Tracker; the chapter “All the formulas” gives the exact calculation of each indicator. Terms are grouped by letter of their French name (shown in italics where it differs from the English), and within each group they are listed in alphabetical order of the English name.

Glossary: A

Annualisation (*annualisation*): conversion of a figure over a period into a “per year” figure. A total return is compounded over 365 days; a volatility is multiplied by √252.

Buy and hold (*achat-conservation*): backtest strategy that buys the securities once and never touches them again; weights drift with prices, and the winners grow.

Import assistant (*assistant d’import*): four-step screen used to tell the software how to read a file: sheet and header row, columns, transaction types and tickers, then result.

Jensen’s alpha (*alpha de Jensen*): annual performance that is not explained by exposure to the market, according to the CAPM. When positive, stock selection created value relative to the index.

Life insurance (*assurance-vie*): tax wrapper compared in the Taxation tab: social contributions of 17.2%, tax of 12.8% before 8 years, then 7.5% after an annual allowance of €4,600 (€9,200 for a couple).

Look-through analysis (*analyse en transparence*): method that splits each ETF according to the composition of the index it tracks (countries, sectors, currencies). A €10,000 MSCI World ETF thus counts for about €7,300 of American shares.

Net contributions (*apports nets*): the sum of all the money put in through purchases, minus the money taken back through sales and dividends. It is the money “from your own pocket” that is still invested.

Refresh prices (*actualiser les cours*): button at the bottom of the sidebar that clears the results kept in memory (for one hour) and downloads the prices again. An Internet connection is required.

Skewness (*asymétrie*): measure of the asymmetry of daily returns. Zero for a normal distribution; when negative, large falls are more frequent or more violent than large rises.

Trade confirmation (*avis d’opéré*): document sent by the bank after an order is executed (date, direction, ISIN, quantity, price, fees). The software can read it in PDF format.

Glossary: B

Backtest: test of a strategy on real history. The **Strategy backtest** tab compares, with the same securities and the same starting weights, buy and hold against regular rebalancing, then investing all at once against investing gradually.

Base 100: series that starts at 100 and tracks performance (100 × product of 1 + r). It makes it possible to compare the portfolio and the index on one chart.

Benchmark: see "Benchmark index".

Beta (*bêta*): sensitivity of the portfolio to movements in the index: covariance of returns divided by the variance of the index. A beta of 1.2 means that a 1% move in the index is accompanied on average by a 1.2% move in the portfolio.

Bootstrap: simulation method that draws at random, with replacement, from real past returns. The "Historical (bootstrap)" projection builds each month from 21 real days of the portfolio.

Brinson-Fachler: performance attribution model (1985) that breaks the gap with the index down into allocation, selection and interaction effects, region by region.

Correlated blocks (*blocs corrélés*): groups of securities whose average correlation exceeds 0.7. Each block counts as a single "bet", whatever the number of holdings.

Local securities database (*base locale de titres*): files delivered with the software (data/base folder) that contain the record and daily prices of several thousand securities, indices and exchange rates. It allows you to work offline.

Glossary: C

Asset class (*classe d'actifs*): broad family of an investment: Equities, Bonds, Gold or Money market. A security of unknown class is treated as an equity, as a precaution.

Cache: local copy of the latest downloaded prices (cache_*.csv files in the data folder). If Yahoo Finance does not respond, the software reads it again and displays "Cached prices (offline)".

Cariño (smoothing) (*Cariño, lissage de*): method that links monthly attribution effects so that their sum lands exactly on the compounded performance gap over the whole period.

Check digit (*clé de contrôle*): last digit of an ISIN code, calculated from the others. The software only accepts an ISIN if its check digit is correct.

Closing price (*cours de clôture*): last price of a trading session. The software uses the closing price not adjusted for dividends.

Cornish-Fisher: correction of the normal distribution quantile for skewness and kurtosis, used for a more realistic VaR. It is not calculated ("n/a") when these moments are too extreme.

Correlation (*corrélation*): measure, between −1 and +1, of the link between the daily returns of two securities. +1: they always move together; 0: no linear link; −1: in opposite directions.

Covariance: measure of how two securities vary together; it combines their correlation and their volatilities. The covariance matrix is the basis of Markowitz and of the risk budget.

Currency hedging (*couverture de change*, *EUR Hedged*): technique used by an ETF to neutralise the effect of currencies. An ETF whose name contains “Hedged” or “couvert” is counted in euros in the currency exposure.

CVaR (*Expected Shortfall*): average loss on the days when the historical VaR is exceeded. It answers “when things go badly, how badly?”.

Risk contribution (*contribution au risque*): share of the portfolio’s volatility contributed by a holding (Euler decomposition). The sum of the contributions equals the total volatility.

Securities account (CTO) (*compte-titres ordinaire*): a tax wrapper with no tax advantage: dividends and capital gains are taxed at the single flat-rate levy (PFU) of 31.4% (2026 rules).

Glossary: D to E

Allocation effect, selection effect, interaction effect (*effet allocation*, *effet sélection*, *effet interaction*): the three parts of the gap with the index according to Brinson-Fachler: choice of regions, choice of securities within each region, and the cross effect of the two.

DCA (*Dollar Cost Averaging*), or gradual investing (*investissement progressif*): placing a sum of capital in several equal monthly instalments rather than all at once. The backtest compares the two, with money waiting to be invested earning the risk-free rate.

Diversification: spreading across investments that do not all move together, to reduce risk without reducing return as much. The software measures it through correlations and the diversification ratio.

Dividend (*dividende*): sum paid by a company (or a distributing ETF) to its shareholders. It is recorded with the type DIVIDENDE, a quantity of 0 and the total amount received in the price.

Drawdown: fall from the highest level reached so far. The **max drawdown** is the worst of these falls over the period.

Duration: sensitivity of a bond to interest rates, in years. A duration of 7 years means a fall of about 7% in the price if rates rise by 1 point.

ETF (*Exchange Traded Fund*), or listed index fund: fund that replicates an index and is bought on the stock market like a share. **Accumulating** (*capitalisant*): it reinvests dividends in its price; **distributing** (*distribuant*): it pays them out.

Exposures (*expositions*): what the portfolio is really exposed to (countries, sectors, currencies, rates), ETFs included on a look-through basis. It is also the name of the tab that presents them with a diagnosis.

Standard deviation (*écart-type*): measure of the dispersion of returns around their mean. Annualised, it gives the volatility.

Trading currency (*devise de cotation*): currency in which a security is quoted (dollar for Apple, pence for a London share). The price of a purchase is entered in this currency; the software converts it into euros.

Glossary: F to I

Benchmark index (*indice de référence*): index to which the portfolio is compared (beta, alpha, tracking error, base-100 chart). It is chosen in **Settings**; by default, the MSCI World represented by the CW8 ETF.

Cash flows (*flux*): money put in (purchase, counted positively) or taken back (sale, dividend, counted negatively) on a given day. Cash flows are removed from the calculation of daily performance.

Composite index (*indice composite*, or blended index): index calculated by the software, a mix of an equity portion and a bond portion (20/80, 60/40 or 80/20), reset to its target weights at the end of each month.

Efficient frontier (*frontière efficiente*): set of portfolios that offer, for each level of expected return, the lowest possible volatility (Markowitz). No randomly drawn portfolio goes beyond it.

Fees (*frais*): brokerage fees on a transaction, always in euros. They are included in the cost basis (PRU) on a purchase and deducted from the capital gain on a sale.

Fernet: encryption method (Python `cryptography` library) that protects saved portfolios, with a key derived from your password.

Flat tax: see "Single flat-rate levy (PFU)".

GBp (pence): quotation unit for London shares. 1,250 GBp is worth £12.50; the software applies a factor of 0.01.

Herfindahl index (*indice de Herfindahl*): sum of the squares of the weights. Its inverse gives the **effective number of holdings**: $1 / \Sigma \text{ weight}^2$.

ISIN: 12-character international code that identifies a security (for example FR0013380607). Two country letters, nine characters and a check digit.

Total gain (*gain total*): what the portfolio has earned in euros since the start: unrealised gains + realised gains + dividends, net of fees.

Glossary: J to M

Capital loss (*moins-value*): loss that is realised (sale below the cost basis) or unrealised (price below the cost basis).

CAPM (*MEDAF*, capital asset pricing model): model that links a portfolio's return to its beta. It is used to calculate Jensen's alpha.

Excess kurtosis (*kurtosis en excès*): measure of the "tails" of the distribution. Zero for a normal distribution; when positive, extreme days, in both directions, are more frequent than expected ("fat tails").

Jarque-Bera test: statistical test that checks whether returns follow a normal distribution, based on skewness and kurtosis. A p-value below 5% leads to normality being rejected.

Localhost: address that refers to your own computer. The dashboard runs at the address http://localhost:8501 (or a following port); nobody else on the network can connect to it.

Log-normal distribution (*loi log-normale*): distribution of a quantity whose logarithm follows a normal distribution. The final value of a projection follows one: its mean exceeds its median.

Markowitz optimisation (*optimisation de Markowitz*): method (1952) that looks for the allocations offering the best return-risk pairing, using the correlations between securities. Optimisation tab.

Max drawdown: worst fall suffered, from the peak to the following trough, measured on the base-100 performance curve.

Money market (*monétaire*): asset class of overnight investments, close to the risk-free rate (€STR ETFs, for example).

Monte Carlo method (*méthode de Monte-Carlo*): simulation of thousands of randomly drawn possible futures (5,000 in the software), to study the distribution of outcomes rather than a single forecast.

Normal distribution (*loi normale*): Gauss's "bell-shaped" distribution, symmetric, described by its mean and standard deviation. It is used for the parametric VaR and for the projection, but underestimates crashes.

Securities memory (*mémoire des titres*): file data/base/memoire.csv that retains the matches found online between an ISIN or a name and a ticker, so they can be recognised afterwards without the Internet.

Glossary: N to O

Bond (*obligation*): debt security that pays interest (coupons). In the software, bonds are tracked through listed funds, with their duration when the reference file gives it.

Confidence level (*niveau de confiance*): probability used for the VaR: 90, 95 (default) or 99%. A 95% VaR is exceeded only about one day in twenty.

Effective number of bets (*nombre effectif de paris*): same idea applied to shares of risk: $1 / \Sigma$ (share of risk)^2. It says how many independent holdings the portfolio really represents.

Effective number of holdings (*nombre effectif de lignes*): number of equal-weight holdings that would give the same concentration: $1 / \Sigma$ weight^2. Four holdings of 40, 30, 20 and 10% are worth about 3.3.

Gold (*or*): a separate asset class, tracked through a listed fund. In look-through analysis, it is counted as “no country” and separately from currencies.

n/a (*n.d.*): “not available”. The calculation makes no sense or did not succeed (for example the Cornish-Fisher VaR with too high a kurtosis, or an IRR with no solution).

OCR (optical character recognition, *reconnaissance de caractères*): reading text from an image. The software uses it for scanned PDFs, if an engine (RapidOCR or Tesseract) is installed.

Optimisation: searching, by calculation, for the weights that minimise volatility or maximise the Sharpe ratio, under constraints (positive weights, sum of 100%, maximum weight per security).

Glossary: P

Cost basis (PRU) (*prix de revient unitaire*): average cost of a security held, purchase fees included. It changes with each purchase and stays fixed on a sale.

PBKDF2: method that turns the password into a fingerprint and an encryption key, recomputing it 600,000 times to slow down hacking attempts.

PEA (*plan d’épargne en actions*, French equity savings plan): wrapper reserved for European equities and eligible funds. After 5 years, gains bear only social contributions (18.6% in 2026).

Percentile: value below which a given percentage of the observations lies. The 5th percentile of the scenarios is the unfavourable scenario of the projection.

Proxy: index or fund used in place of a security with no history, for example the index of its region in a stress test.

p-value: probability of observing a result at least as extreme if the hypothesis tested (here, normality) were true. Below 5%, the hypothesis is rejected.

Realised gain (*plus-value réalisée*): gain cashed in on a sale: quantity × (sale price − PRU) − fees.

Risk parity (*parité des risques*): allocation in which each holding contributes the same share of risk. It uses only volatilities and correlations, not expected returns.

Single flat-rate levy (PFU, flat tax) (*prélèvement forfaitaire unique*): taxation of investment income: 12.8% income tax and 18.6% social contributions, i.e. 31.4% in 2026.

Social contributions (*prélèvements sociaux*): the social part of taxation on investment income: 18.6% in 2026 (CTO, PEA), 17.2% for life insurance.

Tangency portfolio (*portefeuille tangent*): maximum-Sharpe portfolio; combined with the risk-free investment, it offers the best return-risk line.

Trading venue (*place de cotation*): stock exchange where a security is listed, shown in the Yahoo ticker by a suffix: .PA for Paris, .DE for Frankfurt, .L for London, none for the United States.

Unrealised gain (*plus-value latente*): gain not yet cashed in on a holding: quantity × (price − PRU).

Weight (*poids*): share of a holding in the portfolio's total value. The **maximum weight per security** limits this share in the optimisation (30% by default).

Glossary: Q to R

Currency risk (*risque de change*): effect of currency movements on the euro value of a foreign security. An American share can rise in dollars and fall in euros.

Daily return (*rendement quotidien*): one day's change, after removing contributions and withdrawals: (value of the day − flow of the day) / value of the previous day − 1.

Diversification ratio (*ratio de diversification*): sum of the holdings' volatilities, weighted by their weight, divided by the portfolio's volatility. It equals 1 with no diversification at all and increases as holdings offset each other.

Information ratio (*ratio d'information*): annualised average return gap with the index, divided by the tracking error. It says whether the gap with the index has been paid for; above 0.5, it is considered good.

Quantile: synonym of percentile, expressed as a fraction. The 5% quantile of daily returns gives the 95% historical VaR.

Rebalancing (*rééquilibrage*): periodic return to target weights, by selling what has risen and buying what has fallen. The backtest tests it each month, quarter or year.

Reference file (*référentiel*): the project's data/referentiel.csv file that describes securities by hand (country, region, sector, asset class, duration). It takes precedence over the local database.

Turnover (*rotation*): share of the portfolio to be moved to go from one allocation to another (sum of weight gaps divided by 2), or, in the backtest, amounts traded per year.

Glossary: S

Overweight, underweight (*surpondération*, *sous-pondération*): weight of a country, sector or region that is higher (or lower) in the portfolio than in the benchmark index.

Scenarios: unfavourable, median and favourable (*scénarios défavorable*, *médian et favorable*): 5th, 50th and 95th percentiles of the value simulated by the projection. There is a one-in-twenty chance of doing worse than the unfavourable scenario.

Sector (*secteur*): a company's field of activity (technology, healthcare, finance, etc.), according to the translated GICS classification.

Semi-deviation (*semi-déviation*): measure of dispersion that keeps only the days when the return is below the risk-free rate, annualised by √252.

Sharpe ratio (*ratio de Sharpe*): annual excess return (above the risk-free rate) divided by the volatility. It measures whether the risk taken was well rewarded: above 1, very good.

Sortino ratio (*ratio de Sortino*): variant of the Sharpe that replaces volatility with the semi-deviation, so as to penalise only falls.

Streamlit: Python library that displays the dashboard in a browser window.

Stress test: estimate of what the current portfolio would lose if a past crisis happened again (2008, Covid, etc.) or if a hypothetical shock occurred (fall in equities, in the dollar, rise in rates).

Glossary: T to U

Exchange rate (*taux de change*): price of a currency in euros. The software uses Yahoo Finance rates, expressed as the number of currency units per 1 euro (EURUSD=X = dollars per €1).

IRR (*TRI*, internal rate of return): annual return on the money actually invested, which takes into account the amount and date of each contribution. It depends on the timing of your payments.

Risk-free rate (*taux sans risque*): return on a risk-free investment in euros, used for the Sharpe, the Sortino and alpha. 2.50% by default (ECB deposit facility rate), constant over the period, editable in **Settings**.

Ticker: short code for a security at Yahoo Finance, with its trading venue: MC.PA for LVMH in Paris, AAPL for Apple.

Tracking error: annualised volatility of the daily return gap between the portfolio and the index. Low: the portfolio sticks closely to the index.

Transaction (*opération*): purchase, sale or dividend recorded in the portfolio, with its date, security, quantity, price and fees.

TWR (*time-weighted return*): product of daily returns neutralised for contributions. It measures the quality of choices, independently of contributions; it is the fund managers' measure.

UCITS (5/10/40 rule): European fund rule: a holding at most 10% of the fund, and holdings above 5% at most 40% in total. The software uses it as a concentration benchmark.

Glossary: V to Y

Historical VaR (*VaR historique*): VaR read directly from past returns (quantile), with no assumption about the distribution.

Minimum-variance portfolio (*portefeuille de variance minimale*): the least volatile possible allocation of the current securities, under the optimisation constraints.

Normal VaR (parametric) (*VaR loi normale*): VaR calculated by assuming that returns follow a normal distribution with the same mean and the same volatility. It often underestimates crashes.

VaR (*Value at Risk*): loss on a bad day at a given confidence level. A 95% VaR of €2,000 means the loss exceeds this amount only about 5% of days.

Variance: square of the standard deviation; measure of dispersion used in Markowitz calculations.

Volatility (*volatilité*): annualised standard deviation of daily returns (× √252). It is the most common measure of risk: 15% means a year's return typically deviates by 15 points from its mean.

Yahoo Finance: free stock market information service from which prices, exchange rates and the recognition of unknown securities come.

yfinance: Python library used by the software to query Yahoo Finance.

13. Frequently asked questions

This chapter gives short answers to the questions most often asked about Portfolio Tracker, including those from a jury or a teacher during a thesis defence: reliability, sources, confidentiality, methodological choices and limitations. Each answer refers, where needed, to the chapter that covers the subject in detail.

Is the software reliable?

Yes for its **calculations**, with reservations about its **data**.

What is solid

- **The formulas** are those of the financial literature (TWR, IRR, Sharpe, VaR, Markowitz, Brinson-Fachler…) and are described exactly in the "All the formulas" chapter.
- **The calculations are tested** automatically on examples whose result is known (see the fiche "How do you know the formulas are right?").
- **The checks** stop the analysis rather than display a wrong figure: sale of securities not held, missing price or exchange rate, unknown transaction type.
- **The source** of prices and the date of the data are always displayed.

The reservations

- Prices come from **a single source**, Yahoo Finance, with no cross-checking.
- The result depends on **your transactions**: a forgotten dividend or an unentered stock split distorts the analysis.
- Some data are **fixed and dated** (ETF composition, risk-free rate, classification of securities).

In short

It is a reliable educational tool for analysing and understanding a portfolio. It replaces neither the official statements of your institution nor investment advice.

How do you check that prices are right?

What the software does

1. It uses the **raw closing price** (not adjusted for dividends): anyone can compare it with that of the exchange or their broker.
2. **At import**, with Internet access, each purchase and sale price in your file is compared with the Yahoo Finance closing price of the same day. A gap of more than 25% is flagged. This cross-check reveals an entry error just as well as a wrong security or an abnormal price.
3. A ticker without a listing venue is only kept if its price matches yours (median gap below 15%).

4. A missing price, history or exchange rate **stops** the analysis, with a message naming the security.

What it does not do

It does not compare prices with a second source and does not look for outliers in the history. A suspended security keeps its last known price.

What you can say

“The prices are the daily closing prices from Yahoo Finance, not adjusted for dividends; transaction prices were checked against these prices, with an alert threshold of 25%; prices were not cross-checked with a second source.” For an important figure, check it on the exchange’s website or your broker’s.

Why Yahoo Finance and not Bloomberg?

The reasons for the choice

- **Free and no subscription**: the software is an educational tool, usable by any student; a Bloomberg terminal or Refinitiv access is paid and reserved for subscribers.
- **No account or access key**: the software queries Yahoo Finance through the Python yfinance library, with no login.
- **Wide coverage**: stocks and ETFs from the world’s main exchanges, indices and exchange rates, with a daily history of several years.
- **Security recognition**: its search engine makes it possible to find a ticker from an ISIN or a name.

The trade-offs

Yahoo Finance does not guarantee the quality of its data and may be temporarily unavailable. That is why the software keeps a cache and a local database, and always displays the source used.

Changing source

The software offers no setting for this. In the code, all access to prices is isolated in a single file (src/market_data.py): changing provider would only require modifying that file.

Is my data confidential?

With the installed version, yes.

Everything stays on your computer

- The dashboard runs at the localhost address: it accepts no connection from any other device.
- Your files are read and processed on your computer.

- Portfolios saved in your space are **encrypted** with a key derived from your password (PBKDF2 and Fernet): neither other users nor an administrator can read them.

What goes out over the Internet

Only the **security codes** (tickers) to obtain prices and, to recognise an unknown security at import, its ISIN or name. Quantities, amounts and the portfolio composition are never sent.

The exceptions

- **The online version** runs on a remote server: your files are processed there. Prefer the installed version for real data.
- A few shared local files are not encrypted (price cache, security memory, log of unanswered questions): they contain no quantity or amount, but reveal which securities have been analysed.

The details are in the chapter “Your account and the security of your data”.

Can the software’s creator see my portfolio?

No. With the installed version, the creator receives nothing:

- there is **no server** for the software: your account exists only on your computer;
- Streamlit’s usage statistics collection is **disabled**, in the project configuration and at launch;
- questions put to the assistant stay in a local file;
- the software does not even check whether updates exist.

Even with your computer in someone’s hands

Your saved portfolios are encrypted with your password. Without it, nobody, creator included, can read them or reset that password.

Your teacher

They only see your portfolios if you yourself send them a file (CSV export, PDF report) or show them your screen.

On the online version

The software runs there on a server administered by the person who published the site: files you send there are processed on that server.

Does the software work without Internet?

Yes, largely. The software is delivered with a local database of several thousand securities, indices and exchange rates, with their daily prices (since 2015 for securities, 2007 for indices and currencies).

What works offline

Opening the software, signing in, analysing a portfolio whose securities are in the database or the cache, importing a CSV or Excel file, all the analyses, the PDF report, the assistant and the manual.

What requires Internet

Today's prices, a security never seen before, recognition of an unknown ISIN, the **Refresh prices** button, and the world map if its background has never been saved.

How to tell

The banner displays "Cached prices (offline)" with an orange dot, and the sidebar shows the date of the latest known prices. The calculations remain correct, but stop at that date.

Tip for a defence

The day before, analyse your portfolio with Internet: its prices will be added to the database and the cache, and on the day, the software will work even without a network.

What are the limitations of the normal distribution in the software?

Where the software uses the normal distribution

- the **normal-distribution VaR** (parametric);
- the Monte Carlo projection, **"Normal distribution"** method (geometric Brownian motion);
- the comparison curve on the returns histogram.

Its main limitation

Real returns have **fat tails** and often **negative skewness**: crashes are more frequent than the normal distribution predicts. It therefore underestimates extreme losses, especially at 99%.

What the software offers to go beyond it

1. **It measures the gap**: skewness, excess kurtosis, share of days beyond 3 standard deviations (against 0.27% expected) and the Jarque-Bera test, in the Risk tab.
2. **It compares several VaRs**: historical (no distribution assumption), normal distribution, Cornish-Fisher (corrected for skewness and kurtosis), and the CVaR.
3. **It offers a "Historical (bootstrap)" projection**, which draws real days from the portfolio and keeps its actual crashes.
4. **It complements this with stress tests** that replay real crises.

Key point

The normal distribution remains a simple, standard benchmark; the software always presents it next to measures that do not assume it.

TWR or IRR: which one should I look at?

Both are correct: they answer two different questions.

	TWR	IRR
Question	Were the security choices good?	How much has my money earned?
Effect of contributions	neutralised	taken into account
Comparable to an index	yes	no
Used by	fund managers (GIPS standard)	the investor, for their own money

Example

€10,000 gains 20% in the first year; you then add €50,000, and the portfolio then loses 10%. The TWR is +8% over two years (+3.9% a year): the security choices were good on average. The IRR is −6.05% a year: the fall hit a much larger sum than the rise, and you lost money.

Which one to look at

- To **judge the management** or compare it with an index: the TWR.
- To know **what your money has really earned**: the IRR.
- An IRR above the TWR means your contributions were well timed.

The exact formulas are in the “All the formulas” chapter.

How good is Markowitz optimisation really?

It is an **excellent teaching tool** and a poor autopilot.

What it shows well

- the value of diversification: combining weakly correlated securities reduces risk;
- the efficient frontier: no randomly drawn portfolio does better, for equal risk;
- the gap between your allocation and a more efficient allocation, with the amounts to buy or sell.

Its limitations, known and acknowledged

- **Expected returns are past averages**: the optimiser over-exploits the securities that have done best, with no guarantee for the future. A small error on these returns changes the weights considerably.
- **Concentration**: with no limit, it often puts almost everything in 2 or 3 securities. Hence the **Maximum weight per security**, 30% by default.
- **No fees, no taxation**: adjustments are calculated excluding fees and without the tax on a sale.
- **A single period**: volatilities and correlations are assumed to be stable.

What the software offers alongside

Risk parity and **minimum variance**, in the **Risk budget** tab, do not use expected returns, the main source of error.

The screen reminds you: “Academic exercise, not investment advice.”

Is this investment advice?

No. The footer of every screen says so: “Educational tool developed as part of the Master G2C — not investment advice.”

Why

- The software knows neither your situation, nor your objectives, nor your horizon, nor your risk tolerance.
- Its calculations rely on the past, which is no guide to the future.
- The diagnostic **suggestions** (Exposures tab) derive from simple rules with documented thresholds; the screen states: “Educational analysis based on simple rules and past data: it is not investment advice.”
- Optimisation is presented as an “academic exercise”.
- Taxation is simplified.

Proper use

Understanding, measuring and discussing a portfolio: it is a support for learning and dialogue. An investment decision is up to you, or to an authorised professional who knows your situation.

Can I manage several accounts and several portfolios?

Several accounts

Yes: each person using the computer creates their own account, with their own password. Everyone sees only their own portfolios.

Several portfolios per account

Yes: save as many portfolios as you like (for example a PEA (French equity savings plan) and a securities account). They appear in the list in the “Data” section, preceded by “My space”. The **My account** page lets you rename, download or delete them.

A consolidated view?

The software analyses **one portfolio at a time**: there is no view that automatically adds several portfolios together. For a global analysis:

1. download one of the portfolios from **My account** (CSV file);
2. open the other one, click **Add transactions** and upload that file;
3. check the table, then **Save transactions**.

Identical transactions are recognised as duplicates. To keep the two portfolios separate as well, work on a copy: first add the downloaded file as a new portfolio in **My account**, then add the other one's transactions to it.

For an adviser

One portfolio per client works the same way, but the software has no client record and no regulatory profile.

Can the software connect to my bank?

No. The software connects to no bank and no broker, and places no orders. This is also a confidentiality guarantee: it never asks for your banking credentials.

How to get your transactions into it

1. **An export** from your bank or broker, as CSV or Excel: upload it with **Upload a file (CSV, Excel, PDF or image)**; the automatic import recognises most formats, otherwise the import wizard guides you.
2. **A PDF statement or trade confirmation** (several at once if you wish, even password-protected): it is read automatically, whatever the broker (text PDF, or image if character recognition is installed); if it is not recognised, the **Complete the transaction** form suggests the values found in the document.
3. **Manual entry**: **Add transactions**, **Manual entry** tab.

Updating afterwards

No need to send everything again: with **Add transactions**, upload only the new movements. Transactions already present are recognised and are not added twice.

Are bonds, gold and money-market funds supported?

Yes, if they are listed and have a Yahoo Finance ticker, which is the case for bond, gold or money-market ETFs.

How they are treated

- **Asset class**: Equities, Bonds, Gold or Money market, from the project's reference database, otherwise the local database. A security of unknown class is counted as an equity.
- **Bonds**: their **duration** (if the reference database gives it) is used for interest-rate sensitivity and the rate shock.
- **Gold**: counted separately, "without country", in the look-through analysis.
- **Suitable benchmark indices**: euro-zone government bonds, corporate bonds, €STR money market, or mixed 20/80, 60/40 and 80/20.
- **Coupons**: to be recorded as dividends (DIVIDENDE type).

The sample portfolio "Diversified portfolio (multi-asset)" illustrates these classes.

What is not supported

Any product without a daily price on Yahoo Finance: life insurance euro funds, savings accounts, directly held bonds with no quotation, structured products. They cannot be valued.

Note

Performance attribution covers only the **equity portion**; bonds and gold are excluded from it.

How are currencies handled?

Everything is expressed **in euros**.

On entry

- The **price** is entered in the security's **quote currency** (dollars for Apple, pence for a London stock).
- **Fees** are always in euros.
- At import, the software compares each price with that day's price and recognises a price already converted into euros, in currency or in pence; it brings it back to the quote unit.

The conversion

price in euros = price in currency × factor / EURcurrency rate (factor 0.01 for pence), at the rate of the day of each transaction for purchases, sales and dividends, at each day's rate for the history, at the last known rate for the current value. Rates come from Yahoo Finance and are shown at the bottom of the sidebar.

Currency risk

The performance of a foreign security includes the change in its currency. The Exposures tab measures the **real exposure** to currencies, ETFs included: a world ETF quoted in euros remains exposed to the dollar, unless it is hedged ("EUR Hedged"). The stress tests include a 10% fall in the dollar.

How are fees taken into account?

Brokerage fees

Entered in euros in the frais column of each transaction, they count everywhere:

- **on a purchase**, they increase the PRU (unit cost basis);
- **on a sale**, they reduce the realised gain;
- **on a dividend**, they are deducted from the amount received;
- **in performance**, they are part of the flows: the return on the purchase day includes them.

Their total appears in the "Brokerage fees" card of the Overview.

ETF management fees

They are taken within the fund and are therefore **already included in its price**: there is nothing to enter.

Custody fees and other account fees

These are not securities transactions: at import, custody or transfer fee lines are ignored. They are therefore not deducted from performance.

In the advanced analyses

- **Optimisation**: the amounts to buy or sell are calculated excluding fees.
- **Backtest**: adjustable transaction fees (0.10% by default) apply to each purchase and rebalancing.
- **Taxation**: life insurance contract management fees are ignored.

How are dividends taken into account?

They count, provided they are entered

The software does **not** download dividends: only those in your transactions count. A dividend line is written with the type DIVIDENDE, a quantity of 0 and the **total amount received** in the price column (in the quote currency), with any fees separate.

Their effect

- they add to the **total gain** (net of fees);
- they count in **performance** (TWR, IRR) as money taken out;
- they are shown per holding in the Holdings tab and in total in the “Dividends and coupons” card.

Why this is essential

The software uses the **unadjusted** price: on the ex-dividend date, the price falls by roughly the amount of the dividend. If the dividend is not entered, this fall appears as a loss and performance is underestimated.

ETFs

- **Accumulating**: dividends are reinvested in the price; nothing to enter.
- **Distributing**: to be entered as for a stock.

Coupons from a distributing bond fund are entered in the same way.

Can I get a report to hand in or print?

Yes. At the bottom of the sidebar, click **PDF report**, then **Download the PDF report**.

Its content

An A4 document that covers the whole analysis: summary and key figures, holdings, exposures and diversification, performance, risk, optimisation, projection, wealth advisory (taxation, stress tests), asset management (attribution, risk budget, backtest) and a **methodology** section that defines each indicator and its limitations.

Its particulars

- It is always **in French** and on a light background, even if the screen is in English or in night mode.
- It uses the portfolio, benchmark index, risk-free rate and VaR level chosen, but **fixed** settings for the optimisation (maximum weight of 30%) and the projection (10 years, 5,000 scenarios, normal distribution, no contributions).
- Each page reminds you that it is an educational tool, not investment advice.

Other exports

- the transaction history as CSV, in the Transactions tab;
- each portfolio in your space as CSV, from **My account**.

Does the software run on Mac?

Yes, on recent Macs.

Computer	Solution
Apple Silicon Mac (M1, M2, M3, M4…), macOS 12 or later	Portfolio_Tracker_Mac.dmg application
Intel Mac (sold before the end of 2020)	online version of the dashboard
Windows 10 or 11, 64-bit	Installer_Portfolio_Tracker.exe installer
Linux	launch from the source code (Python 3.11 or later)

On Mac, good to know

- On first launch, macOS blocks the application, which does not come from the App Store: you have to allow it once (see the “Getting started” chapter).
- A Terminal window opens: leave it open, closing it stops the software.
- The dashboard opens in Chrome, Edge or Brave in “application” mode if they are installed, otherwise in Safari.
- Your accounts are stored in ~/Library/Application Support/Portfolio Tracker and kept during updates.

The screens, calculations and report are identical on Windows and on Mac.

How do I know whether an update exists?

The software **does not itself check** whether a new version exists (that is also why it sends nothing to its creator).

Where to look

On the project's releases page: https://github.com/DieuUssop/Python/releases/latest. Each version carries a number made up of its build date (for example "2026.10.05").

Installing the new version

Close the software, then install the new version **over** the old one, without uninstalling. Your accounts and saved portfolios are kept. As a precaution, still download a copy of your portfolios from **My account**.

Prices, for their part, update themselves

Each analysis with Internet downloads recent prices and adds them to the local database. A new version also brings a more recent securities database and, where relevant, revised reference data (risk-free rate, ETF composition).

What are the software's known limitations?

Data

- a single price source (Yahoo Finance), with no cross-checking or outlier detection;
- dividends not downloaded: only those entered count;
- stock splits and reverse splits adjusted only from the broker's notice; mergers and other corporate actions not handled;
- ETF composition, classification of securities and risk-free rate fixed and dated;
- country of a stock in the local database deduced from its listing venue.

Method

- risk-free rate constant over the whole period;
- normal distribution for the parametric VaR and one of the projection methods;
- Markowitz based on past returns;
- stress tests without currency effect and on prices excluding dividends;
- simplified taxation (no progressive scale, no contract fees);
- attribution limited to the equity portion, by major regions.

Operation

- only three transaction types: buy, sell, dividend; no short selling;
- no bank connection;
- one portfolio analysed at a time;
- forgotten password = lost portfolios;

- no Intel Mac; temporary accounts on the online version;
- no automatic update.

The project itself cites, as a possible improvement, the use of the historical €STR series for the risk-free rate.

What should I do if the assistant cannot find my answer?

The assistant searches this manual for the entry closest to your question and displays it as written. It is not an artificial intelligence writing text: it cannot invent anything, but it may fail to find an answer.

Try in this order

1. **Rephrase** with other words, simpler ("remove a purchase") or more technical ("delete a transaction").
2. **One question at a time**, short.
3. **The exact name** of an indicator or a button: "VaR", "PRU", "Refresh prices".
4. **The related entries** suggested under the answer.
5. **The contents**: choose a chapter and open its entries.
6. **This chapter**, the "Error messages and problems" chapter and the **glossary**.

Getting the manual completed

Your unanswered question is recorded on this computer. In the "Manual and help" area, the section for unanswered questions lets you export them with **Export the questions (CSV)** and send them to the software's creator. Do not type any personal information there: these questions are visible to all users of the computer.

Why do my figures differ from my bank's?

Several causes, often combined.

Difference	Explanation
Value	Yahoo Finance closing price, not real-time price; another listing venue is possible
Foreign securities	Yahoo Finance exchange rate, not your bank's
PRU	the software includes purchase fees; some banks do not
Capital gains	depend on the PRU, and therefore on fees and exchange rates
Total gain	includes the dividends entered; without them, it is lower
Performance	the software calculates TWR and IRR; your

Difference	Explanation
	bank may show another measure (gain in %, performance since 1 January…)
History	a missing transaction or an unentered stock split changes everything

How to check

1. First compare the **quantities** held (Holdings tab).
2. Then the **prices** and the "Data as of" date in the banner.
3. Then the **transactions** (Transactions tab), in particular dividends and old purchases.

For official amounts, especially tax ones, your institution's documents are what count.

Can I use the software for my tax return?

No. The figures are reliable for analysis and comparison, not for a tax return.

Why

- The software calculates the tax on a **hypothetical full sale**, to compare CTO (securities account), PEA (equity savings plan) and assurance-vie (life insurance), with 2026 rates (flat tax or PFU of 31.4%, PEA after 5 years, life insurance after 8 years).
- It **simplifies**: option for the progressive scale ignored, life insurance premiums assumed below €150,000, contract fees ignored, dividends assumed to be kept within the wrapper.
- Its prices and exchange rates come from Yahoo Finance, not from your institution.
- It only knows the transactions you give it.

What counts

The tax documents and statements sent by your bank or broker.

What the Taxation tab is for

Understanding the effect of the wrapper and the holding period: it shows the net gain according to the year of exit, with the 5-year (PEA) and 8-year (life insurance) thresholds, and the share of the portfolio eligible for a PEA.

How do you know the formulas are right?

Automatic tests

The project contains a `tests` folder with about 150 automatic tests, run by the developer with the command `python -m pytest`. They check the calculations on examples whose result is known: PRU, capital gains, TWR, IRR, volatility, VaR, currencies, optimisation, simulation, file import, encrypted accounts…

Cross-checks

Some mathematical properties must always be true; the tests check them:

- the gain calculated day by day (value – net contributions) lands exactly, on the last day, on the total gain calculated line by line;
- the sum of the risk contributions equals the portfolio volatility (Euler property);
- the sum of the attribution effects equals the performance gap with the index (Cariño smoothing).

Transparent formulas

Each formula is written and commented in the code (src folder) and reproduced in the "All the formulas" chapter, with a worked example that anyone can redo with a calculator or in a spreadsheet.

What the tests do not cover

They do not check Yahoo Finance's data themselves, nor your transactions.

What is the software programmed with?

The software is written in **Python** (version 3.11 or later from the source code).

The libraries

Role	Library
Dashboard	Streamlit
Calculations	pandas, NumPy, SciPy (optimisation)
Interactive charts	Plotly
Report charts, PDF report	Matplotlib, ReportLab
Stock prices	yfinance (Yahoo Finance)
Account encryption	cryptography (PBKDF2, Fernet)
File reading	openpyxl (Excel), pdfplumber (PDF), RapidOCR (image PDF, optional)
Tests	pytest

The organisation

The **calculations** are in the src folder (one file per subject: metrics.py, portfolio.py, optimisation.py…), separate from the **display** (app.py and the vues_*.py files). They can thus be tested without launching the interface.

Distribution

The Windows (.exe) and Mac (.dmg) installers are built automatically by GitHub Actions. They embed their own Python and all the libraries: nothing else to install.

Does the software use artificial intelligence?

No, no generative artificial intelligence.

The manual assistant

It writes nothing. It compares the words of your question with those of the manual entries (title, common phrasings, synonyms, text), tolerating typos and missing accents, then displays the closest entry **as it was written**. It therefore cannot invent anything. If it finds nothing, it says so. It works offline and sends nothing.

The other automatic features

- **Column recognition** in an imported file relies on rules: known column names, cell contents, consistency of the figures.
- **Character recognition** for image PDFs uses a specialised engine (RapidOCR or Tesseract), which reads letters from an image.
- **The diagnostic** in the Exposures tab applies simple rules with documented thresholds.

All figures come from explicit formulas, described in the “All the formulas” chapter.

Is the software available in English?

Yes for the screen. At the top of the sidebar, the **FR | EN** selector switches the dashboard to English: menus, buttons, indicators, charts and most messages.

What stays in French

- **The PDF report**, written in French whatever the screen language.
- **The manual**, as long as its English translation is not provided: the software then displays the French version.
- The names of your securities and the content of your files, as they are.

The preference

The language is kept during the session. The “The screen and navigation” chapter details this setting.

Are stock splits and corporate actions handled?

Partly. The software only knows three transaction types: buy, sell and dividend, and it does not detect a split by itself. But it can adjust your transactions from the split or reverse split notice sent by your broker.

The problem

After a stock split, Yahoo Finance retroactively corrects its prices. Your file, however, still shows the old number of securities: the calculated value becomes wrong. Example: 10 shares bought

at €800, then a 10-for-1 split; Yahoo shows about €80 for the purchase date, and the software would value 10 × €80 instead of 100 × €80.

The signal

At import, the price check flags the gap (“check the ticker, currency or stock split”). A sudden loss of about 50, 67 or 90% should also raise an alert.

The fix

Express the transaction in securities as they are after the split, **without changing the amount**: here, 100 shares at €80. The simplest way: drop the split notice (PDF) on the **Add transactions** page, then click **Apply to earlier transactions** (see the chapter on import, fiche “Stock split or reverse split”). Otherwise, do it by hand in the Transactions tab with **Edit transactions**.

Other corporate actions

Mergers, spin-offs, ticker changes or bonus share allocations: to be translated by yourself into purchases and sales.

Why is the MSCI World the default benchmark index?

The reasons

- **A global market**: most diversified equity portfolios are compared with global equities.
- **Reinvested dividends**: it is represented by the Amundi MSCI World ETF (CW8), which is **accumulating**. As your performance includes your dividends, the comparison is fair.

Why not the CAC 40?

The CAC 40 offered is the price index, **excluding dividends**: it puts the index at a disadvantage, by about 3% a year according to the project. It is only relevant for a portfolio of French equities, keeping this bias in mind.

Choosing the right index

In **Settings**, **Benchmark index** offers equity, bond, money-market and mixed indices (20/80, 60/40, 80/20). Choose the one that resembles your allocation: a balanced portfolio compares better with a 60/40 index than with the MSCI World. An unsuitable index distorts beta, alpha and tracking error; against a bond or money-market index, the Performance tab warns that beta and alpha have little meaning.

Is the Monte Carlo projection a forecast?

No. The screen reminds you: “A projection is not a forecast: it assumes the assumptions hold, which is never guaranteed.”

What it does

It simulates 5,000 possible futures, all consistent with a chosen return and volatility (by default, those of the portfolio's past), and shows their distribution: adverse, median and favourable scenarios, probability of loss, probability of reaching a target.

How to read it

- The **median** is not a forecast: it is the middle of the possibilities.
- The gap between scenarios **grows with the horizon**: uncertainty accumulates.
- The mean exceeds the median (log-normal distribution): the median is the most representative marker.

Its assumptions

- **constant** return and volatility over the whole horizon;
- "Normal distribution" method: crashes underestimated; the "Historical (bootstrap)" method keeps the portfolio's real crashes;
- historical return often flattering: reducing the **Assumed annual return (%)** gives a more cautious projection.

Why several VaRs, and which one should I use?

Each method relies on a different assumption; comparing them is part of the analysis.

Measure	Assumption	Strength	Limitation
Historical VaR	the past repeats itself	no distribution assumed	blind to crises absent from the history
Normal-distribution VaR	normal returns	simple, standard	underestimates crashes
Cornish-Fisher VaR	corrected normal distribution	takes skewness and kurtosis into account	"n/a" if they are too large
CVaR	the past repeats itself	measures the severity beyond the threshold	relies on few days

Which one to use

- **The historical VaR** is the software's main measure: it is the one on the euro card of the Risk tab and in the report.
- **The CVaR** complements it: it says what a day beyond the threshold costs.
- **The gap between historical VaR and normal VaR** is information in itself: the automatic reading comments on it.

Why over one day?

The software's VaRs cover the loss over **one trading day**, calculated on daily returns. For longer horizons, see the max drawdown, the stress tests and the projection.